POWER ELECTRONICS

PE TRANSFORMER

PE Power Grids • PE Grid Components • Power Electronics US

ENTERPRISE ERP — UNIFIED BUILD SPECIFICATION

Order Management • Engineering & BOM • CAD/PDM Linkage • Material Management • MRP & Finite Planning

Production Management • Quality Management • Barcoding & Traceability • RMA & Warranty • Procurement

Controlling (Cost & Profit Centers) • Costing & Pricing • Workflow Automation • Cross-Module Linkages

Integration & Security Architecture • Resilience & Operations • Test, Migration, Cutover & Acceptance

Nine-book consolidated build specification. Functional cores, linkage contracts, enterprise architecture, design governance, visual atlas, screen implementation plan, go-live readiness and the design execution handbook

Legal Entities	PE Power Grids • PE Grid Components • PE Transformer • Power Electronics
Benchmark Standard	SAP S/4HANA (SD, PP, QM, MM, CO) • Oracle Cloud SCM • NetSuite • Microsoft Dynamics 365
Companion Volumes	Inventory & Warehouse Module Spec V3.0 (Mar 2026) • Accounting Module Implementation Spec V2.0 (Mar 2026)
Document Version	1.0
Date	July 2026
Classification	CONFIDENTIAL — Engineering Use Only

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How This Document Is Organized

This edition consolidates the controlled build package into eight Books. Content is unchanged from the released versions except for this front matter and the Workflow Automation section; where a Book's text says "this volume" or "companion volume", read it as the corresponding Book of this document. Cross-references of the form MFG/MM followed by a Part or section number resolve inside Books I and II respectively; INV and ACC remain external released specifications.

Book	Content	Role
Front Matter	Unified revision history, module index, Workflow Automation specification (Section W)	Orientation and shared services
Book I — Manufacturing Core	Parts A–K: inventory gap closure (RMA, Sales Tab, Packing, Inspection Plans, Count Documents, Cost Variance Analysis, Fiscal Periods, Safety Stock, Tax), Order Management, Engineering & BOM & CAD, Production Management & MRP, Quality Management, Barcoding & Traceability, Procurement, Cost & Profit Centers, Costing & Pricing, GL integration map, data model & roadmap	Field level functional authority
Book II — Material Management	Parts A–H: material model & Material Master views, creation & change governance, valuation & automatic account determination, cross-module linkage contracts, special stock & ownership, batch & serial profiles, mass maintenance & migration & data quality, MM data model	Field level functional authority
Book III — Enterprise Completeness	Parts A–H: cross-volume flow closure book (eight end-to-end flows), integration & transaction architecture, security & IAM, resilience & operations, finite scheduling & capable-to-promise, multi-plant & intercompany & global model, CAD/PDM/PLM integration, test & migration & cutover & acceptance	Enterprise architecture authority
Book IV — Design & Development Completion	Controlling baseline and precedence, scope decision register DEC-01…10, conflict register, consolidated flows FL-09/10/11, executable pricing condition model, production versions and planning algorithm baseline, module completion additions, architecture decision records, design and development gates, authoring templates, action register	Governance and delivery
Book V — Designer's Guide & Visual Atlas	SAP S/4HANA capability map and plain-language glossary, structure and flow diagrams, master data atlas, module design briefs with screen inventories, barcoding hardware and label specification, CAD linkage visuals, design system starter	Governance and delivery
Book VI — Screen by Screen Implementation Plan	Seventy eight implementation entries across fifteen module sections with routes, layouts, actions, arrivals, outputs, postings, scans, build waves and field authorities. Seven wave build sequence. Consolidated Costing and Pricing section. Linkage spine table	Governance and delivery
Book VII — Go-Live Readiness Supplement	Gap register closure crosswalk, supplemental screen entries, multi plant planning setup screens and network behaviors, controlled build package manifest, data migration package governance, complete barcode sample book with usage matrix, go live readiness statement and execution directive	Governance and delivery
Book VIII — Design Execution Handbook	Designer operating manual: the eight screen archetypes with wireframes, the lead to delivery business flow sheet, BPMN swimlanes and sequence diagrams for the five governing flows, consolidated inventory, warehousing and accounting detail,	Designer and program operating manual

Book	Content	Role
	target state capability specifications, the reconciliation control register and the implementation package	
Book IX — Target State Module Specifications	Module grade specifications for the nine phased capabilities. Periodic actual costing and the material ledger, the advanced scheduling optimizer, full global multi plant ATP and CTP, supplier and customer portals, full EDI, direct machine integration with OEE and downtime, full field service management, plant maintenance and EAM, broader international localization. Objects, screens, calculations, postings and activation evidence per module	Target state module authority

Module Index — Where Each Capability Lives

Capability	Primary Specification	Linkages Specified In
Order Management (MTS, MTO, ETO, service, replacement)	Book I Part B	Flow closure FL-01/FL-02 (Book III Part A); pricing linkage Book I Part A §4; material Sales view Book II §3.12
Engineering & BOM (EBOM, MBOM, ECR/ECO, placeholders)	Book I Part C	Material linkage Book II §6; MCR placeholder path Book II §4.3; change flow Book III Part G §4
CAD linkage (documents, transmittals, PDM connector)	Book I Part C §5; Book III Part G	Ownership rules Book III Part G §2; interface IF-17 Book III Part B §3
Workflow automation (approvals, notifications, jobs)	Front Matter Section W (below)	Workflow catalog W.4 maps every governed action across all Books
Material Management (types, views, governance, valuation)	Book II	Linkage contracts Book II Part C to every module
MRP planning (netting, exceptions, conversion)	Book I Part D §3; planning fields Book II §3.6	Finite scheduling & CTP Book III Part E; sources Book II §7.5
Production Management (WCs, routings, WOs, floor flow, terminal)	Book I Part D	Flow FL-04; postings Book I Part J; constraint model Book III Part E §2
Quality Management (gates, plans, tests, NCR/CAPA/SCAR, calibration)	Book I Part E; plans Part A §6	Material Quality view Book II §3.10; recall FL-08 Book III Part A
Barcoding & Traceability (21 scan points, genealogy)	Book I Part F	Batch/serial profiles Book II Part E; recall population Book III FL-08
RMA & Warranty	Book I Part A §3	Flow FL-05; postings #17–24; rework Book I Part D §9
Procurement (vendors, AVL, PR, RFQ, PO, subcontract, expediting)	Book I Part G	Info records & source lists Book II §7; flow FL-03
Controlling (cost centers, profit centers, allocations)	Book I Part H	Derivation rules consumed by every posting; budgets ACC 21
Costing & Pricing (standards, rates, estimates, price lists, surcharges, margins)	Book I Part I	Valuation & account determination Book II Part B; CVA Book I Part A §8
Inventory gap items (Packing, Counts, Safety Stock, Fiscal Periods, Tax, CVA, Inspection Plans, Sales Tab)	Book I Part A	Periods matrix governs all postings; tax feeds all documents

Capability	Primary Specification	Linkages Specified In
Integration, security, resilience, scheduling, global, delivery	Book III	Applies across every module above
Design and development completion (decisions, consolidated flows FL-09/10/11, pricing conditions, production versions, ADRs, gates, action register)	Book IV	Precedence over Books I–III on conflicts (Book IV §1.1)
Designer's guide: diagrams, SAP capability map, screen inventories, device and label specs, design system	Book V	The visual atlas and vocabulary
Screen by screen implementation plan with per screen linkages and build waves	Book VI	The build order and the epic list
Go live readiness: gap crosswalk, multi plant setup, package manifest, migration workbook, complete label samples	Book VII	The nine gap items resolve here to their Part A sections
Design execution: archetype wireframes, BPMN and sequence diagrams, lead to delivery sheet, warehousing and accounting consolidation, target state specifications, reconciliation register, implementation package	Book VIII	The designer and program operating manual — start here
Target state modules in build ready depth: material ledger, optimizer, global ATP/CTP, portals, EDI, machine integration and OEE, field service, plant maintenance, localization	Book IX	Module grade specifications with activation evidence per capability

W. Workflow Automation

Every governed action named in this document — approvals, task routing, escalations, alerts and scheduled jobs — executes on one shared automation layer. ACC Section 22 built the approval engine for finance documents; this section generalizes it as the platform service all Books use, and catalogs every workflow so none exists as bespoke code. No module may implement its own approval or notification mechanism.

W.1 Approval Workflow Engine

Element	Specification
Definition model	Per document type: trigger (submit, threshold breach, field-change class), condition rules (value bands, return class, margin floor delta, warranty flag, entity — any document field via the rule grammar), approver chain (sequential steps; each step = role, named user, or derived owner such as Cost Center Manager), parallel steps where flagged (CCB votes), and terminal actions (status transition, posting release, document generation)

Element	Specification
Versioning	Workflow definitions are effective-dated and versioned; in-flight instances complete on the version they started; changes require the owning role in W.4 plus Controller when the workflow gates a posting
Delegation and absence	Per-user delegate with date range; delegation logged on every action taken; standing delegation of financial approvals above $250k prohibited (SoD rule)
Escalation and SLA	Per step: SLA hours, reminder cadence, escalation target (role manager) on breach; breaches feed the exception dashboards (e.g. MCR SLA per Book II §4.2, RMA aging per Book I Part A §3.3)
Recall and resubmit	Initiator may recall before first approval; rejection returns with mandatory comments to the responsible step; resubmission preserves history
Acting surface	In-app queue (ACC 22 pattern), email actions with signed links, and mobile-responsive approval view; every action records user, timestamp, channel and comment on the immutable audit trail
Evidence	Approval history renders on the document Activity tab and exports with it; Part H acceptance evidence (Book III) cites these records

W.2 Notification and Alert Engine

Generalizes the alert rules screen (INV 15.11): rule = event or scheduled condition + scope filter + severity + recipients (roles/users) + channels (in-app, email; SMS reserved) + dedupe window. All alerts named across the Books register here: below safety stock, calibration due, unconfirmed POs, placeholder aging, promise-at-risk, DLQ depth, SLO burn, certificate expiry, count SLA, close checklist blockers. Alert fatigue control: per-rule dedupe, digest option for non-critical severities, and a monthly noise review owned by each rule's owner.

W.3 Scheduled Job Framework

All background processing runs on the batch framework specified in Book III Part D §5 (run history, SLA timers, dependency chains, retries, escalation). The standing job catalog: nightly MRP; ADU and ABC/XYZ classification (monthly); batch expiry sweep (daily); vendor rating rollup (daily); economic nexus monitor (daily); allocation cycles, warranty accrual, E&O run (period-end); costing runs (per policy); recalc queue for tax failover (15 min); data-quality KPI computation (daily); access recertification generation (quarterly). Jobs declare their close-criticality so the period-close gate (FL-06) can hold on failures automatically.

W.4 Workflow Catalog (binding register)

Workflow	Trigger / Chain Summary	Specified In	Owner
PO approval ladder	Value bands: Buyer ≤ $25k → Purchasing Mgr ≤ $250k → Controller → CFO > $1M	Book I Part G §1	Purchasing Mgr
PR approval (manual/expense)	Cost-center + budget check routing	Book I Part G §3; ACC 21	Controller
Credit release	Credit hold → Finance queue (release / reject / raise limit)	Book I Part B §2.2	Controller
Price floor breach	Below floor: Pricing Mgr → Sales Director → CFO (floor −5pts)	Book I Part I §6.3	Pricing Mgr
Quotation internal review	Margin below target → estimator + pricing approval	Book I Part B §3.3	Sales Director

Workflow	Trigger / Chain Summary	Specified In	Owner
Sales change order	Post-confirm commercial changes; re-credit-check on increase	Book I Part B §4.5	Sales Director
RMA approval matrix	Class/value/warranty routing: CS Mgr → Quality Mgr → Controller	Book I Part A §3.10	CS Mgr
RMA disposition & warranty adjudication	Inspector proposes → authorized role approves; goodwill → Controller	Book I Part A §3.6	Quality Mgr
ECR / CCB decision	Impact analysis → CCB votes (parallel step); emergency 24 h SLA	Book I Part C §4.1	Engineering Mgr
ECO task set & release	Task checklist; atomic effectivity release	Book I Part C §4.2	Engineering Mgr
EBOM / routing / plan / document release	Draft → In Review → Released with role approvals	Book I Parts C–E	Engineering/Quality Mgr
MCR data-completion	Per-department task chain with 2-day SLAs → Materials Mgr activation	Book II §4.2	Materials Mgr
MCH governed change / mass change	Class 2 field rules; simulation → approval → scheduled apply	Book II §4.4/§16.1	Materials Mgr
Supersession	Predecessor run-out policy approval	Book II §4.6	Materials Mgr
Valuation method / class change	Period-boundary procedure, Controller approval	Book II §5.4	Controller
Account determination matrix release	Mark → release with completeness check, step-up MFA	Book II §5.3	Controller
Count variance approvals	Tolerance bands: Supervisor → Finance → Controller	Book I Part A §7.5	Warehouse Mgr
Costing run mark/release; WC rate release	Two-step release; revaluation posting	Book I Part I §2–3	Controller
Safety stock change > 25%	Planner edit → Planning Mgr approval	Book I Part A §10.1	Planning Mgr
Fiscal period reopen	Controller + CFO dual control, tagged postings	Book I Part A §9.3	CFO
Vendor onboarding	Docs → review → Active (no PO before Active)	Book I Part G §2.1; ACC 22	Purchasing Mgr
SCAR escalation	Vendor response SLA → AVL Restricted on 2 overdue	Book I Part E §6	Quality Mgr
Recall campaign notifications	Dual approval (Quality Mgr + executive) before external send	Book III Part A FL-08	Quality Mgr
Promise-date change / at-risk	At-risk record → recovery decision → customer notification per class	Book III Part E §5	Production Scheduler
De-peg SO stock	Sales Director approval with reason, logged both orders	Book II §14.4	Sales Director
SoD exception grant	CFO approval with expiry	Book III Part C §3	CFO
Access recertification	Quarterly attestation; failed = auto-revoke	Book III Part C §3	ISO

Build Note: *Build rule mirroring the interface catalog: a governed action not present in this register is either added through change control or it does not ship. The register is configuration seed data; each row becomes a workflow definition in the engine with its tests in the TC-WF range.*

BOOK I

MANUFACTURING CORE

Order Management • Engineering & BOM & CAD • Production & MRP • Quality • Barcoding & Traceability • Procurement • RMA • Controlling • Costing & Pricing • Gap-Register Closure

Order management, engineering, production, quality, traceability, procurement, returns, controlling, costing and pricing at field level

Distribution List

Role	Access Level	Entity
Engineering Lead	Full Access	PE Transformer
Development Partner Team	Full Access	PE Transformer
QA Lead	Review & Comment	PE Transformer
Controller / Finance	Review & Comment	PE Power Grids
Project Management	Read Only	PE Transformer

1. Purpose, Scope and Document Set

This volume is the definitive, field-by-field implementation specification for the PE Transformer ERP manufacturing core. It is written to the same standard, structure and depth as the two companion volumes already in the development partner's hands, and the volumes of the PE Transformer ERP Build Package together form the controlled build baseline (Phase 1 functional scope with an architected target-state path; see Enterprise Completeness Spec V1.0):

Volume	Covers	Status
Inventory & Warehouse Module Spec V3.0	Material Master, warehousing, transactions, serial/batch tracking, counting, valuation screens, inventory GL map (Sections 1–15)	Released — build in progress
Accounting Module Implementation Spec V2.0	AR, AP, GL, bank reconciliation, tax engine, intercompany, budgeting, workflows, RBAC, ASC 606, batch payments, period close (Sections 2–26)	Released
This Book — Manufacturing Core	Order Management, Engineering & BOM, Production Management, Quality Management, Barcoding & Traceability, Procurement, RMA, Cost/Profit Centers, Costing & Pricing, and closure of every item in the Post-Inventory Design Review gap register	This document

Every screen in this volume is specified with: exact fields with data types, requirement level and validation rules; every button and its enabled/disabled conditions; status lifecycles with allowed transitions and colors; GL posting rules with the account numbers already established in the CoA (Inventory Spec Screen 11.4 and Accounting Spec Section 13); cost center and profit center derivation for every financially relevant posting; barcode scan points; role access; error and exception handling; and integration callouts to the other two volumes. Where a screen from a companion volume is referenced, the reference is given as "INV" (Inventory Spec V3) or "ACC" (Accounting Spec V2) plus the section number.

> Dev Note: Nothing in this volume changes INV or ACC. Where this volume assigns new behavior to an existing screen (for example, the RMA receipt reusing the Check-In engine), the change is specified here as a delta and flagged "DELTA to INV x.x". Build against all three volumes together.

1.1 Gap Register Coverage Matrix

The Post-Inventory Design Review (Jun 2026) identified nine navigation items that lacked functional specifications. Each is closed in full in this volume:

#	Gap Register Item	Closed In	Specification Delivered
1	Return from Customer	Part A, Section 3 (RMA)	Complete RMA process: authorization, receipt, inspection, disposition, credit, repair, replacement; screens, fields, statuses, GL, exceptions
2	Sales Tab (Material Master)	Part A, Section 4	Field-level tab spec, price list assignment, customer-specific pricing, inheritance/override engine, downstream document interaction
3	Packing	Part A, Section 5	Handling Units, pack station workflow, packing list & labels, statuses, carrier fields, integration with shipping and invoicing
4	Inspection Plan	Part A, Section 6	Plan master with types, versioning, statuses, characteristics, sampling (ANSI/ASQ Z1.4), approval workflow, no-plan exception policy
5	Count Documents	Part A, Section 7	Document lifecycle, role-based access, blind rules, reconciliation logic, tolerance and approval rules, posting rules

#	Gap Register Item	Closed In	Specification Delivered
6	Cost Variance Analysis	Part A, Section 8	All variance formulas, drill-down chain, report layouts, GL posting rules by variance type, edge cases
7	Fiscal Period Control	Part A, Section 9	Open/close sequence, module-by-module behavior matrix, back-posting policy, roles, standard error messages
8	Safety Stock	Part A, Section 10	Planning-run interaction, formulas, per-warehouse and per-customer overrides, alerts, lead time interplay
9	Tax Configuration	Part A, Section 11	Determination precedence, region handling, rate maintenance workflow, customer/vendor tax profiles, retroactive change policy

1.2 Reading Conventions

- Screen IDs follow the established pattern: module prefix + area + number (OM-SO-001, PP-WO-001, ENG-BOM-001, QM-PLAN-001, BC-SCAN-001, PUR-PO-001, RMA-001, CO-CC-001).
- Field tables use the columns Field / Type / Req / Specification. “Cond.” means conditionally required and the condition is stated in the specification column.
- GL posting tables use the chart of accounts already defined: 1310 Raw Materials, 1320 WIP, 1330 Finished Goods, 1350 In-Transit, 2110 GR/IR, 1200 AR, 2100 AP, 4100 Revenue, 5110/5120 COGS, 5210 PPV, 5220 Material Usage Variance, 5310 Inventory Adjustment, 5320 Scrap & Waste, 5330 Revaluation, 2300 Sales Tax Payable, 2600 Contract Liability, 1250 Contract Asset. New accounts introduced by this volume are listed in Section 2.4.
- Every listing screen inherits the standard table pattern (checkbox column, sortable headers, status badges, saved filters, column config, pagination, bulk actions, export) and every create screen inherits the standard layout (breadcrumb, entity context bar, form body, action buttons top-right) exactly as established in INV Section 2 and ACC Section 2. These global behaviors are not repeated per screen.
- Every screen carries the Entity Selector, immutable Audit Trail and Document Attachment components defined in ACC Section 2. No exceptions.

2. Global Architecture Additions

2.1 Application Navigation (Full System)

The sidebar grows from the current Inventory-centric structure to the full enterprise structure below. Existing entries are unchanged; new top-level modules introduced by this volume are marked NEW.

Top-Level Module	Children	Source
Home / Dashboards	Executive, Inventory (INV 1.1), Finance (ACC 17), Production (Part D), Quality (Part E), Sales (Part B)	Mixed
Sales — NEW	Customers, Quotations, Sales Orders, Deliveries & Shipments, Packing, RMA / Returns, Sales Reports	This volume, Part B (Packing and RMA specified in Part A)
Engineering — NEW	Engineering Items, Engineering BOMs, CAD Documents, Change Requests (ECR), Change Orders (ECO), Where-Used, Engineering Reports	Part C
Production — NEW	Production Dashboard, MRP Workbench, Planned Orders, Production Orders, Dispatch List, Shop Floor Terminal, Work Centers, Routings, Rework Orders, WIP Report, Production Reports	Part D
Quality — NEW (promoted to top level)	Quality Dashboard, Inspection Plans, Incoming Inspection (INV 6.1), In-Process Inspection (INV 15.2), Final Test, Pre-Shipment (INV 15.3), Stock Hold/Release (INV 6.2), NCR (INV 15.4), CAPA, SCAR, Calibration, CoC (INV 15.5)	Part E + INV
Inventory	As built per INV V3 navigation, unchanged	INV
Purchasing — NEW	Vendors, Approved Vendor Lists, Purchase Requisitions, RFQs, Purchase Orders, Expedite Board, Vendor Performance	Part G
Costing & Pricing — NEW	Cost Estimates, Costing Runs, Work Center Rates, Price Lists, Customer Price Agreements, Commodity Surcharges, Margin Analysis	Part I
Controlling — NEW	Cost Centers, Profit Centers, Hierarchies, Allocation Cycles, Cost Center Reports, Profit Center P&L	Part H
Accounting	As built per ACC V2, unchanged	ACC
Configuration	Existing INV/ACC configuration plus: Barcode & Label Settings, Scan Point Policies, RMA Policies, Sales Settings, Production Settings, Engineering Settings, Tax Configuration (Part A §11)	Mixed

2.2 Document Number Series Additions

The following series extend the Number Series Configuration screen (INV 11.3). All are entity-aware where the companion accounting series are entity-aware.

Document Type	Prefix / Format	Example
Quotation	QT-{YYYY}-{SEQ:5}	QT-2026-00311
Sales Order	SO-{YYYY}-{SEQ:5}	SO-2026-01184
Delivery / Shipment	SHP-{YYYY}-{SEQ:5}	SHP-2026-00092
Handling Unit (Packing)	HU-{SEQ:8}	HU-00004417
RMA	RMA-{YYYY}-{SEQ:4}	RMA-2026-0038

Document Type	Prefix / Format	Example
Credit Memo Request	CMR-{YYYY}-{SEQ:4}	CMR-2026-0021
Production Order	WO-{YYYY}-{SEQ:5}	WO-2026-00266
Rework Order	RWO-{YYYY}-{SEQ:4}	RWO-2026-0007
Planned Order	PLO-{SEQ:6}	PLO-018223
Routing	RTG-{TYPE}-{SEQ:4}	RTG-PAD-0012
Engineering BOM	EBOM-{SEQ:5}	EBOM-00341
Manufacturing BOM	BOM-{TYPE}-{YYYY}-{SEQ:4}	BOM-TX-2026-0034 (existing pattern, unchanged)
Engineering Change Request	ECR-{YYYY}-{SEQ:4}	ECR-2026-0102
Engineering Change Order	ECO-{YYYY}-{SEQ:4}	ECO-2026-0059
CAD / Controlled Document	DOC-{TYPE}-{SEQ:5}	DOC-GA-00873
Purchase Requisition	PR-{YYYY}-{SEQ:5}	PR-2026-00944
Request for Quotation	RFQ-{YYYY}-{SEQ:4}	RFQ-2026-0116
Purchase Order	PO-{YYYY}-{ENTITY}-{SEQ:5}	PO-2026-PT-00532
Inspection Plan	IP-{TYPE}-{SEQ:4}	IP-INC-0027
Final Test Report	FTR-{YYYY}-{SEQ:4}	FTR-2026-0201
CAPA	CAPA-{YYYY}-{SEQ:4}	CAPA-2026-0013
Supplier Corrective Action	SCAR-{YYYY}-{SEQ:4}	SCAR-2026-0006
Cost Estimate	CE-{YYYY}-{SEQ:5}	CE-2026-00187
Costing Run	CR-{YYYY}-{SEQ:3}	CR-2026-004
Allocation Cycle Run	ALC-{YYYY}{MM}-{SEQ:2}	ALC-202607-01
Count Document	CD-{CountPlan}-{SEQ:3}	CD-CP-2026-001-014

2.3 Master Data Flow

The single most important architectural rule in this system: **the Material Master is the hub**. Engineering creates and revises items and BOMs; Purchasing sources them; Production consumes and produces them; Sales prices and sells them; Inventory stores and tracks them; Costing values them; Accounting posts them. The flow below governs which module owns which attribute of an item. Ownership means only that module's screens may edit the attribute; all others render it read-only with a link to the owning screen.

Attribute Group (Material Master tab)	Owning Module / Role	Consumed By
General Information, Dimensions, Tracking config	Inventory Admin (INV 2.2/2.3)	All modules
BOM tab (read-only view)	Engineering (Part C) — edits happen in EBOM/MBOM screens	Production, Costing, Purchasing (for shortage checks)
Purchasing tab, AVL, lead times	Purchasing (Part G)	MRP, Reorder planning, Costing
Sales tab, price lists, customer prices	Sales Admin (Part A §4)	Quotations, Sales Orders, Invoices

Attribute Group (Material Master tab)	Owning Module / Role	Consumed By
Quality tab, inspection plans, test parameters	Quality (Part E / Part A §6)	GRN, Production, Shipping
Pricing & Costing tab: valuation, GL mapping	Finance / Cost Accountant (INV 2.3.3)	All posting transactions
Standard cost (current released)	Costing (Part I costing run only — never manual edit)	Valuation, PPV, WO settlement
Cost Center / Profit Center defaults	Controlling (Part H)	Every financially relevant posting

2.4 New GL Accounts Introduced by This Volume

These accounts extend the GL Account Mapping screen (INV 11.4) and the CoA (ACC Section 13). They follow the existing numbering logic.

GL Account #	Account Name	Type	Used By
1315	Subcontract Material at Vendor	Balance sheet (asset)	Subcontract POs (Part G §6)
1325	WIP — Rework	Balance sheet (asset)	Rework orders (Part D §9)
1335	Return / RMA Inventory (Quarantine)	Balance sheet (asset)	RMA receipts (Part A §3)
1360	Packaging Materials on Hand	Balance sheet (asset)	Packing (Part A §5)
2150	Customer Refunds Payable	Balance sheet (liability)	RMA cash refunds
2610	Warranty Reserve	Balance sheet (liability)	Warranty accrual & warranty RMA work
4150	Service & Repair Revenue	P&L (revenue)	Billable repair RMAs
4910	Restocking Fee Income	P&L (revenue)	RMA restocking fees
5130	COGS — Service & Repair	P&L (expense)	Billable repair cost
5230	Labor Rate Variance	P&L (expense)	WO settlement (Part I §4)
5240	Labor Efficiency Variance	P&L (expense)	WO settlement
5250	Overhead Absorption Variance	P&L (expense)	WO settlement
5260	Subcontract Price Variance	P&L (expense)	Subcontract receipt
6300	Warranty Expense	P&L (expense)	Warranty RMA repair/replacement cost
6410	Freight Out	P&L (expense)	Shipments, RMA return freight when PE pays
7500	Cost Center Allocation Clearing	P&L (contra)	Allocation cycles (Part H §5)

Dev Note: Account numbers above are proposals consistent with the existing ranges (13xx inventory, 2xxx liabilities, 4xxx revenue, 5xxx cost variances, 6xxx operating expense). Controller confirms or remaps them once in GL Account Mapping; every posting rule in this volume references the mapping keys, not hard-coded numbers, so remapping requires no code change.

PART A

INVENTORY GAP CLOSURE — THE NINE OPEN ITEMS

Part A closes, in full, the nine items raised in the Post-Inventory Design Review. Each section is a complete build specification: process flow, screens, fields, buttons, statuses, business rules, exception handling, GL postings with cost/profit center derivation, and role access. Section numbering inside Part A starts at 3 to leave Sections 1–2 for the global front matter of this volume.

3. RMA — Return from Customer

PE Transformer's customer return process is the RMA process. No customer material enters any PE warehouse without an approved RMA number. The RMA module replaces the bare "Return from Customer" transaction stub: the inventory movement specified in the gap register (DR Return Inventory / CR COGS reversal) becomes one step inside a controlled, end-to-end process that also handles warranty determination, inspection, disposition, credit memos, repairs, replacements and field returns of installed transformers.

3.1 Return Classes

Class	Typical Case	Key Differences
RC1 — Component / distribution return	PE Grid Components customer returns bushings, arresters, hardware bought from stock	Standard receipt into quarantine; restocking fee rules apply; credit or replacement; fastest path
RC2 — Finished transformer, not yet energized	Unit refused at delivery, damaged in transit, wrong configuration shipped	Serialized; freight and crating logistics; transit-damage insurance claim fields; full inspection
RC3 — Field return of installed / energized unit	Warranty failure at substation; unit de-energized and returned for teardown or repair	Warranty adjudication from serial record; oil handling and hazmat transport; teardown inspection; repair order or replacement unit; failure analysis feeding NCR/CAPA
RC4 — Return for billable repair / refurbishment	Out-of-warranty unit sent in for repair at customer cost	No credit; generates a billable Repair Order and a service invoice (4150 Service & Repair Revenue)

3.2 End-to-End Process Flow

1. **Request.** Customer service creates the RMA from the originating document (invoice, sales order or shipped serial). System pulls customer, item, serial/batch, ship date, warranty status. If no reference document exists, the No-Reference exception path (§3.9) applies.
2. **Approval.** Approval routing per RMA policy matrix (§3.10): value threshold, return class and warranty claim flag decide the approver chain (Customer Service Manager, then Quality Manager for RC3, then Controller above $50,000 expected credit).
3. **Authorization issued.** RMA number and return instructions (PDF) are emailed to the customer: RMA barcode label to affix, ship-to dock, freight terms (who pays per policy), packaging/oil handling instructions for RC3.
4. **Return receipt.** Receiving scans the RMA barcode; the receipt screen (a Check-In of Receipt Type "Return Receipt", DELTA to INV 5.1) is pre-populated from RMA lines. All returned stock lands in the Quarantine warehouse with quality status QI Hold, GL account 1335 Return/RMA Inventory. Serial status moves to Returned.
5. **Inspection.** A Return Inspection (Quality module, reusing the Incoming Inspection engine INV 6.1 with plan type = RMA) is auto-created per line. For RC3, the teardown checklist applies and photographs are mandatory attachments.
6. **Disposition.** Inspector proposes, authorized role approves: Restock / Repair / Replace / Scrap / Return to Customer As-Is. Warranty claims additionally get Warranty Approved or Warranty Denied.
7. **Settlement.** Depending on disposition: credit memo request → ACC credit note (ACC Section 4); replacement sales order at zero price with reference; repair order (rework production order, Part D §9) either under 2610 Warranty Reserve or billable; scrap posting; or return shipment to customer.
8. **Closure.** RMA closes when every line reaches a terminal disposition and all financial documents are posted. Closure is blocked while any linked document (credit note, WO, replacement SO) is open.

3.3 Screen RMA-001 — RMA Listing

Screen ID	RMA-LIST-001
URL Pattern	/sales/rma
Purpose	Work queue and register of all RMAs across statuses; landing page for customer service and receiving

Column	Type	Specification
RMA #	Link	RMA-YYYY-NNNN; opens RMA Detail
Customer	Link	Customer name; opens customer record
Return Class	Badge	RC1 (blue), RC2 (teal), RC3 (orange), RC4 (purple)
Reference Doc	Link	Invoice #, SO # or Serial # used as basis
Items / Qty	Text	Line count and total quantity
Expected Value	Currency	Sum of line return values at original net price
Warranty Claim	Badge	Yes (amber) / No (gray) / Approved (green) / Denied (red)
Status	Badge	Per lifecycle §3.8
Requested Date	Date	Creation date
Received Date	Date	First receipt date; empty until received
Age (Days)	Number	Days in current status; red if past SLA (config, default 5 days in Received without inspection)
Owner	User	Customer service owner

Filters: status, return class, customer, warranty flag, date range, owner, warehouse. Bulk actions: Assign Owner, Print Return Instructions, Export. Quick stats pills: Open RMAs, Awaiting Receipt, In Inspection, Pending Disposition, Pending Credit, Aged > SLA.

3.4 Screen RMA-002 — Create RMA

Screen ID	RMA-CREATE-001
URL Pattern	/sales/rma/new
Purpose	Authorize a customer return with full reference, warranty and logistics data

3.4.1 Header Fields

Field	Type	Req	Specification
RMA #	Auto	Auto	RMA-{YYYY}-{SEQ:4}; assigned on first save
Customer	Lookup	Yes	From Customer Master; blocked customers (credit hold) show warning banner but RMA creation is allowed — settlement by credit is blocked later until hold clears
Return Class	Dropdown	Yes	RC1–RC4 per §3.1; drives approval chain, inspection plan type and logistics fields
Reference Type	Dropdown	Yes	Invoice, Sales Order, Serial Number, No Reference (exception path §3.9)

Field	Type	Req	Specification
Reference Document	Lookup	Cond.	Required unless Reference Type = No Reference. Lookup filtered to the customer. Selecting it populates lines with shipped items, quantities, serials/batches, net prices, ship dates
Warranty Claim	Toggle	No	Defaulted ON automatically when every referenced serial is inside its warranty window (Serial master Warranty End ≥ today); user may switch ON manually — then adjudication is mandatory at disposition
Reason Code	Dropdown	Yes	Defective — Warranty, Defective — Out of Warranty, Damaged in Transit, Wrong Item Shipped, Wrong Item Ordered, Overshipment, Customer Convenience, Recall, Billable Repair. Reason drives restocking fee default and who pays freight
Reason Detail	Textarea	Yes	Free text, min 20 characters; customer-reported failure description for RC3
Return Warehouse	Dropdown	Yes	Default: Quarantine warehouse of the receiving plant; only warehouses of type Quarantine selectable
Freight Terms	Dropdown	Yes	PE Pays (default for Warranty / Wrong Item Shipped / Recall), Customer Pays (default otherwise), Third Party
Carrier / Instructions	Text + Textarea	No	Carrier preference; RC3 shows additional mandatory sub-fields: Oil Drained (Yes/No), Oil Volume (gal), Nitrogen Blanket (Yes/No), Impact Recorder Fitted (Yes/No)
Restocking Fee %	Number	Cond.	Defaults from reason code policy (Customer Convenience = 15%, Wrong Item Ordered = 10%, all others 0). Editable within role limit; above-limit values route to approval. Posts to 4910 on credit
Expected Return Date	Date	No	For receiving planning; overdue RMAs appear on the expedite view
Owner	User lookup	Yes	Defaults to creator

3.4.2 Line Fields

Field	Type	Req	Specification
Item	Lookup	Yes	Auto-filled from reference; manual add allowed only on No-Reference path
Serial # / Batch #	Lookup	Cond.	Mandatory for serial/batch-managed items; lookup restricted to serials/batches shipped to this customer. Serial selection displays warranty window, install date and site inline
Qty Authorized	Number	Yes	Cannot exceed shipped-less-already-returned quantity on the reference line (validation V-RMA-01)
UoM	Auto	Auto	From item
Original Net Price	Currency	Auto	From reference invoice/SO line; basis for credit
Return Value	Currency	Auto	Qty × original net price − restocking fee
Expected Disposition	Dropdown	No	Customer-service guess (Restock/Repair/Replace/Credit); informational, real disposition set at §3.6
Line Notes	Text	No	—

Buttons: **Save Draft**, **Submit for Approval** (validates all lines), **Cancel**. After approval: **Print Return Instructions** (PDF with RMA barcode, Code128 of RMA #), **Email to Customer**, **Cancel RMA** (allowed until first receipt; requires reason).

3.5 Screen RMA-003 — Return Receipt (DELTA to INV 5.1)

Receiving opens the receipt directly by scanning the RMA barcode on the arriving crate (scan point SP-19, Part F). The Check-In engine is reused with Receipt Type = Return Receipt and these deltas:

Delta	Specification
Header reference	RMA # is mandatory; vendor fields hidden; customer shown instead
Line source	Lines pre-populated from RMA authorized lines; receiving may receive partial quantities (creates back-order on RMA line, status Partially Received)
Over-receipt	Received Qty > Authorized Qty is blocked (hard error E-RMA-03). Unexpected extra material is booked on a new line flagged Unauthorized Return, which forces a supervisor override and its own inspection
Condition on arrival	Per line: dropdown (Undamaged / Damaged Packaging / Damaged Product / Leaking Oil) + mandatory photo attachment when not Undamaged
Destination	Location restricted to the Quarantine warehouse from the RMA header; putaway rules engine (INV 15.6) proposes bin inside quarantine zone
Quality status	Always QI Hold; a Return Inspection is auto-created per received line
Serial handling	Scanned serial must equal authorized serial (E-RMA-04 on mismatch); serial lifecycle status set to Returned; location updated

Accounting Entry — RMA Receipt (stock item, credit expected)

Debit	Credit	Amount Basis	Notes
1335 Return / RMA Inventory	5120 COGS (reversal)	Received Qty × valuation rate at original issue	Implements and supersedes the gap-register journal; “Return Inventory” = 1335. Cost center: none; Profit center: from returned item’s product line

Accounting Entry — RMA Receipt (warranty repair expected, unit remains customer property)

Debit	Credit	Amount Basis	Notes
No entry	No entry	—	Customer-owned unit held for repair is tracked as non-valuated customer stock (quantity only, value $0). Only RC3/RC4 lines flagged Customer Property take this path

Dev Note: Distinguish ownership at receipt: if the commercial outcome will be credit or replacement, PE takes ownership back and the value entry above posts. If the unit is in for repair and returns to the same customer, ownership never transfers; the system books a non-valuated receipt (quantity tracking, zero value) so no COGS reversal ever posts. The Expected Disposition field defaults this flag: Credit/Replace/Restock → PE ownership; Repair → customer property. The flag is editable at receipt by Receiving Supervisor only.

3.6 Screen RMA-004 — Inspection & Disposition

The inspection reuses the Incoming Inspection execution screen (INV 6.1.1) with plan type RMA (Part A §6). Additional RMA-specific disposition block:

Field	Type	Req	Specification
Findings Summary	Textarea	Yes	Inspector findings; for RC3 the teardown checklist results table (winding condition, core condition, oil test results, bushing condition, tap changer condition) is embedded
Failure Attributable To	Dropdown	Cond.	Manufacturing Defect, Component Supplier, Transit Damage, Installation Error, Site Conditions, Normal Wear, Undetermined. Mandatory for warranty claims; Supplier selection offers Create SCAR (Part E §8)
Warranty Adjudication	Dropdown	Cond.	Warranty Approved / Warranty Denied + mandatory justification. Mandatory when Warranty Claim = Yes. Denial auto-generates customer letter from template and offers conversion to RC4 billable repair
Disposition (per line)	Dropdown	Yes	Restock, Repair — Warranty, Repair — Billable, Replace, Scrap, Return to Customer As-Is
Disposition Approved By	User + date	Auto	Role per policy matrix §3.10; captured on approval action
NCR Required	Toggle	Auto	Forced ON when Failure Attributable To = Manufacturing Defect or Component Supplier; creates NCR (INV 15.4) linked to the RMA

3.6.1 Disposition Outcomes — System Actions and GL

Disposition	Automatic System Actions	GL Posting
Restock	Stock transfer Quarantine → designated warehouse; quality status Unrestricted; credit memo request created for return value less restocking fee	DR 1330 FG (or 1310 by item type) / CR 1335 at valuation rate. Restocking fee posts with the credit note: DR 4100 Revenue reversal full, CR 1200 AR net, CR 4910 Restocking Fee Income
Repair — Warranty	Rework order (RWO, Part D §9) auto-created, settlement receiver = 2610 Warranty Reserve; unit stays customer property (no inventory value)	Repair costs settle DR 2610 Warranty Reserve / CR 1325 WIP-Rework at RWO close. If reserve insufficient, overflow to 6300 Warranty Expense
Repair — Billable	Rework order created with settlement receiver = COGS-Service; service quote generated for customer approval before work starts; service invoice on completion	Costs: DR 5130 COGS-Service / CR 1325 at close. Invoice: DR 1200 AR / CR 4150 Service & Repair Revenue (+2300 tax)
Replace	Replacement SO created at $0 net price referencing RMA; original line credited in full; defective unit scrapped or repaired-to-stock per secondary disposition	Credit note per ACC Sec 4. Replacement shipment: DR 6300 Warranty Expense (warranty) or 5120 COGS (goodwill, needs Controller approval) / CR 1330 FG at standard
Scrap	Goods issue to scrap from quarantine; serial → Scrapped; certificate of destruction attachment optional	DR 5320 Scrap & Waste / CR 1335 at valuation rate; cost center = Quality (CC-1500)
Return As-Is	Return shipment to customer; no credit; freight per terms	DR 6410 Freight Out / CR 2100 AP for freight if PE pays; inventory: reverse the receipt entry (DR 5120 / CR 1335) since ownership returns to customer

3.7 RMA Detail — Tabs

Tab	Content
Overview	Header data, status timeline visual, linked documents panel (invoice, credit note, WO/RWO, replacement SO, NCR, SCAR) each as clickable chips
Lines	Line grid with authorized/received/inspected/disposed quantities per line and per-line status
Receipts	All return receipts with condition-on-arrival and photos
Inspection	Inspection results, teardown checklist, warranty adjudication
Financials	Expected vs actual credit, restocking fee, repair cost accumulation (live from RWO), freight charges, running GL entries with journal links
Documents	Return instructions PDF, customer correspondence, photos, certificate of destruction
Activity	Immutable audit trail

3.8 RMA Status Lifecycle

Status	Color	Allowed Actions	Transitions
Draft	Gray	Edit, Submit, Delete	Pending Approval
Pending Approval	Yellow	Approve, Reject, Recall	Authorized, Rejected, Draft
Authorized	Blue	Print/Email instructions, Cancel, Receive	Partially Received, Received, Cancelled
Partially Received	Teal	Receive remainder, Close Short (supervisor)	Received, Closed Short
Received	Purple	Inspect	In Inspection
In Inspection	Purple	Record results, Disposition	Disposition Pending
Disposition Pending	Amber	Approve disposition per line	Settling
Settling	Amber	System executes outcomes; user monitors linked docs	Closed
Closed	Green	View, Reopen (Controller, 30-day window)	Reopened
Rejected / Cancelled	Red	View	Terminal

3.9 Exception Handling

Exception	System Behavior
Return without reference document	Reference Type = No Reference requires Customer Service Manager approval regardless of value. Credit basis falls back to current price list less 20% (config) and Controller approval is mandatory for any credit. Lines are keyed manually; serial lookup validates the serial exists and belongs to the customer — if the serial is unknown to the system the line is flagged Unverified Serial and warranty is automatically Denied pending manual verification
Partial return	Authorized qty may be received across multiple receipts; RMA line tracks authorized / received / open. Close Short action (supervisor) cancels the open remainder and documents why
Non-stock / non-inventory item returned	Items of type Service or non-stock: no inventory posting; credit-only path with Controller approval; receipt step records the physical arrival for evidence but posts nothing

Exception	System Behavior
Unauthorized extra material in the crate	Booked as Unauthorized Return line (supervisor override), inspected, then either appended to the RMA via a change (re-approval required) or returned to customer at customer cost
Customer on credit hold	RMA and receipt proceed; credit memo creation is blocked with message E-RMA-07 “Customer is on credit hold — credit requires Controller release”
Damaged-in-transit with PE freight	Condition = Damaged triggers freight claim sub-record (carrier, claim #, claim amount, status) tracked on Financials tab; claim recovery posts DR 1010 Bank / CR 6410 Freight Out when received
Wrong serial arrives	E-RMA-04 blocks the line; receiving books it as Unauthorized Return; customer service contacts customer

3.10 RMA Policy Matrix (Configuration)

Parameter	Default	Notes
Approval — RC1/RC2 ≤ $10,000	Customer Service Manager	Single step
Approval — any RC3 or warranty claim	CS Manager → Quality Manager	Two sequential steps
Approval — expected credit > $50,000	+ Controller	Appended step
Approval — goodwill replacement (out of warranty)	+ Controller (any value)	Because it posts to COGS without revenue
Restocking fee defaults	Customer Convenience 15%, Wrong Item Ordered 10%, else 0%	Editable per RMA within role limits
Receipt-to-inspection SLA	5 business days	Breach alerts Quality Manager (alert rules INV 15.11)
Reopen window after Closed	30 days, Controller only	Full audit

Role access: Customer Service (create/edit draft, submit), CS Manager (approve, close short, cancel), Receiving Clerk (receipts), Quality Inspector (inspection), Quality Manager (disposition approval, warranty adjudication), Controller (credit release, reopen, goodwill approval), Read-Only (view).

4. Material Master — Sales Tab

Screen ID	INV-ITEM-DETAIL-SALES-001
URL Pattern	/inventory/items/{itemId} → Sales tab
Purpose	Complete definition of how an item is sold: sales data, price list membership, customer-specific pricing, and the price determination contract used by every downstream sales document

4.1 Sub-section A — Sales Data

Field	Type	Req	Specification
Sales UoM	Dropdown	No	From UoM master; conversion path to inventory UoM must exist (INV 2.6) else save blocked
Sellable	Toggle	Yes	OFF blocks the item from quotations and sales orders entirely (hard error at document entry). Default ON for Finished Goods, OFF for all other categories
Minimum Sales Order Qty	Number	No	Validated at SO line entry; under-minimum requires CS Manager override
Sales Lead Time (Days)	Number	No	Quoted delivery lead time for make-to-stock lines; ETO items derive dates from production scheduling instead
Warranty (Months)	Number	No	Default warranty written to serials at shipment (Warranty End = ship date + months). Item-level default; SO line may override upward with approval
Customer Part # Cross-Reference	Table	No	Rows: Customer, Customer Part #, Customer Description. Used on order entry (type the customer's number, system resolves the item) and printed on order confirmations, packing lists and invoices
Revenue GL Override	GL Lookup	No	Overrides 4100 for this item (e.g., 4150 for service items). Blank = category default
Tax Category	Dropdown	Yes	Taxable / Tax-Exempt / Zero-Rated (existing field, relocated here from General tab; single source of truth). Feeds tax determination §11
Avalara Tax Code	Text	Cond.	Product tax code sent on every Avalara call (ACC 19.1); required when integration active. Default P0000000 tangible personal property
Commission Group	Dropdown	No	Phase 3 placeholder for rep commissions; stored, not yet consumed

4.2 Sub-section B — Price List Assignments

An item may belong to any number of price lists. Price lists themselves are mastered in Costing & Pricing (Part I §6); this grid shows and edits the rates of this item across lists.

Column	Type	Specification
Price List	Link	Wholesale, OEM, Government, Distributor + any custom lists; each is entity- and currency-specific
Unit Price	Currency	Rate for this item on that list, in list currency, per Sales UoM
Valid From / To	Dates	Effective dating; overlapping validity for the same list is blocked (V-SAL-02). Blank To = open-ended
Min Qty Break Table	Expand	Optional quantity breaks: rows of From Qty → Unit Price; system applies the deepest satisfied break

Column	Type	Specification
Margin @ Standard	Percent (calc)	(Price − current standard cost) ÷ Price; red when below the product line margin floor (Part I §7)
Status	Badge	Draft / Active / Expired
Approved By	User	Price changes route through pricing approval when the change exceeds ±5% or drops below margin floor (config)

4.3 Sub-section C — Customer-Specific Prices

Column	Type	Specification
Customer	Lookup	One row per customer/date-range; a customer may appear multiple times with non-overlapping validity
Unit Price	Currency	Contracted price; currency of the customer
Valid From / To	Dates	Non-overlap enforced per customer (V-SAL-03)
Source Agreement	Link	Optional link to Customer Price Agreement (Part I §6.3) or uploaded contract document
Margin @ Standard	Calc	Same rule as B; below-floor rows require Controller approval, not just Pricing Manager
Status	Badge	Draft / Pending Approval / Active / Expired

4.4 Price Determination — the Contract with Downstream Documents

Every quotation, sales order and invoice line resolves price by the same deterministic sequence. This sequence is implemented once as the Pricing Service and called by all documents; it must never be re-implemented per document.

1. Manual override on the document line, if entered and within the user's override authority (below floor → approval workflow), always wins and is stamped Price Source = Manual.
2. Active customer-specific price for the document's customer, valid on the pricing date (order date by default; configurable to requested ship date). Price Source = Customer Agreement.
3. Active rate from the price list assigned to the customer (Customer Master → Price List field), applying quantity breaks. Price Source = Price List.
4. Item pricing method fallback (INV 2.3.3 Sub-section C): Fixed Price, or Markup over current cost. Price Source = Item Default.
5. No price found → line blocks with E-SAL-01 "No valid price for item {code} / customer {name} on {date}" — the document cannot be submitted with unpriced lines.

Rule	Specification
Inheritance & override display	The SO line shows the resolved price plus a Price Source badge; hovering shows the full waterfall (which levels existed and which one won). Overrides display both original and override values
Repricing	Quotation → SO copy carries quoted prices frozen (Price Source = Quotation); explicit Reprice button re-runs determination with confirmation diff dialog. Invoice always inherits SO price — never reprices silently
Currency	Rates convert only at document level using the daily rate table (ACC Sec 3); price masters never auto-convert

Rule	Specification
Surcharges	Commodity surcharges (copper/CRGO escalators, Part I §6.4) compute as separate line-level surcharge components added after base price resolution, printed as distinct lines on the confirmation
Change protection	Editing a price master never touches existing documents; only new determinations see new rates

Buttons on the tab: **Add Price List Assignment**, **Add Customer Price**, **Mass Update** (percentage adjustment across selected rows → creates new dated rows rather than editing history), **Export**, **Price Simulation** (enter customer + qty + date → shows the waterfall result exactly as a document would resolve it). All edits audit to Change History with old/new values.

5. Packing

Packing converts picked stock into shippable Handling Units (HUs) with weights, dimensions, labels and a packing list. For PE this ranges from a carton of gaskets to a 40-ton transformer on a dedicated flatbed with separate oil totes and accessory crates. Packing sits between Picklist (INV 5.7) and Shipment (Part B §5); goods remain in Finished Goods stock (no GL impact) until post-goods-issue at shipment.

5.1 Handling Unit Model

Concept	Specification
Handling Unit (HU)	A numbered physical package: HU-{SEQ:8} with Code128/SSCC barcode. Types: Carton, Crate, Skid/Pallet, Reel, Tote, Flatbed Load, Container. An HU contains item quantities and/or other HUs (one level of nesting: cartons on a pallet)
Contents	Lines of item + qty + batch/serial. Serialized FG: exactly one transformer serial per primary HU
Attributes	Tare weight, net weight, gross weight (validated: gross ≈ tare + net ± 2%), dimensions L×W×H, packaging material item (consumes 1360/packaging stock via backflush), seal numbers, hazmat flags (oil = UN3082 where applicable), nitrogen pressure (transformers shipped gas-filled), impact recorder serial
Statuses	Open (packing) → Closed (contents locked, label printed) → Loaded (assigned to shipment) → Shipped. Reopen allowed until Loaded, logs reason

5.2 Screen PCK-001 — Packing Queue

Screen ID	OM-PACK-LIST-001
URL Pattern	/sales/packing
Purpose	Everything picked and awaiting packing, and all in-progress pack orders

Column	Specification
Pack Order #	PK-{YYYY}-{SEQ:5}; auto-created when a picklist completes (config: per picklist / per shipment)
Source	Picklist #, Sales Order #, Shipment # links
Customer / Ship-To	Name and destination city
Lines / Qty	Outstanding lines and quantities to pack
Required Ship Date	From SO; red when today or past
Status	Not Started (gray), In Progress (yellow), Packed (green), Cancelled (red)
Packer	Assigned user
HUs	Count of HUs created

5.3 Screen PCK-002 — Pack Station

Screen ID	OM-PACK-STATION-001
URL Pattern	/sales/packing/{packOrderId}
Purpose	Scan-driven packing execution at the pack bench or shipping bay

Layout: left panel = items remaining to pack (item, qty open, batch/serial); right panel = HU tree being built; center = scan input with focus lock. Keyboard/scanner-first design; every action available without mouse.

Step / Element	Specification
1. Create HU	Button or scan a pre-printed HU license plate. Select HU type; packaging material item defaults from type (e.g., Crate-Type-C → item CRATE-C); tare auto-fills from packaging item master
2. Scan item	Scan item/batch/serial barcode (SP-16, Part F). Validation: item belongs to open pack lines; serial must be Picked status; batch must match picked batch. Wrong scan → audible reject + error banner E-PCK-02
3. Enter qty	Qty defaults to remaining; numeric pad entry; cannot exceed remaining (E-PCK-03)
4. Close HU	Requires gross weight entry (from scale integration or manual with supervisor flag), dimensions (default from HU type, editable). Prints HU label + contents label on close
5. Complete Pack Order	Enabled when all lines fully packed or supervisor short-packs (reason: Short Pick, Damage, Hold). Sets status Packed and notifies shipping
Ancillary actions	Reopen HU, Move contents between HUs, Split line across HUs, Print Packing List (draft watermark until pack order complete), Attach photos (mandatory for transformers: 4 corner shots + nameplate, config per item category)

5.4 Documents & Labels

Output	Content Specification
HU Label (4×6 in, ZPL)	PE branding block; HU # (Code128 + human readable); SSCC-18 (GS1-128, AI 00) when customer requires GS1; ship-to; gross/net weight; HU x of y; hazmat diamond when flagged; handling pictograms (this-way-up, keep-dry) per item category
Contents Label	Item, description, qty, batch/serial list (or “see packing list” beyond 10 serials), PO # of customer, SO #
Packing List (PDF, letter)	Header: PE entity letterhead, SO #, customer PO #, ship-to, shipment #, date, Incoterms. Body grouped by HU: HU #, type, dims, tare/net/gross, contents lines (item, description, customer part #, qty, UoM, batch/serial). Footer: totals (HUs, net, gross), special instructions, packer signature block. Language: English; unit system dual (lb/kg) always
Transformer shipping addendum	For RC “Flatbed Load” HUs: nitrogen pressure at dispatch, impact recorder serial + threshold, center-of-gravity diagram attachment, tie-down plan attachment, oil shipped separately note with tote HU references

Integration	Specification
Shipment	Closed HUs are assigned to a Shipment (Part B §5); Loaded status set by loading scan (SP-18). Shipment cannot post goods issue while any assigned HU is Open
Inventory	Packing consumes packaging materials by backflush at HU close: DR 5120 COGS-packaging cost element (or 6410 per policy) / CR 1360 Packaging on Hand. FG itself does not move: it remains in FG stock, bin = STAGE-OUT, until PGI
Invoicing	Invoice trigger = PGI (config per customer: PGI or delivery confirmation). Packing data (weights, HU count) prints on the invoice supplement
Carrier	Phase 3: rate shopping / tracking API. Phase 1 captures carrier, PRO/BOL # manually

6. Inspection Plans

Screen ID	QM-PLAN-001 (listing), QM-PLAN-002 (detail/editor)
URL Pattern	/quality/inspection-plans
Purpose	Master data defining what is tested, how, against which specification, with which sampling — for every inspection type in the system. The Test Results table in an active inspection (INV Screen 6.1.1) pulls its rows from here

6.1 Plan Types and Where They Fire

Plan Type	Trigger	Executed In
Incoming (INC)	GRN line for QI-required item	Incoming Inspection (INV 6.1)
In-Process (INP)	Production order operation flagged QC Checkpoint (Part D §5)	In-Process Inspection (INV 15.2)
Final Test (FIN)	Production order reaches Test operation	Final Test screen (Part E §4)
Pre-Shipment (PSI)	Picklist/shipment creation for FG	Outgoing Inspection (INV 15.3)
RMA (RMA)	RMA return receipt (Part A §3)	Return Inspection

6.2 Plan Header Fields

Field	Type	Req	Specification
Plan #	Auto	Auto	IP-{TYPE}-{SEQ:4}, e.g. IP-INC-0027
Plan Name	Text (200)	Yes	e.g. “CRGO Incoming — Core Loss & Chemistry”
Plan Type	Dropdown	Yes	INC / INP / FIN / PSI / RMA; locked after first release
Assignment Scope	Dropdown	Yes	Item, Item Group, Item + Vendor (incoming only), Item + Operation (in-process only). Most specific active plan wins at runtime; ties broken by newest effective date
Assigned Item / Group / Vendor / Operation	Lookups	Cond.	Per scope. An item may carry many plans of different types but only one Active plan per (type + scope combination + date) — overlap blocked, V-QM-01
Revision	Auto	Auto	Rev A, B, C…; new revision required for any change after release (plans are immutable once Active)
Effective From / To	Dates	Yes/No	Runtime selection uses inspection creation date
Sampling Method	Dropdown	Yes	100% / AQL Sampling / Skip-Lot / Fixed Sample Size
AQL Settings	Group	Cond.	When AQL: Inspection Level (S1–S4, G1–G3; default G2), AQL value (0.65 / 1.0 / 1.5 / 2.5 / 4.0), Switching Rules ON/OFF (Normal↔Tightened↔Reduced per ANSI/ASQ Z1.4: tighten after 2 of 5 consecutive lots rejected; reduce after 10 consecutive accepted with approval). Sample size auto-derives from lot size + level at runtime; inspector may only increase it
Skip-Lot Settings	Group	Cond.	Skip frequency (inspect 1 of N lots, default N=5), qualification rule (vendor quality rating ≥ 98% and ≥ 10 consecutive accepted lots), auto-revoke on any rejection or NCR
Status	Badge	Auto	Lifecycle §6.4

6.3 Characteristics (Test Rows)

The plan body is an ordered table of characteristics. Exactly these rows materialize as the Test Results table in the executed inspection.

Column	Type	Specification
Seq	Number	Execution order; drag to reorder in Draft
Test Name	Text	e.g. Core Loss Test, Dielectric Strength, Turns Ratio
Test Method / Standard	Text	e.g. IEEE C57.12.90 §8, ASTM A343, internal WI-QA-014 (link to controlled document, Part C §5)
Characteristic Type	Dropdown	Quantitative (numeric result vs min/max) or Attributive (Pass/Fail/NA)
Parameter & UoM	Text + UoM	Measured property and unit (W/kg, kV, %, Ω)
Spec Min / Target / Max	Numbers	Quantitative only; at least one bound required; auto Pass/Fail from bounds
Critical	Toggle	Fail on a critical characteristic = automatic overall Reject regardless of other results (rule R-QM-02)
Sample Instruction	Text	Where/how to draw the sample (e.g. “one strip per coil, mid-width”)
Instrument / Gauge Class	Lookup	Links to Calibration register (Part E §9); execution blocks if the selected instrument’s calibration is expired (E-QM-05)
Certificate Required	Toggle	Incoming: vendor test certificate upload mandatory before Accept when ON

6.4 Status Lifecycle and Approval

Status	Color	Actions	Transitions
Draft	Gray	Edit freely, Delete, Submit for Review	In Review
In Review	Yellow	Quality Manager: Approve / Reject with comments	Active, Draft
Active	Green	Use at runtime; Create New Revision; Set Obsolete (QM only, blocked if it would leave a QI-required item with no plan and policy = Block)	Superseded (auto when next rev activates), Obsolete
Superseded	Blue	View, referenced by historical inspections	Terminal
Obsolete	Red	View only	Terminal

6.5 No-Plan Exception Policy

When a trigger fires and no applicable Active plan exists, behavior follows the Quality Policy setting (Configuration → Quality Settings), per plan type:

Policy Option	Behavior
Block (default for INC on QI-required items and for FIN)	The triggering transaction cannot complete: GRN line cannot post to stock (stays Draft with error E-QM-01 “No active inspection plan for {item}/{vendor} — contact Quality”), WO cannot pass the Test operation. Alert to Quality Manager
Manual Inspection	Inspection record is created with an empty characteristics table; inspector adds ad-hoc test rows (each flagged Ad-Hoc and reportable); Quality Manager sign-off mandatory regardless of result
Auto-Accept with Warning	Stock passes to Unrestricted; a warning event is logged and appears on the Quality dashboard exception panel. Permitted only for non-QI-required items; never for FIN/PSI

Buttons: Copy Plan (deep copy to new item/type), Compare Revisions (side-by-side diff of characteristics), Where-Used (all items/vendors resolving to this plan today), Export PDF. Roles: Quality Engineer (create/edit drafts), Quality Manager (approve/obsolete), all others read-only.

7. Count Documents

Screen ID	INV-COUNT-DOC-001 (listing), INV-COUNT-DOC-002 (detail)
URL Pattern	/inventory/counting/count-documents
Purpose	The controlled work papers of a physical or cycle count: generation, assignment, entry, recount, reconciliation, approval and posting. Extends INV 7.1/7.2 with the full document lifecycle the gap register found missing

7.1 Document Model

“Generate Count Sheets” on a Count Plan (INV 7.1) creates one Count Document per counting unit — by default one per warehouse zone per counter team (config: per zone / per aisle / per item group). Each document CD-{plan}-{SEQ:3} carries the line columns already specified (Location, Item, Material Code, Batch #, UoM, Expected Qty, Count Qty, Recount Qty) plus document-level state.

7.2 Lifecycle

Status	Color	Meaning / Allowed Actions	Transitions
Generated	Gray	Sheets created, expected quantities frozen (snapshot timestamp stored). Assign counters, print, void	Assigned, Void
Assigned	Blue	Counter(s) locked in; mobile entry opens for those users only	Counting
Counting	Yellow	Count entry in progress; supervisor may reassign; entries autosave	Submitted
Submitted	Purple	Counter finished; lines with variance beyond recount threshold auto-flag Recount Required	Recount, Variance Review
Recount	Amber	Second count by a different counter (enforced: recounter ≠ original counter, rule R-CNT-02); blind to first count	
Variance Review	Amber	Supervisor reconciles per §7.4; approves or rejects lines	Approved, Counting (rejected lines reopen)
Approved	Green	All variances approved; ready to post	Posted
Posted	Dark Green	Stock adjustments created and posted; document immutable	Terminal
Void	Red	Cancelled before counting; reason mandatory	Terminal

Dev Note: Inventory freeze: while any count document over a location is in Generated→Variance Review, transactional movements on that location are blocked (E-CNT-01 “Location {loc} is frozen for counting under {doc}”) unless the plan was created with Soft Freeze = OFF blocking disabled, in which case movements post but the document line is flagged Moved During Count and forces recount.

7.3 Role-Based Access

Role	Rights
Counter	Sees only documents assigned to them, only in Counting/Recount; enters counts; never sees Expected Qty when plan is Blind (server-side enforcement, not CSS hiding); cannot see variance columns
Count Supervisor (Warehouse Manager)	Generate, assign, reassign, view expected and variances, run reconciliation, approve within tolerance, reject lines

Role	Rights
Finance Approver (Finance Manager)	Second approval for variances beyond value tolerance (§7.5); views valuation impact
Inventory Admin	Everything incl. void; cannot count documents they generated when SOD control is ON

7.4 Reconciliation Logic

1. Final Qty per line = Recount Qty when a recount exists, else Count Qty. If recount differs from first count by more than the recount threshold again, a third count is forced with supervisor + finance jointly present (rule R-CNT-03); third count is final.
2. Variance = Final Qty − Expected Qty; Variance Value = Variance × current valuation rate; both computed at reconciliation, not entry, so late valuation changes do not skew entered documents.
3. Movements posted between snapshot and entry (Soft Freeze mode) are replayed: Expected Qty is adjusted by the net of interim movements before variance is computed; the adjustment is shown as a separate “Interim Movements” column for audit.
4. Batch/serial reconciliation: for batch lines the count is per batch; a batch found that is not on the sheet is added as an Unlisted Batch line (requires supervisor confirmation and creates batch record correction). Serials are scanned; missing serials list and found-but-unexpected serials list are generated automatically.
5. Zero-count confirmation: a count of 0 against non-zero expected requires an explicit “Confirm Empty Location” action, not just a blank.

7.5 Tolerance and Approval Rules

Rule	Default	Behavior
Recount threshold	±2% or ±$500 per line	Beyond either → recount mandatory
Supervisor approval band	Variance value ≤ $2,000 per line and ≤ $10,000 per document	Supervisor approves alone
Finance approval band	Above supervisor band, or any ABC class-A item variance	Sequential approval: Supervisor → Finance Manager
Controller band	Document net variance > $25,000	Third approval by Controller
ABC-driven tolerances	A: 0.5% / B: 2% / C: 5% qty tolerance	Within tolerance lines auto-approve at reconciliation (flagged Auto-Approved)

7.6 Posting Rules

On Post, the system creates one Stock Adjustment (INV 5.8) per count document, adjustment source = Physical Count / Cycle Count, lines for every non-zero variance, and posts it immediately (approvals already satisfied at the count layer — no second approval loop).

Accounting Entry — Count Variance — Surplus

Debit	Credit	Amount Basis	Notes
1310/1320/1330 Inventory (by item type)	5310 Inventory Adjustment	Surplus Qty × valuation rate	Cost center: warehouse cost center; Profit center: item product line

Accounting Entry — Count Variance — Shortage

Debit	Credit	Amount Basis	Notes
5310 Inventory Adjustment	1310/1320/1330 Inventory (by item type)	Shortage Qty × valuation rate	Same dimensions

Posted documents write back to the Count Plan: plan reaches Posted only when all its documents are Posted or Void. The Real-Time Stock Ledger (INV 15.12) shows the adjustment with reference CD → SA → JE for a complete drill path.

8. Cost Variance Analysis

Screen ID	INV-VALUATION-CVA-001
URL Pattern	/inventory/valuation/cost-variance
Purpose	Single workbench for every cost variance the system generates: purchase price, invoice price, material usage, labor rate, labor efficiency, overhead absorption, subcontract. Referenced by the Pricing & Costing tab (INV 2.3.3 E); this section defines the formulas, drill-downs, layouts, postings and edge cases the gap register found undefined

8.1 Variance Types and Formulas

Code	Variance	Formula	Posted When	GL (DR if unfavorable)
PPV	Purchase Price Variance	(PO unit price − standard cost) × received qty	GRN completion (standard-cost items)	5210 vs 2110 GR/IR path per INV Sec 12 #3
IPV	Invoice Price Variance	(Invoice price − PO price) × invoiced qty	Three-way match posting (ACC 6)	5210 (sub-tagged IPV) vs 2110
MUV	Material Usage Variance	(Actual component qty issued − standard qty per released MBOM × good output) × component standard cost	Production order settlement (Part D §8)	5220 vs 1320 WIP
LRV	Labor Rate Variance	(Actual labor rate − work center standard rate) × actual hours	WO settlement	5230 vs 1320
LEV	Labor Efficiency Variance	(Actual hours − standard hours for good output) × standard rate	WO settlement	5240 vs 1320
OAV	Overhead Absorption Variance	Applied overhead (std rate × std driver for output) − actual overhead charged to order	WO settlement	5250 vs 1320
SPV	Subcontract Price Variance	(Subcontract invoice − standard subcontract cost) × qty	Subcontract GRN/invoice	5260 vs 2110
RVV	Revaluation	On-hand qty × (new std − old std)	Standard cost release (INV 8.2)	5330 vs 1310/1330

Sign convention everywhere: **positive = unfavorable** (actual exceeded standard). Favorable variances post as credits to the same accounts. All variance postings carry cost center (from work center for LRV/LEV/OAV, from warehouse for PPV/RVV, from production cost center for MUV) and profit center (product line of the parent item).

8.2 Workbench Layout

Zone	Content
Filter bar	Period (month, range), Entity, Variance types (multi), Item / Item Group, Vendor (PPV/IPV/SPV), Work Center (labor/overhead), Production Order, Cost Center, Profit Center, only-above-threshold toggle
Summary tiles	One tile per variance type: period total with favorable/unfavorable color, % of standard cost base, trend sparkline (6 periods)

Zone	Content
Main grid	Grouped by variance type → item (or work center for labor). Columns: Period, Item/WC, Standard basis, Actual, Variance $, Variance %, Transaction count, GL entry link. Column sets switch per variance type automatically
Drill-down chain	Tile → type; row → transaction list (each GRN/WO/invoice with its contribution); transaction → source document; GL link → journal entry. Every number on the screen is reachable to the posting level in ≤ 3 clicks
Charts	Pareto of top 10 items by absolute variance; monthly trend line per type

8.3 Report Layouts (Export)

Report	Layout
CVA-R1 Variance Summary by Type	Period columns (13), rows = variance types, totals row; Excel + PDF
CVA-R2 PPV by Vendor	Vendor rows; received value at std, at actual, PPV $, PPV %, top contributing POs expandable
CVA-R3 Production Variances by Order	WO rows: output qty, std cost of output, actual cost, MUV/LRV/LEV/OAV split, total variance, settled date; subtotals by product line and by work center
CVA-R4 Standard vs Actual by Item	The Pricing & Costing tab sub-section E table (Period, Standard, Actual Avg, Variance, %, GL posting ref) across all items with filters

8.4 Edge Cases

Case	Rule
Missing standard cost	Receipt/settlement does not block; system uses last released standard; if none has ever existed it posts at actual with variance $0 and writes the item to the CVA Exception List (“No standard — costing run required”), alert to Cost Accountant
Back-dated transactions	A transaction dated into a Closing/Closed inventory period follows Fiscal Period Control (§9): blocked, or posted to first open period with original date stored as reference date. Variance always posts to the same period as its source transaction — never split
Zero/negative quantities	Return-to-vendor and reversals generate mirror-image variance postings referencing the original variance line (flag Reversal)
Currency	Variance computed in entity currency using the source document’s booked rate; no retrospective FX restatement (FX effects live in ACC 7100, never in 52xx)
Partial deliveries / partial invoices	PPV recognized per receipt; IPV per invoice match on matched quantity only
Standard change mid-period	Receipts before release use old standard, after use new; RVV posts once at release for on-hand stock (INV 8.2 rule) — the workbench shows both segments of the period split by effective date

9. Fiscal Period Control (Inventory & Operations)

Screen ID	INV-CONFIG-PERIOD-001
URL Pattern	/inventory/configuration/fiscal-periods
Purpose	Operational period gate for all stock-relevant modules, aligned one-to-one with the finance fiscal periods (ACC 26.1). Defines exactly which transactions each period status blocks, the open/close sequence, back-posting policy, permissions and error messages

9.1 Model

- Periods are owned by Finance (ACC 26.1: Open / Soft Close / Hard Close / Year-End Closed). The inventory statuses Open / Closing / Closed shown on INV 8.3 map to Open / Soft Close / Hard Close respectively — one underlying period record per entity per month, two presentations. No independent inventory calendar exists.
- Every posting transaction in every module validates the period of its **posting date** at save and again at post (race-safe: validation inside the posting transaction).
- Subledger-before-GL sequence: the inventory period must reach Closed (Hard) before the finance period can Hard Close — enforced by ACC 26.2 checklist step 6 (subledger reconciliation) plus a new step 6b “Inventory period closed”.

9.2 Module Behavior Matrix

Module / Transaction	Open	Closing (Soft)	Closed (Hard)
GRN / Check-In, RMA receipt	Post	Post with warning banner	Blocked
Check-Out / issues, transfers, scrap	Post	Post with warning	Blocked
Stock adjustments, count postings	Post	Post with warning; count postings for the closing period allowed (that is their purpose)	Blocked
Production confirmations, WO settlement	Post	Settlement allowed (period-end activity); new confirmations warned	Blocked
Standard cost release / revaluation	Post	Blocked (must land in an open period)	Blocked
Sales PGI / shipment	Post	Post with warning	Blocked
Invoices / credit notes (ACC)	Post	Per ACC soft-close rule	Blocked per ACC
Master data changes	Always allowed — masters are not period-controlled	—	—
Reservations, picklists, packing (no GL)	Always allowed — logistics documents without value posting are not period-blocked; only their posting steps are	—	—

9.3 Open/Close Sequence

1. Day −2: Finance sets period to Soft Close. All operational posting continues with warnings; teams clear WIP documents (Draft GRNs, open counts, unsettled WOs) driven by the Pre-Close Validation list (INV 8.3 step 1).
2. Day +1 to +3: cutoff cleanup — receipts dated correctly (goods physically received before month-end but keyed after are posted with posting date = actual receipt date, allowed during Soft Close with warning).
3. Inventory close run (INV 8.3): pre-close validation must return zero blockers (list with document links; each row has an owner and a Resolve button), valuation snapshot generated, closing JE posted, status → Closed.
4. Finance completes ACC 26.2 checklist; period Hard Closes. Next period was already opened automatically at first Soft Close (always exactly one Open and at most one Soft Close period per entity — rule R-PER-01).
5. Reopen: Closed → Soft Close requires Controller + CFO (ACC roles); the reopen reason and every posting made during a reopen window are tagged Reopened Posting on the audit trail and listed in the close report.

9.4 Back-Posting Policy

Scenario	Policy
Posting date in current Soft Close period	Allowed for all operational roles, warning banner “Period {p} is closing — posting requires month-end review”, logged
Posting date in prior Closed period	Blocked for everyone. No exceptions at the transaction layer. Corrections happen as current-period postings with Reference Date = original date (field available on adjustments, GRN and JE) so operational reports can restate while GL integrity holds
Posting date in future period	Blocked beyond next Open period (E-PER-04) to prevent fat-finger year typos

9.5 Permissions and Standard Errors

Item	Specification
Set Soft Close / reopen to Open	Controller
Set Hard Close	Controller after checklist green; system enforces prerequisite completion
Reopen Hard Closed	Controller + CFO dual action (two distinct users, sequential confirm)
E-PER-01	“Posting date {date} falls in period {period}, status Closed. Post to the current open period and set Reference Date if needed.”
E-PER-02	“Period {period} is in Soft Close. Your posting will be included in month-end review.” (warning, not blocking)
E-PER-03	“Inventory period cannot Hard Close: {n} unresolved documents. Open the Pre-Close Validation list.”
E-PER-04	“Posting date {date} is beyond the next open period.”

10. Safety Stock

Screen ID	INV-PLAN-SAFETY-001
URL Pattern	/inventory/planning/safety-stock
Purpose	Definition, maintenance and planning behavior of buffer stock, per item per warehouse, including overrides, alerts and the exact arithmetic used by replenishment and MRP

10.1 Where Safety Stock Lives and Who May Edit It

Safety Stock (Qty + UoM) exists per item **per warehouse** (the single item-master field from INV 2.2 becomes the default seeded into each warehouse row). The maintenance screen is a grid — one row per item/warehouse — editable inline by Production Planner and Inventory Admin; changes above 25% delta route to Planning Manager approval. Every change is audited with reason code (Demand Change, Lead Time Change, Service Level Change, New Item, Annual Review).

10.2 Planning Arithmetic

Rule	Specification
Reorder point	ROP = (Average Daily Usage × Replenishment Lead Time Days) + Safety Stock. ADU from trailing 90-day consumption (INV 9.1 field); lead time = vendor lead time for buy items, production lead time for make items
Replenishment trigger	Available (On-hand − Reserved − QI Hold − Blocked) + On-Order ≤ ROP → reorder proposal for (Max Stock Level − Available − On-Order), minimum = Reorder Qty, rounded to order multiple
MRP treatment	Safety stock is a floor on projected available balance: net requirement in bucket t = max(0, gross requirement + safety stock − projected available). MRP never plans to consume below safety stock; a projected dip below SS raises exception message SS-BREACH on the MRP Workbench (Part D §3) instead
Static vs statistical	Phase 1: static values maintained here. Phase 2 (flagged, not in current build): suggested SS = Z(service level) × σ(daily demand) × √(lead time); screen already reserves read-only columns Suggested SS and Service Level for that engine
Expiry-managed items	For FEFO shelf-life items, SS proposals cap at (shelf life ÷ 2) × ADU to avoid buffering into expiry write-offs; warning W-SS-02 when a manual value exceeds the cap

10.3 Overrides

Override	Specification
Per-warehouse	The grid row itself; blank row inherits the item default and shows it grayed with “(default)”
Per-customer reservation buffer	Optional rows: Customer + Item + Warehouse + Reserved Buffer Qty + Valid From/To. Implemented as a standing Material Reservation (INV 5.5, source type = Customer Buffer) auto-maintained by the system: the reservation renews as stock is consumed against it. Used for utility frame agreements guaranteeing spare availability. Reservation reduces Available like any other
Seasonal profiles	Optional 12 monthly multipliers per row (default 1.0); effective SS in month m = base SS × multiplier[m]

10.4 Alerts, Exceptions and Reporting

- Alert “Below Safety Stock” (extends INV 15.11 triggers): Available < effective SS → Critical severity, notify Planner + Purchasing real-time; distinct from Below Reorder Point which remains a Warning.
- Stock Projection Report (INV 15.10) draws the safety stock line using the effective (seasonal) value; projected crossings listed under the chart with dates.
- Exception report SS-EX-01: items whose SS > Max Stock Level, SS with zero ADU (dead buffer), SS never breached in 12 months (over-buffered candidates), SS breached ≥ 3 times in 90 days (under-buffered).
- Dashboard tile on Inventory Dashboard: count of active SS breaches with drill-down.

11. Tax Configuration

Screen ID	INV-CONFIG-TAX-001
URL Pattern	/inventory/configuration/tax (sidebar Configuration → Tax Configuration)
Purpose	The operational tax determination layer that connects Material Master tax data, customer/vendor tax profiles and the Avalara engine (ACC Section 19). ACC 19 owns tax codes, GL mapping and filing; this screen owns determination policy and region handling for documents created in logistics modules

11.1 Determination Precedence

Every taxable document line (quotation, SO, invoice request, vendor bill from PO) resolves tax input parameters in this order — each step only fills parameters the previous steps left empty:

1. Document-line manual tax override (role-limited: Finance only; reason mandatory; flagged on the document and in the Avalara Transaction Log as Override).
2. Customer exemption certificate (ACC 19.2): a Verified, unexpired certificate for the ship-to state sets exemption code → Avalara returns $0 for exempt lines.
3. Customer/vendor tax profile (§11.3): default tax code, nexus-relevant registration numbers, use-tax accrual flag for vendors.
4. Item tax data: Tax Category (Taxable / Tax-Exempt / Zero-Rated) and Avalara product tax code from the Sales tab (§4.1); HSN/HTS code for customs classification.
5. Jurisdiction: ship-from (warehouse address) + ship-to (document address) sent to Avalara for rate lookup; PE nexus configuration (§11.4) decides whether tax is collected at all.
6. Failover: Avalara unreachable → tax code override rate (ACC 19.1.1 Rate Override) with Override badge; documents post and are queued for recalculation (background retry every 15 min; discrepancies on recalc create tax adjustment JEs and appear in the Tax Reconciliation report ACC 19.3).

11.2 Configuration Screen Fields

Field / Panel	Specification
Tax Engine Mode	Avalara (default) / Manual Tables. Manual mode exposes a rate table (Jurisdiction, Rate %, Valid From/To) for pre-integration operation and test systems
Default Tax Codes	Per document family: Sales (S1), Purchases (U1 use tax), Intercompany (E-IC exempt), RMA credits (inherit original invoice's codes — hard rule, not configurable)
Calculation Trigger Points	Quote save (estimate), SO save (estimate), Invoice post (commit), Credit note post (ReturnInvoice), Void (DocVoided) — mirrors ACC 19.1.2; checkboxes allow disabling estimates (not commits) for performance
Rounding	Line-level rounding, half-up, to currency minor unit; document total = sum of rounded lines (matches Avalara)
Freight Taxability	Per state matrix maintained by Avalara; manual override list for Manual mode

11.3 Customer / Vendor Tax Profile (new tab on both masters)

Field	Type	Req	Specification
Tax Registration # (per jurisdiction)	Table	No	State + registration/VAT number; printed on invoices where required

Field	Type	Req	Specification
Default Tax Code	Dropdown	No	Fallback when line-level determination yields nothing
Exemption Certificates	Embedded view	—	Read-only embed of ACC 19.2 records for this customer; Add button deep-links there
Use Tax Accrual (vendor)	Toggle	No	ON: bills from this vendor with no tax charged auto-accrue use tax (DR 7400 or inventory / CR 2300-use) per ACC 19.3 Use Tax report
Tax Address Role	Dropdown	Yes	Which address drives jurisdiction: Ship-To (default) / Bill-To (services)

11.4 Nexus and Region Handling

Element	Specification
Nexus register	Grid: State, Entity, Nexus Type (Physical / Economic), Effective From, Registration #, Filing Frequency. Only states present here collect tax; sales into non-nexus states post tax $0 with reason Non-Nexus (visible in Avalara log)
Economic nexus monitor	Background job tracks rolling 12-month sales $ and transaction counts per state vs thresholds table (maintained annually); crossing 80% of a threshold raises an alert to Controller to register
International	Phase 1 scope is US + Canada. Canadian GST/HST handled as jurisdictions with registration in the nexus register; all other exports = Zero-Rated with export documentation checklist on the shipment

11.5 Rate Maintenance Workflow and Retroactive Changes

- Avalara mode: rates are never maintained in PE ERP; the Manual rate table is read-only and grayed. Monthly, the Tax Reconciliation report (ACC 19.3) is reviewed; unexplained GL-vs-Avalara differences become tax adjustment JEs (approval: Controller).
- Manual mode: rate rows are effective-dated; edits create new rows (history immutable); changes route to Controller approval; documents always use the rate valid on their tax point date (invoice date).
- Retroactive policy — the rule the gap register asked for: **posted documents are never restated by configuration changes.** A rate/nexus/exemption change applies only to documents whose tax point date is on or after the change effective date. Corrections to already-posted documents are made exclusively by credit note + rebill (which recalculate at their own dates with reference to the original), or by a tax-only adjustment JE for remittance corrections. The configuration audit trail stores effective date, changed by, and the first document affected.

PART B

ORDER MANAGEMENT (SALES)

Order Management carries a transformer sale from first inquiry to shipped, invoiced and recognized revenue, and carries component/spares sales through a fast make-to-stock path. It owns the Customer Master commercial view, Quotations, Sales Orders, Deliveries & Shipments, and the handoffs: to Engineering for ETO design, to Production for order-pegged manufacturing, to Inventory for reservation/pick/pack/ship, to Accounting for invoicing (ACC 3) and revenue recognition (ACC 24), and to RMA for returns (Part A §3).

1. Order Types

Order Type	Path	Characteristics
MTS — Make/Sell from Stock	SO → reservation → pick → pack → ship → invoice	Components, spares, standard distribution transformers held in FG. ATP-checked, fastest cycle
ETO — Engineer to Order	Quote (with engineering estimate) → SO → design release (Part C) → production order pegged to SO line → final test → ship → milestone invoicing	Power and custom pad-mount transformers; every unit serialized; drawings approved by customer before winding starts (approval milestone on the order)
MTO — Make to Order (standard design)	SO → production order pegged, no new engineering	Catalog designs manufactured on demand
Service	SO with service lines (no inventory) → work performed → invoice	Field service, testing services, billable repair invoicing from RMA RC4
Replacement (no charge)	Zero-price SO created by RMA disposition Replace	Carries mandatory RMA reference; blocks manual creation without RMA link (rule R-OM-04)

2. Customer Master — Commercial View

Screen ID	OM-CUST-001 (listing), OM-CUST-002 (detail)
URL Pattern	/sales/customers
Purpose	Single customer record consumed by Sales, Accounting (AR), Tax and RMA. Accounting fields (payment terms, AR balances) render from ACC data; this screen owns the commercial attributes

2.1 Detail Tabs and Fields

Tab	Fields (Field / Type / Req / Specification)
General	Customer # (auto CUST-{SEQ:5}); Legal Name (text, req); DBA; Customer Class (dropdown, req: Investor-Owned Utility, Municipal Utility, Co-op, EPC Contractor, Distributor, OEM, Industrial, Government); Market Segment; Parent Customer (lookup — corporate families roll up exposure); Status (Prospect / Active / On Hold / Blocked / Inactive); Owner (sales rep)
Addresses	Unlimited addresses typed Bill-To / Ship-To / Both; one default each. Ship-To rows carry: site name, address, delivery instructions, dock constraints (crane on site Y/N, max vehicle), tax jurisdiction preview. Validations: at least one Bill-To and one Ship-To before status Active (V-OM-01)
Contacts	Name, title, role tags (Purchasing, Engineering, AP, Receiving), email, phone; primary flag per role
Commercial	Price List (lookup, req for Active); Currency; Payment Terms (from ACC 14 terms master); Incoterms default (EXW/FOB/DAP/DDP + named place); Freight Terms default; Invoice Delivery (email PDF / portal / EDI-phase 3); Invoice consolidation rule (per shipment / per order / monthly)
Credit	Credit Limit (currency, Controller-owned field); Credit Exposure (calc = open AR + open orders not yet invoiced + in-transit shipments); Available Credit; Risk Class (A–D); Credit Hold flag (manual) + auto-hold rule result; D&B / rating reference fields; Last Review date + reviewer. All edits here restricted to Finance roles; Sales sees read-only
Tax	Tax profile per Part A §11.3 (registration numbers, exemption certificates embed, tax address role)
Sales History	Read-only: orders, revenue by year, margin by year, open quotes, RMAs, on-time delivery % to this customer
Documents / Activity	Contracts, frame agreements, W-9 requests; immutable audit trail

2.2 Credit Management Rules

- Auto credit check runs at: quotation → order conversion, SO submit, any SO change increasing value, and shipment release (final gate).
- Check: Exposure + new order value ≤ Credit Limit, and no invoices > 30 days past due (config). Fail → order enters Credit Hold status; Finance queue (ACC 22 approval queue, type Credit Release) can Release, Reject, or Raise Limit (Controller).
- Shipment-release gate cannot be bypassed by Sales roles; Controller release is logged on the order Activity tab.
- Government/municipal class customers can be flagged Exempt from Auto Hold (Controller only).

3. Quotations

Screen ID	OM-QT-001 (listing), OM-QT-002 (create/detail)
URL Pattern	/sales/quotations
Purpose	Priced, versioned offers; for ETO, the container for the engineering estimate and technical configuration that later drive design and costing

3.1 Header Fields

Field	Type	Req	Specification
Quotation #	Auto	Auto	QT-{YYYY}-{SEQ:5}; version suffix -V2, -V3 on revisions
Customer / Prospect	Lookup	Yes	Prospect status customers allowed on quotes (not on orders)
Customer RFQ Reference	Text	No	Customer bid/RFQ number; printed on quote
Quote Type	Dropdown	Yes	Budgetary (±25%, watermarked BUDGETARY, no ATP), Firm (binding, full costing)
Valid Until	Date	Yes	Default +30 days (config per quote type); expiry auto-status
Requested Delivery	Date	No	Drives feasibility check against production capacity (Part D §11) for ETO
Incoterms / Freight	Dropdowns	Yes	Default from customer; quote-level override
Sales Rep / Estimator	Users	Yes/Cond.	Estimator mandatory on ETO quotes
Win Probability %	Number	No	Pipeline reporting
Loss Reason	Dropdown	Cond.	Mandatory when status → Lost: Price, Lead Time, Technical, Relationship, No-Bid, Other + notes

3.2 Line Fields

Field	Type	Specification
Line Type	Dropdown	Catalog Item / ETO Configured / Service / Text-only
Item	Lookup	Catalog & service lines. ETO lines may reference a template item (e.g. TX-PAD-TEMPLATE) before a real item exists
ETO Configuration Panel	Structured attributes	Visible for ETO lines: kVA/MVA rating, primary voltage, secondary voltage, phases, frequency, impedance %, cooling class (ONAN/ONAF…), tap range, BIL, altitude, temperature rise, K-factor, efficiency standard (DOE 2016/CSL), bushing config, accessories checklist. Attributes save as structured JSON, print on the quote technical data sheet, and copy forward to the SO and the engineering request (Part C §2)
Qty / UoM	Number	—
Unit Price	Currency	From Pricing Service (Part A §4.4); ETO lines from Cost Estimate + margin (below)
Cost Estimate Link	Link	ETO/Firm: mandatory link to a Cost Estimate CE-# (Part I §3). Quote margin panel shows estimate cost, price, margin %, floor check
Commodity Surcharge	Calc lines	Copper/CRGO escalator lines per Part I §6.4 with index basis and date shown

Field	Type	Specification
Lead Time (Wks)	Number	Quoted production + delivery lead time per line
Discount %	Number	Role-limited; below-floor → approval

3.3 Statuses, Versioning, Outputs

Element	Specification
Lifecycle	Draft (gray) → Internal Review (yellow: estimator + pricing approval when margin below target) → Issued (blue, PDF sent, content locked) → statuses Won (green) / Lost (red) / Expired (gray) / Superseded (when revised)
Revisions	Revise button on Issued quote → full copy as -V(n+1) in Draft; predecessor → Superseded, chain visible; diffs viewable side-by-side
Quote PDF	Entity letterhead; commercial page (lines, prices, surcharges, totals, terms, validity); technical data sheet per ETO line from the configuration attributes; standard terms & conditions appendix (versioned document, Part C §5); signature block
Convert to Order	Won → Create Sales Order: copies lines, frozen prices (Price Source = Quotation), configuration, estimate links; quote locks with SO cross-reference

4. Sales Orders

Screen ID	OM-SO-001 (listing), OM-SO-002 (create/detail)
URL Pattern	/sales/orders
Purpose	The commercial contract of record. Every downstream document — reservation, production order, delivery, invoice, revenue contract — hangs off an SO line

4.1 Header Fields

Field	Type	Req	Specification
SO #	Auto	Auto	SO-{YYYY}-{SEQ:5}
Customer	Lookup	Yes	Active customers only (E-OM-02 otherwise)
Customer PO #	Text	Yes	Mandatory (utility buyers always issue POs); uniqueness warning per customer
Quote Reference	Link	Cond.	Mandatory when converted from quote; manual orders above $250k without a quote require Sales Director approval (rule R-OM-02)
Order Type	Dropdown	Yes	MTS / MTO / ETO / Service / Replacement (§1)
Bill-To / Ship-To	Lookups	Yes	From customer addresses; ship-to editable per line for multi-site orders
Requested Ship Date	Date	Yes	Header default; per-line override
Incoterms / Freight Terms / Carrier	Dropdowns	Yes	Defaults from customer
Payment Terms	Dropdown	Auto	From customer; edit requires Finance role
Billing Plan	Dropdown	Yes	On Shipment (default MTS) / Milestone (default ETO — §4.4) / On Delivery Confirmation
Currency / Rate	Auto	Auto	Customer currency; booked rate at order date displayed
Status	Badge	Auto	§4.6

4.2 Line Fields

Field	Type	Specification
Line #	Auto	10, 20, 30… (renumber-safe references)
Item / Customer Part #	Lookup	Either enters; cross-reference resolves (Part A §4.1). Sellable=OFF blocks (E-SAL-02)
Description Override	Text	Prints instead of item description when set
ETO Configuration	Panel	Copied from quote; editable until Design Release milestone, then locked — changes only via Sales Change Order (§4.5)
Qty / UoM	Number	Min-order-qty check (Part A §4.1)
Unit Price / Surcharges / Discount	Currency	Per Pricing Service; frozen from quote when referenced
Requested / Promised Ship Date	Dates	Promised set by ATP (MTS) or scheduling (MTO/ETO); customer-visible commitment

Field	Type	Specification
Warehouse	Dropdown	Fulfilling plant/warehouse
Warranty Months Override	Number	Item default → upward override with CS Manager approval
Serial Assignment	Panel	ETO/MTO: production serial(s) attach automatically at WO release; MTS serialized: picked serials attach at PGI
Revenue Contract Link	Link	Auto-created ACC 24 contract for Milestone billing plans; line maps to performance obligations
Line Status	Badge	Open / In Engineering / In Production / Ready to Ship / Partially Shipped / Shipped / Invoiced / Closed / Cancelled

4.3 Availability & Supply Pegging

Order Type	Behavior at Submit
MTS	ATP check per line: Available + scheduled receipts before requested date − prior commitments. Full → auto Material Reservation (INV 5.5, source = SO). Partial → dialog: accept later promised date / split line (ship available now, balance later) / backorder. Promised Ship Date written back
MTO / ETO	No ATP; line spawns demand: ETO → Engineering Request (Part C §2) then production order at design release; MTO → planned order → WO (Part D). WO carries SO line peg; peg shown both sides; FG receipt from a pegged WO auto-reserves to the SO line
Service	No supply; optionally links a service work record

4.4 Milestone Billing Plan (ETO)

Milestone plans generate invoice requests into ACC 3 and map one-to-one onto the revenue contract (ACC 24) performance obligations. Default template (editable per order, must total 100%):

Milestone	Default %	Trigger	Evidence Attachment
Order Acceptance / Down Payment	20%	SO status Confirmed	Signed order confirmation
Design Approval	10%	Customer drawing approval milestone complete (Part C §4)	Approved drawing transmittal
Major Materials Procured	20%	All long-lead POs (core steel, copper, bushings, tap changer) confirmed	PO confirmations report
Production Complete / Tested	30%	Final Test passed (FTR issued)	Certified test report
Shipment	15%	PGI posted	BOL
Delivery / Commissioning Support	5%	POD or site acceptance	POD

> Dev Note: Milestone completion creates an Invoice Request in a Finance queue rather than auto-posting the invoice — Finance reviews evidence and posts (ACC 3 flow). Revenue recognition follows the ASC 606 contract (ACC 24), not the billing plan; deferred/unbilled mechanics per ACC 24.2 apply automatically.

4.5 Order Change Management

Change	Rule
Before Confirmed	Free edit in Draft

Change	Rule
After Confirmed, commercial fields (qty, dates, prices)	Sales Change Order dialog: captures old/new, reason, initiator; reprices if needed; re-runs credit check when value increases; re-promises dates; both versions retained; customer-facing Change Confirmation PDF
After Confirmed, ETO technical configuration	Locked once Design Release milestone passes. Change requires ECR (Part C §4) with cost/schedule impact quote to customer; accepted impact flows back as a Sales Change Order referencing the ECO
After any shipment	Shipped lines immutable; only new lines or cancellation-of-remainder (with restocking/cancellation fee logic per contract terms)
Cancellation	Full/line cancellation captures cancellation fee % (contract-based); fee invoices through a service line; pegged WOs get Cancel signal — WO in progress requires Production Manager disposition (finish to stock / stop and scrap / stop and hold) before SO line closes

4.6 Status Lifecycle

Status	Color	Meaning / Exits
Draft	Gray	Editing; → Submitted
Submitted	Yellow	Validations + credit check; → Confirmed / Credit Hold / Rejected
Credit Hold	Red	Awaiting Finance release; → Confirmed / Cancelled
Confirmed	Blue	Contract in force; supply pegged; milestones live; → In Fulfillment
In Fulfillment	Purple	Any line in engineering/production/picking; → Partially Shipped / Shipped
Partially Shipped	Teal	At least one PGI; remainder open
Shipped	Green	All lines PGI'd; → Invoiced when billing complete
Invoiced / Closed	Dark Green	All billing posted, revenue contract settled; terminal (Reopen: Controller)
Cancelled	Red	Terminal; cancellation fee logic applied

5. Deliveries & Shipments

Screen ID	OM-SHP-001 (listing), OM-SHP-002 (detail)
URL Pattern	/sales/shipments
Purpose	Groups SO lines into physical outbound events; owns pick/pack orchestration, pre-shipment QC gate, post goods issue and shipping documents

5.1 Flow and Fields

1. Create shipment from one or more SO lines (same ship-to, same date window). Header: SHP-#, SO refs, ship-to, planned ship date, carrier, Incoterms, vehicle/equipment type (flatbed, step-deck, container), permits required flag (over-dimensional loads: state permits checklist for large transformers).
2. Release to warehouse → generates Picklist(s) (INV 5.7) → Packing (Part A §5) → HUs assigned to the shipment.
3. Pre-Shipment Inspection gate (INV 15.3) for FG transformers: shipment cannot post while PSI is open or failed (E-OM-07). Passing PSI auto-generates the CoC (INV 15.5).
4. Credit gate re-check (final), then **Post Goods Issue**: stock leaves FG, serials → Shipped with customer + site written, HUs → Shipped, SO lines update, invoice request fires per billing plan.
5. Documents produced: Bill of Lading (VICS format), Packing List (Part A §5.4), CoC, test reports bundle (FTR), country-specific export docs when applicable. POD capture field (date, name, upload) closes the delivery leg and can trigger the final milestone.

Accounting Entry — Post Goods Issue (shipment of finished goods)

Debit	Credit	Amount Basis	Notes
5120 COGS — Finished Goods	1330 Finished Goods Inventory	Shipped Qty × standard cost (or valuation rate for non-standard items)	Cost center: none (COGS is PC-level); Profit center: product line of item. Serial-level cost detail retained for margin reporting

Accounting Entry — Freight accrual at PGI (PE-paid freight)

Debit	Credit	Amount Basis	Notes
6410 Freight Out	2100 AP (accrual, vendor = carrier)	Quoted freight cost	Reversed/replaced by carrier invoice at 3-way match

6. Sales Reports & Role Access

Report	Content
Open Order Book	All open SO lines: value, promised date, status, pegged supply status, late flags; grouped by customer / product line / month
Bookings / Billings / Backlog	Monthly BBB with product line and entity dimensions; the management headline report
On-Time Delivery	Promised vs actual PGI; OTD % by month, customer, product line
Margin by Order	Price vs standard cost at PGI vs actual cost at WO settlement (three-margin view: quoted, standard, actual)
Quote Funnel	Quotes by status/age, win rate by segment, loss reasons pareto
Late & At-Risk Lines	Promised date within 7 days without Ready-to-Ship, or pegged WO behind schedule (from Part D dispatch data)

Role	Rights
Sales Rep	Customers (commercial read/write), quotes, orders create/edit own; no credit fields; no cost visibility toggle (config per company: reps see margin Y/N)
Sales Admin / CS	All orders, change orders, RMA creation, shipment creation
Sales Director	Approvals (no-quote orders, discounts beyond rep limit), all reports
Finance (Controller/AR)	Credit fields, credit release, price-floor overrides, billing plan edits
Production Planner	Read orders, edit promised dates via scheduling only
Read-Only	View all, export none

PART C

ENGINEERING — BOM, CAD AND CHANGE CONTROL

Engineering owns the technical definition of every product: the engineering item, the Engineering BOM (EBOM), the controlled drawing set, and all change control. Production consumes a released Manufacturing BOM (MBOM) and routing; Costing rolls the MBOM; the Material Master BOM tab (INV 2.3.9) is a read-only window into what this Part produces. The design goal: an engineer can take an ETO order from Sales configuration to released drawings, EBOM and MBOM without leaving the module, and nothing unreleased can ever be bought, built or shipped.

1. Engineering Workspace

Screen ID	ENG-HOME-001
URL Pattern	/engineering
Purpose	Engineer’s landing page: work queue and design pipeline

Panel	Content
My Design Queue	Engineering Requests assigned to me: SO line, configuration summary, due date (from order promise back-scheduled), status, customer-approval flag
Open ECRs / ECOs	Change pipeline with aging and status
Awaiting My Approval	BOMs, drawings, ECOs where I am an approver
Recently Released	Last 20 released BOMs/drawings with links
Exceptions	Items used in released BOMs that lost their vendor (AVL empty), drawings referenced but not Released, placeholder parts older than 14 days

2. Engineering Request (ETO Intake)

Screen ID	ENG-REQ-001
URL Pattern	/engineering/requests
Purpose	The handoff object from a confirmed ETO sales order line to design

Field	Type	Req	Specification
Request #	Auto	Auto	ER-{YYYY}-{SEQ:4}
SO Line Reference	Link	Yes	Auto-created at SO Confirm for ETO lines; carries the full configuration attribute set (Part B §3.2) read-only
Design Type	Dropdown	Yes	New Design / Variant of Existing (select base design — copies its EBOM and drawing set as the starting point) / Repeat (released design, no engineering: request auto-completes and MBOM releases immediately)
Assigned Engineer(s)	Users	Yes	Electrical + mechanical designer roles
Design Due Date	Date	Auto	Back-scheduled: promised ship date − production lead time − procurement long-lead time; editable by Engineering Manager with schedule-impact warning to the SO
Customer Approval Required	Toggle	Yes	ON for utility customers (default from customer class): inserts the Drawing Approval milestone (transmittal → customer review → approved/returned-with-comments loop)
Deliverables Checklist	Checklist	Auto	Generated from design type: GA drawing, nameplate drawing, electrical schematic, core & coil design sheet, winding diagrams, tank/fabrication drawings, accessories list, test plan, EBOM, MBOM, cost estimate. Each row links to its object and shows status; request completes only when all rows done
Status	Badge	Auto	Open → In Design → Customer Review (optional loop) → Released → Closed; Blocked (with reason) available from any state

3. EBOM Editor — How the Engineer Fills the BOM

Screen ID	ENG-BOM-001 (list), ENG-BOM-002 (editor)
URL Pattern	/engineering/boms/{ebomId}
Purpose	Multi-level BOM authoring: structure, quantities, references to drawings, alternates, effectivity. This is the screen the gap instructions call “how the engineer will fill up the BOM”

3.1 Editor Layout

Zone	Specification
Left — structure tree	Multi-level tree (assembly → sub-assembly → component), drag-to-move with level revalidation, expand/collapse, level badges (1, 1.1, 1.1.1), per-node status dot (component item status)
Center — line grid	Children of the selected node as an editable grid (fields §3.2); inline add row; keyboard-first (Enter = next row, Tab = next cell)
Right — context panel	For the selected line: item card (stock, cost, AVL, lead time), linked drawing preview thumbnail, where-used, alternates editor
Header bar	EBOM # (EBOM-{SEQ:5}), Parent Item, Revision, Status, Base Design reference, buttons: Validate, Compare Revisions, Import, Copy From, Release Package

3.2 BOM Line Fields

Field	Type	Req	Specification
Find #	Number	Auto	Sequential position number; printed on drawings and pick documents; stable across edits
Component	Lookup + create	Yes	Search the Material Master; or **Create Placeholder Part** inline: engineer enters description, category, UoM, est. cost, target vendor → a Draft item is created (status Draft, flagged Engineering Placeholder) and an Item Creation Request task goes to Inventory Admin/Purchasing to complete sourcing data. Release of the EBOM is blocked while any placeholder remains Draft (V-ENG-03)
Qty Per / UoM	Number	Yes	Quantity per one parent unit; UoM from item; dual display when purchase UoM differs
Scrap %	Number	No	Expected process loss; consumed by MRP and costing (e.g. copper winding scrap 3%)
Reference Designators	Text	No	Position references (e.g. bushing H1/H2/H3, tap positions)
Phantom	Toggle	No	Phantom assemblies explode into parent demand (e.g. “hardware kit” groupings); phantoms cannot be stocked
Consume At Operation	Dropdown	Cond.	Routing operation where the component is issued (populated after routing exists; mandatory before MBOM release — V-ENG-05). Drives staged material issue on the floor
Drawing Reference	Lookup	Cond.	Link to controlled document (per §5); mandatory for fabricated/made parts (V-ENG-04)
Alternates	Sub-table	No	Approved alternate components with preference rank and conditions (e.g. alternate CRGO grade with loss delta note); MRP may not auto-substitute — alternates are planner-selected
Effectivity From / To	Dates or Serial	No	Line-level effectivity for running changes (see ECO §4.5)

Field	Type	Req	Specification
Make/Buy	Auto	Auto	From item; shown as badge (Make items must themselves carry a released EBOM before parent release — V-ENG-06)
Notes	Text	No	Line note; prints on MBOM and traveler when flagged Print

3.3 Authoring Aids

Function	Specification
Copy From	Copy an entire released EBOM (or subtree) of a similar design, then edit deltas; provenance stored (Base Design field) for design-reuse reporting
CAD BOM Import	CSV/XLSX import from CAD parts list: wizard maps columns (find #, part #, description, qty, refdes); matches items by material code / manufacturer part #; unmatched rows become placeholder-part proposals reviewed line-by-line before commit; import is transactional (all-or-nothing) with downloadable error log
Validate	Runs all V-ENG rules + circular-reference check + UoM conversion check + duplicate-component warning + qty=0 error; results panel with click-to-line
Compare Revisions	Side-by-side rev diff: added (green), removed (red), changed qty/fields (amber); exportable — this is the change-evidence document for ECO review
Where-Used	For any component: all EBOMs/MBOMs using it, with revision and status — mandatory pre-check surface for ECR impact analysis
Mass Update	Replace component A with B across selected BOMs (creates ECO automatically — cannot be done silently; §4)

3.4 EBOM Status Lifecycle

Status	Color	Rules
Draft	Gray	Free edit by assigned engineers
In Review	Yellow	Locked; reviewers comment (threaded per line); Approve → Released, Reject → Draft with comments
Released	Green	Immutable. Feeds MBOM generation (§6). Any change requires ECO. Approvers: Engineering Manager + (for designs with customer approval) evidence of customer drawing approval attached
Superseded	Blue	Replaced by later released revision; historical WOs keep their revision reference
Obsolete	Red	Design retired; blocked from new MBOMs/orders

4. Engineering Change Control (ECR / ECO)

4.1 ECR — Engineering Change Request

Screen ID	ENG-ECR-001
URL Pattern	/engineering/change-requests
Purpose	Anyone proposes a change; engineering evaluates impact; CCB decides

Field	Type	Req	Specification
ECR #	Auto	Auto	ECR-{YYYY}-{SEQ:4}
Requested By / Source	User + dropdown	Yes	Source: Engineering, Production, Quality (NCR link), Purchasing (obsolescence/AVL), Sales (customer change), RMA/Field Failure
Affected Objects	Multi-lookup	Yes	Items, EBOMs/MBOMs, drawings, routings; validated against Where-Used
Change Description / Reason	Textareas	Yes	What and why; classification dropdown: Safety/Regulatory, Cost Reduction, Quality/CAPA, Supply Continuity, Customer Request, Documentation Only
Impact Analysis Panel	Auto + manual	Auto	System computes and displays: open WOs using affected revisions (count + list), on-hand stock of affected components (qty + value), open POs (qty + value), open SO lines, cost delta (current rollup vs simulated rollup with proposed change — one-click simulation via costing engine Part I §3), affected customers (designs shipped). Engineer adds narrative + risk assessment
Priority	Dropdown	Yes	Emergency (safety — 24h CCB SLA), Urgent, Standard
CCB Decision	Panel	Auto	Change Control Board = Engineering Manager + Quality Manager + Production Manager + (cost delta > $10k) Controller. Votes recorded; outcome Approve → spawns ECO; Reject / Defer with reason

4.2 ECO — Engineering Change Order

Field / Panel	Specification
ECO #, ECR link	ECO-{YYYY}-{SEQ:4}; one ECO may implement several ECRs
Task Checklist	Generated from affected objects: revise EBOM X to Rev n+1, revise drawing DOC-…, update routing, update inspection plan, run costing simulation, update MBOM, notify purchasing, customer re-approval (when the design was customer-approved — mandatory task auto-added). Each task has owner + due date + status; ECO cannot close with open tasks
Effectivity Decision	§4.5 — date, serial, or immediate
Old-Stock Disposition	Mandatory decision per affected component with on-hand > 0: Use-Up (consume remaining then switch — system watches stock and flips revision automatically at zero), Rework (creates RWO), Scrap (posts 5320 with approval per INV thresholds), Return to Vendor (RTV)
Open-WO Disposition	Per open WO on old revision: Complete As-Is / Incorporate Change (WO gets change task + updated pick list deltas) / Stop & Rework
Release Action	Atomic cutover: all new revisions (EBOM, MBOM, drawings, routing, plans) release together at effectivity; partial release impossible

4.5 Effectivity Rules

Mode	Specification
Date effectivity	New revision governs all documents created on/after the date; in-flight documents keep old revision unless WO disposition says otherwise
Serial effectivity	ETO/MTO: “effective from serial PE-TX-2026-0142 onward”; the WO for that serial and later pull the new revision; traceability reports show revision per serial
Immediate (Emergency)	All open WOs blocked instantly pending disposition; used for safety/regulatory changes; requires Quality Manager + Engineering Manager joint release

5. CAD & Controlled Document Management

Screen ID	ENG-DOC-001 (register), ENG-DOC-002 (detail)
URL Pattern	/engineering/documents
Purpose	Controlled storage, revisioning and linking of drawings and technical documents. This is the system of record the shop floor and customers see; local copies are uncontrolled by definition

5.1 Document Master Fields

Field	Type	Req	Specification
Document #	Auto/manual	Yes	DOC-{TYPE}-{SEQ:5}; types: GA (general arrangement), NP (nameplate), SCH (schematic), WND (winding diagram), COR (core), TNK (tank/fab), TP (test plan), WI (work instruction), TC (T&C / templates), CERT (certificate templates)
Title	Text	Yes	—
Revision	Auto	Auto	Rev A, B…; every revision keeps its own files and approval record
Native File(s)	Upload	Yes	DWG/DXF/SLDDRW/SLDPRT/STEP/PDF/DOCX; max 500 MB; checksum stored; virus-scanned
Viewable File	Upload/auto	Yes	PDF rendition mandatory for release (auto-generated from native where converter available; else engineer uploads). The floor and customers only ever receive the PDF rendition
Title Block Metadata	Group	Yes	Drawn By, Checked By, Approved By, Scale, Sheet x of y, Drawing Size (A–E), Project/SO reference; entered at check-in (metadata extraction from title block is Phase 3)
Linked Objects	Multi-link	No	Items, EBOM lines, routings, inspection plans, customer orders (ETO variant drawings link to the specific SO line)
Security Class	Dropdown	Yes	Internal / Customer-Shareable / Restricted (export-controlled: visible only to users with EAR flag)

5.2 Check-In / Check-Out and Lifecycle

Rule	Specification
Check-out	Locks the document to the borrower; others see who holds it and since when; Engineering Manager may force-release a lock (logged)
Check-in	Uploads new working files against the same in-work revision; nothing changes for consumers until release
Lifecycle	WIP (gray) → In Review (yellow: redline comments layer on the PDF, reviewer approvals) → Released (green: files frozen, rendition stamped “RELEASED Rev X + date” watermark) → Superseded (blue) → Obsolete (red: stamped OBSOLETE, excluded from all pick-up points)
Where drawings surface	WO traveler prints the drawing list with revisions; shop floor terminal shows current released PDFs for the operation (never a superseded one — resolved at print/view time); purchasing attaches fabrication drawings to POs automatically from the BOM line link; transmittals (below) send to customers
Transmittals	Customer drawing packages: pick documents → generate transmittal record TR-{SEQ} (recipient, date, revs sent, purpose For Approval / For Construction / As-Built) → email/portal; customer response captured (Approved / Approved as Noted / Revise & Resubmit) which drives the Design Approval milestone (Part B §4.4)

6. EBOM → MBOM Transformation

The MBOM is what Production and Costing consume (the BOM-TX-… objects already referenced by INV 2.3.9). Transformation is a guided screen, not a copy-paste:

1. Source: a Released EBOM revision. Target: MBOM for a specific plant (multi-plant ready; single plant today).
2. Phantom resolution preview: phantoms flatten into parent per policy; engineer confirms.
3. Operation mapping: every line's Consume At Operation validated against the chosen routing (Part D §4); unmapped lines listed and must be resolved.
4. Packaging & consumables append: MBOM adds manufacturing-only items (crating per packing spec, oil top-up allowance, welding consumables) that the EBOM deliberately excludes; these lines are flagged MFG-Add and never sync back to the EBOM.
5. Scrap % review: costing-relevant; defaults from EBOM, production may adjust with Engineering approval.
6. Release: MBOM status Active; prior Active revision → Superseded; cost rollup invalidation flag set for the parent (Part I §3 picks it up in the next costing run); Material Master BOM tab now renders this revision.

> Dev Note: Sync guard: if the source EBOM gets a newer released revision, all MBOMs derived from older revisions show an "EBOM ahead" banner and appear in the Engineering exceptions panel until an ECO task regenerates them. MBOMs are never edited structurally outside this transformation — structural change always starts at the EBOM (single source of truth).

7. Engineering Reports & Roles

Report	Content
Design Reuse	New designs vs variants vs repeats per quarter; average design hours by type
Change Cycle Time	ECR→ECO→released duration percentiles; emergency change count
Placeholder Aging	Draft placeholder parts by age and owning engineer
Drawing Approval Status	Transmittals awaiting customer response with age; approval loop counts per order
Where-Used Explosion	Full multi-level where-used for any item (audit and obsolescence tool)

Role	Rights
Design Engineer	Requests, EBOMs, drawings (draft/check-out), ECR create, costing simulation read
Engineering Manager	Release approvals, force unlock, ECO release, effectivity decisions
CCB members	ECR decisions per §4.1
Production / Purchasing / Quality	Read released objects; ECR create; task execution on ECOs
Customer-portal (Phase 3)	Transmittal receive/respond only

PART D

PRODUCTION MANAGEMENT

Production Management plans, releases, executes, confirms and settles all manufacturing. It consumes released MBOMs and routings (Part C), reserves and draws materials from Inventory, employs Quality gates at defined operations, generates the serialized genealogy that traceability depends on, and settles every order’s cost to the GL with variances split by cause (Part A §8). This Part also specifies the physical flow of a transformer through the plant and the exact screen the operator touches at each workstation.

1. Plant Model — Work Centers and Floor Flow

1.1 Standard Work Center Set (Transformer Plant)

Work centers are the scheduling and costing resources. Initial set for the PE plant — each row becomes a Work Center master record (§2); the development partner must treat this as seed data, not hard-coding:

WC Code	Work Center	Function	Cost Center
WC-050	Tank Fabrication & Welding	Steel cutting, forming, welding, pressure test of tanks, radiators fit-up	CC-1150
WC-100	Core Cutting	CRGO slitting, cropping, mitred cutting (step-lap)	CC-1100
WC-110	Core Stacking	Lamination stacking, core clamping, core loss pre-check	CC-1100
WC-200	LV Winding	Foil/strip LV coils	CC-1200
WC-210	HV Winding	Wire/CTC HV coils, layer/disc winding	CC-1200
WC-220	Coil Assembly & Sizing	Phase assembly, sizing press, brazing	CC-1200
WC-230	Vapor Phase Drying / Oven	Insulation drying cycles (batch resource)	CC-1250
WC-300	Core & Coil Assembly	Core-coil marriage, cleating, lead dressing, tap lead connection	CC-1300
WC-310	Tanking	Active part into tank, gasketing, cover fit	CC-1300
WC-320	Vacuum & Oil Filling	Vacuum cycle, degassed oil fill, standing time	CC-1300
WC-400	Final Assembly & Fittings	Bushings, tap changer, radiators, conservator, cabinet, wiring, accessories	CC-1400
WC-500	Test Lab	Routine tests per IEEE C57.12.90; type tests when ordered	CC-1500 (Quality)
WC-600	Paint & Finishing	Blast, prime, topcoat, nameplate mounting	CC-1600
WC-700	Packing & Shipping Prep	Drain/gas-fill for transport, accessory crating, loading	CC-1700

1.2 Product Flow Through the Floor

Reference flow for a padmount/power transformer; the routing (§4) encodes this per design. Parallel branches: the tank (WC-050) and the active part (WC-100→WC-300) progress concurrently and merge at Tanking. Every operation start/complete is a barcode scan event (Part F, SP-10/SP-11); every material issue is a scan (SP-08/SP-09); every QC gate is an inspection record.

1. Core: WC-100 → WC-110: CRGO batches issued by heat number (scan captures batch → serial genealogy); stacked core weighed and core loss pre-checked (in-process inspection IP-INP “Core Assembly”).
2. Windings: WC-200 / WC-210 draw copper (batch-scanned, conductivity certs on file); winding data sheet (turns, tap positions) from drawing set on-screen; QC gate “Winding” (turn count, resistance, dimensions).
3. Coil assembly & drying: WC-220 assembles phases; WC-230 dries per cycle recipe — batch resource: multiple orders share an oven run (the terminal supports multi-order batch confirmation with cycle chart attachment).
4. Tank line: WC-050 fabricates/receives tank; pressure/leak test recorded as inspection; blast & prime may run at WC-600 before assembly per routing variant.

5. Assembly merge: WC-300 core & coil into tank at WC-310; WC-320 vacuum + oil fill (oil batch scanned — traceability to oil supplier batch); QC gate “Pre-Test” checklist.
6. Final assembly WC-400: serialized components installed and scan-linked as child serials (bushings, tap changer, relays — SP-12 parent-child link scan). Accessory checklist from SO configuration.
7. Test WC-500: Final Test per plan IP-FIN (Part E §4); FTR generated; failures route to Rework (§9).
8. Finish WC-600 → WC-700: paint, nameplate (nameplate data printed from system — SP-13 verification scan of nameplate serial barcode), pack per Part A §5, FG receipt (§8) or direct staging to shipment.

2. Work Center Master

Screen ID	PP-WC-001 (list), PP-WC-002 (detail)
URL Pattern	/production/work-centers
Purpose	Capacity, cost rates and scheduling parameters per resource

Field	Type	Req	Specification
WC Code / Name	Text	Yes	Per §1.1
Plant / Warehouse Link	Dropdown	Yes	Issuing warehouse default for materials consumed here
Cost Center	Lookup	Yes	Every WC maps to exactly one cost center (Part H); labor/overhead postings derive from this link
Capacity Model	Group	Yes	Shifts per day, hours per shift, working days calendar (plant calendar link), number of parallel stations/machines, efficiency % (default 85), utilization target %
Batch Resource	Toggle	No	ON for oven/vacuum: capacity consumed per cycle, not per unit; cycle size (units), cycle hours
Labor Rate — Standard	Currency/hr	Yes	From Part I §2 rate calculation (read-only here, link to rate screen)
Machine Rate — Standard	Currency/hr	No	Optional separate machine costing
Overhead Rate / Basis	Currency/hr or %	Yes	Overhead applied per labor hour (default) or % of labor cost; from Part I §2
Queue / Move Time (h)	Numbers	No	Scheduling buffers before/after operations at this WC
Default Crew Size	Number	No	Labor hours = run hours × crew size when ON
Alternate WC	Lookup	No	Overflow routing target with efficiency factor
Status	Badge	Yes	Active / Maintenance (blocks new dispatch, warns scheduler) / Inactive

3. MRP Workbench & Planned Orders

Screen ID	PP-MRP-001
URL Pattern	/production/mrp
Purpose	Net-requirements planning across all demand and supply; the planner's daily control desk. Extends the read-only INV 15.9 view into the acting workbench

3.1 MRP Run

Element	Specification
Scope & schedule	Nightly full run + on-demand run (item / item group / plant); regenerative (full re-plan) with net-change option
Demand inputs	Confirmed SO lines (MTS ship dates), WO component requirements, forecast (INV 15.8, consumed by actuals per period), safety stock floors (Part A §10), quarantine exclusions
Supply inputs	On-hand available by warehouse, open POs (confirmed dates), open WOs (scheduled completion), planned orders, open transfers, subcontract POs
Logic	Low-level-code explosion through MBOMs with scrap %; lot sizing per item (Lot-for-Lot default; Fixed Qty; Min/Multiple rounding); lead-time offsetting (purchased: vendor lead time + GR processing days; made: routing-based lead time); make/buy split by item flag
Outputs	Planned orders (PLO- for make, PR proposals for buy), reschedule-in/out messages for existing supply, exception messages

3.2 Exception Messages

Code	Message	Suggested Action Button
EX-01	Shortage — no supply covers requirement on date	Create PR / Create WO
EX-02	Reschedule In — supply later than need	Expedite (writes to PO Expedite Board, Part G §7)
EX-03	Reschedule Out — supply earlier than need	Push out (PO change proposal / WO reschedule)
EX-04	Excess — supply exceeds demand horizon	Cancel proposal
EX-05	SS-BREACH — plan dips below safety stock	Raise supply proposal per Part A §10.2
EX-06	Past-due supply	Expedite / re-date
EX-07	BOM/routing missing or unreleased for make item	Open Engineering exception

3.3 Planned Order Conversion

Planner selects planned orders → **Convert**: make → Production Order draft (WO) with pegs preserved; buy → PR (Part G §3) grouped by vendor per lot-sizing. Mass-convert with preview; conversions log the MRP run ID for audit. Firmed planned orders (planner locks date/qty) survive the next regenerative run.

4. Routings

Screen ID	PP-RTG-001/002
URL Pattern	/production/routings
Purpose	Operation sequences with times and QC gates; the source of scheduling and of labor/overhead standards

Field (Operation Row)	Type	Specification
Op #	Number	10, 20, 30…; parallel branches via branch labels (A10, A20 tank line) merging at a defined op
Work Center	Lookup	From §2
Description	Text	Operation text; prints on traveler
Setup Hours / Run Hours / Basis	Numbers	Run basis: per unit or per lot; batch WCs use per-cycle
Crew Size	Number	Overrides WC default
QC Checkpoint	Toggle + plan link	ON inserts an in-process inspection gate: operation cannot complete until linked IP-INP inspection passes (or Quality Manager conditional-pass with restriction note)
Work Instructions	Doc links	Controlled WI documents (Part C §5); terminal renders current released PDF
Components At This Op	Auto view	MBOM lines mapped here (Consume At Operation); read-only from BOM
Milestone Confirmation	Toggle	ON = operation confirmation reports progress % for the order (used by ETO milestone “Production Complete” logic and WIP valuation)

Routing lifecycle mirrors EBOM: Draft → In Review → Released → Superseded, changes via ECO. Standard hours from released routings feed the cost rollup (Part I §3) and LEV calculation (Part A §8).

5. Production Orders

Screen ID	PP-WO-001 (list), PP-WO-002 (detail)
URL Pattern	/production/orders
Purpose	The execution and cost-collection object for making one item (one serial for ETO transformers; lot quantity for components)

5.1 Header Fields

Field	Type	Req	Specification
WO #	Auto	Auto	WO-{YYYY}-{SEQ:5}
Item / MBOM Rev / Routing Rev	Lookups	Yes	Revisions frozen at release; displayed prominently (rev governs pick list, ops, drawings)
Order Qty / UoM	Number	Yes	ETO: 1 per serialized unit (multi-unit orders auto-split one WO per serial at release)
SO Line Peg	Link	Cond.	Mandatory for MTO/ETO; drives priority and reserves output
Serial #	Auto	Auto (serialized)	Generated at release per item-type format PE-{TYPE}-{YYYY}-{SEQ:4}; nameplate serial for transformers; parent for genealogy
Start / Due Dates	Dates	Yes	Scheduled by backward scheduling from due date (forward from start on demand); Gantt view per WC
Priority	Number/badge	No	Dispatch ordering; late pegged orders auto-escalate
Production Cost Center	Auto	Auto	Derived per operation from WC; header shows primary
Profit Center	Auto	Auto	From item product line
Status	Badge	Auto	§5.3

5.2 Release Checks (hard gates, run at Release action)

1. MBOM and Routing revisions Released (E-PP-01).
2. Material availability: full-kit check against reservations; shortages listed with supply dates; release with shortages requires Planner override flag Release Short (order carries Shortage badge; short items appear on the Expedite Board).
3. All drawings on the deliverables list Released (ETO) and customer approval milestone complete when required (E-PP-03).
4. Inspection plans Active for every QC checkpoint op (E-PP-04, per Part A §6.5 policy).
5. Standard cost exists for the item (else warning + CVA exception list entry).
6. On release: material reservations firm to the WO (source type = Production Order), serial generated, traveler printable, dispatch entries created per operation.

5.3 Status Lifecycle

Status	Color	Meaning
Created	Gray	Draft; editable; scheduling sandbox
Released	Blue	Frozen revisions; on floor; costs collect

Status	Color	Meaning
In Progress	Purple	First operation started
Testing	Amber	At Final Test operation
Completed	Green	FG received; no further confirmations; awaiting settlement
Closed (TECO)	Dark Green	Technically complete — settlement run performed, variances posted; order locked
On Hold	Red	Blocked (quality hold, ECO immediate effectivity, SO cancellation pending disposition); reason mandatory
Cancelled	Red	Terminal; collected costs settle to scrap/variance per Controller decision

5.4 Traveler / Shop Packet

Content	Specification
Cover	WO barcode (Code128 + QR), item, serial, SO/customer (ETO), qty, dates, revision block
Operation sheets	Per op: op barcode, WC, description, std hours, WI document list with revisions, QC gate marker, sign-off box (belt-and-suspenders paper trail alongside digital confirmations)
Pick list	Components by operation with quantities, locations, batch/serial requirement flags
Drawing list	All linked drawings with released revisions

6. Shop Floor Terminal

Screen ID	PP-SFT-001
URL Pattern	/production/terminal (kiosk mode, badge login)
Purpose	The operator's single screen at every workstation: dispatch, start/stop, material scan, labor, quantity and quality capture. Touch + scanner first; zero typing for the standard path

Element	Specification
Login	Badge scan (operator barcode SP-07); WC selected per kiosk config; multi-operator crew clock-in supported
Dispatch list	Operations queued at this WC in priority order: WO, item/serial, op, qty, std hours, material-ready flag (green when all op components staged), drawing/WI buttons
Start Operation	Scan traveler op barcode (SP-10) → status In Progress, labor clock opens for the crew
Material Consumption	Per op component list; scan item/batch/serial label (SP-08); qty confirm (defaults to standard × order qty); wrong-material scan hard-blocks with E-PP-07 (item not on this op's list) — supervisor override creates an NCR automatically
Data Capture	Op-specific structured fields when configured (e.g. winding: turns count, measured resistance; oven: cycle #, chart upload) — defined per routing op as capture templates; stored on the confirmation record
QC Gate	When op is a checkpoint: Request Inspection button → creates IP-INP inspection; op Complete disabled until result posted
Complete Operation	Scan op barcode again (SP-11) + enter Good / Scrap / Rework quantities (must sum to started qty); scrap requires reason code (drives 5320 posting and scrap pareto); labor clock closes; next-op material staging signal to warehouse
Hold / Andon	Hold button with reason (material, machine down, quality, drawing question) — pauses labor, flags dispatch, notifies supervisor; machine-down auto-notifies maintenance list
Rework path	Rework qty spawns/extends a Rework Order (§9) linked to this WO and op

6.1 Confirmation Posting Logic

Event	System Postings
Material issue (manual issue policy items)	Check-Out to Production (INV 5.2): DR 1320 WIP (order) / CR 1310 RM at valuation rate; batch/serial genealogy written
Op completion (backflush policy items)	Backflush components mapped to the op at standard qty × confirmed good qty: same GL as above; negative-stock guard per INV policy
Labor confirmation	Actual hours (crew × elapsed or manual within ±20% guardrail): DR 1320 WIP / CR 7500-series absorption credit on the WC cost center at standard labor rate; simultaneous overhead application at WC overhead rate: DR 1320 / CR 7500 absorption
Scrap at operation	Component value scrapped: DR 5320 Scrap & Waste (cost center of WC) / CR 1320 WIP

7. WIP Visibility

Screen	Content
WIP Board (PP-WIP-001)	Kanban by work center: cards = WO operations currently at each WC with age, priority, hold flags; drag disabled (status changes only via confirmations — the board is a mirror, not a control)
WIP Valuation Report	Per open WO: materials issued to date, labor applied, overhead applied, scrap, total WIP value; reconciles to GL 1320 balance (subledger check in ACC 26.2 step 6 extends to WIP)
Order Progress	Per WO: op-by-op timeline actual vs scheduled; milestone % complete; projected completion vs promise (feeds Sales at-risk report)

8. Production Receipt & Order Settlement

8.1 FG Receipt

Final operation completion (after Final Test pass and paint/pack ops per routing) enables **Receive to Stock**: a Check-In of type Production Receipt (INV 5.1) auto-filled with WO, item, serial, qty good. Location: FG warehouse or STAGE-OUT for direct-ship pegged orders (auto-reservation to the SO line either way).

Accounting Entry — Production Receipt (FG at standard)

Debit	Credit	Amount Basis	Notes
1330 Finished Goods	1320 WIP	Good Qty × released standard cost of the FG item	Profit center: product line. Matches INV Sec 12 #6

8.2 Settlement (at Close/TECO)

Close is allowed when: all ops confirmed or cancelled, all inspections closed, FG received or scrap-disposed. The settlement run (per order, or batch month-end run screen PP-SETTLE-001 with preview) clears residual WIP to variance accounts split by cause:

Accounting Entry — Order Settlement — variance split

Debit	Credit	Amount Basis	Notes
5220 Material Usage Variance	1320 WIP	Actual material − standard material for good output	Unfavorable shown; favorable posts as credit. CC: production cost center of primary WC; PC: product line
5230 Labor Rate Variance	1320 WIP	(Actual rate − std rate) × actual hours	Same dimensions
5240 Labor Efficiency Variance	1320 WIP	(Actual hours − std hours) × std rate	Same
5250 Overhead Absorption Variance	1320 WIP	Applied − actual overhead share	Same

Dev Note: After settlement the order's WIP balance is exactly zero by construction; the settlement document stores the variance math per Part A §8 formulas and links each amount to its JE. Reopening a settled order requires Controller and reverses the settlement JE first.

9. Rework Orders

Element	Specification
Creation	From op completion (rework qty), from Final Test failure disposition, from NCR disposition Rework, or from RMA disposition Repair (Part A §3.6). RWO-{YYYY}-{SEQ:4}, linked to source object
Structure	Free-form routing (ops added as needed from any WC) + ad-hoc material list; no standard cost — actuals only, collected in 1325 WIP-Rework
Settlement receivers	By origin: production rework → parent WO (costs merge before its settlement); RMA warranty → 2610 Warranty Reserve (overflow 6300); RMA billable → 5130 COGS-Service; NCR vendor-caused → 1355 vendor claim receivable when a recovery agreement exists, else 5320
Statuses	Open → In Progress → Completed → Settled; quality re-inspection mandatory before Complete when origin is test failure or RMA

10. Subcontract Operations

For operations sent outside (e.g. specialty machining, galvanizing): routing op flagged Subcontract with vendor + PO link. Material issued to vendor posts DR 1315 Subcontract at Vendor / CR 1310; receipt back posts CR 1315 / DR 1320 plus vendor service cost DR 1320 / CR 2110 GR/IR at PO price (variance to 5260). Vendor-side quantity visible on the item Stock Overview as “At Subcontractor”. Full spec of the PO side in Part G §6.

11. Scheduling & Capacity (Phase Scope)

Capability	Phase	Specification
Backward/forward scheduling per order	1	Ops placed on WC calendars with queue/move buffers; infinite capacity with overload flags
WC Load View	1	Weekly load vs capacity bar chart per WC; drill to orders; overload highlighted
Interactive re-scheduling	2	Drag operations within WC with dependency validation; what-if mode
Finite scheduling / optimizer	3	Out of current scope; data model (op durations, calendars, alternates) already supports it

12. Production Reports

Report	Content
Order Status / Late Orders	All open WOs vs schedule; late and at-risk with pegged SO exposure
Scrap Pareto	Scrap value by reason code / WC / item; month trend
Efficiency	Std hours earned vs actual hours by WC and crew; utilization vs capacity
Throughput	Units and standard value produced per period per line
WIP Aging	Open order value by age bucket; stalled orders (no confirmation in N days)
Genealogy (per serial)	Complete build record: batches consumed by op, child serials, operators, inspections, test results — the production slice of the Material Traceability Report (INV 10.4)

PART E

QUALITY MANAGEMENT

Quality Management is promoted to a top-level module unifying the screens already specified in INV (Incoming Inspection 6.1, Stock Hold/Release 6.2, In-Process 15.2, Outgoing/Pre-Shipment 15.3, NCR 15.4, CoC 15.5) with the pieces this volume adds: Inspection Plans (Part A §6), Final Test, CAPA as a standalone register, Supplier Corrective Action (SCAR), Calibration, and the Quality dashboard. The through-line: every inspection everywhere executes from an Inspection Plan, every failure can escalate to NCR → CAPA/SCAR, and every quality event is permanently attached to the batch/serial it concerns.

1. Quality Process Map (Gate by Gate)

Gate	Trigger	Plan Type	Executed In	Pass Effect	Fail Effect
Incoming (IQC)	GRN line, QI-required item	INC	INV 6.1	QI Hold → Unrestricted	Blocked → NCR/RTV/Scrap disposition; vendor rating hit
In-Process (IPQC)	Routing op flagged QC Checkpoint	INP	INV 15.2	Operation may complete	Op blocked; rework/NCR; Emergency ECR path for design causes
Final Test (FQC)	Test operation (WC-500)	FIN	§4 below	FTR issued; order proceeds to finish	Disposition: rework loop and full retest; NCR mandatory
Pre-Shipment (OQC)	Shipment for FG	PSI	INV 15.3	CoC generated; PGI enabled	Shipment blocked; NCR
Return (RQC)	RMA receipt	RMA	Part A §3.6	Disposition path	Disposition path

2. Quality Dashboard

Screen ID	QM-DASH-001
URL Pattern	/quality/dashboard
Purpose	Quality team landing page

Widget	Content
Open Inspections	Counts by gate with age coloring; SLA breaches on top; click-through queues
First Pass Yield	FPY by gate and by work center, monthly trend
NCR Funnel	Open NCRs by status/severity; overdue CAPA count
Vendor Quality	Acceptance rate by vendor (rolling 90d); vendors below threshold flagged; SCAR count
Cost of Quality	Scrap (5320) + rework (1325 settlements) + warranty (6300/2610 usage) by month — pulled from GL by account + cost center
Hold Inventory	Value in QI Hold / Blocked / Quarantine by warehouse with age

3. Test Data Capture — Transformer Routine Tests

The Final Test plan (IP-FIN) for transformers standardizes these characteristics; capture templates render instrument-friendly entry forms with automatic pass/fail against design values pulled from the ETO configuration (Part B §3.2) and the design sheet:

Test	Method Ref	Captured Values	Pass Criteria Source
Winding Resistance	IEEE C57.12.90 §5	R per phase per tap (mΩ), winding temp	Design sheet ±tolerances
Turns Ratio & Polarity	§6	Ratio per tap per phase, deviation %	±0.5% of nameplate ratio
Insulation Resistance	—	MΩ HV-LV, HV-G, LV-G, PI	Plan minimums
No-Load Loss & Excitation Current	§8	W at 100%/110% V, %I	Design guarantee ± IEEE tolerance (loss ≤ guarantee +10% single unit)
Load Loss & Impedance	§9	W at rated current, %Z at 85°C ref	Guarantee tolerance; %Z ± 7.5% (two-winding)
Applied Voltage (Hipot)	§10	kV, duration, result	Withstand — attributive
Induced Voltage	§10.7	kV, frequency, duration, PD where required (pC)	Withstand / PD limits
Oil Tests	ASTM D877/D1533	BDV (kV), moisture (ppm)	Plan minimums
Temperature Rise (type test, when ordered)	§11	Top-oil and winding rise (K)	Rating (65K std)

Instrument integration: Phase 1 = manual entry with instrument ID (calibration-checked); Phase 2 = CSV import from test-set software with template mapping; Phase 3 = direct interface. Every captured set stores: instrument IDs, ambient conditions, tester badge, timestamps — all print on the FTR.

4. Final Test Screen & FTR

Screen ID	QM-FIN-001
URL Pattern	/quality/final-test
Purpose	Execute IP-FIN for a WO serial; produce the certified Final Test Report

Element	Specification
Queue	WOs at Testing status; sequence by promised date
Execution	Characteristics from IP-FIN plan; capture templates per §3; partial save and multi-session (tests span days); retest handling: failed characteristic → rework loop → affected characteristics re-run, prior results retained as Run 1, Run 2… (full history, no overwrite)
Result & FTR	All pass → Final Result Pass; FTR-{YYYY}-{SEQ:4} PDF generated: PE letterhead, nameplate data block, all test sections with measured vs guaranteed vs tolerance, run history, instrument list with cal dates, tester + Quality Manager signatures (digital), customer name/SO. FTR attaches to the serial record, the WO, and the shipment document set; milestone “Production Complete/Tested” fires (Part B §4.4)
Failure path	Fail → mandatory NCR; disposition Rework (RWO with retest), Engineering Review (ECR link), or Scrap (rare; Controller + Eng Manager). WO status stays Testing until Pass or scrap disposition

5. CAPA Register (standalone)

Screen ID	QM-CAPA-001
URL Pattern	/quality/capa
Purpose	Corrective/preventive actions as first-class records — sourced from NCRs (INV 15.4 Section C graduates here), audits, RMA failure analyses, and management review; one CAPA can span many NCRs

Field	Type	Req	Specification
CAPA #	Auto	Auto	CAPA-{YYYY}-{SEQ:4}
Sources	Multi-link	Yes	NCRs, RMAs, audit findings, SCARs
Problem Statement	Textarea	Yes	Consolidated statement
Root Cause / Method	Textarea + dropdown	Yes	5-Why, Fishbone, 8D, FMEA; analysis attachments
Actions Table	Rows	Yes	Type (Containment / Corrective / Preventive), description, owner, due date, status, evidence attachment per action
Effectiveness Check	Group	Yes	Method, check date (≥ 30 days after implementation, config), result Pass/Fail; Fail reopens actions
Status	Badge	Auto	Open → Analysis → Actions In Progress → Implemented → Effectiveness Check → Closed / Reopened; Quality Manager owns transitions to Closed

6. SCAR — Supplier Corrective Action

Element	Specification
Creation	From vendor-attributed NCR or RMA failure (Failure Attributable To = Component Supplier); SCAR-{YYYY}-{SEQ:4}
Content	8D-format sections completed jointly: PE fills D1–D3 (team, problem, containment expected), vendor responds D4–D8 via emailed form (portal Phase 3); response due date with escalation
Enforcement	Vendor with ≥ 2 open overdue SCARs → AVL status Restricted (new POs require Purchasing Manager override, Part G §2); resolution restores
Recovery	Cost recovery agreement fields (amount, credit note ref) tie to RWO settlement receiver logic (Part D §9)

7. Calibration Register

Screen ID	QM-CAL-001
URL Pattern	/quality/calibration
Purpose	Instruments and gauges with calibration schedules; hard interlock with test execution

Field	Specification
Instrument ID / barcode	CAL-{SEQ:4} label; scanned at test entry (SP-14)
Type / Range / Location / Owner	Master data; linked work center
Calibration Interval / Last / Next Due	Interval in days; due-soon alerts at 30/14/3 days (INV 15.11 rules)
Certificates	Upload per calibration event with external lab reference
Status	In Cal (green) / Due Soon (amber) / Overdue (red) / Out of Service
Interlock	Any inspection or test referencing an Overdue/OOS instrument blocks with E-QM-05; results already captured with an instrument later found out-of-cal are flagged for review (report QM-CAL-R2: “measurements at risk” listing every result since last good calibration)

8. Roles

Role	Rights
Quality Inspector	Execute inspections/tests, create NCRs, view plans
Quality Engineer	Author plans, capture templates, CAPA analysis, SCAR authoring
Quality Manager	Plan approval, dispositions, conditional releases, CAPA closure, FTR countersign, calibration overrides (logged)
Production roles	Request inspections, view results, execute rework
Read-Only / Auditor	View everything incl. full history exports

PART F

BARCODING & TRACEABILITY

Barcoding is the data-capture backbone: every physical movement, consumption, build step and quality event is confirmed by scan, which is what makes the Material Traceability Report (INV 10.4) trustworthy end to end. This Part specifies symbologies, label formats, hardware, the complete scan-point matrix (which step scans what, validates what, and posts what), error behavior, and the genealogy data model.

1. Standards & Symbology

Object	Symbology	Encoded Content	Label
Item (shelf/product label)	Code128	Material Code	2×1 in thermal; item name, code, UoM human-readable
Batch	Code128 + QR (optional detail)	Batch # (BAT-…); QR adds JSON: item, vendor batch, expiry	4×2 in; item, batch, vendor batch/heat #, mfg/expiry dates, qty at print
Serial	Code128	Serial # (PE-…)	4×2 in poly (durable) for components; engraved/rivet nameplate + barcode plate for transformers
Location / Bin	Code128	Location code	4×1 in reflective for rack labels; floor signs for zones
Handling Unit	Code128 + GS1-128 SSCC when customer requires	HU # / SSCC-18 (AI 00); AI 02+37 contents optional	4×6 in per Part A §5.4
WO Traveler / Operations	Code128 per op + QR on cover	WO # and WO#|OP# composite	Laser letter sheets in packet
RMA	Code128	RMA #	On customer return instructions PDF
Operator Badge	Code128	User ID	Badge card
Instrument	Code128	CAL ID	1×0.5 in asset label

Dev Note: One symbology (Code128) for everything internal keeps scanner config trivial; GS1-128 appears only on outbound HU labels for customers demanding GS1 compliance. QR usage is limited to human-assist payloads (a phone can read the traveler) — transactional scans always use the 1D code. Barcode format default per INV 11.1 policy remains Code128; this Part fixes the content standards.

2. Hardware & Printing

Component	Requirement
Printers	ZPL-compatible thermal transfer (Zebra ZT411 class) at Receiving, Quarantine, Pack, Serial/nameplate station; laser for travelers. Print service: server-rendered ZPL templates, printer registry per station, reprint with reason + audit (reprints watermark “DUPLICATE” except location labels)
Scanners	Rugged 2D imagers (handheld + hands-free at fixed stations); keyboard-wedge over WebSocket wedge service so the web terminal receives scans as events, not keystrokes into random fields (focus-lock pattern)
Mobile	Android rugged devices running the responsive web app for count entry, putaway, picking; camera fallback scanning permitted for supervisors only (slower, logged)
Offline	Terminals queue scan events locally for up to 4 hours of network loss; queued events post in order with original timestamps; conflicts (e.g. stock moved meanwhile) land in a Scan Exception queue for supervisor resolution — never silently dropped

3. Scan Point Matrix

The definitive list. “Validates” failures always produce an on-device error with the E-code and block the transaction (supervisor overrides where noted create audit + NCR entries).

SP	Process Step	Scans (in order)	Validates	Posts / Records
SP-01	Receiving — start GRN line	PO barcode or select; item label or vendor barcode	Item on PO; open qty	GRN line begin (INV 5.1)
SP-02	Receiving — batch capture	New batch label printed & scanned back	Label printed = record created	Batch record w/ vendor batch/heat keyed
SP-03	Receiving — serial capture	Each serial (vendor or PE-printed)	Format; uniqueness	Serial records, status Available/QI Hold
SP-04	Putaway	Item/batch label + destination bin	Bin allowed (putaway rules INV 15.6); capacity	Bin transfer
SP-05	QI sample pull	Batch + inspector badge	QI Hold status	Sample qty logged to inspection
SP-06	Stock transfer ship/receive	Item/batch/serial + from-bin / to-bin	Transfer doc lines	INV 5.3 movements
SP-07	Terminal login	Operator badge	Active user; WC assignment	Session; labor attribution
SP-08	Material issue to WO op	Component batch/serial + (bin)	Component on op list (E-PP-07); batch Unrestricted; FEFO/FIFO suggestion honored or override-logged	Issue posting; genealogy row (WO, op, batch/serial, qty, operator, ts)
SP-09	Kit/stage delivery to WC	Picklist + staging location	Kit complete flags	Staged status per op
SP-10	Operation start	Traveler op barcode	Predecessor ops complete; QC gates clear; WO Released	Op In Progress; labor clock on
SP-11	Operation complete	Traveler op barcode + qty entry	Good+scrap+rework = started	Confirmation; backflush; labor; next-op signal
SP-12	Serialized component install	Parent serial + child serial	Child on BOM; child Available/Issued to this WO; not already linked (E-BC-05)	Parent-child link (INV 2.3.4 genealogy); child status In Production
SP-13	Nameplate verification	Nameplate serial barcode at WC-600	Scanned = WO serial (E-BC-06 on mismatch — wrong nameplate mounted)	Verification event on serial
SP-14	Test execution	Instrument CAL labels + serial	Calibration current (E-QM-05)	Instruments bound to test record
SP-15	Pick confirmation	Bin + item/batch/serial + picklist	Right batch per FEFO strategy; serial in Reserved status	Pick line confirm (INV 5.7 scan-to-pick)
SP-16	Pack — add to HU	HU + item/batch/serial	On pack order; picked status	HU content line (Part A §5.3)
SP-17	HU close & label	New HU label scan-back + scale weight	Weight sanity ±2%	HU Closed; label printed
SP-18	Load & ship	Each HU at truck + shipment doc	HU on this shipment (E-BC-08); PSI passed; credit gate	HU Loaded; completeness check enables PGI
SP-19	RMA receipt	RMA barcode + serial/batch on crate	Authorized line/serial (E-RMA-04)	Return receipt lines (Part A §3.5)
SP-20	Cycle/physical count	Bin + item/batch (+ serials)	On count doc; frozen state	Count entry (Part A §7)

SP	Process Step	Scans (in order)	Validates	Posts / Records
SP-21	Scrap bin confirmation	Item/batch + scrap location	Disposition approved	Scrap issue posting

4. Error Handling & Governance

Rule	Specification
Wrong-scan messaging	Every failed validation shows: what was scanned, what was expected, E-code, and the next action (“Scan a component listed for op 30”) — with audible + color feedback on device
Override discipline	Overridable validations (batch suggestion, staging shortcuts) require supervisor badge scan inline; hard validations (wrong item, wrong serial, cal expired, HU not on shipment) have no override — the correcting transaction must be done properly
Label reprint	Any label reprint requires reason code; serial/nameplate reprints additionally require Quality approval (duplicate nameplate risk)
Unlabeled stock exception	Found stock without label: supervisor flow “Identify & Relabel” — resolves against expected inventory at that bin, prints label, logs exception; feeds count accuracy metrics
Scan telemetry	All scan events (incl. failures) stored 24 months: device, user, ts, payload, result — the forensic layer beneath the audit trail

5. Genealogy Data Model & Traceability Output

Entity	Key Fields	Notes
genealogy_event	event_id, type (ISSUE/INSTALL/TEST/PACK/SHIP/RETURN), wo_id, op_id, parent_serial_id, child_serial_id, batch_id, item_id, qty, operator_id, station, ts, source_scan_id	Append-only; one row per SP-08/11/12/13/14/16/18/19 event with stock impact or linkage
serial_link	parent_serial_id, child_serial_id, installed_event_id, removed_event_id	Current + historical structure (removal during rework recorded, not overwritten)
batch_consumption	wo_id, op_id, batch_id, qty, event_id	Aggregated view for the traceability tree

The Material Traceability Report (INV 10.4) is extended to render from these tables: input a FG serial → tree of ops with consumed batches (vendor, PO, QI result, certs), child serials, operators and inspections per node, forward to customer/site/warranty; input a raw batch (e.g. suspect CRGO heat) → every WO, serial, customer, and shipped location it reached, with one-click customer notification list export. Target: complete tree in < 10 seconds for a 5-level transformer build.

PART G

PROCUREMENT & PURCHASING

Purchasing turns demand (MRP proposals, reorder triggers, manual requests) into on-time, on-spec, on-price supply. It owns the Vendor Master operational view, Approved Vendor Lists, Purchase Requisitions, RFQs, Purchase Orders (including subcontract), confirmations and expediting. Receiving (GRN), three-way match and payment are already specified (INV 5.6, ACC 6/8/25); vendor quality feedback loops in from Part E.

1. How Purchasing Orders Parts and Materials — The Daily Operating Loop

1. **Demand intake.** The buyer's workbench opens on: PR queue (MRP-generated + reorder auto-PRs + manual), MRP exception messages assigned to buy items (EX-01/02/03/06), and the Expedite Board. Everything is bucketed by buyer code (items carry a Buyer assignment).
2. **Source determination.** For each PR line the system proposes the source: AVL preferred vendor → last PO price and lead time → open contract/blanket if one exists. Buyer accepts or overrides (override reason logged). No AVL vendor for a QI-critical item = hard block (E-PUR-02) → start AVL qualification or RFQ.
3. **RFQ when needed.** New items, expiring price agreements, or value > $25k without contract (policy) route through RFQ (§4): multi-vendor invite, sealed responses, comparison grid, award writes the new AVL price basis.
4. **PO creation.** PRs group into POs per vendor per lot-sizing (one PO, many lines; delivery schedules per line where staged deliveries are wanted). Approval workflow by value (ACC 22 engine): Buyer ≤ $25k, Purchasing Manager ≤ $250k, + Controller above, + CFO above $1M.
5. **Transmission & confirmation.** PO PDF (entity letterhead, terms, quality clauses, drawing attachments for fabricated parts) emails to vendor; vendor confirmation (date + qty per line) is captured — unconfirmed POs older than 3 business days surface on the workbench.
6. **Expedite / de-expedite.** MRP reschedule messages write suggestions onto the Expedite Board (§7); buyer actions become dated change requests to the vendor with full history.
7. **Receipt & closure.** GRN against PO (INV 5.1/5.6), QI per plan, three-way match (ACC 6), PPV/IPV visible in CVA (Part A §8). Lines close at full receipt + invoice; short-close with reason allowed (Purchasing Manager).

2. Vendor Master (Operational View) & AVL

Screen ID	PUR-VEND-001/002
URL Pattern	/purchasing/vendors
Purpose	Single vendor record; financial fields (bank details, 1099, payment terms) are owned by ACC screens and render read-only here

2.1 Fields (Operational)

Field	Type	Req	Specification
Vendor # / Legal Name / DBA	Auto + text	Yes	VEND-{SEQ:5}
Vendor Class	Dropdown	Yes	Raw Material Mill, Component OEM, Distributor, Subcontractor, Freight, MRO, Services
Addresses / Contacts	Tables	Yes	Order-from, remit-to (ACC-owned), ship-from sites (multiple; each with country — feeds Buy America)
Default Currency / Incoterms / Payment Terms	Dropdowns	Yes	Terms read-only from ACC
Buyer	User	Yes	Owning buyer
Quality Approval Status	Badge	Auto	Approved / Conditional / Restricted / Disqualified — driven by Part E §6 SCAR enforcement and vendor rating; Restricted blocks new POs without Purchasing Manager override; Disqualified hard-blocks
Compliance Panel	Group	Cond.	W-9 on file (link ACC), insurance cert + expiry, ISO 9001 cert + expiry, conflict minerals statement, **Buy America capable** flag + supporting docs per ship-from site; expiring docs alert at 30 days
Onboarding Status	Badge	Auto	Draft → Docs Pending → Under Review → Active (ACC 22 vendor approval workflow; no PO before Active — fraud control per ACC gap register)

2.2 Approved Vendor List (AVL)

Column	Specification
Item + Vendor	The AVL row key; surfaced on Item Purchasing tab (INV 2.3.6 A) and here by vendor
Preferred Rank	1..n; MRP source determination uses rank 1
Vendor Part # / Lead Time / MOQ / Price Basis	Operational sourcing data; price basis = last award or contract
Qualification Status	Qualified (green) / Conditional (first-article pending; receipts forced 100% inspection) / Suspended
First Article	FAI record link for made-to-print parts: FAI required flag, report, approval date
Quality Rating	Rolling acceptance % from IQC results (auto)

3. Purchase Requisitions

Screen ID	PUR-PR-001
URL Pattern	/purchasing/requisitions
Purpose	Demand records awaiting sourcing

Field	Type	Req	Specification
PR # / Source	Auto + badge	Auto	PR-{YYYY}-{SEQ:5}; source MRP / Reorder / Manual / ECO / Project
Line: Item, Qty, UoM, Need Date	Mixed	Yes	Need date from MRP requirement; manual PRs validate date ≥ today + lead time (warning)
Suggested Vendor / Price	Auto	Auto	From source determination
Requested By / Cost Center	User + lookup	Yes	Cost center mandatory for expense/MRO purchases (drives GL + budget check ACC 21: shows Budget remaining on the cost center + account combination; over-budget flags approval)
Peg Reference	Link	Auto	WO / SO / ECO that created the demand
Status	Badge	Auto	Open → In Approval (manual/expense PRs only; MRP PRs pre-approved by policy) → Approved → Converted (PO link) / Rejected / Cancelled

4. RFQ

Element	Specification
Create	RFQ-{YYYY}-{SEQ:4} from PR(s) or blank; lines = item, qty bands (e.g. 1k/5k/10k lbs), target date; attachments auto-pull item drawings (released revisions only)
Invite & respond	Vendor list (AVL + prospects); emailed response form; buyer keys or imports responses (per line per qty band: price, lead time, validity, MOQ, tooling/NRE, freight terms)
Comparison Grid	Vendors as columns, lines as rows; normalized landed cost column (price + est. freight + duty %); lead time, quality rating, Buy America flag displayed; lowest-landed highlighted, buyer picks with justification note when not lowest
Award	Creates/updates AVL rows (price basis, lead time), optionally creates PO(s) directly; losing vendors get regret notice template
Statuses	Draft → Issued → Responses In → Awarded / Cancelled; sealed until Issued closes (responses hidden from other vendors always, from buyer until due date if Sealed toggle ON)

5. Purchase Orders

Screen ID	PUR-PO-001/002
URL Pattern	/purchasing/orders
Purpose	The commercial commitment to the vendor; anchor for GRN, QI, 3-way match, PPV

5.1 Header Fields

Field	Type	Req	Specification
PO #	Auto	Auto	PO-{YYYY}-{ENTITY}-{SEQ:5}
Vendor / Order-From / Ship-From Site	Lookups	Yes	Ship-from site pins country of origin default for Buy America accumulation
Ship-To Warehouse	Dropdown	Yes	Receiving plant
Currency / Rate / Incoterms / Freight Terms	Mixed	Yes	Booked rate display; landed-cost elements estimated at PO, trued at GRN (INV 5.6.1 landed cost allocation)
Payment Terms	Auto	Auto	From vendor (ACC)
Quality Clauses	Multi-select	Cond.	Clause library: Q1 Mill certs required per heat, Q2 FAI required, Q3 100% inspection, Q4 CoC each shipment, Q5 PPAP-lite; auto-set from item flags; prints on PO
Buy America Required	Toggle	Auto	ON when any peg traces to a BA-flagged SO/project: restricts source country and requires vendor BA cert per shipment
Status	Badge	Auto	Draft → In Approval → Approved → Sent → Confirmed → Partially Received → Received → Matched/Closed; Cancelled; Change Pending overlay during change orders

5.2 Line Fields

Field	Specification
Item / Vendor Part # / Description	Cross-reference printed both ways
Qty / UoM / Purchase UoM conversion	Dual display (buy in lbs, stock in kg — INV 2.6 conversion mandatory)
Unit Price / Extended	From AVL basis; deviation > 5% from basis flags approval note
Delivery Schedule	Sub-rows: date + qty (staged deliveries); each sub-row receivable independently; MRP consumes schedule dates
Drawing / Spec Attachments	Released revisions frozen onto the PO line; ECO on the drawing after PO issue raises a PO Change task (Part C §4.2 checklist)
Inspection Requirement	Auto from item QI flag + plan preview link; drives receipt QI Hold
Peg	WO/SO references (visible to receiving and QI for priority)
Tax Code	Vendor/use-tax logic per Part A §11.3
Line Status	Open / Partial / Received / Closed / Cancelled

5.3 PO Change Orders

Any edit after Sent creates PO Revision (Rev 1, 2…) with a change summary table (field, old, new), re-approval when value increases beyond threshold, and re-transmission; vendor confirmation resets per changed line. Full revision history renders on the PO detail. Cancellations after confirmed date require reason; cancellation exposure (restock/cancel fees) captured as a note for AP.

6. Subcontract Purchase Orders

Element	Specification
PO Type = Subcontract	Lines reference a service item (the operation) + a components-provided list (from the WO subcontract op, Part D §10)
Material provision	Issue to vendor movement prints a provision shipping doc; stock visible “At Subcontractor {vendor}” per item
Receipt	GRN receives the processed item/op completion; consumes provided components from 1315

Accounting Entry — Issue materials to subcontractor

Debit	Credit	Amount Basis	Notes
1315 Subcontract Material at Vendor	1310 Raw Materials	Issue Qty × valuation rate	PC: product line of parent

Accounting Entry — Subcontract receipt (service + material return)

Debit	Credit	Amount Basis	Notes
1320 WIP (order)	2110 GR/IR	Service qty × PO price	Service cost to order; variance vs standard subcontract → 5260
1320 WIP (order)	1315 Subcontract Material at Vendor	Provided material at valuation	Material returns into order WIP

7. Expedite Board & Vendor Performance

Screen	Content
Expedite Board (PUR-EXP-001)	All EX-02/EX-06 suggestions + Release-Short shortages (Part D §5.2) + past-due confirmed lines; columns: item, PO/line, vendor, confirmed vs needed date, peg (WO/SO at risk), action log (calls/emails with dates), promise history; buyers work top-down by pegged-order priority; aging SLA coloring
Vendor Performance (extends INV 10.8)	OTD % (received vs confirmed date), quality acceptance % (IQC), price compliance (invoice vs PO), confirmation responsiveness, SCAR history; quarterly scorecard PDF per vendor; ratings feed AVL status proposals

Role	Rights
Buyer	PR/RFQ/PO create within limit, expedite actions, AVL edit proposals
Purchasing Manager	Approvals to limit, AVL approval, restricted-vendor override, short-close
Quality	AVL qualification status, FAI approvals, quality clauses
Controller	High-value approvals, vendor onboarding financial release
Receiving / AP	Read POs; their own transactions per INV/ACC

PART H

COST CENTERS & PROFIT CENTERS (CONTROLLING)

Controlling gives every dollar in the GL two management dimensions: **where it was incurred** (cost center — responsibility for spend) and **which business it belongs to** (profit center — responsibility for margin). ACC Section 10 already carries the dimensions on journal items; this Part supplies what was still missing per the accounting gap analysis: the masters, the hierarchies, the sources and derivation rules, allocations, and the reports — and wires the dimensions into every logistics and production posting specified in this volume.

1. Sources — Where Cost Centers and Profit Centers Come From

Dimension	Source of Truth	Governance
Cost Centers	The organizational chart: one cost center per department or supervisory area that owns budgetary responsibility. Seeded from the org structure workshop with Finance; created/changed only by Controlling (Controller approval), effective-dated, never deleted (deactivated)	Annual review with budget cycle (ACC 21); mid-year changes via change request with mapping rules for history
Profit Centers	The management view of the business: product lines and service lines per legal entity, defined by the CFO. Stable small set; changes are rare and CFO-approved	P&L segmentation must reconcile to entity totals — every P&L-relevant posting carries exactly one PC

1.1 Seed Cost Center Hierarchy (PE Transformer plant)

Node	Cost Center	Contains / Maps To
CC-1000 Manufacturing	summary node	All production cost centers
— CC-1100 Core Shop	postable	WC-100, WC-110
— CC-1150 Fabrication	postable	WC-050
— CC-1200 Winding	postable	WC-200, WC-210, WC-220
— CC-1250 Drying	postable	WC-230
— CC-1300 Assembly	postable	WC-300, WC-310, WC-320
— CC-1400 Final Assembly	postable	WC-400
— CC-1600 Paint & Finish	postable	WC-600
— CC-1700 Pack & Ship	postable	WC-700 + shipping dock
CC-1500 Quality	postable	WC-500 Test Lab + inspectors + calibration
CC-2000 Warehousing	postable	Warehouses (each warehouse master links here — INV 3.2 Cost Center field)
CC-3000 Engineering	postable	Design engineering
CC-4000 Purchasing & Materials	postable	Buyers, planners
CC-5000 Sales & Customer Service	postable	—
CC-9000 G&A	summary	CC-9100 Executive, CC-9200 Finance, CC-9300 IT, CC-9400 HR, CC-9500 Facilities

1.2 Seed Profit Center Hierarchy

Profit Center	Scope
PC-100 Distribution Transformers	Pad-mount & pole-type product families
PC-200 Power Transformers	Medium/large power units

Profit Center	Scope
PC-300 Grid Components	PE Grid Components merchandise business
PC-400 Service & Repair	Billable repair, field service, testing services
PC-900 Corporate / Unassigned	Items and postings pending classification — target zero balance; exception report weekly

2. Master Data Screens

Screen ID	CO-CC-001/002 and CO-PC-001/002
URL Pattern	/controlling/cost-centers, /controlling/profit-centers
Purpose	CRUD + hierarchy management for both dimensions

Field	Type	Req	Specification
Code	Text (10)	Yes	CC-#### / PC-###; immutable after first posting
Name / Description	Text	Yes	—
Entity	Dropdown	Yes	Each CC/PC belongs to one legal entity; intercompany allocations use paired centers
Parent Node	Lookup	Yes	Position in the standard hierarchy; summary nodes are not postable (validation V-CO-01)
Manager / Owner	User	Yes	Receives budget variance alerts (ACC 21.3 integration) and approval routings scoped to the center
Valid From / To	Dates	Yes/No	Postings validated against validity at posting date
Default Profit Center (CC only)	Lookup	Cond.	For expense postings that arrive with CC but no PC, PC derives from this mapping (derivation rule 6 below)
Allowed GL Ranges	Rule	No	Optional guard: e.g. production CCs may not receive 9xxx G&A accounts
Blocked for Posting / Planning	Toggles	No	Soft retirement path before deactivation
Status	Badge	Yes	Active / Blocked / Inactive

Hierarchy screen: tree editor with drag-move (effective-dated moves; history preserved for comparative reporting), where-used check before deactivation (open WOs, budget lines, allocation cycles, WC links), and alternate hierarchies (reporting-only groupings) in Phase 2.

3. Derivation Rules — How Every Posting Gets Its Dimensions

Priority-ordered engine, evaluated at posting time by the shared Posting Service; the first rule that yields a value wins per dimension. Every financially relevant document in all three volumes passes through it. Rule evaluations are stored on the journal item (rule # that fired) for auditability.

#	Rule	Cost Center	Profit Center
1	Explicit entry on the document line (role-permitted)	As entered	As entered
2	Production confirmation / WO posting	Work center → CC link (Part D §2)	Parent item product line
3	Inventory movement (receipt, issue, transfer, adjustment, count, scrap)	Warehouse master CC (INV 3.2)	Item product line (item master PC field, new — added to INV 2.3.3 B GL section)
4	Sales documents (PGI/COGS, revenue, freight out, RMA credit)	— (revenue/COGS carry PC only; freight/warranty expense: shipping/quality CC)	SO line item product line; header PC for order-level charges
5	Purchasing expense lines (MRO/services)	PR/PO line cost center (mandatory, Part G §3)	CC default PC
6	Expense JE with CC only	—	CC master Default Profit Center
7	Fallback	CC-9000 G&A + exception report row	PC-900 + exception report row

Dev Note: Validation posture: for P&L accounts both dimensions are mandatory at posting (rule 7 guarantees a value but writes the posting to the CO Exception Report — target zero rows). Balance sheet accounts carry PC where meaningful (inventory, WIP by product line) and skip CC except where specified. The Posting Service exposes a simulation endpoint so any screen can preview derived dimensions before posting — surfaced as the “GL Preview” panel already specified on Stock Adjustments (INV 5.8) and extended to all posting screens.

4. Budgets by Cost Center

ACC Section 21 budgeting already supports Cost Center as a dimension. This volume binds it: every postable cost center must have a budget row set for the fiscal year (completeness check in the budget cycle); the Budget vs Actual report gains a Cost Center Manager self-view (their centers only, RBAC-scoped); over-budget indicator on GL account fields (ACC 21.1.1) evaluates at CC + account granularity.

5. Allocation Cycles (Period-End)

Screen ID	CO-ALC-001 (cycle definitions), CO-ALC-002 (run)
URL Pattern	/controlling/allocations
Purpose	Distribute shared costs from service/G&A cost centers to consuming cost centers on defined drivers — the Allocation Engine flagged missing in the accounting gap analysis

Element	Specification
Cycle definition	Ordered segments; each segment: sender CC(s) + sender account range, receiver CC set, driver (Fixed %, Headcount, Sq Ft, Machine Hours, Labor Hours, Revenue Share), driver value source (statistical key figures table maintained quarterly, or live from confirmations for hours)
Statistical key figures	Screen CO-SKF-001: per CC per period: headcount, sq ft, machine hours…; import from XLSX; history kept
Run	Per period per entity: preview (sender balances, computed distribution, receiver amounts, rounding to cents with remainder-to-largest rule) → post. Rerun replaces prior run for the period (reversal + new posting, both retained in GL with run ID ALC-YYYYMM-NN)
Sequence	Cycles run in defined order (e.g. Facilities → IT → Finance → to production CCs) before cost-center reporting and before work-center rate review; step 5b added to ACC 26.2 close checklist: “Allocation cycles posted”

Accounting Entry — Allocation posting (per segment)

Debit	Credit	Amount Basis	Notes
Receiver CC — 7500 Allocation Clearing (or original account mirroring per policy)	Sender CC — 7500 Allocation Clearing	Driver-proportional share of sender balance	Net zero at entity level; both sides carry their CC; PC follows CC default mapping

6. Reporting

Report	Content
Cost Center Report (CO-R1)	Per CC (respecting hierarchy rollups): actuals by account group, budget, variance $/%, YTD, allocations received/sent shown separately; drill to journal items (ACC 10 filters)
Profit Center P&L (CO-R2)	Full P&L by PC columns: revenue, COGS at standard, production variances, gross margin, allocated SG&A (optional toggle), operating margin; reconciliation column proves sum = entity P&L
Work Center Cost & Rate Review (CO-R3)	Per production CC/WC: actual costs vs absorbed (labor + overhead credits from confirmations), over/under absorption — the input to annual rate setting (Part I §2)
CO Exception Report (CO-R4)	Rule-7 fallback postings, postings to blocked centers, P&L rows missing dimensions (should be structurally zero), PC-900 balance

Role	Rights
Controlling Analyst	Masters draft, SKF maintenance, cycle definition, run preview
Controller	Master approval, cycle posting, hierarchy changes, exception clearance
Cost Center Manager	Own-center reports, budget entry (ACC 21) for own centers
CFO	PC structure changes, all reports

PART I

COSTING & PRICING

Costing computes what things cost — standards built from MBOM + routing + rates, WIP and settlement math, and the estimate engine that prices ETO quotes before a design exists. Pricing computes what things sell for — price lists, agreements, commodity surcharges and margin control. The two meet in the margin floor: no quote or price master survives approval below the floor without the defined escalation.

1. Cost Components

Component	Source	Into Standard Cost Of
Material	Σ (component qty per × (1+scrap%) × component standard cost) over the released MBOM	Every make item, rolled level by level
Material Overhead	% on material value (procurement/handling burden), rate per item group	Make and buy items
Labor	Σ (op std hours × crew × WC labor rate) over released routing	Make items
Machine	Σ (op machine hours × WC machine rate) where used	Make items
Production Overhead	Applied per labor hour (or % labor) at WC overhead rate	Make items
Subcontract	Subcontract op PO price basis	Make items with external ops
Buy item base	AVL price basis (rank-1 vendor) + freight/duty estimate %	Purchased items

2. Work Center Rates

Screen ID	CST-RATE-001
URL Pattern	/costing/work-center-rates
Purpose	Annual (and interim) computation and release of labor/machine/overhead rates per work center

Element	Specification
Rate math	Labor rate = planned cost-center labor cost ÷ planned productive hours (from capacity model × efficiency). Overhead rate = planned CC overhead pool (incl. received allocations, Part H §5) ÷ same hours. Machine rate where configured = machine pool ÷ machine hours
Screen	Grid per WC: planned costs (pull from approved budget, ACC 21, by CC), planned hours (from WC capacity), computed rates, prior-year rates, delta %; simulation column (what-if hours/costs); Release action versions the rates with effective date
Governance	Rates released annually with the budget; interim change requires Controller approval and triggers costing-run guidance (mid-year rate changes revalue nothing retroactively — they apply to confirmations from effective date; standards keep the rate frozen at last rollup)
Feedback	CO-R3 over/under-absorption report is the annual sanity check on rate quality

3. Cost Estimates & Costing Runs

3.1 Cost Estimate (single item / quote support)

Screen ID	CST-CE-001
URL Pattern	/costing/estimates
Purpose	Cost one item or one ETO configuration: current-standard simulation, what-if, and quote-backing estimates (CE-# links on quotes, Part B §3.2)

Element	Specification
Basis selector	Released MBOM+routing (standard simulation) / Draft EBOM (engineering what-if) / **Parametric ETO model**: for quotes before design, estimator enters driver quantities — core weight (kg), copper weight (kg), oil volume (gal), tank steel weight, bushing/tap changer selections, est. labor hours per WC — against a maintained driver-rate table (e.g. $/kg CRGO at current commodity basis); template per product family seeded from historical actuals
Itemization view	Cost tree: every component and operation with qty, rate, value; component prices show source (standard / AVL basis / manual override with note); subtotals by cost component
Comparison	Estimate vs current standard vs last actual (settled WOs of same/similar design) side by side
Status & reuse	Draft → Released (locks values, quotable) → Superseded; quote margin panel consumes Released estimates only

3.2 Costing Run (mass standard rollup)

Screen ID	CST-RUN-001
URL Pattern	/costing/runs
Purpose	Periodic (annual + on-demand) recalculation and release of standard costs across the item base — the engine behind Standard Cost Maintenance (INV 8.2), which remains the approval/impact UI

1. Scope: plant, item selection (all standard-cost items / item groups / changed-BOM-flag only — the invalidation flags set by MBOM releases, Part C §6).
2. Explode & roll: low-level-code order; errors collected, not fatal (missing rate, missing AVL basis, BOM loop) into an error list with owners; items with errors are excluded from release.
3. Review: per item old standard vs proposed, delta %, value impact on current on-hand (qty × delta); sort by absolute impact; drill to itemization.
4. Mark & release: two-step per INV 8.2 (mark = staged, release = effective). Release posts revaluation per INV Sec 12 #15 (1310/1330 vs 5330) and stamps the new standard + effective date on items; PPV baselines shift from that moment.
5. Frequency policy: full run annually at budget; interim runs for new items weekly (auto, new-items-only scope); ad-hoc runs Controller-approved.

4. WIP Accounting & Settlement Math (Worked Example)

One place to see the whole cost flow of a production order, consistent with Part D §6.1/§8 and Part A §8. Example: WO-2026-00266, one 1500 kVA padmount, standard cost $61,000 (material 42,000 / labor 8,000 / overhead 11,000).

Event	DR	CR	Amount
Issue CRGO, copper, oil, components (actual qty)	1320 WIP	1310 RM	$43,150
Labor confirmations 410 h @ std $48	1320 WIP	7500 absorption (CC credits)	$19,680 → of which labor $ per rate split; overhead applied same event
Overhead applied 410 h @ $27	1320 WIP	7500 absorption	$11,070
Scrap at winding (damaged copper)	5320 Scrap	1320 WIP	$600
FG receipt at standard	1330 FG	1320 WIP	$61,000
Settlement — MUV (43,150 − 42,000 = 1,150 unfav)	5220	1320	$1,150
Settlement — labor+OH variance (30,750 applied? vs std 19,000) split LRV/LEV/OAV per formulas	5230/5240/5250	1320	$11,150 total, split per Part A §8
Result	—	—	WIP balance for the order = exactly $0

Dev Note: Numbers are illustrative; the settlement engine must reproduce the invariant, not the example: FG receipt at standard + variances posted = total actual charges − scrap already expensed, leaving order WIP at zero. The 7500 absorption credits sit on the sending cost centers and are the counterweight measured by CO-R3 over/under absorption.

5. Actual Costing Note (Scope Boundary)

The system is standard-cost primary for make items with full variance capture; moving average remains available for purchased items per INV valuation methods. A parallel actual-costing ledger (periodic actual cost per unit) is explicitly out of Phase 1–2 scope; the data to enable it later (actual charges per order, per-batch receipt costs) is already captured, so no schema change is expected.

6. Pricing Masters

6.1 Price Lists

Screen ID	PRC-PL-001
URL Pattern	/pricing/price-lists
Purpose	Header master for the lists consumed by the Material Master Sales tab (Part A §4.2)

Field	Specification
List Code / Name	WHOLESALE, OEM, GOVT, DISTRIB + customs
Entity / Currency	One currency per list
Basis	Manual rates / Derived (base list + formula: ±% or margin-over-standard) — derived lists recompute on demand with preview, never silently
Rounding Rule	e.g. nearest $1 / $5 / two decimals
Validity, Status, Approval	Draft → Pending Approval → Active → Archived; mass rate changes ride the same approval (Pricing Manager; Controller when any row breaches floor)

6.2 Customer Price Agreements

Contract container: customer, validity, scope (items / groups), rates or discount matrix off a base list, volume rebate terms (accrual note to ACC), attachment of signed contract. Rows surface on the item Sales tab §4.3 and resolve at precedence step 2 of the Pricing Service.

6.3 Margin Floors

Level	Floor Source	Breach Handling
Product line (PC)	Annual plan: floor % per PC (e.g. PC-100 22%, PC-300 18%)	Quote/SO/price rows below floor route: Pricing Manager → Sales Director → (below floor−5pts) CFO; every approval stores justification; breach register report monthly
Item override	Optional stricter floor per item	Same chain

6.4 Commodity Surcharges (Copper / CRGO Escalators)

Element	Specification
Index master	Indexes: COMEX copper $/lb, CRGO basis $/kg (published internal basis), monthly values maintained by Costing with source note; history immutable
Surcharge rule per item/family	Base index value (the value embedded in the list price), content per unit (e.g. lbs copper per unit — from design sheet or estimate), formula: surcharge = (index at pricing date − base index) × content × pass-through % (default 100)
Document behavior	Quote/SO lines show surcharge as separate component line with index, date and math printed; re-fires at defined repricing events only (quote issue, order confirm — per contract term Fixed at Order vs Ship-Date Adjusted; the latter recalculates at PGI and prices the delta on the invoice as a surcharge adjustment line)

7. Margin Analytics

Report	Content
Quote Margin Register	All quotes: estimate cost, price, margin, floor status, approval trail
Order Margin — three-view	Per SO line: quoted margin (vs estimate), standard margin (vs standard at PGI), actual margin (vs settled WO actuals) — the closed loop from Part B §6
Price Realization	Average realized price vs list by item/customer; discount and surcharge decomposition
Floor Breach Register	Every below-floor approval with ageing and win/loss outcome

Role	Rights
Cost Accountant	Estimates, costing runs (mark), rates draft, driver tables
Controller	Rate release, costing release, floor policy, interim runs
Pricing Manager	Price lists, agreements, surcharge rules, first-line floor approvals
Estimator	Parametric estimates, quote costing
Sales roles	Consume prices; margin visibility per company toggle (Part B §6)

PART J

CROSS-MODULE ACCOUNTING INTEGRATION MAP (EXTENDED)

This table extends the inventory map (INV Section 12) and the accounting map (ACC Section 28) with every posting introduced in this volume. Together the three tables are the complete journal universe of the system. CC = cost center, PC = profit center, derived per Part H §3 unless stated.

#	Transaction	Debit	Credit	Amount Basis	Trigger / Dimensions
17	RMA receipt (PE ownership)	1335 RMA Inventory	5120 COGS reversal	Qty × valuation at original issue	RMA receipt post; PC item line
18	RMA restock	1330/1310 Inventory	1335 RMA Inventory	Valuation	Disposition Restock
19	RMA credit note	4100 Revenue (+2300 tax reversal)	1200 AR; 4910 restocking fee portion	Credit value	ACC 4 credit note from CMR; PC item line
20	RMA scrap	5320 Scrap & Waste	1335 RMA Inventory	Valuation	Disposition Scrap; CC quality
21	Warranty repair settlement	2610 Warranty Reserve (overflow 6300)	1325 WIP-Rework	Actual repair cost	RWO close (warranty)
22	Billable repair	5130 COGS-Service / 1200 AR	1325 / 4150 + 2300	Cost; invoice value	RWO close + service invoice; PC-400
23	Replacement shipment (warranty)	6300 Warranty Expense	1330 FG	Standard cost	PGI on $0 replacement SO
24	Warranty reserve accrual	6300 Warranty Expense	2610 Warranty Reserve	Accrual % × FG shipments (policy rate per product line)	Month-end recurring JE (ACC 9.2); PC product line
25	PGI — sales shipment	5120 COGS	1330 FG	Qty × standard	Shipment post; PC item line
26	Freight out accrual	6410 Freight Out	2100 AP accrual	Quoted freight	PGI (PE-paid terms); CC shipping
27	Packing materials consumption	5120 COGS (packaging element)	1360 Packaging on Hand	Backflush at standard	HU close
28	Material issue to WO	1320 WIP	1310 RM	Qty × valuation	SP-08/backflush; CC from WC, PC parent item
29	Labor + overhead applied	1320 WIP	7500 Absorption (CC credit)	Hours × std rates	Op confirmation
30	Operation scrap	5320 Scrap & Waste	1320 WIP	Scrapped value	Confirmation scrap qty; CC of WC
31	FG production receipt	1330 FG	1320 WIP	Good qty × standard	Receive to stock
32	WO settlement variances	5220/5230/5240/5250	1320 WIP	Per Part A §8 formulas	Order close; CC primary WC, PC product line

#	Transaction	Debit	Credit	Amount Basis	Trigger / Dimensions
33	Rework (production-caused)	1325 WIP-Rework → parent 1320	various	Actuals	RWO settlement
34	Subcontract provision	1315 Subcontract at Vendor	1310 RM	Valuation	Issue to subcontractor
35	Subcontract receipt	1320 WIP	2110 GR/IR + 1315	PO price + provided material	GRN subcontract; variance → 5260
36	Allocation cycle	Receiver CC 7500	Sender CC 7500	Driver share	Period-end run; net zero
37	Milestone invoice	1200 AR	2600 Contract Liability (or 4100 per ASC 606 timing)	Milestone %	Invoice request post; rev rec per ACC 24
38	Surcharge adjustment at PGI	1200 AR	4100 Revenue (surcharge element)	Index delta math	Ship-date-adjusted contracts

Dev Note: Every posting above flows through the shared Posting Service: single implementation of account mapping, dimension derivation (Part H §3), period validation (Part A §9), and journal creation with source-document back-links. No module posts to the GL directly.

Period Close Additions

The ACC 26.2 twelve-step checklist gains these operational sub-steps (auto-validated like the rest):

Step	Task	Validation
5b	Allocation cycles posted	All active cycles have a posted run for the period
6b	Inventory period closed	INV 8.3 status = Closed for the period
6c	WIP reconciliation	Σ open WO WIP (Part D §7) = GL 1320 + 1325 balances
6d	Production orders settled	No WO in Completed status older than 5 days unsettled
6e	RMA aging review	No RMA in Settling with posted-document gaps
9b	Warranty reserve adequacy	Reserve balance vs open warranty RWO exposure; top-up JE proposed

PART K

DATA MODEL, INTEGRATION MATRIX & ROADMAP

1. New Core Entities

Entity	Key Fields	Relationships
Customer	customer_id, class, status, price_list_id, credit_limit, hold_flag	→ Addresses, Contacts, Tax Profile, SOs, RMAs
Quotation / QuoteLine	qt_id, version, status, valid_until; line: item_id, config_json, ce_id, price, source	→ Customer, Cost Estimate, SO (conversion)
SalesOrder / SOLine	so_id, type, status, billing_plan; line: item_id, qty, price, promised_date, peg refs	→ Customer, Quote, WO, Shipments, Invoices, Revenue Contract
Shipment / HandlingUnit / HULine	shp_id, status; hu_id, type, weights, sscc; contents	→ SOLines, Picklists, PackOrders
RMA / RMALine	rma_id, class, status, warranty_flag; line: item, serial, qty auth/recd, disposition	→ Customer, Invoice, CreditNote, RWO, NCR
EngineeringRequest	er_id, so_line_id, design_type, due, status, checklist	→ SOLine, EBOM, Documents
EBOM / EBOMLine	ebom_id, parent_item, rev, status; line: component, qty, scrap, op_map, effectivity	→ Items, MBOM, Documents, ECOs
Document / DocRevision / Transmittal	doc_id, type, security; rev, files, status; tr recipients & responses	→ Items, BOM lines, Routings, WOs, SOs
ECR / ECO / ECOTask	ecr_id, impact snapshot, ccb votes; eco_id, effectivity, dispositions; tasks	→ affected objects
WorkCenter / Routing / RoutingOp	wc_id, cc_id, capacity, rates; rtg_id, rev, status; op: hours, qc_flag, wi docs	→ CostCenter, WOs
ProductionOrder / WOOperation / Confirmation	wo_id, item, serial, revs, status, pegs; op status; conf: qtys, hours, operator	→ MBOM, Routing, SOLine, Inspections, GL
ReworkOrder	rwo_id, origin, receiver, status	→ WO/RMA/NCR
GenealogyEvent / SerialLink / BatchConsumption	per Part F §5	→ Serials, Batches, WOs
PR / RFQ / RFQResponse / PO / POLine / POSchedule	standard procurement chain with revisions	→ Vendors, Items, WO/SO pegs, GRNs
AVLEntry	item_id, vendor_id, rank, status, price_basis, fai_ref	→ Item, Vendor
CostCenter / ProfitCenter / Hierarchy / SKF / AllocationCycle	per Part H	→ every journal item
CostEstimate / CostingRun / WCRateVersion	per Part I	→ Items, Quotes
PriceList / PriceListRate / CustomerPrice / SurchargeRule / IndexValue	per Part I §6	→ Items, Customers
InspectionPlan / PlanCharacteristic	per Part A §6	→ Items, Vendors, Operations, Inspections
CountDocument / CountLine	per Part A §7	→ CountPlan, Locations, SAs
CAPA / SCAR / Instrument / CalibrationEvent	per Part E	→ NCRs, Vendors, Tests

2. Integration Matrix (New Flows)

Source	Target	Trigger	Payload / Effect
SO Confirm (ETO)	Engineering Request	Auto-create	Configuration attributes, due date
EBOM Release	MBOM transform	Guided action	Structure + op mapping
MBOM Release	Costing	Invalidation flag	Next run re-rolls item
MRP Run	PRs / Planned WOs	Nightly + on-demand	Proposals with pegs
WO Release	Inventory	Reservation firm-up + serial generation	INV 5.5 source Production
Op Confirmation	GL + WIP + genealogy	Real-time	Postings #28–30, events SP-08…SP-13
QC gate	WO	Inspection result	Op complete enable / block
Final Test pass	SO milestone + FTR + CoC chain	Event	Billing request, documents
PGI	GL + AR trigger + serial status	Event	Postings #25–27; invoice request
RMA disposition	Credit note / RWO / replacement SO / NCR	Per §3.6	Linked documents
IQC result	Vendor rating + AVL	Rolling calc	Quality %, SCAR triggers
Budget approval (ACC 21)	WC rates (Part I §2)	Annual cycle	Planned cost pools
Allocation run	CC reports + rates review	Period-end	Posting #36

3. Implementation Roadmap (This Volume)

Sequenced to extend the inventory build (INV Section 13 phases, weeks 1–12) and the accounting plan (ACC Section 27, 24 weeks). Dependencies: Controlling masters precede any posting-bearing module here; Engineering precedes Production; Production precedes full Costing; the nine gap items ride with their host modules.

Phase	Weeks	Scope	Key Deliverables
M1 — Controlling & Foundations	1–4	Part H masters + derivation engine; Posting Service extension; number series; Part A §9 period matrix; Part A §11 tax determination	CC/PC live on all existing postings; CO exception report at zero on test data
M2 — Sales Core & Gap Items	3–8	Customers, quotes, SOs (MTS path), pricing service + Sales tab (§4), safety stock (§10), count documents (§7)	Order-to-cash MTS demo incl. reservation→pick→pack (§5)→PGI→invoice
M3 — Engineering	6–12	Part C complete: requests, EBOM, documents, ECR/ECO, MBOM transform	ETO design cycle demo with customer approval loop
M4 — Production Core	10–18	WCs, routings, WOs, terminal, confirmations, WIP, receipt; barcode scan points SP-07…SP-13; MRP workbench	Full build of a pilot transformer in-system with genealogy
M5 — Quality & RMA	14–20	Part E (plans already from M2 config), final test & FTR, CAPA/SCAR, calibration; Part A §3 RMA end-to-end	Quality gates enforced; RMA classes RC1–RC4 demo
M6 — Procurement & Costing	16–22	Part G full loop incl. subcontract; Part I rates, estimates, costing runs, surcharges; CVA workbench (§8)	Standards released system-wide; PPV/settlement variances reconciling
M7 — Integration Hardening & UAT	20–26	Extended close checklist, traceability performance, load tests at target volumes, security/SOD review, training	Parallel-run month-end close; go-live readiness sign-off

4. Non-Functional Requirements

Area	Requirement
Volume assumptions	50k items, 5k active BOM structures avg 120 lines × 4 levels, 300 WOs/month, 2,000 SO lines/month, 40k scan events/day, 1.2M journal lines/year
Performance	Scan round-trip < 400 ms P95; MRP full run < 30 min; traceability tree < 10 s; costing run (full) < 60 min; dispatch list refresh < 2 s
Availability	Shop-floor path (terminal, scans, printing) 99.5% during production hours; offline scan queueing per Part F §2
Auditability	Every posting reachable from its source document in ≤ 3 clicks and vice versa; scan telemetry 24 months; document files WORM after release
Security	RBAC per ACC 23 extended with roles named in each Part; SOD checks extended: WO confirm vs settlement approve; PO create vs GRN post; price edit vs price approve

5. Document Sign-Off

Metric	Count
Parts	11 (A–K)
Gap-register items closed	9 of 9 (100%)
New screens specified	75+ across Sales, Engineering, Production, Quality, Purchasing, Controlling, Costing
Field-level tables	190+
GL posting rules added	22 new journal templates (#17–38) with dimensions
Status lifecycles	18 complete workflows
Barcode scan points	21 (SP-01…SP-21) with validations and postings
Seed data sets	Work centers, cost/profit center hierarchies, number series, quality test catalog, RMA policies

This volume, together with Inventory & Warehouse Specification V3.0 and Accounting Implementation Specification V2.0, constitutes the PE Transformer ERP Build Package — Phase 1 functional scope with an architected target-state path per the Enterprise Completeness Specification V1.0. Any deviation during development must be documented and approved by the PE ERP project lead before implementation.

Role	Name	Signature	Date
Project Sponsor			
Product Owner			
Technical Lead (PE)			
Development Partner Lead			
Engineering Manager			
Production Manager			
Quality Manager			
Controller			
CFO			

BOOK II

MATERIAL MANAGEMENT MODULE

Material Model • Material Master Views • Creation & Change Governance • Valuation & Account Determination
Cross-Module Linkage Contracts • Special Stock • Batch & Serial Profiles • Migration & Data Quality

Distribution List

Role	Access Level	Entity
Engineering Lead	Full Access	PE Transformer
Development Partner Team	Full Access	PE Transformer
Materials Manager / Master Data Steward	Full Access	PE Transformer
Purchasing Manager	Review & Comment	PE Transformer
QA Lead	Review & Comment	PE Transformer
Controller / Finance	Review & Comment	PE Power Grids
Project Management	Read Only	PE Transformer

1. Purpose, Scope and Position in the Document Set

This volume is the definitive, field-by-field implementation specification for Material Management in the PE Transformer ERP. It is the fourth controlled volume of the build package and is written to the same standard, structure and depth as the three volumes already in the development partner's hands:

Volume	Covers	Status
Inventory & Warehouse Module Spec V3.0	Warehousing, stock transactions, serial/batch tracking, counting, valuation screens, inventory GL map. Owns the Material Master record physically (screens INV 2.2/2.3).	Released — build in progress
Accounting Module Implementation Spec V2.0	AR, AP, GL, bank, tax engine, intercompany, budgeting, workflows, RBAC, ASC 606, period close.	Released
Manufacturing Core Spec V2.0	Order Management, Engineering & BOM, Production, Quality, Barcoding & Traceability, Procurement, RMA, Controlling, Costing & Pricing, plus the nine gap-register closures.	Released
This Book — Material Management	The material object itself and every contract it has with the rest of the system: material types and numbering, all Material Master views to field level, creation and change governance, valuation and automatic account determination, purchasing info records and source lists, planning parameters, special stock ownership, batch/serial profiles, mass maintenance, migration and data quality.	This document

1.1 Why a Separate Material Management Volume Exists

The material hub principle is stated in Book I (MFG Core §2.3, the Material Master is the hub), and the hub itself must be specified as a buildable object. Attributes were defined where each module needed them: the Sales tab in MFG Core Part A §4, purchasing data in Part G, planning data split between INV 2.2 and Part D, valuation data in INV 2.3.3. What was missing, and what this volume supplies, is:

- A single material type model with a behavior matrix that tells the development team exactly which views, validations, postings and defaults apply to each kind of material PE handles, from CRGO coils to nameplates to transformer oil to MRO consumables to service items.
- The complete Material Master screen specification: every view, every field, every validation, every button, ownership and edit rights per field group, and what each downstream module reads from it.

- Governed lifecycles: how a material comes into existence (including the engineering placeholder path already sketched in MFG Core Part C §3.2), which departments must complete which data before it can be used, how it is changed under control, and how it is retired without stranding stock, open orders or history.
- Automatic account determination: the deterministic matrix that turns any stock-relevant transaction into GL postings using valuation class and event keys, replacing the per-screen posting notes with one engine specification.
- Purchasing master data in full: vendor-material info records, source lists and their interaction with AVL, MRP source determination and PO pricing.
- Special stock and ownership types: vendor consignment, customer property, subcontract stock, sales-order-pegged stock and in-transit stock, each with quantity, value and posting treatment.
- Data governance, mass maintenance, migration and completeness measurement, covering data governance, migration and the master ownership register.

Build Note: *Nothing in this volume moves a screen out of INV, ACC or MFG Core. Where an existing screen gains fields or behavior, the change is specified here as a delta and flagged "DELTA to INV x.x" or "DELTA to MFG x". Where an existing specification already covers a view in full (for example the Sales tab, MFG Core Part A §4), this volume references it and specifies only what is new. Build against all four volumes together.*

1.2 Enterprise Requirements Addressed by This Book

The following enterprise requirements are specified or advanced in this Book. Finite scheduling, CAD and PLM connectors, intercompany, platform architecture, testing and cutover are owned by Book III and Book IV.

Ref	Requirement	Specified In
F-04 (partial)	Plant-specific master data readiness for multi-plant manufacturing	Section 3.3 (plant-level data split), Section 18 (material_plant_data entity)
F-11 (partial)	Consignment/VMI foundations within advanced procurement	Section 14.2 vendor consignment, Section 7.4 source list
T-01 (partial)	Physical data model for master data: keys, effectivity, audit	Section 18 with constraints, effectivity and audit rules for all MM entities
T-10	Master-data stewardship, quality rules, lineage, retention	Part F: governance roles, completeness KPIs, change logging, archive rules
D-01	Cross-volume master ownership register	Section 3.2 field-group ownership matrix, Section 6–13 linkage contracts
D-04 (partial)	Migration: source inventory, mapping, cleansing, mock loads	Section 16.3 migration specification for material, batch, serial and info-record data

1.3 Reading Conventions

- Screen IDs follow the established pattern with the module prefix MM (MM-MAT-001, MM-MCR-001, MM-IR-001, MM-SL-001, MM-VAL-001). Existing screens keep their INV/MFG IDs.
- Field tables use the columns Field / Type / Req / Specification. "Cond." means conditionally required and the condition is stated in the specification column.
- GL posting tables use the chart of accounts already defined in INV 11.4, ACC Section 13 and MFG Core §2.4. New accounts introduced by this volume are listed in Section 2.6 and follow

the same numbering logic; every posting rule references mapping keys, never hard-coded numbers.

- Every listing screen inherits the standard table pattern and every create screen inherits the standard layout exactly as established in INV Section 2 and ACC Section 2. Every screen carries the Entity Selector, immutable Audit Trail and Document Attachment components defined in ACC Section 2. These global behaviors are not repeated per screen.
- The abbreviations INV, ACC and MFG refer to the three companion volumes with their section numbers.

PART A

MATERIAL MODEL AND MASTER DATA

Part A defines what a material is in this system: the type model that drives all behavior, the numbering and classification schemes, the unit-of-measure model, and the complete view-by-view Material Master specification.

2. Material Type Model

2.1 Material Types

Every material carries exactly one Material Type, assigned at creation and immutable afterwards (conversion is a controlled supersession, §4.6, never an edit). The type decides which Material Master views exist, which defaults apply, whether the material is inventory-managed and valuated, and which valuation classes are allowed. The current item categories in INV 2.2 map onto these types one-to-one at migration (Section 16.3).

Type	Code	Typical PE Examples	Key Behavior
Raw Material	RAW	CRGO coil by heat, copper wire/CTC/foil, insulation kraft, tank steel plate, transformer oil in bulk	Inventory-managed, valuated (1310 range), batch-managed by default, purchased only (no BOM as parent), QI-required by default
Semi-Finished	SEMI	Wound coils, stacked cores, fabricated tanks, sub-assemblies	Inventory-managed, valuated (1320/1310 per policy), made in-house, must carry released EBOM/MBOM and routing before use, serial or batch per profile
Finished Good	FIN	Padmount transformers, power transformers, catalog distribution units	Inventory-managed, valuated (1330), sellable default ON, serialized default ON, standard cost mandatory before first receipt
Trading Good	TRADE	PE Grid Components merchandise: bushings, arresters, connectors bought and resold	Inventory-managed, valuated (1330 or 1310 per mapping), purchased and sellable, no BOM as parent
Packaging	PACK	Crates, skids, oil totes, desiccant, VCI wrap	Inventory-managed, valuated (1360), consumed by backflush at HU close (MFG Part A §5.4)
Spare Part	SPARE	Service spares for field and warranty work: gaskets, gauges, valves, radiator fins	Inventory-managed, valuated, sellable optional, links to service BOMs (Phase 2 field reserved)
Consumable / MRO	CONS	Welding wire, gloves, blades, shop supplies	Inventory-managed optional (config per material), expensed at receipt or issue to cost center; no product costing relevance
Service	SERV	Testing services, subcontract operations, freight, engineering services	Never inventory-managed, never valuated; PO lines post to expense or WIP per context; sellable for service revenue (4150)
Non-Stock	NSTK	One-time buys direct to a WO or cost center, customer-supplied reference items	No stock record; quantity tracked on the consuming document only; expensed on receipt

2.2 Type Behavior Matrix

This matrix is configuration seed data, not code. Each cell is enforced by the Material Master save validations and by document entry (a PO line for a SERV material can never create stock; an SO line for a RAW material is blocked unless Sellable was explicitly allowed by an authorized role).

Behavior	RAW	SEMI	FIN	TRADE	PACK	SPARE	CONS	SERV	NSTK
Inventory-managed	Yes	Yes	Yes	Yes	Yes	Yes	Config	No	No
Valuated	Yes	Yes	Yes	Yes	Yes	Yes	Config	No	No
Purchasable	Yes	Exception	Exception	Yes	Yes	Yes	Yes	Yes	Yes
Producible (BOM parent)	No	Yes	Yes	No	No	Config	No	No	No
Usable as BOM component	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Subcontract op only	No
Sellable default	Off	Off	On	On	Off	Config	Off	On	Off
Batch-managed default	On	Config	Off	Config	Off	Config	Off	n/a	n/a
Serialized default	Off	Config	On	Config	Off	Config	Off	n/a	n/a
QI at receipt default	On	On (prod.)	On (prod.)	Config	Off	Config	Off	n/a	n/a
Costing relevance	Std/MAV	Standard	Standard	MAV	MAV	MAV	Expense	Rate only	Expense

> **Build Note:** *“Exception” for purchasable SEMI/FIN covers buy-complete scenarios (a fabricated tank bought outside during capacity crunch). The purchase of a make item requires the Special Procurement key on the Planning view (§3.6) so MRP and costing know the sourcing; it is not a free-for-all flag at PO entry.*

2.3 Material Numbering

Material numbers are system-assigned from type-specific series, extending the Number Series Configuration screen (INV 11.3). Legacy codes migrate into the Legacy Code field and remain searchable everywhere a material lookup exists.

Material Type	Format	Example
RAW	RM-{GROUP:3}-{SEQ:5}	RM-CRG-00114 (CRGO), RM-CU-00021 (copper)
SEMI	SA-{GROUP:3}-{SEQ:5}	SA-COIL-00092
FIN	TX-{FAMILY:3}-{SEQ:5}	TX-PAD-00034, TX-PWR-00007
TRADE	GC-{GROUP:3}-{SEQ:5}	GC-BSH-00220
PACK	PK-{SEQ:5}	PK-00017
SPARE	SP-{GROUP:3}-{SEQ:5}	SP-GSK-00456
CONS	MRO-{SEQ:5}	MRO-00873
SERV	SV-{SEQ:5}	SV-00041

Material Type	Format	Example
NSTK	NS-{SEQ:5}	NS-00202

- The {GROUP} token is the first-level material group code (§2.4), fixed at creation. Renumbering is prohibited; if the group was wrong the material is superseded (§4.6).
- Manual numbers are not permitted. The Engineering Placeholder path (§4.3) reserves the number at placeholder creation so drawings and BOM lines never reference a temporary code.
- Material numbers never encode revision. Revision-controlled materials (fabricated parts to drawing) carry revision on the linked document set and BOM effectivity, per MFG Part C.

2.4 Material Groups and Classification

Material Groups are a three-level hierarchy used for search, reporting, approval routing, material overhead rates (MFG Part I §1) and default assignment of valuation class, buyer and inspection plans. The hierarchy is maintained in Configuration → Material Settings → Material Groups (Materials Manager approval; effective-dated; nodes deactivate, never delete).

Level 1 (examples)	Level 2 (examples)	Level 3 (examples)
CRG Core Steel	CRGO Standard Grades, CRGO HiB	M-4, M-5, 23ZDKH85
CU Copper & Conductors	Wire, CTC, Foil, Busbar	By gauge/spec family
INS Insulation	Kraft/Crepe, Pressboard, DDP, Resins	By spec
OIL Fluids	Mineral Oil, Ester Fluid	By standard (D3487 Type II…)
STL Steel & Fabrication	Plate, Structural, Hardware	By grade/size family
BSH Components	Bushings, Tap Changers, Gauges, Relays, Radiators	By voltage class / vendor family
PKG Packaging	Crates, Totes, Dunnage	—
MRO Consumables	Weld, PPE, Shop	—

In addition to the group hierarchy, each material carries the classification attributes needed for compliance and trade: HTS/HS code (customs), Schedule B (export), country of origin default (overridden by batch actuals at receipt), Buy America relevance flag (feeds MFG Part G §5.1), and hazardous material profile (UN number, packing group, hazard class) used by Packing (MFG Part A §5) and shipping documents. A free characteristic panel (typed key-value attributes defined per material group, e.g. voltage class, kVA, conductor gauge) supports parametric search; characteristics are search and print data only in Phase 1 and become configuration model inputs in the product-configuration phase flagged by the review (F-02).

2.5 Units of Measure

The UoM model extends INV 2.6 and is restated here as the binding contract because every module depends on it:

- Base UoM: the unit in which stock quantities and valuation are kept. Immutable after the first stock-relevant posting (hard rule; enforced at save, E-MM-04).
- Alternative UoMs with fixed conversion factors: Purchase UoM (per vendor overridable in the info record, §7.2), Sales UoM (MFG Part A §4.1), Issue UoM (shop-floor convenience, e.g. copper issued by kg while stocked by kg but bought by lb), and up to 6 additional conversions. Every conversion is a rational pair (numerator/denominator) to avoid floating drift: 1 lb = 45359237/100000000 kg is stored, not 0.45359237.

- Dimension classes (mass, length, volume, count, area) prevent nonsense conversions; cross-dimension conversions (e.g. length of conductor to mass) are permitted only via a material-specific conversion with both UoMs declared on the material and an engineering-approved factor.
- Catch weight is out of scope; oil is bought and stocked by volume (gal) with density held as a characteristic for freight calculations.

2.6 New GL Accounts Introduced by This Volume

These accounts extend GL Account Mapping (INV 11.4) and the CoA (ACC Section 13). As with the MFG Core additions, the numbers are proposals consistent with existing ranges; the Controller confirms or remaps them once in GL Account Mapping, and every posting rule references mapping keys.

GL Account #	Account Name	Type	Used By
1311	Raw Materials — Consignment Offset Memo (statistical)	Memo / statistical	Vendor consignment quantity visibility (§14.2); never in trial balance
1340	Spare Parts Inventory	Balance sheet (asset)	SPARE type materials when the Controller elects separate disclosure; else maps to 1310
2120	Consignment Payable Clearing	Balance sheet (liability)	Consignment withdrawal to settlement (§14.2)
5215	Consignment Price Variance	P&L (expense)	Consignment settlement price vs info record price (§14.2)
5340	Inventory Reserve — Excess & Obsolete	Balance sheet (contra-asset)	E&O reserve postings (§17.3)
6320	Excess & Obsolete Expense	P&L (expense)	E&O reserve build/release

2.7 Document Number Series Additions

Document Type	Prefix / Format	Example
Material Creation Request	MCR-{YYYY}-{SEQ:4}	MCR-2026-0117
Material Change Request (mass or controlled single)	MCH-{YYYY}-{SEQ:4}	MCH-2026-0031
Purchasing Info Record	IR-{SEQ:6}	IR-004215
Supersession Record	SUP-{YYYY}-{SEQ:4}	SUP-2026-0009
E&O Reserve Run	EOR-{YYYY}{MM}-{SEQ:2}	EOR-202607-01

3. Material Master — Complete View Specification

The Material Master remains physically hosted where INV built it (INV 2.2 item listing, INV 2.3 item detail). This section is the binding specification of every view on that record, including the views INV stubbed and this volume completes. All additions are DELTA to INV 2.2/2.3.

3.1 Material Listing (DELTA to INV 2.2)

Screen ID	MM-MAT-001 (extends INV-ITEM-LIST)
URL Pattern	/inventory/items
Purpose	Single register of all materials across types and plants; the working queue for master data stewards, buyers and planners

Added columns and filters:

Column	Type	Specification
Material Type	Badge	RAW (brown), SEMI (teal), FIN (blue), TRADE (purple), PACK (gray), SPARE (green), CONS (gray), SERV (outline), NSTK (outline)
Material Group	Text	Level 1 / Level 2 path; filterable by any level
Lifecycle Status	Badge	Per §4.1: Draft, Data Completion, Active, Restricted, Blocked, Obsolete
Data Completeness	Percent bar	Computed per §16.4: share of mandatory-for-use fields populated across required views; red < 100% for Active materials (should be structurally impossible; report row if it occurs)
Plants Extended	Chips	Plant codes where a plant-data record exists (§3.3); Phase 1 shows the single PE plant
Buyer / Planner / Steward	Users	Owning roles; filter “My Materials” per role
Std Cost / MAV	Currency	Current released standard or moving average per valuation method
Legacy Code	Text	Migrated identifier; searchable

Bulk actions: Assign Buyer/Planner/Steward, Set Block (function-specific, §4.5), Export, Add to Change Request (mass maintenance, §16.1). Quick stat pills: Draft, In Data Completion, Active with Missing Info Record, Restricted, Blocked, Obsolete with Stock.

3.2 Field-Group Ownership (supersedes MFG Core §2.3 table)

Ownership means only the owning role’s screens may edit the field group; every other view renders it read-only with a link to the owning screen. The Posting Service and all documents consume the fields regardless of ownership. This table is the master ownership register required by review item D-01.

View / Field Group	Owner (edit rights)	Primary Consumers
General, classification, UoM, tracking config	Master Data Steward (Inventory Admin role group)	All modules
Planning (MRP) view §3.6	Production Planner	MRP, reservations, ATP, shortage checks
Purchasing view §3.7, info records §7.2, source list §7.4	Buyer / Purchasing Manager	MRP source determination, PR/PO pricing, expediting
Production view §3.8	Manufacturing Engineer	WO creation, backflush, staging, scheduling
Warehouse view §3.9	Warehouse Manager	Putaway/pick strategies, counting
Quality view §3.10	Quality Engineer	GRN QI routing, plan resolution, certificates
Sales view (MFG Part A §4)	Sales Admin	Quotes, SOs, invoices

View / Field Group	Owner (edit rights)	Primary Consumers
Accounting & Costing view §3.11	Cost Accountant / Controller	Valuation, account determination, costing runs
Standard cost (released)	Costing run only (MFG Part I §3) — never manual edit	Valuation, PPV, settlement
BOM view	Engineering (read-only render of MFG Part C output)	Production, Costing, Purchasing
CC/PC defaults	Controlling (MFG Part H)	Every financially relevant posting

3.3 Global vs Plant-Level Data

To satisfy review item F-04 without redesign later, the material record is split now into global data (one per material) and plant data (one per material per plant). Phase 1 operates one plant, so screens show the plant block inline; the data model (Section 18) keys plant data separately from day one. Extending a material to a future plant is an Extend action that copies plant defaults and routes through the same data-completion workflow as creation (§4.2).

Scope	Field Groups
Global (material level)	Type, number, descriptions, material group, classification/characteristics, UoM set, batch/serial profile, hazmat profile, documents, sales-global data (Sellable, warranty default), HTS/origin defaults
Plant level	MRP/planning parameters, purchasing data (buyer, GR processing time), production data (issue warehouse, backflush, tolerances), warehouse assignments and strategies, quality settings (QI flag, plan scope), valuation (class, method, standard cost, MAV) — valuation is per plant per entity

3.4 View: General (DELTA to INV 2.3.1)

Field	Type	Req	Specification
Material #	Auto	Auto	Per §2.3 series; shown with copy button and barcode preview (Code128)
Short Description	Text (80)	Yes	Uppercase-normalized for search; uniqueness warning (not block) on 90% similarity per §4.4
Long Description	Rich text (2000)	No	Prints on POs and travelers when flagged
Material Type	Badge	Auto	From creation; immutable (§2.1)
Material Group L1/L2/L3	Cascading dropdowns	Yes	Drives valuation class default, buyer default, overhead group, search
Legacy Code	Text (40)	No	Migration identifier; indexed for lookup
Manufacturer / Mfr Part #	Table	No	Rows: Manufacturer, MPN, preferred flag. Used by CAD BOM import matching (MFG Part C §3.3) and PO print
Characteristics	Panel	Cond.	Typed attributes per material group class (§2.4); mandatory set configurable per group (e.g. voltage class mandatory for BSH group)
HTS Code / Schedule B / Country of Origin	Lookups	Cond.	Mandatory before first international PO or export shipment (validation fires at document, not at save)

Field	Type	Req	Specification
Hazmat Profile	Group	Cond.	UN #, class, packing group, proper shipping name; mandatory when Hazardous toggle ON; consumed by HU labels and BOL
Buy America Relevant	Toggle	No	ON: origin accumulation and vendor BA cert requirements activate on POs (MFG Part G §5.1)
Lifecycle Status	Badge	Auto	Per §4.1 with function-block chips (§4.5)
Steward / Buyer / Planner	User lookups	Yes (Active)	Owning users; mandatory before Active

3.5 View: Units & Dimensions (DELTA to INV 2.3.2)

Field	Type	Req	Specification
Base UoM	Dropdown	Yes	Immutable after first posting (E-MM-04)
Alternative UoMs	Table	No	Rows: UoM, numerator, denominator, usage tag (Purchase / Sales / Issue / Other); rational factors per §2.5; delete blocked while referenced by an open document
Net / Gross Weight + Weight UoM	Numbers	Cond.	Mandatory for FIN, SEMI, RAW (freight, HU gross-weight sanity ±2% per MFG Part F SP-17)
Volume + UoM	Number	No	Oil and fluids: mandatory
Dimensions L×W×H	Numbers	Cond.	Mandatory for FIN (permits and load planning, MFG Part B §5.1)

3.6 View: Planning (MRP) — NEW

This view consolidates and supersedes the scattered planning fields (INV 2.2 reorder fields, Safety Stock per MFG Part A §10). It is the single input surface the MRP Workbench (MFG Part D §3) reads. All fields are plant-level.

Field	Type	Req	Specification
MRP Type	Dropdown	Yes	MRP (deterministic net-requirements; default for RAW/SEMI/FIN in BOMs), Reorder Point (consumption-based; default for CONS/PACK/SPARE), None (no planning; NSTK/SERV forced). Drives which engine plans the material — an item is never planned by both (V-MM-07)
Procurement Type	Dropdown	Yes	Buy / Make / Both (Both requires planner quota note; MRP defaults to Make and flags for review)
Special Procurement	Dropdown	No	Subcontracting (component provision per MFG Part D §10 / G §6), Consignment (§14.2), Buy-Complete for make item (§2.2 note), Stock Transfer (reserved for multi-plant)
Lot Sizing	Dropdown	Yes	Lot-for-Lot (default), Fixed Qty (+ field), Min Qty / Order Multiple (+ fields), Period (weekly bucket consolidation + field)
Safety Stock	Link + grid	No	Per-warehouse grid per MFG Part A §10 (renders here; same data)

Field	Type	Req	Specification
Reorder Point / Max Stock	Numbers	Cond.	Mandatory when MRP Type = Reorder Point; arithmetic per MFG Part A §10.2
Planned Delivery Time (days)	Number	Cond.	Buy items: vendor-independent default; the info record (§7.2) overrides per vendor at source determination
GR Processing Time (days)	Number	No	Dock-to-stock time incl. inspection; MRP adds it to supply availability dates
In-House Production Time (days)	Number	Cond.	Make items without released routing (early ETO); once a routing is released, scheduling uses routing times and this field grays out with “from routing”
Planning Time Fence (days)	Number	No	Inside the fence MRP does not create or re-date proposals automatically; exceptions EX-08 “inside fence” raised instead. Answers the review’s planning-fence gap at the material level
Scrap % (assembly)	Number	No	Parent-level scrap applied on top of component scrap from BOM lines
Backflush	Toggle	No	ON: components of this material’s orders consume at confirmation per MFG Part D §6.1; per-component override on the MBOM line wins
Issue Method	Dropdown	Yes (make)	Staged Kit (picklist to WC staging per SP-09), Direct Issue (scan at op, SP-08), Backflush; default per material type, overridable per MBOM line
Strategy	Dropdown	Yes (FIN/SEMI)	MTS (plan from forecast + SO consumption), MTO/ETO (plan only from SO peg; no anonymous stock), Assembly-to-Order reserved Phase 2. Governs whether forecast rows (INV 15.8) are consumed for this material
ABC / XYZ Class	Badges	Auto	ABC by value (annual usage × cost), XYZ by demand variability; computed monthly by job; drives count frequency (INV 7) and buyer review lists
Discontinuation	Group	Cond.	Run-out date + Follow-Up Material (supersession §4.6); MRP switches new proposals to the follow-up after stock of the old is consumed

Build Note: *MRP reads exactly this view plus BOM, routing, info records and source list. Any field MRP consumes that is not on this view (or its linked grids) is a build defect. The MRP Workbench exception EX-07 (missing/unreleased BOM or routing) extends with EX-08 (inside planning fence) and EX-09 (no valid source of supply, §7.4).*

3.7 View: Purchasing — NEW (consolidates MFG Part G references)

Field	Type	Req	Specification
Buyer	User	Yes (Buy items)	Workbench bucketing (MFG Part G §1)
Purchase UoM	Dropdown	No	Default order unit; info record may override per vendor; conversion to base mandatory

Field	Type	Req	Specification
Min Order Qty / Order Multiple	Numbers	No	Applied to PR→PO grouping and MRP lot rounding
Over/Under Delivery Tolerance %	Numbers	No	GRN validation: over-receipt beyond tolerance requires Purchasing Manager override (E-MM-11); under-delivery closes line as short-closed when flagged final
GR-Based Invoice Verification	Toggle	Yes	Default ON: three-way match at ACC 6 requires GRN; OFF only for SERV with Purchasing Manager approval
Source Determination Policy	Dropdown	Yes	AVL Rank (default; MFG Part G §2.2), Source List Fixed (§7.4 fixed vendor per period), RFQ Always (forces RFQ for every PR, high-value engineered parts)
Default Quality Clauses	Multi-select	No	Pre-selects PO quality clauses Q1–Q5 (MFG Part G §5.1) for this material
Purchasing Text	Rich text	No	Prints on PO below the line (packaging instructions, mill cert wording)
Tax Category (purchase side)	Dropdown	Yes	Feeds use-tax logic per MFG Part A §11.3
Info Records	Embedded grid	—	All IR rows for this material (§7.2) with vendor, price basis, lead time, validity; Add opens MM-IR-002 pre-filtered
Source List	Embedded grid	—	Per §7.4

3.8 View: Production — NEW

Field	Type	Req	Specification
Production Scheduler	User	Yes (Make)	Dispatch escalations and WO exception routing
Default Issue Warehouse	Dropdown	Yes (Make)	Where components stage from; per-WC override via WC master (MFG Part D §2)
Production Batch Size (default)	Number	No	Default WO qty for components made in lots; ETO transformers fixed at 1 (enforced by serialization rule, MFG Part D §5.1)
Over/Under Delivery Tolerance % (production)	Numbers	No	FG receipt beyond WO qty + tolerance blocked (E-MM-12); underdelivery allowed with WO short-close reason
Serialization at	Dropdown	Cond.	WO Release (default FIN; serial exists during build — required for genealogy) / FG Receipt (components serialized only when stocked)
Batch at	Dropdown	Cond.	WO Release (batch per order) / FG Receipt / Per Shift (repetitive, reserved)
Active MBOM / Routing	Read-only links	—	Current released revisions with “EBOM ahead” banner logic per MFG Part C §6
Standard Text / Work Notes	Rich text	No	Prints on traveler cover when flagged

3.9 View: Warehouse & Storage (DELTA to INV 2.3.5)

Field	Type	Req	Specification
Default Warehouse / Bin	Lookups	No	Putaway proposal seed; rules engine (INV 15.6) can override
Putaway / Picking Strategy	Dropdowns	Yes	Putaway: Fixed Bin, Next Empty, Zone by Group. Picking: FIFO, FEFO (forced ON when shelf-life managed, V-MM-09), Serial-Directed
Storage Conditions	Multi-select	No	Indoor Dry, Climate Controlled, Outdoor Allowed, Flammable Cabinet, Oil Containment; validated against warehouse capabilities at putaway (E-MM-13)
Min / Max Bin Qty	Numbers	No	Replenishment of fixed pick bins from bulk (INV movement type Bin Replenishment)
Count Class Override	Dropdown	No	Overrides ABC-derived count frequency for this material
Shelf Life (days) / Remaining Life for GR %	Numbers	Cond.	Shelf-life managed materials: GRN rejects receipts with less remaining life than the threshold (E-MM-14); expiry from vendor batch data at receipt

3.10 View: Quality (DELTA to INV 2.3.7)

Field	Type	Req	Specification
QI Required at Receipt	Toggle	Yes	ON routes GRN to QI Hold and requires an Active INC plan per MFG Part A §6.5 policy
QI Required from Production	Toggle	Yes (Make)	Final/in-process gates from routing QC checkpoints; this flag adds receipt-side inspection for SEMI stocked into inventory
Certificate Requirements	Multi-select	No	Mill Cert per Heat, CoC per Shipment, CoA, FAI Report; GRN blocks Accept until required certificates attach (E-QM-06)
Vendor Qualification Required	Toggle	No	ON: only AVL vendors with Qualified status may appear on POs (hard block E-PUR-02 already specified, MFG Part G §1)
Inspection Plan Scope Preview	Read-only panel	—	Shows which Active plans currently resolve for this material by plan type, with effective dates (runtime resolution per MFG Part A §6.2)
Batch Record Requirements	Multi-select	No	Heat # mandatory, Vendor batch mandatory, Expiry mandatory — enforced as GRN field requirements per material

3.11 View: Accounting & Costing (DELTA to INV 2.3.3)

Field	Type	Req	Specification
Valuation Class	Dropdown	Yes (valuated)	Per §5.1; defaulted from material type + group, editable by Cost Accountant only while no stock and no open documents exist; afterwards changes via §5.4 procedure
Valuation Method	Dropdown	Yes	Standard (mandatory for Make: SEMI, FIN) / Moving Average (default for Buy: RAW,

Field	Type	Req	Specification
			TRADE, PACK, SPARE, CONS). Change procedure per §5.4
Price Unit	Number	Yes	Cost per 1 / 100 / 1000 of base UoM (copper and steel priced per 100 kg avoid rounding drift)
Standard Cost (current)	Read-only	—	Released standard with effective date and costing run link; never manually editable (MFG Part I §3)
Moving Average (current)	Read-only	—	Live MAV with last-movement drill
Future / Previous Standard	Read-only	—	Marked (staged) cost from current run; previous released cost for reference
Profit Center	Lookup	Yes (valuated)	Item product line; feeds derivation rule 3 (MFG Part H §3). Formalizes the “item master PC field” noted there
Overhead Group	Dropdown	No	Material overhead % differentiation per group (MFG Part I §1)
Costing Lot Size	Number	Yes (Make)	Lot size for which the standard is computed; setup amortization uses it
Do Not Cost	Toggle	No	Excludes from costing runs (obsolete or customer-property materials); requires reason
E&O Reserve %	Read-only	—	Current reserve rate applied by the E&O run (§17.3)
GL Determination Preview	Panel	—	Simulates the account determination matrix (§5.2) for this material: shows the accounts each event key resolves to today, via the Posting Service simulation endpoint (MFG Part H §3 Dev Note)

3.12 View: Sales

The Sales view is specified in full in MFG Core Part A §4 (sales data, price list assignments, customer-specific prices, price determination contract) and is not repeated. Deltas introduced by this volume:

- The Sellable toggle is now governed by the type behavior matrix (§2.2): switching it ON for types where the default is OFF requires Sales Admin plus Materials Manager approval and records a governance entry (§4.4).
- Sales view completeness (Sales UoM path, tax category, Avalara code when active) joins the data-completion checklist gating Active status for sellable materials (§4.2).

3.13 Reference Tabs

Tab	Content / Source
BOM	Read-only render of the active MBOM revision (MFG Part C §6); Where-Used button; “EBOM ahead” banner passthrough
Documents	Controlled documents linked to the material (MFG Part C §5): drawings, specs, MSDS/SDS for hazmat (mandatory when Hazardous ON — V-MM-11), photos

Tab	Content / Source
Stock Overview	Live stock by warehouse/bin/status incl. special stock buckets (§14.1): At Subcontractor, Consignment (vendor-owned), Customer Property, SO-Pegged, In-Transit
Info Records / Sources	Purchasing grids per §3.7
Where-Used	Multi-level where-used across EBOMs/MBOMs/inspection plans/price lists (MFG Part C §7 engine)
Change History	Immutable field-level audit: field, old, new, user, timestamp, source (screen / mass change MCH- / migration / API), and governance reference where applicable

3.14 Buttons and Global Actions (Material Detail)

Button	Enabled When	Behavior
Save / Submit for Completion	Draft or Data Completion	Runs view-level validations; Submit routes per §4.2 workflow
Activate	All completion tasks done	Materials Manager action; runs full validation set V-MM-01…; sets Active
Set Block…	Active/Restricted	Function-specific block dialog per §4.5 with reason code
Supersede…	Active/Restricted/Blocked	Starts supersession record §4.6
Extend to Plant…	Active; multi-plant enabled	Copies plant defaults; data-completion workflow for the new plant
Copy From…	Create mode	Deep copy of a template or similar material minus stock/history; copies flagged field groups only (checklist dialog)
Simulate GL	Valuated	Opens §3.11 GL Determination Preview
Print Label	Always	Item shelf label per MFG Part F §1 with reprint governance

3.15 Validation Register (Material Master)

Code	Rule
V-MM-01	Activation requires: descriptions, group, UoM set, steward/buyer/planner per type, all mandatory views complete per type matrix, valuation class + method for valuated types, PC assigned
V-MM-02	Material type immutable; group L1 immutable (encoded in number); L2/L3 changes allowed with governance entry
V-MM-03	Batch-managed and Serialized flags immutable once any stock or genealogy record exists; changes only via supersession
V-MM-04 / E-MM-04	Base UoM immutable after first stock-relevant posting
V-MM-07	Exactly one planning engine per material (MRP Type)
V-MM-09	Shelf-life managed forces FEFO picking
V-MM-11	Hazardous ON requires SDS document link before Active
E-MM-11 / E-MM-12	Receipt tolerance violations (purchase / production)

Code	Rule
E-MM-13	Storage-condition vs warehouse capability mismatch at putaway
E-MM-14	Remaining shelf life below GR threshold at receipt

4. Material Lifecycle and Governance

4.1 Lifecycle Statuses

Status	Color	Meaning	Allowed Transitions
Draft	Gray	Skeleton exists (number reserved); free edit by creator; invisible to documents	Data Completion, Deleted (only status where physical delete is allowed, and only with zero references)
Data Completion	Yellow	MCR workflow running (§4.2); each department completes its view; material usable nowhere except as EBOM placeholder line (MFG Part C §3.2)	Active, Draft (recall), Rejected
Active	Green	Fully usable per type behavior matrix	Restricted, Blocked, Obsolete (via supersession/run-out)
Restricted	Amber	Use-up: existing stock consumable, open documents finish; NEW demand blocked (no new BOM lines, no new PO/SO/WO except closure of existing)	Active (reinstate, Materials Manager), Blocked, Obsolete
Blocked	Red	All movements and documents blocked except the correcting transactions named in the block reason (§4.5)	Active, Restricted, Obsolete
Obsolete	Dark Red	Terminal for use; record retained for history/traceability; stock must be zero and all documents closed before entry (V-MM-20)	None (Archive per retention policy, §18.4)

4.2 Material Creation Request (MCR) Workflow

Screen ID	MM-MCR-001 (queue), MM-MCR-002 (detail)
URL Pattern	/materials/creation-requests
Purpose	Controlled multi-department creation of a material; nothing becomes Active by a single person

Creation paths all converge on the same MCR object:

Path	Initiator	Notes
Engineering Placeholder	Design Engineer from EBOM editor (MFG Part C §3.2 Create Placeholder Part)	Number reserved instantly; MCR auto-created with engineering data pre-filled; EBOM release stays blocked while the MCR is open (V-ENG-03 unchanged)
Purchasing Initiated	Buyer (new vendor part, MRO, packaging)	Purchasing view pre-filled; engineering task skipped for non-engineered types
Sales Initiated	Sales Admin (new trading good / service)	Sales view pre-filled

Path	Initiator	Notes
Migration	Data load (Section 16.3)	MCRs auto-approved by load policy but completeness KPIs still computed

The MCR carries a task checklist generated from the material type. Each task = one view to complete + its owning role + SLA. Default matrix:

Task (view to complete)	Owner	RAW	SEMI/FIN	TRADE	PACK/CONS	SERV
General + classification + UoM	Master Data Steward	Yes	Yes	Yes	Yes	Yes
Planning (MRP)	Production Planner	Yes	Yes	Yes	Yes	No
Purchasing + first Info Record	Buyer	Yes	If buy-complete	Yes	Yes	Yes
Production	Manufacturing Engineer	No	Yes	No	No	No
Warehouse	Warehouse Manager	Yes	Yes	Yes	Yes	No
Quality (+ inspection plan check)	Quality Engineer	Yes	Yes	Config	No	No
Accounting & Costing (valuation class, method, PC)	Cost Accountant	Yes	Yes	Yes	Yes	Yes (rates)
Sales (when sellable)	Sales Admin	No	FIN yes	Yes	No	Yes

- Tasks run in parallel except Accounting & Costing, which requires Planning and Purchasing done first (valuation method and first price basis depend on them).
- SLA default 2 business days per task; breach escalates to the role's manager and appears on the Data Quality dashboard (§16.4).
- Activation: when all tasks complete, the Materials Manager reviews the assembled record and Activates (runs V-MM-01). Rejection returns to the responsible task with comments.
- The MCR stores the originating demand (EBOM line, PR, quote) and unblocks it automatically at activation: the EBOM placeholder flag clears, the PR becomes sourceable, the quote line resolves.

Build Note: *Do not implement per-department "mini masters". There is one material record; the MCR only orchestrates who fills which view and in what order, with field-group edit rights per §3.2 enforced the same during and after creation.*

4.3 Engineering Placeholder Path — Exact Behavior

- Create Placeholder Part (in the EBOM editor) captures: description, material group, UoM, est. cost, target vendor note, drawing link. The system reserves the real material number (§2.3) and creates the material in Draft plus the MCR in Data Completion.
- The BOM line references the real number from minute one; there is no renumbering event later, so CAD drawings annotated with the number never go stale.
- Est. cost from the placeholder becomes the costing simulation basis until the Accounting task releases a real basis (info record award or first standard). Estimates consuming placeholder costs flag "contains placeholder pricing" on the quote margin panel (MFG Part B §3.2).
- Placeholder aging report (MFG Part C §7) reads MCR ages; the 14-day exception remains.

4.4 Change Governance and Duplicate Prevention

Control	Specification
Field change classes	Class 1 — free edit by owner role (texts, notes, non-controlling attributes). Class 2 — governed edit: owner role + approval by the named second role, recorded with reason (Sellable override, valuation class, method, batch/serial flags pre-stock, MRP type, procurement type, PC). Class 3 — never editable (type, number, base UoM after posting); change = supersession. The class of every field is part of the field dictionary (Section 18) — the UI derives editability from it, not from per-screen code
Material Change Request (MCH)	Class 2 single changes and all mass changes (§16.1) run as MCH documents: proposed rows (material, field, old, new), approval per class rules, scheduled apply (immediate or effective-dated), full audit. MCH links to ECO when the change originates from engineering change control (MFG Part C §4.2 task “update material”)
Duplicate check	On create and on description edit: fuzzy match (trigram similarity ≥ 0.9) against descriptions + manufacturer part numbers within the same material group L1. Matches show side-by-side compare; creating anyway requires a “not a duplicate because…” note stored on the MCR. Hard block only on identical MPN + manufacturer (E-MM-02)
Reference-data protection	Deactivation/blocks run a where-used sweep (open POs, WOs, SOs, BOM lines, price rows, info records, source lists, inspection plans) and show the impact list before confirmation; the same sweep gates Obsolete (must be empty) per V-MM-20

4.5 Function-Specific Blocks

Blocks are additive chips on an Active/Restricted material, each with reason code, set-by and date; each blocks a specific document family rather than the whole record:

Block	Blocks	Typical Reason	Set By
Purchasing Block	New PRs/RFQs/POs (open POs receivable)	Quality escalation on the only vendor; commercial dispute	Purchasing Manager
Sales Block	New quotes/SO lines (open SOs shippable unless Shipping Block too)	Recall candidate; margin review	Sales Director
Production Block	New WOs; existing WOs continue unless Immediate-effectivity ECO says otherwise	Design problem under ECR	Engineering Manager
Shipping Block	PGI for this material	Quality hold at product level (batch/serial holds stay in INV 6.2)	Quality Manager
Posting Block	All stock movements (count correction transactions exempt)	Valuation investigation; migration freeze	Controller

4.6 Supersession (Follow-Up Material)

Screen ID	MM-SUP-001
URL Pattern	/materials/supersessions
Purpose	Controlled replacement of one material by another with stock run-out and document cutover

- Supersession record SUP-#: predecessor, successor, relationship (One-for-One / One-for-Many note), run-out policy (Use-Up stock then switch / Immediate switch + stock disposition), effective date, initiating ECO/MCH link.
- MRP behavior: with Use-Up, demand pegs to predecessor stock until exhausted, then proposals flip to the successor automatically (mirrors ECO use-up logic, MFG Part C §4.2). With Immediate, remaining predecessor stock routes to the disposition chosen (rework/scrap/RTV) via the ECO disposition engine.
- Document behavior: open POs on the predecessor raise buyer tasks (change or receive-and-close); BOM lines flip per ECO effectivity; sales cross-references (customer part numbers) carry to the successor with a supersession note printed on order confirmations.
- The predecessor moves Active → Restricted at record approval and → Obsolete when V-MM-20 conditions are met; the successor's record links back for history. Traceability (MFG Part F) is unaffected: genealogy always stores the material that was actually consumed.

PART B

VALUATION AND AUTOMATIC ACCOUNT DETERMINATION

This Part specifies the engine that turns every stock-relevant transaction in all four volumes into GL postings without any screen naming account numbers. It formalizes and supersedes the per-screen posting notes: those tables remain correct, but they are now outputs of this matrix, not independent rules.

5. Valuation and Account Determination

5.1 Valuation Classes

The valuation class is the single attribute that connects a material to its balance-sheet account and its offset accounts. Defaults derive from material type + group and are confirmed in the MCR Accounting task.

Valuation Class	Default For	Inventory Account (event key STOCK)
VC-3000 Raw Materials	RAW (all groups except OIL if Controller elects split)	1310 Raw Materials
VC-3010 Raw Materials — Fluids	RAW group OIL (optional split, Controller decision at config)	1310 (or split account if remapped)
VC-7900 Semi-Finished	SEMI	1310 per current policy (INV valuation) with WIP flow via 1320 during orders
VC-7920 Finished Goods	FIN	1330 Finished Goods
VC-3100 Trading Goods	TRADE	1330 (or 1310 per Controller mapping)
VC-3060 Packaging	PACK	1360 Packaging Materials on Hand
VC-3040 Spares	SPARE	1340 Spare Parts (or 1310 if not split)
VC-3080 Consumables	CONS (when inventory-managed)	1310 or direct expense per config

5.2 Event Keys and the Determination Matrix

Every posting the Posting Service creates resolves accounts by (event key, valuation class) and, where noted, transaction qualifiers. Event keys are closed-list system constants; new business behavior means a new row in the matrix, never code that names an account.

Event Key	Fires On	Resolves To (default mapping)
STOCK	Any valuated stock increase/decrease at inventory account level	13xx per valuation class (§5.1)
GRIR	Goods receipt / invoice receipt clearing	2110 GR/IR
PPV	Purchase price variance at GRN (standard-cost materials) and IPV at match	5210
USAGE	Production issue offset (component to WIP)	1320 WIP (order)
ABSORB	Labor/overhead absorption credit	7500 series per MFG Part H

Event Key	Fires On	Resolves To (default mapping)
VARSET	Settlement variance split	5220/5230/5240/5250 by variance type (MFG Part A §8)
SUBCON	Subcontract provision and receipt legs	1315 / 5260 per MFG Part G §6
SCRAP	Scrap and waste disposals	5320
ADJ	Count and stock adjustments	5310
REVAL	Standard cost release revaluation	5330
COGS	PGI and COGS recognition	5110/5120/5130 per order/item context
RETQ	RMA quarantine stock	1335
CONSIG	Consignment withdrawal liability	2120 Consignment Payable Clearing (§14.2)
CONSPV	Consignment settlement price variance	5215
EORES	Excess & obsolete reserve	5340 / 6320 pair (§17.3)
INTRANS	Stock in transit between own locations	1350 In-Transit

Build Note: *Implementation contract: the matrix is a configuration table (event_key, valuation_class nullable, qualifier nullable, dr_account_key, cr_account_key, effective dates, approval trail). Resolution order: exact (key+class+qualifier) → (key+class) → (key). The Controller edits it through a governed screen MM-VAL-001 with simulation before release (same two-step mark/release pattern as costing runs). Every journal line stores the matrix row ID that resolved it, giving finance a complete lineage from posting back to configuration.*

5.3 Screen MM-VAL-001 — Account Determination Workbench

Screen ID	MM-VAL-001
URL Pattern	/materials/account-determination
Purpose	Controller-governed maintenance and simulation of the determination matrix
Element	**Specification**
Matrix grid	Rows per §5.2 with filter by event key / valuation class; effective-dated versions; diff view between versions
Simulation panel	Pick material + transaction (GRN, issue, PGI, settlement, adjustment…) + qty/value → full journal preview with dimensions (derivation per MFG Part H §3) before saving any change
Release workflow	Draft rows → Mark → Controller Release with effective date; postings always use the version valid at posting date; historical postings never restate (same retroactivity doctrine as MFG Part A §11.5)
Completeness check	Validation: every (event key × valuation class in use) resolves; gaps listed with the materials affected; release blocked while gaps exist (E-MM-21)

5.4 Price Control Rules and Change Procedures

Rule	Specification
Method by type	Make items (SEMI, FIN): Standard, mandatory. Buy items: Moving Average default; Standard allowed for high-volume raws (CRGO, copper) where PPV visibility is wanted — Controller decision per material group
MAV mechanics	Receipt at PO price re-averages; issues at current MAV; invoice price differences post to stock when coverage exists, else to 5210 (matches INV Section 12 logic; restated as the single rule)
Standard mechanics	Receipts at standard with PPV to 5210; revaluation only at costing-run release (REVAL); no mid-period manual standard edits ever
Method change (MAV → Standard)	Allowed at period boundary only: requires released standard, zero open GRN-without-invoice on the material (warning list), Controller approval; conversion posts REVAL for on-hand delta between MAV and new standard
Method change (Standard → MAV)	Controller approval; on-hand revalues to current standard as opening MAV; PPV baseline note recorded
Price unit changes	Allowed with automatic factor-consistent recompute preview; blocked mid costing-run mark window (E-MM-22)

PART C

CROSS-MODULE LINKAGE CONTRACTS

This Part is the answer to the question the review raised about material management linkages: for each neighboring module, exactly which material data it reads, which events flow in each direction, which validations bind the two sides, and which failure behaviors apply. Each contract is written so a developer can implement either side against it without reading the other module's code. Events named here join the integration matrix in Section 19.

6. Linkage: Material ↔ Engineering

6.1 Data Contract

Direction	Data	Rule
MM → ENG	Material number, description, group, UoM, make/buy, lifecycle status, characteristics	EBOM editor lookups (MFG Part C §3.2) may select only materials whose status permits new BOM usage: Active, or Data Completion for placeholder lines. Restricted/Blocked/Obsolete are excluded from component search (E-ENG-08)
ENG → MM	Placeholder creation (description, group, UoM, est. cost, target vendor)	Creates Draft material + MCR per §4.3; number reserved immediately
ENG → MM	Drawing links per material	Document links (MFG Part C §5.1 Linked Objects) render on the material Documents tab; fabricated/made parts require a released drawing before MCR activation of a SEMI (task-level check in the Quality/Production tasks)
ENG → MM	ECO effects	ECO tasks may raise MCH rows (governed field changes, e.g. new revision-relevant characteristics), supersessions (§4.6) and status changes (Production Block during Immediate effectivity). ECO release applies them atomically with the BOM/routing cutover (MFG Part C §4.2)
MM → ENG	Where-used and stock/impact data for ECR impact analysis	The ECR Impact Analysis Panel (MFG Part C §4.1) reads on-hand, on-order and open-document data through the same where-used sweep as §4.4 — one implementation, two consumers

6.2 Events

Event	Producer → Consumer	Payload / Effect
MaterialPlaceholderCreated	ENG → MM	Draft material + MCR; EBOM line reference
MaterialActivated	MM → ENG	Clears placeholder flags on EBOM lines; revalidates affected EBOMs (V-ENG-03)
MaterialSuperseded	MM → ENG	Raises ECO task proposals for BOMs using the predecessor
ECOReleased	ENG → MM	Applies linked MCH/supersession/status changes at effectivity

7. Linkage: Material ↔ Purchasing

7.1 Data Contract

Direction	Data	Rule
MM → PUR	Purchasing view (§3.7), quality clause defaults, tax category, tolerances, Buy America flag	PR/PO line entry reads these at document creation; changes never restate open documents
MM → PUR	Info records and source list (below)	Source determination order per §7.5
PUR → MM	Info record updates from RFQ awards and PO history	Award writes price basis + lead time to the info record (was "AVL price basis" in MFG Part G §4; the info record is the formal container, AVL keeps qualification data)
PUR → MM	Vendor qualification state	AVL status transitions (Qualified/Conditional/Suspended) gate source determination; driven by Quality per MFG Part E §6

7.2 Purchasing Info Records — NEW

Screen ID	MM-IR-001 (list), MM-IR-002 (detail)		
URL Pattern	/purchasing/info-records		
Purpose	The vendor-material contract of record: prices, lead times, order data per vendor per material. Complements the AVL (which owns qualification); replaces the loose "price basis" fields		
Field	**Type**	**Req**	**Specification**
Info Record #	Auto	Auto	IR-{SEQ:6}; one active IR per (material, vendor, entity) — V-MM-30
Material / Vendor	Lookups	Yes	Vendor must be Active (onboarding complete, MFG Part G §2.1)
Vendor Part # / Vendor Description	Text	No	Prints on PO; feeds order-entry cross-reference
Purchase UoM Override	Dropdown	No	Overrides material default for this vendor; conversion path required
Price Basis	Currency + unit	Yes	Net price per price unit in vendor currency; source badge: RFQ Award / Contract / Manual (Manual requires Purchasing Manager approval above 5% delta vs last basis)
Price Validity From / To	Dates	Yes/No	Effective-dated rows; history immutable; overlapping validity blocked (V-MM-31)
Scale Prices	Sub-table	No	Qty-break rows (From Qty → Price); deepest satisfied break applies at PO pricing
Surcharge Linkage	Lookup	No	Commodity index rule (MFG Part I §6.4) applied on the purchase side: PO price = basis ± index delta × content; printed as separate PO condition line
Planned Delivery Time (days)	Number	Yes	Vendor-specific; overrides material Planning view default in MRP offsetting
Min Order Qty / Rounding	Numbers	No	Vendor-specific constraints; MRP lot rounding uses the winning source's values

Screen ID	MM-IR-001 (list), MM-IR-002 (detail)		
Incoterms / Freight Estimate %	Mixed	No	Landed-cost estimation inputs (INV 5.6.1 true-up unchanged)
Tax Code Default	Dropdown	No	Per MFG Part A §11.3 vendor logic
Last PO / Last GRN / Last Invoice Price	Read-only	—	Rolling actuals with document links; deviation of invoice vs IR basis > 5% raises buyer alert
Status	Badge	Auto	Active / Expired / Blocked (vendor blocked or AVL Suspended cascades here)

7.3 PO Price Determination (Purchase Side)

Mirror of the sales Pricing Service, implemented once as the Purchase Pricing Service and called by PR valuation, PO creation and MRP planned-cost estimation:

- Manual price on the PO line within buyer authority (deviation > 5% from IR basis flags approval note per MFG Part G §5.2) — stamped Price Source = Manual.
- Contract / scheduling agreement price when one exists (Phase 2 object; field reserved).
- Active info record price for (material, vendor) valid on the PO date, scales applied — Price Source = Info Record.
- Last PO price for the pair, flagged “basis stale” if older than 12 months — Price Source = Last PO.
- No basis → line enters at zero with hard task: buyer must price manually; PO cannot be Sent with zero-price valuated lines (E-MM-32).

7.4 Source List — NEW

Screen ID	MM-SL-001		
URL Pattern	/purchasing/source-lists		
Purpose	Period-based declaration of valid supply sources per material; the governance layer MRP and buyers obey		
Field	**Type**	**Req**	**Specification**
Material	Lookup	Yes	Grid per material; rows below
Row: Vendor (or Subcontractor)	Lookup	Yes	Must hold an Active info record (V-MM-33) and AVL row where qualification is required
Valid From / To	Dates	Yes	Non-overlapping Fixed rows per period (V-MM-34)
Fixed	Toggle	No	Exactly one Fixed source per period allowed; MRP assigns PRs to it without buyer touch when Source Determination Policy = Source List Fixed (§3.7)
Blocked	Toggle	No	Explicit negative: this vendor may not supply this material in the period regardless of AVL (commercial embargo); hard block at PO entry (E-MM-35)
MRP Relevant	Toggle	Yes	OFF rows are for spot buys only; MRP ignores them

7.5 Source Determination Order (binding)

- Source list Fixed vendor for the period, when policy = Source List Fixed and the row is MRP Relevant.

- AVL rank 1 with Active info record (policy = AVL Rank, the default) — qualification state per MFG Part G §2.2 gates eligibility.
- Any AVL vendor with Active info record, by rank.
- No eligible source: PR flagged EX-09 “no valid source of supply”; QI-critical materials hard-block per E-PUR-02; buyer starts RFQ or AVL qualification.

> **Build Note:** *The AVL (MFG Part G §2.2) keeps: rank, qualification status, FAI, quality rating. The info record keeps: commercial terms. The source list keeps: period governance. These three tables are joined by (material, vendor); do not duplicate fields across them — the MFG Part G AVL columns “Price Basis / Lead Time” become read-through views onto the info record after this volume.*

8. Linkage: Material ↔ Production and MRP

Direction	Data	Rule
MM → PP	Planning view fields (§3.6)	MRP engine inputs; the workbench never reads planning data from anywhere else
MM → PP	Production view fields (§3.8)	WO creation defaults: issue warehouse, tolerances, serialization/batch timing
MM → PP	Batch/serial profile (§15)	Determines scan behavior at SP-08/SP-12 and genealogy granularity
MM → PP	Lifecycle status + Production Block	WO create/release validations (E-PP-01 family extends with E-PP-08 “material blocked for production”)
PP → MM	Consumption actuals	ADU (trailing 90-day) recompute for reorder arithmetic; ABC/XYZ classification job inputs
PP → MM	WO receipt/settlement events	MAV updates (buy items consumed), standard-cost variance data to CVA; no direct field writes on the material

Availability semantics (single definition, used by ATP, MRP, reservations and release checks): Available = On-hand Unrestricted − Reserved − QI Hold − Blocked − Customer Property; On-Order = open PO + open WO + in-transit receipts, dated. Special stock buckets (§14.1) are excluded from anonymous availability except SO-pegged stock, which is available only to its pegged SO line.

9. Linkage: Material ↔ Inventory and Warehouse

Direction	Data	Rule
MM → INV	Warehouse view (§3.9): strategies, storage conditions, shelf-life rules	Putaway proposals (INV 15.6), pick strategy enforcement (FEFO/FIFO/serial-directed at SP-15), GR shelf-life validation (E-MM-14)
MM → INV	Tracking config (batch/serial flags)	Movement screens render batch/serial capture accordingly; immutability per V-MM-03
MM → INV	Lifecycle status + Posting Block	All movement types validate; block error names the block reason and owner
INV → MM	Stock, valuation and movement history	Stock Overview tab, MAV, count classification inputs, E&O aging inputs (§17.3)
INV → MM	Count results	No master-data writes; counting never mutates the material record

10. Linkage: Material ↔ Sales

The Sales view contract is fully specified in MFG Part A §4 (fields, price determination waterfall, repricing, surcharges) and Part B (document usage). Deltas from this volume: lifecycle gating (Sales Block and Restricted stop new quote/SO lines with E-SAL-03 naming the block), supersession cross-reference printing (§4.6), and the completeness gate for sellable activation (§3.12). No other change.

11. Linkage: Material ↔ Quality

Direction	Data	Rule
MM → QM	Quality view (§3.10): QI flags, certificate requirements, batch record requirements	GRN routing to QI Hold, certificate enforcement (E-QM-06), field-level GRN requirements
QM → MM	Inspection plan resolution	Runtime resolution per MFG Part A §6.2 remains in QM; the material's Quality view only previews it. Plans are never stored on the material (assignment scope lives on the plan)
QM → MM	Vendor quality outcomes	IQC results roll vendor quality rating (AVL) and can trigger Purchasing Block proposals on the material (steward task, not automatic)
QM → MM	Holds	Batch/serial-level holds stay in INV 6.2; material-level Shipping/Posting blocks per §4.5 when the whole material is suspect

12. Linkage: Material ↔ Costing and Controlling

Direction	Data	Rule
MM → CST	Accounting & Costing view (§3.11): valuation class, method, price unit, costing lot size, overhead group, Do Not Cost	Costing run scoping and rollup inputs (MFG Part I §3.2)
CST → MM	Released standard cost + effective date	Written by costing-run release only; REVAL posting per §5.2; "never manual edit" restated
MM → CO	Profit center per material	Derivation rule 3 (MFG Part H §3) source field; mandatory for valuated materials (V-MM-01)
CST → MM	Estimate bases	Placeholder est. cost (§4.3) and info-record prices feed estimates with source badges (MFG Part I §3.1 component price sources)

13. Linkage: Material ↔ Accounting

- All GL effects route through the account determination matrix (Section 5) inside the shared Posting Service; the material contributes valuation class, method, PC and price data, never account numbers.
- Period control: material master edits are never period-blocked (masters are not postings, MFG Part A §9.2); valuation-relevant changes (method, class) are restricted to period boundaries per §5.4.

- Reconciliation: the inventory subledger by valuation class must tie to the 13xx accounts at every close; the close checklist step 6 report gains a by-valuation-class breakdown so a mapping error surfaces as a class-level difference, not a plug.
- Audit: every journal line stores material, valuation class version, matrix row ID and price source, closing the lineage loop demanded by review item T-10.

PART D

SPECIAL STOCK AND OWNERSHIP TYPES

Standard stock is PE-owned, anonymous and warehouse-located. Real operations add stock that is somewhere else, owned by someone else, or dedicated to something. This Part gives each special stock type a defined bucket, availability rule and posting treatment. Buckets are visible on the material Stock Overview (§3.13) and the Real-Time Stock Ledger (INV 15.12).

14. Special Stock Types

14.1 Bucket Overview

Bucket	Owner	Valuated by PE	In Anonymous ATP	Established By
Unrestricted / QI Hold / Blocked (standard)	PE	Yes	Unrestricted only	INV as built
At Subcontractor (1315)	PE	Yes	No (visible to the owning WO/PO)	MFG Part D §10 / G §6
Vendor Consignment	Vendor until withdrawal	No (memo qty; §14.2)	Configurable: counts as supply with flag “consignment” in MRP	This volume
Customer Property	Customer	No (non-valuated qty)	No	MFG Part A §3.5 (RMA repair path); generalized here for customer-supplied material in ETO jobs
SO-Pegged Stock	PE	Yes	Only for its SO line	WO receipt of pegged orders (MFG Part B §4.3); formalized here
In-Transit (1350)	PE	Yes	No until receipt	Transfers (INV 5.3)

14.2 Vendor Consignment

Consignment lets a vendor stock material in PE’s warehouse that remains vendor property until PE withdraws it. Target use: high-volume hardware and selected insulation from partner vendors. Activation is per (material, vendor): Special Procurement = Consignment on the Planning view plus a consignment-flagged info record.

- Fill-up: consignment PO type (no commitment value) → GRN books quantity into the Consignment bucket for that vendor. No value posting; the statistical memo (1311) carries quantity × IR basis for insurance and disclosure reporting only.
- Withdrawal: any consuming movement (issue to WO, transfer to own stock, sale) automatically converts ownership at the moment of consumption. The withdrawal posts the value entry below and writes a consignment settlement line for the vendor/period.
- Settlement: monthly run per vendor aggregates withdrawals into a self-billing settlement document to AP (ACC 6 invoice created from settlement, no vendor invoice matching); price = IR basis valid at each withdrawal date; deviations at true-up post to 5215.

- Counting: consignment stock counts like own stock (it is in PE custody); variances adjust the memo quantity and raise vendor reconciliation lines, never 5310.
- Returns to vendor from consignment reverse quantity only. MRP treats consignment on-hand as supply with source tag so planners see the ownership difference.

Accounting Entry — Consignment withdrawal (to production order)

Debit	Credit	Amount Basis	Notes
1320 WIP (order)	2120 Consignment Payable Clearing	Withdrawn qty × IR basis at withdrawal date	Event keys USAGE + CONSIG; CC from WC, PC parent item

Accounting Entry — Consignment settlement (monthly, per vendor)

Debit	Credit	Amount Basis	Notes
2120 Consignment Payable Clearing	2100 AP (vendor)	Sum of period withdrawals at IR basis	Self-billing document; tax per vendor profile
5215 Consignment Price Variance	2120 (or reverse)	True-up delta when a retroactive price correction applies	Requires Controller approval; rare by design

14.3 Customer Property (generalized)

- Beyond RMA repair units (MFG Part A §3.5), ETO contracts sometimes include customer-supplied components (e.g. customer-specified relays). These receive as Customer Property: quantity tracked with full batch/serial and genealogy, zero value, never in ATP, never counted in E&O.
- Consumption into a WO posts no material value; genealogy records the consumption normally so the as-built record is complete. The cost estimate flags customer-supplied lines at zero material cost with a note on the quote (MFG Part I §3.1).
- Return of unused customer property ships on a zero-value delivery with a property return note; the SO closure checklist includes “customer property reconciled” when any was received.

14.4 SO-Pegged Stock

- FG receipt from a pegged WO lands in the SO-Pegged bucket keyed to the SO line (already the behavior in MFG Part B §4.3; the bucket makes it explicit and auditable). Pegged stock is invisible to anonymous ATP and cannot be picked for another order without a formal De-Peg action (Sales Director approval + reason; logged on both SO lines).
- Valuation is standard, same as FG; the bucket is a stock dimension, not a valuation change.

PART E

BATCH AND SERIAL PROFILES

15. Batch and Serial Management (Master-Data Layer)

Execution behavior (scans, genealogy, statuses) is specified in INV and MFG Part F and unchanged. This Part specifies the master-data layer: what a batch record is, which profile drives serialization, and the immutability rules that keep traceability trustworthy.

15.1 Batch Master (DELTA to INV batch records)

Field	Type	Req	Specification
Batch #	Auto	Auto	BAT- series per INV; label per MFG Part F §1
Vendor Batch / Heat #	Text	Cond.	Mandatory per material Batch Record Requirements (§3.10); heat # is the traceability key for CRGO and copper
Manufacture / Receipt / Expiry Dates	Dates	Cond.	Expiry mandatory for shelf-life materials; FEFO reads expiry
Country of Origin (actual)	Dropdown	Cond.	Captured at GRN; overrides material default; feeds Buy America accumulation per shipment
Certificates	Attachments	Cond.	Mill cert / CoA per §3.10; enforced at GRN accept
Vendor / PO / GRN References	Links	Auto	Provenance chain for the traceability tree
Batch Status	Badge	Auto	Unrestricted / QI Hold / Blocked / Consumed / Expired (job flips at expiry date; expired stock auto-moves to Blocked with alert)
Characteristics (measured)	Panel	No	Actuals from incoming inspection (core loss W/kg, conductivity) copied from inspection results for batch search (“find all CRGO batches with loss ≤ X”)

15.2 Serial Number Profiles

Profile	Applies To	Behavior
SN-FIN Transformer	FIN transformers	Serial at WO release; nameplate format PE-{TYPE}-{YYYY}-{SEQ:4}; full genealogy; warranty tracking; customer/site on shipment (all per MFG Part D/F, restated as profile)
SN-COMP Serialized Component	TRADE/SPARE serialized (bushings, tap changers, relays)	Serial at GRN (vendor serial captured or PE label printed); parent-child install links (SP-12); RMA-traceable
SN-SEMI Optional	SEMI where engineering elects	Serial at WO release or FG receipt per §3.8 field
SN-NONE	Everything else	No serialization

Build Note: *Profiles are configuration rows (profile, serial timing, format, warranty relevance, install-link allowed). The material references a profile; documents read the profile, never per-material bespoke logic. Changing a material’s profile follows V-MM-03 immutability once records exist.*

PART F

MASS MAINTENANCE, MIGRATION AND DATA QUALITY

16. Stewardship Toolset

16.1 Mass Maintenance

Screen ID	MM-MCH-001
URL Pattern	/materials/change-requests
Purpose	Governed mass changes across materials (the MCH object from §4.4)

- Selection: filter set (type, group, status, buyer, planner, free field predicates) or uploaded material list; preview grid shows current values of the target fields.
- Change definition: target field(s) + new value or expression (set, adjust %, find/replace within text); Class rules per §4.4 decide the approval chain; Class 3 fields are not selectable.
- Simulation: full before/after grid, validation results per row (rows failing validations are excluded and listed), effect summary (e.g. “124 materials, 3 will lose active source”), export.
- Apply: immediate or effective-dated; transactional per row with error isolation; result log downloadable; every row writes the standard field-level audit with the MCH reference.

16.2 Import Templates

XLSX templates per view (columns = fields with data dictionary validation on upload) support both migration and ongoing bulk creation. Uploads create MCRs/MCHs rather than writing directly, so governance is identical regardless of entry channel. Template versioning follows the field dictionary; an outdated template upload is rejected with a per-column diff.

16.3 Migration Specification (answers review D-04 for MM scope)

Step	Specification
Source inventory	Legacy item list extract with: code, description, type mapping candidate, UoM, on-hand by location, last costs, vendor/price data, open PO/SO/WO references. Extract format fixed in the template set (§16.2)
Mapping & cleansing	Type mapping table (legacy category → material type §2.1) approved by Materials Manager; UoM normalization; duplicate resolution run (§4.4 checks against the whole load); description standardization rules (noun-modifier format) applied and exceptions listed
Mock loads	Minimum two full mock conversions in the test environment; completeness KPIs (§16.4) computed on each; error budget for go-live: zero Class A errors (unloadable rows), < 0.5% Class B (loaded with warnings)
Valuation cutover	Opening stock loads at legacy value as opening MAV, or at released standard where costing runs completed pre-cutover; count-based verification per INV 7 before value release; opening JE ties subledger to GL by valuation class
Batch / serial history	Active batches and serials load with provenance fields; historical genealogy beyond active stock loads into the archive store (read-only traceability), not the operational tables
Info records / source lists	Last-price and lead-time history converts to IR rows (source badge Migration); AVL qualification states carry with evidence links

Step	Specification
Freeze & reconciliation	Legacy master-data freeze at T-5 days; delta capture list for manual re-key; post-load reconciliation reports: counts by type/status, value by valuation class vs legacy trial balance, open-document reference resolution 100%

16.4 Data Quality Dashboard and KPIs

KPI	Definition	Target
Completeness	Mandatory-for-use fields populated ÷ required, per material, rolled up by type and owner	100% for Active (structural); ≥ 95% including Data Completion WIP
MCR cycle time	Creation → Active, percentiles by type	P80 ≤ 5 business days
Duplicate rate	Confirmed duplicates found per 1,000 active materials per quarter	< 2
Stale sourcing	Buy materials whose only IR basis is older than 12 months	< 5%
Governance exceptions	Class 2 changes applied without linked approval (structurally zero; audit query)	0
PC-900 materials	Valuated materials on fallback profit center	0 (weekly exception per MFG Part H §1.2)

PART G

REPORTS AND ANALYTICS

17. Material Management Reports

Report	Content
MM-R1 Master Data Completeness	KPI drill per §16.4: by type, owner, view; row-level missing-field list; export
MM-R2 Where-Used Explosion	Multi-level where-used for any material across BOMs, plans, price rows, IRs, source lists (extends MFG Part C §7)
MM-R3 Sourcing Coverage	Buy materials × (AVL rows, active IR, source list state, lead time); gaps highlighted: no source, single source on A-class, stale basis
MM-R4 Slow-Moving & Dead Stock	No consumption in 6/12/24 months with on-hand value, last movement, disposition suggestion; feeds E&O run
MM-R5 Supersession Status	Open supersessions with run-out progress, remaining predecessor stock and blocked documents
MM-R6 Consignment Position	By vendor: on-hand consignment qty, memo value, period withdrawals, unsettled balance vs 2120
MM-R7 Special Stock Overview	All §14.1 buckets by material/warehouse with document references
MM-R8 Lifecycle Funnel	Materials by status with aging; Restricted/Blocked with reason codes and owner

17.3 Excess & Obsolete Reserve Run

Screen ID	MM-EOR-001
URL Pattern	/materials/eo-reserve
Purpose	Quarterly (config) computation and posting of the E&O inventory reserve — Controller-owned

- Policy table: aging buckets (no consumption 6/12/24+ months) × reserve % per material type; overrides per material with reason (e.g. insurance spares exempt).
- Run: compute per material (on-hand value × bucket %), compare to existing reserve, propose delta; review grid with drill to MM-R4 evidence; Controller posts.
- Posting: DR 6320 E&O Expense / CR 5340 Inventory Reserve for builds; reverse for releases (consumption resumed or stock scrapped — scrap first relieves the reserve, then 5320 takes only the unreserved remainder).
- The reserve is a valuation adjustment account, never a quantity change; traceability and availability are unaffected.

PART H

DATA MODEL, INTEGRATION MATRIX AND ROADMAP

18. Data Model

18.1 Core Entities

Entity	Key Fields / Constraints	Relationships
material	material_id (PK), material_no (unique), type, group_l1/l2/l3, base_uom, status, batch_flag, serial_profile_id, hazmat_json, created/updated audit. Type and material_no immutable (DB trigger + app rule)	→ every module
material_text	material_id, lang, short_desc, long_desc, purchasing_text, work_notes	→ material
material_uom	material_id, uom, numerator, denominator, usage_tag; unique (material_id, uom)	→ material
material_plant_data	material_id, plant_id (composite unique), all §3.3 plant-level groups as typed columns, status per plant. Exists from day one even with one plant	→ material, plant
material_valuation	material_id, plant_id, entity_id, valuation_class, method, price_unit, std_cost, std_effective, mav, effective-dated history rows (no updates in place)	→ account determination
material_characteristic	material_id, class_id, attr_key, typed value columns; class per group	→ material, characteristic_class
mcr / mcr_task	mcr_id, material_id, origin_ref, status; task: view, owner_role, sla_due, status	→ material, users
mch / mch_row	mch_id, status, approval refs; row: material_id, field, old, new, applied_at, error	→ materials, ECO
supersession	sup_id, predecessor_id, successor_id, policy, effective, status	→ materials, ECO
info_record	ir_id, material_id, vendor_id, entity_id (unique active), price rows child table effective-dated, scales child table	→ material, vendor, PO pricing
source_list_row	material_id, vendor_id, valid_from/to, fixed, blocked, mrp_relevant; overlap constraints per V-MM-34	→ material, vendor
acct_determination_row	event_key, valuation_class NULLABLE, qualifier NULLABLE, dr_key, cr_key, valid_from/to, version, approval ref	→ Posting Service
special_stock_bucket	stock rows gain owner_type (OWN/VENDOR/CUSTOMER), owner_ref, peg_type (NONE/SO/WO), peg_ref — extends INV stock tables, no parallel table	→ INV stock
batch (extended)	adds vendor_batch, heat_no, coo, expiry, measured characteristics json	→ INV batch
serial_profile	profile_id, timing, format, warranty_flag, install_link_flag	→ material

Entity	Key Fields / Constraints	Relationships
eo_reserve_run / row	run_id, period, policy version; row: material_id, bucket, value, reserve_delta, je_ref	→ GL

18.2 Physical Rules (applies to every MM entity)

- Surrogate integer/UUID PKs; business keys unique-indexed; foreign keys enforced in the database, not only the app.
- Effective-dated tables never update rows in place: new row + closed predecessor, both retained (valuation, IR prices, account determination, source list).
- Field-level audit via the shared audit stream (screen, MCH, migration, API as sources); audit rows are append-only and excluded from purge.
- Concurrency: optimistic locking (version column) on the material and children; the MCR/MCH apply steps take row locks per material to prevent cross-workflow races.
- Archive/purge: Obsolete materials archive after the retention period (default 10 years, aligned to genealogy retention) into the read-only store; material numbers are never reused.

18.3 API and Event Surface

Interface	Specification
REST: /api/v1/materials	GET (filter set of §3.1), GET by id (view-shaped payloads), POST creates MCR (never a direct Active material), PATCH creates MCH rows for governed fields / direct write for Class 1 with audit; all responses carry the field dictionary version
REST: /api/v1/info-records, /source-lists, /supersessions	CRUD with the same governance semantics as the screens; idempotency keys on POST
Events (outbox pattern)	MaterialCreated, MaterialActivated, MaterialChanged (field list), MaterialStatusChanged, MaterialSuperseded, InfoRecordPriceChanged, SourceListChanged, StdCostReleased — versioned payloads, ordered per material, consumed by MRP cache, pricing, search index and any future PLM/CAD connector (review F-03 lands on this surface)

19. Integration Matrix (New Flows)

Source	Target	Trigger	Payload / Effect
EBOM editor	MCR	Placeholder creation	Draft material + tasks; number reserved
MCR activation	EBOM / PR / Quote	All tasks complete	Unblocks originating documents
RFQ award (MFG G §4)	Info Record	Award action	Price basis + lead time rows
Info Record / Source List	MRP	Nightly + on change	Source of supply and lead times per §7.5
MRP	PR with source	Proposal creation	EX-09 when no source resolves
Costing run release	Material valuation	Release	New standard + REVAL posting
ECO release	MCH / Supersession / Blocks	Effectivity	Atomic apply with BOM cutover

Source	Target	Trigger	Payload / Effect
Consignment withdrawal	Settlement store + GL	Consuming movement	CONSIG posting + settlement line
Monthly settlement run	AP (ACC 6)	Period end	Self-billing invoice per vendor
E&O run	GL	Quarterly	Reserve build/release JE
ABC/XYZ job	Material plant data	Monthly	Class badges; count frequency feed
Expiry job	Batch status	Daily	Expired → Blocked + alert

20. Roadmap Insertion and Non-Functional Requirements

20.1 Build Sequencing (extends MFG Part K §3 phases M1–M7)

Phase	MM Scope Added	Rationale
M1 (wks 1–4)	Material type model, numbering, field dictionary with change classes, plant-data split, valuation classes and account determination matrix + MM-VAL-001, GL preview	Everything downstream posts through the matrix; it must precede any posting-bearing module
M2 (wks 3–8)	MCR workflow + placeholder path, Planning/Purchasing/Production/Warehouse/Quality/Accounting views, duplicate check, function blocks	Sales core and gap items in M2 need governed materials to exist
M3–M4 (wks 6–18)	Info records, source lists, purchase price determination, serial profiles, batch master extensions, special stock buckets incl. SO-pegged and customer property	Production and procurement loops consume them
M5–M6 (wks 14–22)	Consignment end-to-end, supersession, mass maintenance, E&O run, MM reports	Requires stable GRN/AP and costing foundations
M7 (wks 20–26)	Migration execution per §16.3, data quality dashboard live, completeness gates enforced, UAT scripts for every linkage contract in Part C	Go-live readiness

20.2 Non-Functional Requirements (MM additions)

Area	Requirement
Volumes	50k materials at go-live growing 5k/year; 8k info records; 120k batches/year; 1.5k serials/year (FIN) plus component serials; dictionary of ~450 material fields
Performance	Material detail load < 1.5 s P95 with all views; listing filter response < 2 s at 50k rows; duplicate check inline < 500 ms; account determination resolution < 10 ms in-posting (cached matrix); MCH apply ≥ 50 rows/s
Integrity	Zero postings without a resolvable matrix row (E-MM-21 enforced at release, monitored in production); subledger-by-valuation-class tie-out at every close
Auditability	Any field value answerable “who set this, when, from where, under which approval” in ≤ 3 clicks; matrix row lineage on every journal line
Security	Field-group edit rights per §3.2 enforced server-side; SOD: MCR task completion vs activation; matrix edit vs release; MCH request vs approve

21. Document Sign-Off

Metric	Count
Parts	8 (A–H)
Review items closed or advanced	6 (F-04 partial, F-11 partial, T-01 partial, T-10, D-01, D-04 partial)
New screens specified	12 (MM-MAT delta, MCR ×2, MCH, SUP, IR ×2, SL, VAL, EOR, plus view-level deltas to INV 2.2/2.3)
Field-level tables	30+
New GL accounts / posting rules	6 accounts; consignment, E&O and determination-matrix posting rules
Linkage contracts	8 (Engineering, Purchasing, Production/MRP, Inventory, Sales, Quality, Costing, Accounting)
Validation register entries	V-MM-01…V-MM-34, E-MM-02…E-MM-35 as specified per section

This Book, together with the Inventory & Warehouse and Accounting specifications and Book I, constitutes the material-management-complete build package. Any deviation during development must be documented and approved by the PE ERP project lead before implementation.

Role	Name	Signature	Date
Project Sponsor			
Product Owner			
Materials Manager / Master Data Steward			
Technical Lead (PE)			
Development Partner Lead			
Engineering Manager			
Purchasing Manager			
Quality Manager			
Controller			
CFO			

BOOK III

ENTERPRISE COMPLETENESS

Cross-Volume Flow Closure • Integration & Transaction Architecture • Security & IAM • Resilience & Operations
Finite Scheduling & CTP • Multi-Plant & Global Model • CAD/PDM/PLM • Test, Migration, Cutover & Acceptance

Distribution List

Role	Access Level	Entity
Engineering Lead	Full Access	PE Transformer
Development Partner Team	Full Access	PE Transformer
IT / Infrastructure Lead	Full Access	PE Transformer
QA Lead	Full Access	PE Transformer
Controller / Finance	Review & Comment	PE Power Grids
Project Management	Read Only	PE Transformer

1. Purpose and Coverage Map

This Book supplies the enterprise architecture layer of the specification. Functional depth alone does not constitute an enterprise build package. The flows must be proven closed across modules, and the platform, security, operational, planning, global and delivery requirements must exist as controlled content. This Book carries them. It is deliberately cross-cutting: it specifies behavior that spans the Inventory and Accounting specifications, Book I and Book II rather than repeating any of them.

Ref	Requirement	Specified In
P0-01	Version and change control, closure map	Version and change control governed through the build package manifest; this table is the requirement to section map
P0-02	Cross-volume flow closure matrix	Part A: eight flows, each specified against the fifteen field closure template
P0-03	Integration and transaction architecture	Part B
P0-04	Security architecture	Part C
P0-05	Resilience and operations	Part D
P0-06	Finite scheduling and CTP requirements	Part E, including the explicit go-live boundary statement
P0-07	Multi-plant, intercompany and global model	Part F
P0-08	CAD/PDM/PLM integration specification	Part G
P0-09	Test, migration, cutover and acceptance plan	Part H

Scope statement. This package carries the Phase 1 build scope with an architected path to the target state. Capabilities phased beyond first go live are listed in each Part with their trigger conditions and are specified at module depth in Book IX. External descriptions must use the benchmark language on the cover and nothing stronger.

> **Build Note:** *Executed test evidence and executed sign off cannot be supplied by any document. Part H defines the traceability structure, the test classes and the acceptance gates so that evidence accumulates during the build. The sign off pages must remain blank until the named approvers execute them.*

1.1 Reading Conventions

- Abbreviations: INV (Inventory & Warehouse Spec V3.0, external), ACC (Accounting Spec V2.0, external), MFG (Book I), MM (Book II).
- Journal template numbers #1 to #38 refer to the unified posting map (INV Section 12, ACC Section 28, MFG Part J); MM postings reference the account determination matrix (MM Part B).
- Interface names use the catalog in Part B §3; test case ranges use the scheme in Part H §2.
- Requirement IDs: this volume assigns the package-wide scheme REQ-{VOL}-{SECTION}-{SEQ}; Part H §2 defines how existing volume content maps into it without rewriting the volumes.

PART A

CROSS-VOLUME FLOW CLOSURE BOOK

Each flow below is specified against the full fifteen field closure template. Every field cites the controlling specification section so the closure is verifiable, and every flow carries a reserved test-case range that Part H binds into the traceability matrix. A flow is accepted as closed only when its test range shows passing evidence for every row class: happy path, partial, cancellation, reversal, late posting, concurrency, integration failure, security and reconciliation.

FL-01 — Make-to-Stock Order-to-Cash

Closure Field	Specification
1. Flow and objective	Sell stocked components and standard transformers from FG with credit-safe, tax-correct, reconciled cash closure
2. Trigger and prerequisites	Confirmed SO (MFG Part B §4.6) for MTS lines; customer Active with Bill-To/Ship-To; price resolved (MFG Part A §4.4); credit check passed (MFG Part B §2.2)
3. Source document and owner	Sales Order; owner: Sales Admin until Confirmed, then fulfillment owned by Warehouse (pick/pack/ship) and Finance (invoice to cash)
4. Status transitions and approvals	SO lifecycle per MFG Part B §4.6; shipment statuses per §5; invoice/payment statuses per ACC 3/5; approvals: credit release (Finance), price-floor breaches (MFG Part I §6.3)
5. Master data dependencies	Material Sales view + price rows (MM §3.12/MFG A §4), customer commercial+tax profile, warehouse strategies (MM §3.9), tax nexus config (MFG A §11.4)
6. Inventory and valuation effects	Reservation at confirm; pick/pack no GL; PGI: DR COGS / CR FG at standard (template #25); packaging backflush #27
7. Financial postings	Invoice: DR 1200 AR / CR 4100 + 2300 (ACC 3, Avalara commit per MFG A §11.2); cash application ACC 5; freight accrual #26; discounts/deductions per ACC 4 credit memo rules
8. Quality and traceability gates	Pre-shipment inspection gate blocks PGI (INV 15.3, E-OM-07); CoC auto-generation (INV 15.5); serial ship status + customer/site write (MFG Part F SP-18)
9. Interface contract and boundary	IF-04 sales-order events, IF-07 shipment events, IF-09 invoice-request (Part B §3); PGI is one atomic transaction: stock decrement + serial update + journal + invoice request row via transactional outbox (Part B §2.3)
10. Timeout, retry, duplicate, recovery	Invoice-request consumer idempotent on shipment ID; Avalara failover per MFG A §11.1 step 6 (rate override + 15-min recalc queue); scan offline queue per MFG Part F §2; duplicate PGI structurally impossible (single state transition with optimistic lock)
11. Exceptions and reversals	Partial shipment (line split per MFG B §4.3); cancellation before PGI releases reservation; post-PGI correction only via RMA (FL-05) or credit/rebill (ACC 4); invoice reversal re-opens revenue via ACC rules; bad debt per ACC 5 write-off with Controller approval
12. Reconciliation and closure	Daily: PGI COGS vs FG movement report; monthly: AR subledger = 1200 (ACC 26.2 step 6), revenue vs shipped-not-invoiced report. Closure: all lines Shipped + Invoiced, cash applied or receivable approved open, no open credit requests. Terminal status Invoiced/Closed
13. Security and SoD	Sales cannot release credit holds; shipping cannot edit prices; PGI role ≠ invoice-posting role; per ACC 23 + MFG Part B §6 role table
14. Performance and availability	PGI round-trip < 2 s P95; invoice request visible to Finance < 1 min; shipping path inside the 99.5% shop-floor availability envelope (Part D §2 service tiers)

Closure Field	Specification
15. Test range and evidence	TC-O2C-001…060: happy path, partial, split, cancel, credit hold, tax failover, PGI concurrency (two users one HU), invoice reversal + rebill, cash application, close reconciliation. Evidence per Part H §4

FL-02 — Engineer-to-Order Quote-to-Cash

Closure Field	Specification
1. Flow and objective	Custom transformer from firm quote through design, build, test, ship, milestone billing and revenue closure
2. Trigger and prerequisites	Firm quotation Won → SO with ETO lines (config attributes complete); released cost estimate CE-# linked (MFG B §3.2); Estimator + margin approval trail
3. Source document and owner	SO line is the commercial anchor; Engineering Request owns design phase (MFG C §2); WO owns build; revenue contract (ACC 24) owns recognition
4. Status transitions and approvals	SO §4.6 + ER lifecycle + EBOM/MBOM release approvals + WO §5.3 + milestone approvals (Finance posts each invoice request after evidence review, MFG B §4.4)
5. Master data dependencies	Placeholder-to-active materials via MCR (MM §4.2/4.3); work centers and rates (MFG D §2 / I §2); inspection plans FIN type; drawing set released (E-PP-03)
6. Inventory and valuation effects	Issues to WIP #28; subcontract legs #34/35; FG receipt at standard #31 into SO-pegged bucket (MM §14.4); PGI #25
7. Financial postings	Milestone invoices #37 with ASC 606 recognition per ACC 24 (billing plan ≠ revenue timing); WO settlement variances #32; surcharge adjustment at PGI #38 for ship-date-adjusted contracts
8. Quality and traceability gates	Customer drawing approval milestone (MFG C §5.2 transmittals); in-process gates per routing; Final Test + FTR (MFG E §4) gates the Production Complete milestone; full genealogy per Part F
9. Interface contract and boundary	IF-03 engineering events, IF-05 WO events, IF-06 confirmation postings, IF-09 invoice requests; confirmation posting is atomic: issue + labor + genealogy + WIP journal in one boundary (Part B §2.3)
10. Timeout, retry, duplicate, recovery	Milestone triggers are event-driven with idempotent invoice-request creation (unique milestone ID); scan retries per offline queue; costing simulation calls are read-only and safely retryable
11. Exceptions and reversals	Drawing rejection loops (transmittal Revise & Resubmit); config change after Design Release only via ECR→Sales Change Order (MFG B §4.5); WO cancellation dispositions; milestone reversal only via credit note with contract re-measurement (ACC 24.2); promise-date change control per Part E §5
12. Reconciliation and closure	WIP by order = 1320/1325 (close step 6c); billing plan total = contract value; deferred/unbilled roll per ACC 24.2. Closure: unit delivered, all milestones invoiced, revenue recognized per contract, WO settled to zero WIP, genealogy complete, warranty record active
13. Security and SoD	Engineer cannot approve own EBOM release; milestone evidence reviewer ≠ SO owner; costing release ≠ costing mark (MFG NFR SOD set + Part C §4 role model)
14. Performance and availability	Design-to-release cycle tracked (MFG C §7); scheduling board refresh < 2 s; test-data capture within shop-floor availability tier
15. Test range and evidence	TC-ETO-001…080: incl. drawing rejection, ECO mid-build with serial effectivity, shortage-released WO, failed final test + rework + retest, partial milestone, surcharge true-up, contract modification, full close simulation

FL-03 — Procure-to-Pay

Closure Field	Specification
1. Flow and objective	Demand to paid supplier with three-way-match integrity, PPV capture and supplier-account reconciliation
2. Trigger and prerequisites	MRP PR / reorder PR / manual PR (MFG G §3); source resolved per MM §7.5; vendor Active; material purchasable per MM type matrix
3. Source document and owner	PR → PO (Buyer owns to Confirmed); GRN owned by Receiving (INV 5.1/5.6); invoice and payment owned by AP (ACC 6/8)
4. Status transitions and approvals	PO approval ladder (MFG G §1: Buyer ≤ $25k → PM ≤ $250k → Controller → CFO > $1M); GRN QI routing; match exceptions to AP approval queue (ACC 22)
5. Master data dependencies	Material Purchasing view + info record + source list (MM §3.7/7.2/7.4); AVL qualification (MFG G §2.2); quality clauses; tax profile (MFG A §11.3)
6. Inventory and valuation effects	GRN: DR 13xx (or DR at MAV) / CR 2110 GR/IR with PPV #—per INV Sec 12 #3; QI Hold until inspection; landed cost true-up INV 5.6.1; consignment fill-up posts nothing (MM §14.2)
7. Financial postings	Three-way match: DR 2110 / CR 2100 with IPV to 5210; payment run ACC 8; use-tax accrual per vendor profile; subcontract legs #34/35 with 5260 variance
8. Quality and traceability gates	Incoming inspection per plan (INV 6.1); certificates enforced (E-QM-06); batch provenance captured (heat #, CoO per MM §15.1); vendor rating updates from IQC
9. Interface contract and boundary	IF-01 PO transmission (email PDF Phase 1, EDI Phase 2 field reserved), IF-02 vendor confirmation capture, IF-08 GRN events, IF-10 AP match; GRN is atomic: stock + batch/serial + QI + journal + match-basis row
10. Timeout, retry, duplicate, recovery	Duplicate GRN prevented by PO-line open-quantity lock; duplicate vendor invoice by (vendor, invoice #) uniqueness (ACC 6); match retries idempotent; over-receipt beyond tolerance blocked E-MM-11
11. Exceptions and reversals	Over/under receipt, RTV with GRN reversal and PPV mirror (MFG A §8.4), invoice variance routing, PO cancellation exposure notes, short-close with reason (MFG G §1), consignment true-up 5215
12. Reconciliation and closure	GR/IR aging report: 2110 items > 30 days worked weekly; supplier statement reconciliation monthly (ACC 6); PPV/IPV visible in CVA. Closure: PO lines Received + Matched/Closed, GR/IR cleared, payment executed or scheduled, no open NCR/SCAR against the receipts
13. Security and SoD	PO create ≠ GRN post ≠ invoice approve ≠ payment release (four-way separation, ACC 23 + MFG NFR); vendor bank-detail changes dual-approved (ACC fraud controls)
14. Performance and availability	GRN scan round-trip < 400 ms P95; match job latency < 5 min; payment runs inside close calendar windows
15. Test range and evidence	TC-P2P-001…070: incl. partial receipts across schedules, RTV, price/qty match variances, duplicate-invoice attempt, consignment fill-withdraw-settle, subcontract loop, GR/IR aging clearance, supplier reconciliation

FL-04 — Plan-to-Produce

Closure Field	Specification
1. Flow and objective	Demand netting to settled production with zero residual WIP and complete genealogy

Closure Field	Specification
2. Trigger and prerequisites	MRP run inputs (MFG D §3.1); released MBOM + routing (E-PP-01); materials Active; standards released (warning otherwise + CVA exception)
3. Source document and owner	Planned order (Planner) → WO (Production Scheduler owns to Close); confirmations owned by operators at WCs
4. Status transitions and approvals	WO lifecycle MFG D §5.3 with release gates §5.2; Release Short override (Planner flag); settlement approval at close; reopen requires Controller
5. Master data dependencies	Material Planning + Production views (MM §3.6/3.8); WC capacity + rates; issue methods; serial/batch profiles (MM §15)
6. Inventory and valuation effects	Issues/backflush #28; scrap #30; subcontract #34/35; FG receipt #31; rework via 1325 (MFG D §9)
7. Financial postings	Labor/overhead absorption #29; settlement variance split #32 to 5220–5250; CC/PC derivation rules 2–3 (MFG H §3)
8. Quality and traceability gates	QC checkpoint ops block completion (MFG D §4); wrong-material scans hard-block E-PP-07 with auto-NCR on override; genealogy rows per SP-08…SP-13
9. Interface contract and boundary	IF-05 WO events, IF-06 confirmation postings, IF-11 MRP proposals; MRP run publishes a run-ID-stamped proposal set atomically (all-or-nothing swap, no partial plan visible)
10. Timeout, retry, duplicate, recovery	Confirmation posting idempotent on (WO, op, event UUID) — offline queue replay-safe; MRP run recovery = rerun (regenerative, side-effect free until conversion); reservation races resolved by row locks at firm-up
11. Exceptions and reversals	Shortage release, rework loops, WO cancellation dispositions (finish/stop-scrap/hold per MFG B §4.5), confirmation correction (reversal confirmation with mirror postings, reference to original), settlement reversal on reopen
12. Reconciliation and closure	WIP report = GL 1320+1325 (close 6c); settled orders leave exactly zero WIP (MFG I §4 invariant); unsettled Completed > 5 days breach (close 6d). Closure: TECO with variances posted and genealogy complete
13. Security and SoD	WO confirm ≠ settlement approve; scrap beyond threshold needs supervisor + reason; standards never manually editable
14. Performance and availability	Full MRP < 30 min; dispatch refresh < 2 s; terminal path in top availability tier with 4-hour offline queueing
15. Test range and evidence	TC-PTP-001…070: incl. backflush vs direct-issue mix, batch WC oven multi-order confirm, offline queue replay with conflict, mid-order ECO, scrap and rework, settlement math property test (invariant), WIP reconciliation

FL-05 — RMA and Warranty Resolution

Closure Field	Specification
1. Flow and objective	Authorized return through disposition and financial settlement with linked-document closure (per MFG Part A §3, closure fields added here)
2. Trigger and prerequisites	Approved RMA per policy matrix §3.10; reference document or No-Reference exception path
3. Source document and owner	RMA; Customer Service owns to Authorized; Receiving/Quality own the middle; Finance owns settlement documents
4. Status transitions and approvals	Lifecycle §3.8; warranty adjudication mandatory when claimed; disposition approvals per class; Controller for credits > $50k, goodwill, reopen

Closure Field	Specification
5. Master data dependencies	Serial warranty windows; customer credit state; RMA inspection plans; restocking policies
6. Inventory and valuation effects	Receipt #17 to 1335 (or non-valuated customer property per ownership rule §3.5); restock #18; scrap #20
7. Financial postings	Credit note #19 with restocking fee 4910; warranty repair settlement #21 against 2610 reserve; billable repair #22; replacement shipment #23; reserve accrual #24 monthly
8. Quality and traceability gates	Return inspection with teardown checklist (RC3); NCR forced for manufacturing/supplier causes; SCAR path; genealogy return events SP-19
9. Interface contract and boundary	IF-12 RMA events feeding credit note creation, RWO creation and replacement SO — each an idempotent consumer keyed on RMA line disposition ID
10. Timeout, retry, duplicate, recovery	Disposition executes outcomes as separate retry-safe steps; partial execution visible on Settling status with per-document state; no outcome executes twice (disposition ID uniqueness)
11. Exceptions and reversals	Full register in MFG A §3.9 (no-reference, unauthorized extra, credit hold, wrong serial, freight claims); credit note reversal via rebill; reopen window 30 days Controller
12. Reconciliation and closure	Close checklist 6e: no Settling RMAs with posted-document gaps; warranty reserve adequacy 9b; AR effects reconciled in ACC. Closure: every line terminal disposition + all linked documents posted and closed
13. Security and SoD	Adjudicator ≠ credit approver; receiving cannot disposition; reopen restricted
14. Performance and availability	Receipt-to-inspection SLA 5 days with alerting; settlement documents created < 1 min after disposition approval
15. Test range and evidence	TC-RMA-001…050: all four return classes, warranty approve/deny, every disposition outcome, credit-hold block, unauthorized return, reopen, reserve overflow to 6300

FL-06 — Period Close

Closure Field	Specification
1. Flow and objective	Monthly operational and financial close with subledger-to-GL tie-out across all volumes
2. Trigger and prerequisites	Close calendar (ACC 26.1); soft close at day −2 per MFG A §9.3
3. Source document and owner	Close checklist instance (ACC 26.2, extended by MFG Part J and MM); Controller owns; module owners execute their steps
4. Status transitions and approvals	Period Open → Soft → Hard per matrix MFG A §9.2; reopen dual-control Controller + CFO
5. Master data dependencies	Allocation cycles + SKFs current (MFG H §5); accrual templates; valuation-class mapping current (MM §5)
6. Inventory and valuation effects	Count postings settle; valuation snapshot at inventory close (INV 8.3); E&O reserve run quarterly (MM §17.3)
7. Financial postings	Allocations #36; warranty accrual #24; close JEs per ACC; no back-posting into Closed (E-PER-01)
8. Quality and traceability gates	None blocking; quality-cost reporting feeds dashboards

Closure Field	Specification
9. Interface contract and boundary	IF-13 close-checklist automation events: each auto-validated step queries its module report API; steps are read-only checks plus explicitly posted close JEs
10. Timeout, retry, duplicate, recovery	Checklist steps re-runnable; allocation rerun = reversal + repost with run ID (MFG H §5); failed batch steps alert and hold the gate, never silently pass
11. Exceptions and reversals	Late postings via Soft-Close warnings; corrections in next period with Reference Date; reopen protocol with tagged postings (MFG A §9.3)
12. Reconciliation and closure	Steps 5b, 6, 6b–6e, 9b as specified (MFG Part J); MM adds by-valuation-class tie-out (MM §13). Closure: Hard Close with zero unresolved checklist items and archived close report
13. Security and SoD	Step executor ≠ step approver where flagged; reopen dual-control; allocation posting restricted to Controller
14. Performance and availability	Close-critical batch jobs complete inside the close window; target day+3 hard close by M7 exit
15. Test range and evidence	TC-CLS-001…040: full close simulation (M7 parallel run), late-posting handling, reopen with audit, allocation rerun, tie-out failures surfaced, checklist gate enforcement

FL-07 — Intercompany Supply

Closure Field	Specification
1. Flow and objective	Cross-company sale/transfer between PE entities with transfer pricing, IC AR/AP and elimination-ready postings. Phase: model live at go-live for the PE Grid Components ↔ PE Transformer merchandise flow; full multi-plant manufacturing transfers activate with plant 2 (Part F)
2. Trigger and prerequisites	IC relationship configured (Part F §3); transfer price list active; both entities open periods
3. Source document and owner	Paired documents: selling-entity SO (auto) ↔ buying-entity PO; owner: initiating entity buyer/CSR; Finance owns IC reconciliation
4. Status transitions and approvals	Standard SO/PO lifecycles; IC price deviations from transfer price list require Controller approval on both sides
5. Master data dependencies	Material extended to both entities (MM §3.3 Extend action); transfer price rules (Part F §4); IC partner mapping in CoA (ACC intercompany accounts)
6. Inventory and valuation effects	Seller PGI at transfer price basis; buyer GRN at PO (transfer) price into own valuation; in-transit bucket 1350 between
7. Financial postings	Seller: DR IC AR / CR IC Revenue + tax as applicable; buyer: DR inventory / CR IC AP; profit-in-inventory tracked via transfer-margin field for elimination (ACC consolidation feed); elimination itself executes in ACC group reporting
8. Quality and traceability gates	Genealogy continues across entities (serial retains chain, entity stamped per event); certificates travel with the transfer
9. Interface contract and boundary	IF-14 intercompany pairing: one business event creates both documents atomically or neither (single transaction, both entity contexts); no async pairing that could orphan one side
10. Timeout, retry, duplicate, recovery	Pairing idempotent on source event; mismatch monitor (order pairs whose values or statuses diverge) alerts Finance daily
11. Exceptions and reversals	Partial shipments mirror on both sides; returns create paired reversal documents; FX differences post to ACC 7100 at settlement

Closure Field	Specification
12. Reconciliation and closure	Monthly IC reconciliation: IC AR(seller) = IC AP(buyer) per pair, differences worked before Hard Close (ACC 26.2 IC step); profit-in-inventory schedule ties to elimination entries. Closure: pair settled and reconciled
13. Security and SoD	Transfer price maintenance restricted to Controlling; entity data isolation per Part C authorization model
14. Performance and availability	Pair creation synchronous < 2 s; reconciliation report on demand
15. Test range and evidence	TC-ICO-001…030: pair creation/atomicity, partial ship, return pair, FX difference, mismatch detection, elimination schedule accuracy

FL-08 — Recall and Suspect-Batch Campaign

Closure Field	Specification
1. Flow and objective	Contain, notify, recover and disposition all product affected by a suspect batch or serial population, with regulator-ready evidence
2. Trigger and prerequisites	Quality Manager opens a Recall Campaign from a suspect batch/serial (source: NCR, field failure, supplier notice); campaign object RCL-{YYYY}-{SEQ:3} (new, this volume)
3. Source document and owner	Recall Campaign; Quality Manager owns; executive approver for external notifications
4. Status transitions and approvals	Identify → Contain → Notify → Recover → Disposition → Effectiveness → Closed; external notification text requires dual approval (Quality Manager + executive)
5. Master data dependencies	Complete genealogy (MFG Part F §5); customer/site data on serials; batch provenance (MM §15.1)
6. Inventory and valuation effects	Contain: automatic Blocked status on all on-hand affected stock (batch/serial holds INV 6.2, mass-applied by the campaign); recovered units via RMA class Recall (freight PE-pays per reason code)
7. Financial postings	Recall RMAs post per FL-05 templates; campaign cost accumulation (freight, credits, rework) tagged with campaign ID for total-cost reporting; reserve raise proposal if material (Controller)
8. Quality and traceability gates	Population derivation: forward trace from the suspect batch (< 10 s per tree, MFG F §5) producing the affected list: serials, customers, sites, in-house stock, WIP, in-transit; effectiveness check requires 100% population accounted (recovered / field-corrected / documented-exception)
9. Interface contract and boundary	IF-15 campaign actions: mass-hold command, customer notification list export, RMA batch creation — each executed as an auditable bulk job with per-item result log
10. Timeout, retry, duplicate, recovery	Bulk jobs resumable; per-item idempotency (a serial already held is a no-op, logged); notification list versioned as population refines
11. Exceptions and reversals	Unreachable customer procedure (documented attempts, escalation); partial recovery accepted only with Quality Manager + executive sign-off; campaign scope expansion re-runs population derivation with diff
12. Reconciliation and closure	Closure condition: population fully accounted, dispositions complete, effectiveness check passed, total-cost report issued, CAPA linked and in progress or closed. Campaign record and evidence retained per genealogy retention (10 years minimum, Part D §6)
13. Security and SoD	Campaign creation restricted; external notification dual-approved; population data export logged (contains customer PII, Part C §6)

Closure Field	Specification
14. Performance and availability	Population derivation < 60 s for worst-case multi-level batch; mass-hold execution < 10 min for 10k units
15. Test range and evidence	TC-RCL-001…030: annual recall drill is an operating requirement — mock campaign end-to-end incl. population accuracy audit against seeded genealogy, mass-hold, notification list, recovery RMAs, effectiveness gate

Build Note: *The Recall Campaign object (RCL) is the one net-new functional object introduced by this Book; it is a thin orchestration layer over existing holds, RMA, genealogy and notification exports, specified here so recall closes as a first class flow. Build effort sits in M5 alongside RMA.*

PART B

INTEGRATION AND TRANSACTION ARCHITECTURE

This Part turns the integration matrices of the functional Books into executable contracts. It defines the platform standards every interface must obey, the transaction and consistency model, the interface catalog with ownership and reconciliation, and the operational tooling.

2. Platform Standards

2.1 API Standard

Standard	Specification
Style and versioning	REST over HTTPS, JSON payloads; URI-versioned (/api/v1/...); breaking changes require a new major version; N and N−1 supported concurrently; deprecation announced ≥ 90 days with response-header warnings and consumer-usage reporting before shutdown
Schemas	Every endpoint and event carries a published JSON Schema in the contract registry (source-controlled with the code); payloads validate at the boundary; unknown fields ignored on read, never emitted undocumented
Authentication	OAuth2 client-credentials for system callers; user-context calls carry the OIDC access token (Part C §2); every token scoped to named interfaces; no shared secrets in code (Part C §7)
Authorization	The API layer enforces the same role/entity/plant authorization as screens (single policy engine, Part C §4); no “trusted internal caller” bypass
Idempotency	All mutating endpoints require an Idempotency-Key header (UUID); the platform stores (key, caller, request hash, response) for 7 days and replays the stored response on retry; mismatched hash with same key returns 409
Errors	RFC 7807 problem+json: type, title, status, detail, correlation ID, and the domain error code (the E-codes from the functional volumes travel here); 4xx never retried by clients, 5xx retried per policy below
Pagination, filtering	Cursor-based pagination; filter grammar consistent with listing screens; hard response cap with explicit continuation

2.2 Event Standard

Standard	Specification
Transport and format	Durable log-based broker (Kafka-class) with JSON envelopes: event_id (UUID), event_type + schema version, occurred_at, producer, correlation_id, causation_id, entity context, payload

Standard	Specification
Ordering	Partition key = business object key (material_id, wo_id, so_id…) guaranteeing per-object ordering; consumers must not assume cross-object order
Delivery and idempotency	At-least-once delivery; every consumer keeps a processed-event store keyed on event_id (or a natural idempotency key named in the contract) — duplicate deliveries are no-ops by construction
Retry and dead-letter	Consumer retry with exponential backoff (1 s → 5 min cap) for 24 h, then dead-letter queue; DLQ items alert the owning team and are replayable from the integration console (§4) after cause resolution; DLQ replay preserves original event_id so idempotency still holds
Replay	Topics retain 30 days; full-history rebuild for read models via snapshot + replay procedure documented per consumer
Schema evolution	Additive changes only within a version; consumers tolerate unknown fields; breaking change = new event_type version, dual-published during migration window

2.3 Transaction and Consistency Model

- Atomic business transactions: the boundaries named in Part A field 9 (GRN, PGI, confirmation, IC pairing, disposition outcomes) execute as single database transactions covering stock, tracking, journal and outbox rows together. No business event is ever half-posted.
- Transactional outbox: events and cross-module commands are written in the same transaction as the state change and published by a relay with at-least-once semantics. Direct dual-writes (DB + broker in separate steps) are prohibited.
- Concurrency: optimistic locking (version column) on documents; pessimistic row locks only inside posting transactions (reservation firm-up, stock decrement, number-series allocation). Stock rows use atomic conditional decrements — negative-stock races are structurally impossible where policy forbids negatives.
- Compensating actions: where a saga spans modules (RMA disposition outcomes, IC pairing rollback on partial config error), each step has a named compensator specified in the interface contract; sagas surface in the console with step-level state.
- Clock and sequence: all timestamps UTC server-side; document numbers from gap-free series where finance requires (invoices) and gap-tolerant elsewhere (performance), per series configuration.

3. Interface Catalog

Every integration in the four volumes maps to a cataloged contract. Owner = the team accountable for the contract, its monitoring and its reconciliation. Contracts live in the registry with schemas; this table is the governance index.

ID	Interface	Type	Producer → Consumer	Owner	Reconciliation Control
IF-01	PO transmission to vendor	Email PDF (Phase 1) /	Purchasing → Vendor	Purchasing	Unconfirmed-PO report > 3 days (MFG G §1)

ID	Interface	Type	Producer → Consumer	Owner	Reconciliation Control
		EDI 850 (Phase 2)			
IF-02	Vendor confirmation capture	Screen / EDI 855 (Phase 2)	Vendor → Purchasing	Purchasing	Confirmed-vs-need-date exception board
IF-03	Engineering lifecycle events	Events	ENG → MM/PP/CST	Engineering IT	Exceptions panel: EBOM-ahead, placeholder aging
IF-04	Sales order events	Events	OM → PP/INV/ACC	Sales IT	Open-order book vs pegged supply daily
IF-05	WO lifecycle events	Events	PP → OM/INV/CST	Production IT	WIP report = GL tie (close 6c)
IF-06	Confirmation postings	Internal atomic + events	Shop floor → GL/genealogy	Production IT	Scan telemetry vs posted confirmations daily count match
IF-07	Shipment/PGI events	Events	OM → ACC/INV	Sales IT	Shipped-not-invoiced report
IF-08	GRN events	Events	INV → ACC/QM/PUR	Inventory IT	GR/IR aging (FL-03 field 12)
IF-09	Invoice requests	Outbox command	OM/RMA → ACC	Finance IT	Request-to-invoice latency + orphan report
IF-10	Three-way match basis	Internal contract	INV/PUR → ACC	Finance IT	Match exception queue age
IF-11	MRP proposal set	Batch publish (run ID)	MRP → PUR/PP	Planning IT	Proposal count + exception count per run logged and trended
IF-12	RMA disposition outcomes	Outbox commands	RMA → ACC/PP/OM	Sales IT	Close step 6e (Settling gaps)
IF-13	Close checklist automation	Read APIs	ACC → all modules	Finance IT	The checklist itself
IF-14	Intercompany pairing	Atomic internal	Entity A ↔ Entity B	Finance IT	Monthly IC recon (FL-07)
IF-15	Recall campaign bulk actions	Bulk jobs	QM → INV/OM/RMA	Quality IT	Per-item result log; population audit
IF-16	Tax engine (Avalara)	External REST	OM/ACC → Avalara	Finance IT	Tax reconciliation report (ACC 19.3) + recalc queue
IF-17	CAD/PDM connector	External (Part G)	PDM ↔ ENG	Engineering IT	Sync conflict queue (Part G §4)
IF-18	Identity provider	OIDC/SAML + SCIM	IdP → platform	IT Security	Access recertification (Part C §3)
IF-19	Label/print service	Internal REST	All → printers	Plant IT	Failed-print queue with reprint audit
IF-20	Scale/instrument capture	Station integration (Phase 2)	Devices → QM/Pack	Plant IT	Manual-entry fallback flagged rows

> **Build Note:** *Rule for the build: no module-to-module dependency exists unless it appears in this catalog with a contract in the registry. A developer adding an uncataloged integration is making an architecture change and must extend this table through change control.*

4. Integration Operations Console

Element	Specification
Message monitor	Per interface: throughput, lag, error rate, DLQ depth, oldest unprocessed age; drill to individual message with full envelope and correlation chain
DLQ management	Inspect, edit-and-replay (edit rights restricted + audited), bulk replay after fix, discard with mandatory reason (discard of a financially relevant command requires Controller)
Correlation tracing	correlation_id propagates from originating user action through every API call, event, job and journal line; a single search reconstructs the distributed flow (feeds the ≤ 3 clicks auditability NFR)
Reconciliation reports	The per-interface controls in §3 run scheduled; failures raise incidents per Part D §5, not emails that can be missed
Contract tests	Every contract has producer and consumer tests in CI (schema + behavior incl. idempotent-replay test); a contract change that breaks a consumer test cannot merge

PART C

SECURITY AND IDENTITY ARCHITECTURE

RBAC role tables exist throughout the functional Books. This Part supplies the enterprise identity, authorization, data protection and security-operations architecture those tables plug into.

2. Identity and Authentication

Control	Specification
SSO	OIDC (primary) with SAML 2.0 fallback against the corporate IdP (Entra ID assumed; confirm at kickoff — the design is IdP-agnostic). No local passwords for workforce accounts; local break-glass accounts only per §5
MFA	Enforced at the IdP for all interactive access; step-up authentication (re-prompt) for: payment release, vendor bank changes, period reopen, matrix release (MM §5.3), user-admin actions
Provisioning	SCIM 2.0 from the IdP: joiner/mover/leaver automation; role assignment via IdP groups mapped to ERP roles; leaver deactivation SLA 1 hour from HR trigger; orphan-account report weekly
Shop-floor sessions	Kiosk terminals: device-bound certificate + operator badge scan opens a scoped session (WC-limited, per MFG D §6); inactivity lock 10 min (config per station class); badge-out on operation complete optional per plant policy; no shared browser sessions
Service accounts	OAuth2 client-credentials per integration (one client per IF-nn); secrets in the vault (§7), rotated ≤ 90 days, usage-monitored; no user-account API usage by systems
External parties	Vendors/customers (portal phases) authenticate through a separate external IdP tenant; internal and external identities never share a namespace

3. Authorization Model

- One policy engine, three layers: Role (what functions), Scope (which entity/plant/warehouse/cost-center rows), Field group (which attributes, per MM §3.2 ownership and ACC field rules). Screens, APIs and events all consult the same engine; there is no UI-only hiding of data (the blind-count rule INV/MFG A §7.3 is the model everywhere).
- Roles are the ones named per module in the four volumes, catalogued centrally with their permission sets; new roles or permission changes ride change control with SoD analysis.
- Separation of duties: the SoD rule set consolidates every pair named in the volumes (PO create vs GRN post vs invoice approve vs payment release; WO confirm vs settlement; price edit vs approve; count vs approve; matrix edit vs release; MCR task vs activation). The engine blocks toxic combinations at role-assignment time, not just at transaction time; exceptions require CFO approval with expiry date.
- Row-level scope: every query filters by the user's entity/plant grants server-side; documents render cross-entity only for explicitly granted group roles (Controller group view).
- Restricted documents: export-controlled drawings (MFG C §5.1 Security Class) require the EAR user flag; access logged individually.
- Recertification: quarterly access review per role owner with attestation; failed attestations auto-revoke; report retained as audit evidence.

4. Privileged Access and Break-Glass

Control	Specification
PAM	Infrastructure and database access only through the PAM broker: just-in-time elevation, ticket-linked, session-recorded; standing admin access prohibited in production
Production data access	Developers have no default production access; diagnostic access via time-boxed grants with masked PII views unless the ticket justifies raw access (Controller/ISO approval for finance data)
Break-glass	Two sealed local emergency accounts (app + infra), credentials in escrow, use alarms the security channel immediately and forces post-incident review + rotation
Change execution	Deployments run from CI/CD service identities (Part D §4); humans do not apply schema or config changes by hand in production

5. Data Protection

Control	Specification
Encryption in transit	TLS 1.2+ everywhere including internal service traffic and printer/scanner websocket wedge; no plaintext listeners
Encryption at rest	Storage-level encryption for databases, object storage and backups; column-level encryption for bank details and personal data fields (key separation from storage keys)
Key and secret management	Central vault: application secrets, API clients, encryption keys; automatic rotation schedules; access audited; certificates auto-renewed with expiry alerting ≥ 30 days
PII inventory	Personal data catalog: employee operator data (badges, labor records), customer contacts, vendor contacts; retention and access rules per class; recall exports (FL-08) logged as PII disclosures
Export control	EAR-restricted document class enforced (Part C §3); denied-party screening on new customers/vendors (Phase 2 service via IF catalog extension) with interim manual checklist at onboarding
Data residency	Phase 1 US region deployment; residency constraints re-evaluated at international expansion (Part F trigger)

6. Security Operations

- Security event logging: authentication events, authorization denials, privilege use, SoD exception use, data exports, break-glass, admin changes — shipped to the SIEM with 13-month retention minimum.
- Alerting: brute-force patterns, impossible travel (IdP), privilege anomalies, DLQ tampering, mass-export anomalies; alerts route to the incident process (Part D §5) with security severity mapping.
- Vulnerability management: dependency scanning in CI, monthly platform patching windows, annual penetration test whose findings enter the risk register with owners and dates; criticals block go-live (Part H gate).
- Incident response: documented runbook with roles, evidence preservation, notification decision tree (customers, insurers, counsel); tabletop exercise annually alongside the recall drill.

PART D

RESILIENCE, AVAILABILITY AND OPERATIONS

Targets are set per business service, not as one global number, because the shop floor stopping is a different event from a report being slow.

2. Service Tiers, RTO and RPO

Tier	Business Services	Availability Target	RTO	RPO
T1 — Production-critical	Shop-floor terminal, scanning, label printing, GRN/issue/PGI posting, WO confirmations	99.5% during production hours (existing target, now formalized as T1)	1 hour	0 (committed transactions); ≤ 4 h of offline scan queue replays after recovery per MFG F §2
T2 — Business-day critical	Order entry, purchasing, engineering, quality screens, MRP on-demand, costing	99.0% business hours	4 hours	15 minutes
T3 — Periodic	Nightly MRP, costing runs, allocations, close jobs, reporting/analytics	Job-window commitments (below)	24 hours	24 hours (rerunnable jobs)

Batch windows: nightly MRP completes by 05:00 plant time; label/scan services never share a maintenance window with production shifts; close-critical jobs have priority lanes in the last 3 and first 3 business days of each month.

3. Disaster Recovery and Degraded Modes

Element	Specification
Topology	Primary region multi-zone: database synchronous replication across zones (zero-RPO zone failure); warm standby in a second region with continuous async replication for regional disaster (RPO ≤ 15 min, RTO ≤ 4 h for T1/T2)
Backups	Continuous WAL archiving + daily full snapshots; 35-day point-in-time recovery; quarterly restore test to an isolated environment with checksum and row-count verification — restore evidence filed in the operations log (Part H gate input)
Failover	Zone failover automatic; region failover is a decided action (runbook, two-person rule) with DNS cutover and consumer replay from outbox/topics; annual full region-failover exercise in the staging replica
Degraded modes	Defined per dependency: broker down → outbox accumulates, postings continue, consumers catch up (monitor depth); tax engine down → MFG A §11.1 failover; IdP down → existing sessions continue, break-glass for new critical access; printer down → station failover printer + queued reprints; WAN down at plant → T1 terminals run the 4-hour offline queue, everything else read-only
Data integrity on recovery	Recovery always replays from the committed log; the outbox pattern (Part B §2.3) guarantees no lost events for committed transactions and idempotent consumers absorb replays

4. Deployment and Change

- Environments: DEV → TEST (integrated, contract tests) → UAT (business validation, migration rehearsal target) → PROD; configuration promoted as versioned artifacts, never hand-edited upward.
- CI/CD: trunk-based with mandatory review; pipeline gates: unit, contract (Part B §4), migration dry-run, security scans; artifacts immutable and signed.
- Zero-downtime releases: rolling deploys behind health checks; database migrations expand-and-contract (additive first, code tolerant of both schemas, contraction after soak); feature flags for functional cutover so deploy ≠ release.
- Rollback: every release has a tested rollback path (previous artifact + contract-compatible schema); flags allow functional rollback without redeploy; rollback drills part of M7.
- Release calendar: T1-affecting releases outside production shifts; close-freeze on finance-touching releases during Hard Close days.

5. Observability and Support Operations

Element	Specification
Telemetry	Metrics, structured logs and distributed traces on every service; correlation_id (Part B §4) joins traces to business documents; scan telemetry retained 24 months per MFG F §4
SLOs and error budgets	Per NFR: scan round-trip < 400 ms P95, PGI < 2 s, detail load < 1.5 s, MRP < 30 min, traceability tree < 10 s, dispatch refresh < 2 s. Each is an SLO with burn-rate alerting; sustained budget burn freezes feature releases for the offending service until recovered
Alerting and incidents	Severity matrix (S1: T1 service down or data-integrity suspicion → 15-min response 24×7 during production; S2: T2 degradation → 1 h business hours; S3/S4 queued); on-call rotation; incident timeline, postmortem without blame, action items tracked to closure
Job monitoring	Batch framework with run history, SLA timers, dependency chains, auto-retry policy and failure escalation; failed close-critical jobs page, never just log
Support model	Tier 1 (plant super-users, runbook-driven), Tier 2 (application support, console access per Part C), Tier 3 (the development partner engineering with SLA per support contract); month-end hypercare roster for the first three closes
Runbooks	Written per degraded mode and per S1 scenario before go-live (Part H readiness gate item); stored versioned beside the code

6. Capacity and Retention

- Capacity model: sized for the NFR volumes (MFG Part K §4, MM §20.2) with 3× headroom on scan peaks (shift start bursts) and 5-year growth; load, stress, soak and failover tests against these profiles are Part H test classes, executed in M7 with results filed as evidence.
- Data retention: transactional data online 7 years; genealogy and quality evidence 10 years minimum (recall support, FL-08); released document files WORM per MFG NFR; scan telemetry 24 months; archives queryable read-only; legal hold flag suspends any purge per object.

PART E

FINITE SCHEDULING AND CAPABLE-TO-PROMISE

A contractual ETO promise date computed on infinite capacity is not defensible. This Part defines the constraint model, the scheduling engine requirements, the CTP check, the governance around promises, and an explicit statement of what ships at go-live versus what follows. The data model already carries operation durations, calendars, crews and alternates (MFG D), so this is an engine and policy layer, not a redesign.

2. Constraint Model

Constraint	Source Data	Scheduling Behavior
Work center capacity	WC capacity model: shifts, hours, parallel stations, efficiency (MFG D §2)	Finite buckets per WC per shift; operations consume capacity; overload impossible in the finite plan (only in the released past)
Batch resources	Oven/vacuum cycle size and hours (MFG D §2)	Cycle-level scheduling: orders pack into cycles; a cycle is one capacity consumption regardless of fill
Labor and crews	Crew size per op/WC; workforce calendar (new: crew roster per WC per shift, maintained by Production Scheduler)	Operation needs both machine slot and crew slot; crew shortfall delays exactly like machine shortfall
Skills and certifications	Operator qualification register (new object QAL-: operator, skill, level, expiry; linked to training records)	Ops flagged with required skill schedule only into shifts with a qualified operator; expired certification removes the operator from eligibility automatically (and interlocks the terminal at SP-07 login for flagged ops)
Tooling and fixtures	Tool register (new object TOOL-: tool, quantity, WC compatibility, maintenance status)	Ops declaring a tool requirement consume tool availability; tools in maintenance are unavailable
Material availability	MRP supply picture incl. confirmed PO dates	An operation cannot schedule before its op-mapped components' availability dates (full-kit by operation, consistent with staging SP-09)
Maintenance windows	WC status Maintenance (MFG D §2) + planned maintenance calendar (new)	Blocked capacity; planned windows visible in what-if before commitment
Subcontract capacity	Vendor lead times + declared weekly capacity on the AVL row (new field)	Subcontract ops schedule against vendor capacity declarations; breach raises buyer exception rather than silent slip

3. Scheduling Engine Requirements

- Modes: backward from due date (default), forward from release, and bottleneck-centered (schedule the constraint WC first — Test Lab and Vapor Phase Drying are the expected bottlenecks — then subordinate upstream/downstream).
- Alternates: alternate WC with efficiency factor (existing field) is engine-selectable under load, logged as a decision; alternate routings are planner-selected only (engineering-controlled, never silent).
- Setup families: sequence-dependent setup reduction by grouping (e.g. winding wire gauge families); family definitions per WC maintained by Manufacturing Engineer.

- Freeze horizon: inside N days (default 5, per WC) the schedule does not auto-move; changes inside the freeze require Production Scheduler action with reason. Consistent with the material-level planning fence (MM §3.6).
- What-if: sandbox copies of the schedule for scenario testing (rush order insertion, WC outage); publishing a scenario is an explicit replace action with diff preview; sandboxes never post.
- Rescheduling policy: engine proposes moves as exceptions (EX-02/03 extended to operations); auto-apply only outside the freeze horizon and only within promise-date guardrails (§5); everything else is planner-confirmed.
- Determinism and audit: same inputs produce the same schedule (seeded heuristics); every published schedule stores its input snapshot ID so any promise can be traced to the plan that made it.
- Performance: full plant reschedule < 5 min at NFR volumes; single-order CTP < 10 s (the quote-time budget).

4. Capable-to-Promise

Element	Specification
CTP check	For a quote line or SO line: explode the (template or released) BOM and routing, check material availability dates (supply picture + info-record lead times for unstocked buys) AND finite capacity insertion across constrained WCs, returning the earliest feasible completion + ship date with the binding constraint named (“Test Lab week 34” / “bushing lead time 16 wks”)
Quote integration	Firm ETO quotes must carry a CTP result (replaces the feasibility check reference in MFG B §3.1); Budgetary quotes may quote lead-time bands without CTP, watermarked accordingly
Promise capture	Accepting a CTP result reserves soft capacity (decaying reservation, config 14 days) so parallel quotes do not double-promise the same bottleneck slot; SO confirmation converts to firm capacity load
Revalidation	Promise dates re-validate automatically on: ECO affecting the order, WC outage inside its path, supply slip on a pegged PO, schedule publish that moves its operations. A promise at risk raises the at-risk workflow (§5), never a silent new date
ATP vs CTP boundary	MTS lines keep the existing ATP (stock + scheduled receipts, MFG B §4.3); CTP applies to MTO/ETO. Global multi-plant ATP activates with Part F plant 2

5. Promise-Date Governance

- A customer-visible promise changes only through the Sales Change Order path with reason, impact and (per customer class policy) customer notification; planners move internal schedule, not promises.
- At-risk workflow: projected completion beyond promise − transit buffer creates an at-risk record routed to Production Scheduler + Sales owner with recovery options (overtime shift, alternate WC, partial ship, re-promise); decisions logged against the order.
- Promise reliability is a first-class KPI: promised-vs-actual distribution by product family and by binding constraint, monthly; CTP model calibration (planned vs actual durations) reviewed quarterly and fed back into routing standards.

6. Delivery Boundary Statement

Capability	Go-Live (Phase 1)	Phase 2	Phase 3
Finite capacity per WC + batch resources	Yes — finite buckets, freeze horizon, exceptions	Refinement	—
CTP on ETO/MTO with soft reservation	Yes — single plant	Calibration + broader adoption	Global ATP/CTP (multi-plant, Part F)
Skills/certification constraints + terminal interlock	Yes (register + interlock)	Training-system integration	—
Tooling constraints	Register + availability check	Maintenance-plan integration	—
Setup families and sequence optimization	Manual family grouping	Heuristic optimization	Optimizer/solver-based scheduling
Interactive drag reschedule with validation	View + move via actions	Full interactive board (existing Phase 2 item)	—
What-if simulation	Single-scenario sandbox	Multi-scenario compare	Optimization-assisted

Build Note: *This table is the explicit statement of what is and is not delivered at go live. Contractual promises from day one rest on finite CTP, not infinite scheduling. What is deferred is optimization quality, not promise integrity.*

PART F

MULTI-PLANT, INTERCOMPANY AND GLOBAL OPERATIONS MODEL

PE operates one manufacturing plant today with four legal entities. This Part fixes the enterprise organization model in the data structures now, specifies the flows that are live at go-live (intercompany merchandise, FL-07), and defines the activation packages for plant 2 and international expansion so neither requires redesign.

2. Enterprise Organization Model

Object	Definition	Keyed Into
Legal Entity	PE Power Grids, PE Grid Components, PE Transformer, Power Electronics; owns books, currency, tax registrations, fiscal calendar	Every document and journal (entity-aware series exist); ACC multi-entity as built
Plant	Physical manufacturing/distribution site belonging to one entity; owns work centers, warehouses, plant-level material data (MM §3.3)	material_plant_data, WCs, routings (plant-keyed), MRP scope
Sales Organization	Selling unit per entity (Phase 1: one per selling entity); owns price lists and customer commercial terms	Price lists (entity/currency per MFG I §6.1), SO header
Purchasing Organization	Buying unit (Phase 1: central); owns vendor terms, info records (entity-keyed per MM §7.2)	PO series PO-{YYYY}-{ENTITY}-…, info records
Shipping Point	Dock/yard grouping within a plant (Phase 1: implicit one; field reserved)	Shipments
Calendars	Plant working calendar (exists), entity fiscal calendar (exists), crew calendars (Part E)	Scheduling, MRP offsets, close

Build Note: *Build rule: no table added from this point may assume single-plant or single-entity. The review found entity-aware fields already present; this section makes the remaining implicit assumptions (one shipping point, central purchasing) explicit reserved objects instead of hidden constants.*

3. Intercompany Flows at Go-Live

- Live flow: PE Grid Components sells merchandise sourced from PE Transformer stock and vice versa — the paired SO/PO mechanism, postings, in-transit handling and monthly reconciliation are specified as FL-07 and build in M6.
- IC relationships are configuration: (selling entity, buying entity, transfer price list, IC account pair, tax treatment); undefined pairs cannot trade (hard block at pairing).
- Entity-level financial separation is already carried by ACC; this volume adds the operational pairing and the profit-in-inventory data feed for consolidation.

4. Transfer Pricing

Element	Specification
Method	Cost-plus per product family (markup % table per IC relationship, Controller-owned, effective-dated) or negotiated fixed price rows (transfer price list, same structure as customer price lists per MFG I §6.1)
Documentation	Markup table changes require attached rationale (tax compliance file); annual review with the close calendar
Profit in inventory	Buyer-side stock received via IC carries the transfer margin per unit; the PII schedule (qty on hand × margin) feeds the elimination entry in group reporting (ACC); consumption/sale releases it
Audit	Every IC document stores the price source row (fixed or markup + basis cost version) so any transfer price is reproducible

5. Plant 2 Activation Package (pre-specified)

- Master data: Extend materials to plant 2 (MM §3.3 Extend action + data-completion workflow); plant-level MRP/production/warehouse/valuation data per plant; WCs, routings and rates per plant; production versions (material + plant → MBOM/routing pair) activate as the selector object.
- Planning: network MRP — demand at plant A satisfiable by transfer from plant B via Special Procurement = Stock Transfer (field reserved, MM §3.6); transfer lead times on a plant-lane table; global ATP checks both plants' supply pictures.
- Logistics: stock transport orders with in-transit bucket 1350 (intra-entity) or FL-07 pairing (cross-entity); cross-plant genealogy continues on the same serial/batch chain with plant stamped per event (already in the event model).
- Finance: plant-level valuation already keyed (MM §18.1 material_valuation has plant_id); cost center hierarchy gains a plant dimension node; work-center rates per plant.
- Acceptance: plant 2 go-live requires re-running the FL-04 and FL-07 test packs in the multi-plant configuration plus a cross-plant recall drill (FL-08).

6. International Expansion Package (pre-specified)

Area	Phase 1 Reality	Expansion Design
Tax	US + Canada via Avalara and nexus register (MFG A §11.4)	Additional regimes as jurisdictions in the same determination layer; VAT/GST logic via tax engine; e-invoicing mandates as country adapters on IF-16; no document flow redesign
Currency	Multi-currency documents with daily rates (ACC 3) already live	Additional functional currencies per new entity; group reporting currency in consolidation (ACC scope)
Trade compliance	HTS/Schedule B/CoO on materials (MM §2.4/15.1); Buy America; export doc checklist on shipments	Customs interfaces, export licenses, denied-party screening service (Part C §5), Incoterms-driven landed cost expansion (duty engine on IF catalog)
Localization	English, dual units (lb/kg) on documents	Language layer on document templates (material_text is language-keyed already, MM §18.1); statutory report adapters per country in ACC

Area	Phase 1 Reality	Expansion Design
Governance	—	Each new country enters via a compliance matrix (registrations, statutory books, filing calendar) approved by Controller before first transaction

PART G

CAD, PDM AND PLM INTEGRATION SPECIFICATION

The controlled document repository (MFG Part C §5) remains the ERP system of record for released renditions. This Part specifies the integration that connects native CAD authoring to it. Toolchain decision required at kickoff: this specification is written against SolidWorks + SolidWorks PDM Professional as the working baseline because it matches PE Engineering's current authoring environment; every requirement is stated tool-neutrally so a different selection (Inventor/Vault, Creo/Windchill) changes the connector implementation, not the contract.

2. Ownership Rules (the contract's foundation)

Data	System of Record	Rule
Native CAD files, assembly structure, in-work revisions	PDM vault	ERP never edits native files; check-in/check-out authority lives in PDM once the connector is live (the ERP check-out screens become read views of vault state)
Released drawing renditions (PDF), release stamps	ERP controlled documents (MFG C §5)	Release in PDM triggers rendition publish to ERP; the floor and customers consume only ERP renditions — unchanged
Item identity (material number)	ERP (MM §2.3)	PDM parts carry the ERP material number as an attribute; number creation only via MCR/placeholder path — the connector requests numbers, never invents them
EBOM structure	ERP (MFG C §3)	CAD assembly structure proposes; the engineer confirms in the EBOM editor; ERP EBOM remains the released truth (MBOM derives from it, unchanged)
ECO/change control	ERP (MFG C §4)	PDM change states map to ECO tasks; effectivity and disposition decisions are ERP-only

3. Connector Functional Requirements (IF-17)

Function	Specification
Part attribute sync	PDM part card ↔ ERP material: number, description, material group, UoM, revision; bidirectional field-level mapping table with per-field direction (most fields ERP→PDM read-only; description/mass PDM→ERP proposal); sync on vault check-in and on ERP MCH apply
New part flow	Engineer creates part in CAD → connector detects missing ERP number → raises placeholder request (MM §4.3) pre-filled from the part card → returns the reserved number to the PDM attribute; drawing annotations reference the real number immediately (same guarantee as the manual path)
Assembly-to-EBOM extraction	On demand or at PDM release: extract the assembly structure (find numbers, quantities, reference designators from CAD properties) → stage as an EBOM import proposal in the existing CAD BOM Import wizard (MFG C §3.3) with line-level match/conflict states; commit remains transactional and engineer-confirmed
Structure comparison	Diff view: current ERP EBOM vs current CAD assembly — added/removed/qty-changed lines; runs automatically when a released EBOM's source assembly gets a newer vault revision, feeding the "EBOM ahead"-style exception panel with a CAD-ahead banner
File dependency handling	The connector resolves assembly external references (sub-assemblies, drawing sheets, design tables) and publishes the complete rendition set on release; broken

Function	Specification
	references block release on the PDM side (vault rule) so ERP never receives partial sets
Rendition publishing	On PDM release: generate PDF (and STEP where flagged) → create/replace the ERP document revision with release stamp → link to items/BOM lines per the mapping; checksum verified; failure queues for retry with alert (never a silently missing drawing)
Title-block extraction	Extract drawn-by/checked-by/scale/sheet/size from CAD properties at publish (structured properties, not OCR), populating MFG C §5.1 metadata automatically; mismatch between title block and vault attributes blocks publish
Conflict resolution	Concurrent edits (ERP MCH on a synced field vs PDM check-in) land in a conflict queue: side-by-side values, resolver role per field ownership, resolution audited; no last-write-wins anywhere in the mapping
Offline/cache behavior	Designers work vault-local per PDM standard behavior; the connector is server-side (vault events), so designer offline work syncs on their next check-in; ERP outages queue vault events for replay (at-least-once, idempotent on vault transaction ID)

4. Change Flow Mapping

PDM State/Event	ERP Effect
Work in progress (checked out)	ERP document shows In-Work at vault (informational); released rendition unaffected
PDM change order opened	Proposal to link/create ERP ECR (engineer confirms); PDM CO number cross-referenced
PDM release approved	Requires the linked ERP ECO task set to be release-ready (the connector checks via API); publishes renditions; ERP ECO release then executes the atomic cutover (BOM/routing/docs/effectivity per MFG C §4.2) — the two releases are sequenced, PDM first, ERP effectivity governs the floor
Obsolete in PDM	ERP document → Obsolete with the standard exclusion rules; where-used sweep warns on active usage before allowing it

Build Note: *Sequencing rule to make explicit in the build: the shop floor keys off ERP effectivity, never off PDM state. A PDM release with a future ERP effectivity date is the normal case, not an error. This preserves the single source of execution truth while giving engineering native tools.*

5. Acceptance Test for the Connector

- TC-CAD-001…030 covering: new part with number reservation; assembly extraction to committed EBOM; structure diff after CAD change; full release round-trip with rendition checksum; title-block mismatch block; conflict queue resolution both directions; ERP-outage event replay; broken-reference block; ECO-linked release sequencing; publish-failure retry with alert.
- Exit criterion: an ETO design executed end-to-end (configuration → CAD → EBOM → MBOM → WO → floor drawings) with zero manual file uploads and zero manual number entry.

PART H

TEST, MIGRATION, CUTOVER AND ACCEPTANCE FRAMEWORK

This Part defines the structures that produce evidence during the build. It does not claim the evidence exists today; it makes its accumulation mandatory and auditable, and it gates go-live on it.

2. Requirements Traceability

- Requirement ID scheme: REQ-{VOL}-{SECTION}-{SEQ} generated by parsing the four functional volumes plus this volume: every field table, validation code, posting rule, status transition, button, interface and NFR becomes one or more requirement rows. The parse is mechanical (the volumes are structured for it) and the resulting register is the master RTM backbone, maintained in the ALM tool, exported monthly to the project record.
- Every RTM row links: requirement → design/build ticket → test case ID(s) → latest execution result → defects → acceptance status. Requirements to test traceability for every screen, rule, transition, posting and integration is satisfied structurally: an RTM row without a test case is a visible gap count on the readiness dashboard, target zero at M7 exit.
- Test case ID ranges reserved per flow in Part A field 15, plus module ranges (TC-MM-, TC-VAL-, TC-CAD-, TC-SEC-, TC-OPS-) — IDs are stable so evidence remains traceable across runs.

3. Test Classes (mandatory per flow and per module)

Class	Content	Examples Already Bound
Positive / happy path	Every specified screen behavior and posting	All Part A flows field 15
Negative / validation	Every E-code and V-rule fires as specified	E-RMA-03/04, E-MM-11…14, E-PER-01…04
Concurrency	Documented race scenarios: double PGI, parallel reservations, simultaneous count vs movement, two buyers one PR	TC-O2C concurrency set; reservation firm-up locks
Reversal / correction	Every reversal path posts mirrors with references	GRN reversal, confirmation correction, invoice reversal + rebill, settlement reopen
Integration failure & recovery	Kill the broker/tax engine/printer mid-flow; verify outbox, DLQ, replay, no loss, no duplicates	Avalara failover; offline scan replay with conflict
Security & SoD	Role matrix enforcement, scope isolation, SoD toxic pairs blocked, field-group edit rights	TC-SEC range; recertification evidence
Performance	SLO verification at NFR volumes + peak profiles (Part D §6)	Scan P95, MRP window, traceability tree, CTP < 10 s
Resilience	Zone kill, region failover exercise, restore test, rollback drill	TC-OPS range; quarterly restore evidence
Reconciliation	Every Part A field-12 control produces correct results on seeded discrepancies	WIP=GL, GR/IR aging, IC recon, subledger tie-outs

4. Evidence Standard

- A test execution is evidence only with: test case ID + version, environment + build ID, data set reference, executor, timestamp, result, and artifacts (screens, journal extracts, report outputs) attached in the ALM record.
- UAT sign-off is per business scenario pack (one per Part A flow) signed by the named business owner — these signatures are the content of the acceptance gate, collected as they pass, not at the end.
- Defect policy at gates: zero open S1/S2 on gated scope; S3 with approved workarounds documented; the defect-to-requirement link keeps the RTM honest.

5. Data Migration (package level)

MM §16.3 specifies the material-domain migration in full. Package level adds:

Object Set	Approach	Reconciliation Proof
Open AR/AP and GL balances	ACC opening-balance load at cutover date	Trial balance tie to legacy close; subledger = GL by account
Open sales orders	Load open lines with statuses mapped; historical closed orders stay in legacy read archive	Open order book value = legacy report; per-customer spot checks
Open POs and GR/IR	Open lines + received-not-invoiced positions	2110 opening = legacy accrual schedule
Open WOs	Preferred: drain to minimum by cutover freeze; residual WOs load with WIP value as opening 1320 by order	WIP by order = legacy job cost report
Stock with batch/serial	Count-verified quantities with provenance (MM §16.3)	Value by valuation class = legacy inventory sub-ledger
Genealogy history	Active-stock provenance operational; deeper history to read archive	Sample-based trace equivalence checks (20 serials end-to-end)
Customers/vendors/prices	Full masters with dedup pass; price rows effective-dated	Count + spot checks; no document loads without resolved master

- Two full mock conversions minimum (M7), each timed to prove the cutover window, each producing the reconciliation set above; mock 2 must pass with zero manual fixes to qualify for go-live.
- Freeze and delta: legacy freeze T-5 days for masters, T-1 for transactions; delta list re-keyed under dual control; go/no-go checkpoint reviews reconciliation results before unfreeze in the new system.

6. Cutover Runbook and Rollback

Element	Specification
Runbook	Sequenced task list with owner, planned duration, dependency, verification step and sign-off per task (master load → balance load → open documents → verification → user enablement → first transactions witnessed); rehearsed in mock 2
Checkpoints	Named go/no-go points with criteria: post-master-load reconciliation, post-balance tie-out, smoke-test pack green (a 40-case cross-flow subset), print/scan infrastructure verified live on the floor

Element	Specification
Rollback	Before the first productive posting: rollback = discard and return to legacy (legacy kept fully operational through the first close); after productive postings: roll-forward-only with the incident process — this boundary is stated to the steering committee at go/no-go, no ambiguity
Fallback operation	Paper travelers and manual GRN forms pre-printed for 48 h of T1 outage during hypercare; re-key procedure defined (the offline queue covers terminals; fallback covers total outage)
Hypercare	Three closes with the Part D §5 hypercare roster; exit criteria: close on time twice consecutively, SLOs met for 30 days, defect arrival below threshold

7. Go-Live Readiness Gate and Acceptance

Gate Item	Pass Condition
RTM completeness	Zero requirements without test linkage on go-live scope; coverage report signed by QA Lead
Flow closure evidence	All eight Part A flows: UAT scenario packs signed; reconciliation controls demonstrated on seeded errors
Integration health	Contract tests green; DLQ drill executed; correlation tracing demonstrated end-to-end
Security	Pen-test criticals closed; SoD matrix verified; recertification round-1 complete; break-glass tested
Resilience	Restore test + zone-failure test + rollback drill evidence on file; runbooks complete
Performance	SLO verification report at NFR volumes + peak profiles
Migration	Mock 2 passed clean; cutover window proven inside plan
Operations	Support model staffed; hypercare roster confirmed; monitoring dashboards live
Training	Role-based training completion ≥ 95% for go-live roles; super-users certified per plant area
Formal acceptance	Signatories per the sign-off table below have executed; deviations carry approved waivers with dates

8. Document Sign-Off

This Book, together with the Inventory & Warehouse and Accounting specifications and Books I and II, constitutes the ERP Build Package. Signatures below accept this volume as the controlling specification for enterprise architecture, security, operations, planning, global model, CAD integration and delivery governance. This page is issued blank and is executed through the approval workflow, never pre filled.

Role	Name	Signature	Date
Project Sponsor			
Product Owner			
Technical Lead (PE)			
IT / Infrastructure Lead			

Role	Name	Signature	Date
Information Security Officer			
Development Partner Lead			
QA Lead			
Engineering Manager			
Production Manager			
Controller			
CFO			

BOOK IV

DESIGN & DEVELOPMENT COMPLETION

Controlling Baseline & Decision Register • Flow A: Sales Order to Delivery • Flow B: Design-to-Source • Flow C: Traceability Contract
Pricing Condition Model • Production Versions & Planning Baseline • Module Completions • Architecture Decision Records • Gates & Authoring Standards • Action Register

The controlling design governance layer of this specification

Book IV is the design governance layer of this specification. It records the mandatory scope decisions, resolves cross Book precedence, carries the three consolidated flows (Sales Order to Delivery, Design to Source, and the physical Traceability Contract), supplies the executable pricing condition model, the production version and planning algorithm baseline, the remaining module completion requirements, the architecture decision records, and the design and development gates with authoring standards. Where an artifact must live outside a specification (Figma prototypes, OpenAPI files, executed test evidence), this Book defines the artifact, its owner, its repository and its acceptance criterion, so the specification governs it without pretending to contain it.

1. Controlling Baseline, Precedence and Decision Register

1.1 Version and Precedence Register

Controlled Item	Standing	Precedence Rule
This document — Unified Build Specification	Controlling	The controlling document. Book IV has precedence over Books I–III where they conflict; within a Book, the conflict register §1.3 resolves precedence
Book I — Manufacturing Core	Contained	Functional baseline for its modules
Book II — Material Management	Contained	Controls the material object, valuation and account determination
Book III — Enterprise Completeness	Contained	Controls flows FL-01…FL-08, integration, security, resilience, planning, global model, CAD integration and delivery framework
Inventory & Warehouse Module Spec	V3.0 (Mar 2026) — external	Version-locked into the baseline. Deltas flagged “DELTA to INV x.x” in Books I–II control over the referenced INV section. Cross reference validation executes at design gate DG-1 (§9)
Accounting Module Implementation Spec	V2.0 (Mar 2026) — external	Version-locked. Same delta rule. O2C/P2P/close behavior validated per FL-01/FL-03/FL-06 test packs

1.2 Mandatory Scope Decision Register (DEC-01 … DEC-10)

These decisions are recorded as the controlling scope baseline. Each names its go-live commitment and its phase trigger. Changing any DEC row is a steering-committee change-control action, not an editorial one.

ID	Decision	Recorded Outcome
DEC-01	Scheduling at go-live	RESOLVED — Book III Part E controls and supersedes Book I Part D §11. Go-live delivers finite-capacity buckets, batch resources, freeze horizons and CTP with soft capacity reservation (Part E §6 boundary table). The Book I §11 rows are re-read accordingly: its “Phase 1 infinite capacity” row is void; interactive drag rescheduling remains Phase 2; solver-based optimization remains Phase 3. See conflict C-01
DEC-02	Actual costing	Standard costing is the go-live and Phase 2 method (Book I Part I §5). Periodic actual costing / material ledger is a Phase 3 decision gate; the per-order and per-batch actuals needed for it are already captured, so no schema rework is expected. Until then, margin truth is the three-view report (quoted/standard/actual per order, Book I Part B §6)

ID	Decision	Recorded Outcome
DEC-03	Product configuration	Go-live: structured ETO configuration attributes + template designs + Copy From variant authoring (Books I B §3.2, C §3.3). Configurable 150% BOMs, option rules and configurable routings: Phase 3, architected via the characteristics layer (Book II §2.4). No configurator at go-live is a recorded limitation, not an omission
DEC-04	Carrier / TMS	Go-live: manual carrier selection, PRO/BOL capture, freight accrual #26, POD capture per Flow A §2. Carrier API integration (rating, tendering, tracking, e-POD) is Phase 3 as interface IF-21 (reserved in the catalog). Freight claims process is in scope at go-live (Flow A stage 14)
DEC-05	Supplier / customer electronic integration	Phase 2: supplier EDI 850/855/856/810 (IF-01/02 upgrade) for the top-volume vendors. Phase 3: supplier portal and customer portal incl. drawing responses (Book I C §5.2 transmittals stay email-based until then). All Phase 1 processes function without these channels
DEC-06	Field service and installed base	Installed-base record and warranty entitlement engine are IN at go-live (§7.4 of this Book) because RMA adjudication depends on them. Field service execution (dispatch, technician work orders, service billing beyond RMA RC4) is OUT until a Phase 3 scope decision
DEC-07	Multi-plant / global first delivery	Go-live scope: one manufacturing plant, four legal entities, intercompany merchandise flow FL-07 live. Plant-2 activation and international expansion execute per the pre-specified packages (Book III Part F §5–6); the activation trigger is a signed steering decision plus the FL-04/FL-07/FL-08 multi-plant test packs
DEC-08	Machine and test-equipment integration	Phase 1: manual capture with instrument-ID binding and calibration interlock (Book I Part E §3). Phase 2: CSV import from test sets. Phase 3: direct interfaces and OEE/downtime automation (IF-20). Manual fallback remains specified for every phase
DEC-09	Quality maturity	Phase 2: SPC (control charts, capability, trend alerts on IP-INP/FIN characteristics) and audit management. Control plans/FMEA/APQP/PPAP: implemented only where a customer contract demands them, activated per customer flag; otherwise excluded. Deviation/waiver control is IN at go-live (§7.3)
DEC-10	Application architecture	Modular monolith with enforced bounded contexts, single relational database with schema-per-context ownership, log-based event broker with transactional outbox. Approved through the ADR set in §8

1.3 Conflict Register

ID	Conflict	Resolution	Status
C-01	Book I Part D §11 (infinite-capacity Phase 1) vs Book III Part E §6 (finite CTP at go-live)	DEC-01: Book III Part E controls; Book I §11 superseded as described	Closed
C-03	AVL “Price Basis / Lead Time” columns (Book I G §2.2) vs purchasing info records (Book II §7.2)	Already resolved in Book II §7.5: info record owns commercial terms; AVL columns are read-through views. Recorded here for the register	Closed
C-04	Completeness claims vs scope language	The benchmark language on the cover governs all external descriptions. Nothing stronger may be claimed	Closed

ID	Conflict	Resolution	Status
C-05	Statistical safety stock “Phase 2” (Book I A §10.2) vs any assumption of go-live statistical buffers	Static safety stock at go-live stands; the reserved Suggested SS columns remain read-only until the Phase 2 engine ships	Closed
C-06	Two ATP definitions (Book I B §4.3 vs Book III availability semantics in Book II §8)	The single Availability definition in Book II §8 controls; Book I §4.3 is its order-entry application	Closed

Build Note: *Register discipline: any newly discovered cross-Book contradiction gets a C-row here with an owner and a resolution before the affected module passes design gate DG-1. A build that codes around an unregistered conflict fails review by definition.*

2. Flow A — Sales Order to Delivery (FL-09)

FL-09 consolidates the fulfillment path that FL-01/FL-02 cover commercially into one execution design across MTS, MTO and ETO branches, and adds the objects the fulfillment design must carry: schedule lines, deterministic delivery split/merge, allocation and backorder processing, carrier/POD closure and controlled PGI reversal. Everything here extends Book I Part B and INV 5.7/Part A §5 without changing them.

2.1 Order Schedule Lines (new sub-object of the SO line)

Field	Type	Req	Specification
Sched #	Auto	Auto	10, 20, 30 within the SO line; created automatically (one per line by default; more on split)
Requested Qty / Date	Number + date	Yes	Customer ask at schedule granularity; header/line dates default the first schedule
Confirmed Qty / Date	Number + date	Auto	Written by ATP (MTS) or CTP (MTO/ETO) with the supply/capacity constraint named and the plan snapshot ID stored (Book III Part E §3)
Delivered Qty / Open Qty	Numbers	Auto	Rolled from delivery postings; Open = Confirmed − Delivered − Cancelled
Backorder Flag	Badge	Auto	Set when Confirmed < Requested or a confirmed schedule slips; feeds the Backorder Workbench §2.2
Peg Reference	Link	Auto	Reservation / WO / PR / planned order pegged to this schedule (pegging is schedule-level from this Book forward; line-level pegs in Book I read as schedule 10)
Status	Badge	Auto	Open → Confirmed → Allocated/Pegged → Delivery Due → Partially Delivered → Delivered → Closed / Cancelled

Rules: delivery documents consume schedule lines, never raw order lines; a schedule line is the unit of confirmation, allocation, backorder and closure. Invoice-relevant quantities roll up from schedules to the line so ACC contracts are untouched. Migration: existing open SO lines convert to single-schedule lines automatically.

2.2 Allocation and Backorder Processing

- Allocation applies when available supply of an MTS material is less than the sum of confirmed schedules due in the horizon. Priority order: contractual customer buffers (Book I A §10.3) first,

then customer priority class (A/B/C on the customer master, Sales Director owned), then requested date, then order entry time. Fair-share within a class prorates by open quantity.

- Backorder Workbench (screen OM-BO-001, /sales/backorders): all backordered and at-risk schedules with supply options (incoming POs/WOs with dates, substitutable stock per approved alternates), one-click re-promise proposals, and mass re-confirmation runs after each MRP run. Every reallocation that takes confirmed supply from one order to another requires Sales Director approval and writes an audit pair on both orders.
- Substitution at allocation follows engineering-approved alternates only (Book I C §3.2); customer-visible substitutions additionally require CS confirmation per customer policy flag.

2.3 Delivery Split and Merge Rules (deterministic)

Rule	Behavior
Split — mandatory	Different ship-to sites; different Incoterms; hazmat segregation (oil totes vs electronics per hazmat profile); over-dimensional units needing permits ship alone; customer “no partial” flag forces full-quantity waiting instead of split
Split — planner elective	Date-window separation (schedules > N days apart, default 7); equipment type differences (flatbed vs van); weight/volume over vehicle capacity auto-proposes a split with the load plan
Merge — allowed	Same customer + ship-to + Incoterms + date window across orders when the customer consolidation rule (Book I B §2.1 Commercial tab) permits; hazmat compatibility checked; merged deliveries keep per-order packing list sections
Determinism	The Delivery Creation run (nightly + on-demand from the cockpit) applies these rules identically every time; every split/merge decision stores the rule ID that fired, so an unexpected grouping is explainable (E-DLV-01 messages name the rule)

2.4 Delivery Cockpit and Carrier/POD Closure

Element	Specification
Delivery Cockpit (OM-DLV-001, /sales/deliveries)	Work queue of deliveries by status (Planned → Released → Picking → Packed → Quality Cleared → Loaded → PGI → Delivered/Closed) with readiness flags per Flow A stage 7 (supply, quality, credit, documents) on one object — the single answer to “can this ship today?”
Load planning fields	Vehicle/equipment type, permit checklist for over-dimensional loads (existing Book I B §5.1), loading sequence per HU, dock door and appointment slot (plant calendar); loading scan SP-18 confirms sequence
Carrier and tracking (Phase 1 manual per DEC-04)	Carrier, PRO/BOL #, planned transit days; tracking status manual until IF-21; expected-POD date = ship date + transit
POD / Claim object (OM-POD)	Per delivery: POD status Expected → Received / Exception → Resolved/Closed; receiver name, date, condition notes, signed document upload. Exceptions open a Claim record: type (shortage, damage, refusal), quantity/value, carrier claim reference, resolution (credit via CMR, redelivery, repair) — claims reuse RMA/credit paths, never invent parallel ones. Missing POD after expected date + N days alerts (W.2 rule)
Final milestone	POD receipt (or site acceptance for ETO) fires the Delivery/Commissioning milestone (Book I B §4.4) exactly once, idempotent on delivery ID

2.5 PGI Reversal and Delivery Correction (controlled)

Case	Rule
Before invoice posted	Shipping Manager may reverse PGI within the same fiscal period: atomic mirror of postings #25/#26/#27, serial statuses restored (Shipped → Reserved, customer/site cleared with history retained), HUs → Loaded, delivery → Packed, invoice request withdrawn. Reason mandatory; reversal references the original PGI document
After invoice posted	No direct PGI reversal. Path is credit/rebill (ACC 4) or RMA (FL-05) depending on whether goods physically returned; the delivery record links the correction chain
Partial PGI failure	Structurally impossible (single transaction, Book III Part B §2.3); a failed PGI leaves the delivery at Loaded with the error surfaced — never a half-posted state
Genealogy	Reversal writes reversal events per the Traceability Contract (§4.3); shipped events are never deleted
Tests	TC-O2C-061…070 added: reversal before/after invoice, same-period guard, serial restoration audit, double-reversal block, claim-driven credit

2.6 Flow A Design Deliverables and Gate

- BPMN swimlane set (Sales, Pricing, Credit/Finance, Planning, Engineering, Production, Warehouse, Quality, Shipping, Carrier, Accounting) and sequence diagrams for confirmation, promise, delivery creation, PGI, POD and billing — repository: the design vault, IDs BPMN-FLA-01…; acceptance: every diagram element carries the REQ/IF/TC references it implements.
- UX pack for order workbench, promise dialog, allocation/backorder, delivery cockpit, pick/pack/load, PGI confirmation, POD/claim — Figma library per §9 standards, approved states per the screen template.
- Test pack TC-FLA-001…080 covering the 16 stages incl. every mandatory exception class (duplicate customer PO, blocked customer, tax outage, soft-reservation expiry, split/merge determinism, wrong serial at pick, repack, failed PSI, carrier rejection, double PGI, POD exception, close reconciliation).

3. Flow B — Design-to-Source: Engineering to Purchasing (FL-10)

FL-10 is the dedicated design to source flow. It governs how an engineering need becomes a revision-correct, compliance-checked purchase, and how engineering change hits open purchase documents. It composes existing objects (ER, EBOM, MCR, AVL, IR, RFQ, PO, GRN, ECO) and adds three new controls: release levels, the long-lead purchase release, and the immutable supplier technical package.

3.1 Release Levels (new control on the design package)

Release Level	Rule
Concept	Design exploration; nothing purchasable or buildable; placeholders may exist (MCR running)
Preliminary / RFQ	Package may go to suppliers for quotation only, watermarked NOT FOR MANUFACTURE; RFQs allowed, POs blocked (E-DTS-01)
Long-Lead Purchase Release (LLR)	Named long-lead items purchasable under §3.2 controls before full design release; production use still blocked
Approved for Purchase	Commercial/technical spec final for buying; POs unrestricted per normal rules; production release may still be pending

Release Level	Rule
Released for Production	Existing meaning (Book I C §3.4/§6): MBOM/routing/drawings/plans effective for WOs
Superseded / Obsolete	Existing meanings; open-document and stock disposition per ECO/supersession engines

The release level lives on the Engineering Request deliverables package and stamps every document and EBOM revision it covers; RFQ and PO line validations read it. Levels only move forward except via ECO.

3.2 Long-Lead Purchase Release (LLR object)

Field / Rule	Specification
LLR-{YYYY}-{SEQ:4}	Items (long-lead flagged: core steel, copper, bushings, tap changers per estimate), quantity and value caps, revision basis (exact package version §3.3), risk class (Low: catalog component / Medium: parametric spec stable / High: geometry still moving)
Approvals	Engineering Manager + Purchasing Manager; Controller joins above $100k exposure or risk class High; approval records the accepted cancellation exposure
PO behavior	POs created under an LLR carry the flag Long-Lead — Preliminary Revision and the package version; vendor terms must include the negotiated change/cancellation clause reference
Exposure tracking	Running exposure = open LLR PO value + cancellation fees; visible on the ETO order cost panel (feeds committed cost, §7.6) and on the LLR record; breaching the cap blocks further POs (E-DTS-02)
Change handling	Any ECO touching an LLR item forces the ECO open-document disposition (Book I C §4.2) plus a commercial decision per PO: proceed / change order to vendor / stop with exposure booking; unresolved LLR impacts block ECO release
Closure	LLR closes when final release confirms the purchased revision (no-change path) or every affected PO has a disposition (change path); the ETO order cannot invoice its Materials milestone with an open LLR exposure decision

3.3 Supplier Technical Package (STP object)

Element	Specification
STP-{SEQ:6}	Immutable bundle: drawing IDs + revisions (released renditions only), specifications, characteristics/tolerances extract, quality clauses (Q1–Q5), certificate requirements, origin/compliance requirements, acceptance criteria; SHA-256 checksum over the manifest
Versioning	Any content change = new STP version; old versions never mutate. RFQ and PO lines reference exactly one STP version; a superseded STP cannot attach to new documents (E-DTS-03)
Vendor acknowledgement	Confirmation capture (IF-02) extends with package acknowledgement: vendor confirms qty/date/price AND the STP version; state Acknowledged / Exception (deviation raised) / Rejected. Receipt against a PO whose vendor never acknowledged a revised STP hard-blocks (E-DTS-04)
Deviation requests	Vendor-proposed alternates/deviations create a Supplier Deviation record routed to Engineering + Quality (workflow W.4 addition); approval scope: affected lots/orders, expiry; approved deviations link into genealogy so affected serials are queryable (Traceability rule §4.3)

Element	Specification
Receipt validation	GRN validates the batch/certificates against the STP requirements (extends E-QM-06); wrong-revision documentation at receipt blocks acceptance with the STP diff shown

3.4 FL-10 Stage Contract

Stage	Controlling Specification	Evidence
1 Trigger (ETO order, ECO, obsolescence, NCR, make/buy review)	ER auto-create Book I C §2; ECR sources §4.1	Trigger type, owner, due date, linked SO/WO/ECR
2 Preliminary design + placeholders	Book I C §3; Book II §4.3	Preliminary release level; placeholder MCRs open
3 Material creation (MCR)	Book II §4.2	Activation blocked until mandatory views complete
4 Make/buy + source strategy	Book II §3.6 procurement type; alternates Book I C §3.2	Approved decision, effective date, owner
5 Long-lead release decision	§3.2 LLR	Caps, risk class, exposure acceptance
6 Purchase specification package	§3.3 STP	Immutable package ID + checksum on RFQ/PO line
7 Source determination	Book II §7.5 order; AVL qualification Book I G §2.2	Eligible source + reason; EX-09 exception on none
8 RFQ / technical bid	Book I G §4 + sealed/technical-commercial split scoring (technical pass/fail by Engineering before commercial opens; weighted award grid)	Bid versions, deviation list, award approval, AVL/IR write-back
9 PR + budget approval	Book I G §3; ACC 21/22	Approval + budget evidence
10 PO creation/release	Book I G §5 + STP version + LLR flag where applicable	Released PO revision + transmission record
11 Supplier acknowledgement	§3.3 acknowledgement states	Acknowledged/Exception/Rejected per line
12 Engineering change impact	Book I C §4.2 dispositions + §3.2 LLR commercial decisions + PO revision mechanics Book I G §5.3	Per-line disposition before effectivity
13 ASN/receipt/provenance	GRN + batch provenance Book II §15.1; ASN Phase 2 per DEC-05	Traceability chain starts (T01/T02)
14 Incoming quality	Plans Book I A §6; certificates vs STP §3.3	Accept/conditional/reject/RTV/NCR/SCAR
15 Release to production	Stock status rules; genealogy T03/T04	Unrestricted stock with lineage
16 Closure	PO matched/closed (FL-03), ECO impacts resolved, no open deviation, LLR closed	Reconciliation clean; test evidence filed

Test pack TC-DTS-001…040 must cover, at minimum, long-lead purchase followed by engineering change with supplier work started, vendor alternate proposal with preserved traceability, wrong-revision receipt hard-block, and no-valid-source exception routing.

4. Flow C — End-to-End Traceability Contract (FL-11)

FL-11 turns the scan points (Book I Part F) and genealogy model into a physical contract: a canonical event dictionary, the mandatory schema, immutable correction rules, completeness controls and the required outputs. The genealogy tables of Book I Part F §5 remain the storage design; this section is their binding data contract.

4.1 Canonical Event Dictionary

Event	Captured At (source)	Mandatory Identity and Evidence
T01 Supplier source	PO release (FL-10 stage 10)	Vendor, vendor site, PO line, manufacturer/MPN, STP version
T02 Receipt identity	GRN (SP-01…03)	GRN, internal batch/serial, vendor batch/heat, qty, CoO, dates, certificate checksums
T03 Incoming quality	Inspection close (INV 6.1)	Plan + revision, sample, results, instrument (SP-14), inspector, disposition, NCR/SCAR refs
T04 Stock movement	Every movement (SP-04/06/15/21)	From/to bin, stock status, ownership bucket, batch/serial/HU, qty, user/device
T05 Reservation/issue	Issue/backflush (SP-08/09)	SO/WO/op, material, batch/serial, qty, source bin, operator, event ID
T06 Parent-child install/remove	SP-12 (+ rework removals)	Parent serial, child serial, op, installed/removed, reason
T07 Operation confirmation	SP-10/11	WO/op, MBOM+routing revisions, good/scrap/rework, labor, tool/machine ID, capture-template data
T08 Test/inspection	Final test / IPQC (Book I E §4)	Serial, plan revision, run #, results, instrument calibration state, tester, attachments
T09 Finished receipt	FG receipt (Book I D §8.1)	Serial, WO, standard cost version, location, quality status
T10 Pack/HU	SP-16/17	HU/SSCC, contents, weights, label IDs, packaging item, seals
T11 Ship/deliver	SP-18 + PGI + POD (§2.4)	Shipment, BOL, carrier, customer, site, PGI doc, POD, condition
T12 Return/service	SP-19 + RMA/RWO events	RMA, condition, repair/removal/install events, replacement serial, disposition
T13 Recall/containment	Campaign actions (FL-08)	Campaign, population version, holds, notifications, recovery, exceptions, effectiveness

4.2 Mandatory Event Schema

- Identity: event_id (UUID), event_type + schema_version, business object key, entity, plant, source transaction + line.
- Causality: correlation_id, causation_id, prior/reversal event reference, user/device/integration principal, originating requirement (REQ id) where rule-driven.
- Time: occurred_at UTC, recorded_at UTC, original offline timestamp when replayed (Book I F §2), monotonic sequence within the object.
- Product: material, revision/effectivity actually used (not current — rule §4.3), batch/serial, parent/child, qty + UoM.
- Location/ownership: warehouse/bin/status, owner type/ref, peg type/ref, supplier/customer/site.

- Process references: PO/GRN/SO/delivery/WO/op/inspection/HU/shipment/RMA/campaign as applicable per event type (dictionary defines the required set per type).
- Evidence: certificate/document/test/instrument links with checksums, result/disposition.
- Integrity: append-only insert, row hash chained per object (tamper evidence), correction status (Active / Reversed / Replacement), retention class.

4.3 Correction and Reconciliation Rules (binding)

Rule	Required Behavior
No destructive edit	Genealogy rows are never updated or deleted. Corrections = reversal event + replacement event, both referencing the original (correction UI guides the pair). Database privileges enforce this, not convention
Transaction/event parity	Every traceability-relevant posting produces exactly one accepted event; every event resolves to a valid source transaction or an approved manual-exception record. Daily parity job counts both sides per type
Serial uniqueness	A serial has at most one active location, HU, parent and ownership state; violations are structurally prevented by the active-link constraint (serial_link model, Book I F §5) and monitored
Batch quantity balance	Received + produced + positive adjustments = on-hand + consumed + scrapped + shipped + returned, within UoM rounding tolerance; imbalance rows go to the completeness dashboard with owner
Revision proof	Consumption and shipment events store the engineering/document revision actually effective at execution (from the WO frozen revisions), never resolved-at-query-time
Certificate lineage	Certificate files and checksums are immutable; a replacement certificate links the original with reason; STP references (§3.3) keep supplier package lineage
Recall completeness	Campaign population is versioned; closure requires 100% accounted (recovered / field-corrected / documented exception) per FL-08 field 12
Daily controls	Automated: orphan events, missing events, broken parent-child links, invalid status sequences, source/event count mismatches — reported with assigned owners on the Traceability Completeness dashboard (coverage %, orphan counts, delayed offline events, unresolved corrections)

4.4 Required Outputs and Tests

- Outputs (extending INV 10.4): backward trace, forward trace, as-built record (incl. rework and removed components), recall population report, certificate dossier (customer/audit package: CoC, FTR, material certs, drawing/as-built references, shipment evidence — one-click per serial), completeness dashboard. Performance: full tree < 10 s (existing NFR); archive queries < 60 s (Book III D §6 retention tiers).
- Test pack TC-TRC-001…040 must cover, at minimum: multi-WO forward trace from one heat, backward trace through rework with removed/replaced children, offline scan replay with conflict and no duplicates, wrong-batch correction with explainable history, mock recall end-to-end, archive-and-retrieve at performance target.

5. Executable Pricing Condition Model

The pricing waterfall (Book I Part A §4.4) remains the business rule. This section converts it into condition technique structure so the Pricing Service is configurable data, not narrative.

5.1 Condition Types and Calculation Schema

Step	Condition Type	Source (access sequence)	Behavior
10	PB00 Base Price	1) Manual override (authority-checked) 2) Customer-specific price 3) Price list rate with qty breaks 4) Item default (fixed/markup)	First hit wins; Price Source badge stores which access supplied it; below-floor detection runs after step 40
20	DIS1 Discount %	Document line entry within role limit; contract discount matrix (agreements §6.2 Book I I)	Applies to PB00; stacked discounts prohibited (one DIS1 per line)
30	SURC Commodity Surcharge	Surcharge rule per item/family (Book I I §6.4): index basis, content, pass-through	Separate element, printed separately; re-fires only at defined repricing events
40	FRT1 Freight	Per Incoterms policy: included (no element) / estimated pass-through (accrual basis) / separately billed line	Never buried in PB00; margin calculations exclude pass-through freight
50	TAX	External engine per Book I A §11 precedence	Committed at invoice; estimates at quote/SO save

5.2 Binding Calculation Rules

- Persisted price elements: every document line stores its condition results as rows (type, source row ID, basis, rate, value, currency) at determination time — documents never re-derive prices from masters afterwards. Repricing replaces the element set explicitly with a diff dialog (existing rule, now at element granularity).
- Currency: conversion at document level only, using the daily rate of the pricing date; price masters never auto-convert; the booked rate is stored on the document (existing ACC 3 rule, restated as the condition-model rule).
- Rounding: each condition value rounds half-up to currency minor unit at line level; the document total is the sum of rounded lines (matches the tax engine, Book I A §11.2); price unit arithmetic (per 100/1000) resolves before rounding.
- UoM: condition rates apply per Sales UoM; conversion to base for margin uses the rational factors (Book II §2.5); a missing conversion path blocks determination (existing V-SAL rule).
- Validity and overlap: all condition records are effective-dated, non-overlapping per key (V-SAL-02/03 generalized to every condition table); retroactive master changes never restate documents — corrections go through credit/rebill (the Book I A §11.5 doctrine applies to prices as well as tax).
- Explainability: the simulation endpoint (formalizing the Price Simulation button) is /api/v1/pricing/simulate — input customer, item, qty, date; output the full element set with every access tried, the winning source rows and the floor check; same service the documents call, guaranteeing simulation = reality.

6. Production Versions and Planning Algorithm Baseline

6.1 Production Version (new master object)

Field	Type	Req	Specification
PV #	Auto	Auto	PV-{SEQ:5} per material + plant
Material / Plant	Lookups	Yes	Make items only

Field	Type	Req	Specification
MBOM Revision / Routing Revision	Lookups	Yes	Both must be Released; pairing is the point — no more independent selection
Inspection Plan Set	Auto view	—	INP/FIN plans resolving for this pairing (preview per Book I A §6.2)
Lot Size Range	Numbers	No	From/to qty this version applies to (e.g. prototype vs series routing)
Valid From / To	Dates	Yes/No	Effective dating; exactly one valid PV per (material, plant, lot-size band, date) — V-PV-01 blocks overlaps
Status	Badge	Auto	Draft → Released (Engineering Manager) → Superseded/Obsolete; ECO cutover swaps PVs atomically

Selection rule: WO creation and MRP explosion select the valid production version — never a bare MBOM or routing. Existing single-pair materials migrate to auto-generated PV-1 records, so Book I Part D behavior is unchanged where only one version exists. Plant-2 activation (Book III Part F §5) rides this object as designed.

6.2 MRP Algorithm Baseline and Golden Data

- The deterministic order of the run is binding: (1) low-level-code assignment; (2) per material by LLC ascending: gross requirements aggregation by bucket (daily), (3) netting: net(t) = max(0, gross(t) + safety stock − projected available(t)), projected available rolls supply at confirmed dates; (4) lot sizing per policy with rounding; (5) scrap application (component + assembly, Book I C §3.2 / Book II §3.6); (6) lead-time offsetting (routing-based or field-based per Book II §3.6, calendars respected); (7) source assignment per Book II §7.5; (8) exception generation EX-01…EX-09; (9) firmed proposals untouched; planning fences respected (EX-08).
- Worked example: the specification repository carries the golden dataset GD-MRP-01: a 3-level BOM (transformer, coil sub-assembly, copper), seeded demand and supply, expected planned orders and exceptions per run — the regression anchor for every engine change. Property test: rerunning MRP on unchanged inputs yields byte-identical proposals (determinism NFR, mirrors Book III Part E §3).
- Pegging: full multi-level pegging is stored per run (requirement → supply chain), powering the Backorder Workbench (§2.2), shortage explanations and promise revalidation. De-pegging follows §2.2 approval rules.
- Plan snapshots: each published schedule/MRP run stores its input snapshot ID (existing Part E rule) so every confirmed date names the plan that produced it.

7. Module Completion Additions

The remaining module completion requirements are closed here. Requirements that demand artifacts (diagrams, prototypes, schema files, test evidence) are governed by the §9 gates; requirements already specified in Books I–III are cross referenced in the module index and not repeated.

7.1 Material Duplicate Merge (completes Book II §4.4)

- Matching: trigram similarity on normalized descriptions + exact MPN/manufacturer matches, threshold 0.9 review / 0.97 auto-flag, scoped within material group L1 (existing); the steward review queue shows side-by-side with usage counts.

- Merge = supersession specialization: survivor chosen by steward; loser becomes Restricted with supersession record (Book II §4.6); open references rewrite where safe (BOM lines via ECO task, price rows re-keyed with approval, info records merged keeping best basis with history), stock transfers to survivor batch-by-batch with genealogy continuity events; unresolvable references block the merge with a worklist. Merges are irreversible by design — the audit chain is the undo story.

7.2 Warehouse Task Model and Labor Time Boundary

- Warehouse tasks (receiving, putaway, replenishment, pick, stage, pack, load, transfer) become explicit task records with assignment, device, status (Queued → Assigned → In Progress → Done/Exception) layered on the existing documents (picklists, putaway proposals) — the mobile work queue reads tasks, documents stay the system of record. No behavior change to INV postings.
- Labor time boundary: shop-floor labor capture (Book I D §6) is for costing and genealogy only. It is not a payroll system: no pay rules, no attendance. The time-system boundary exports approved labor confirmations (operator, WO/op, hours) to the payroll/time system as IF-22 (reserved, Phase 2); supervisor corrections follow the confirmation-correction rule (mirror postings) with dual sign-off.

7.3 Quality Deviation / Waiver Control (new object, go-live per DEC-09)

Field / Rule	Specification
DEV-{YYYY}-{SEQ:4}	Scope: material / batch / WO / serial set / customer; deviation description; specification reference (characteristic + released value vs accepted value)
Type	Use-As-Is (waiver) / Temporary Process Deviation / Customer-Approved Concession
Approvals	Quality Manager + Engineering Manager; customer approval evidence mandatory when the customer flag or contract requires (attachment + reference); Controller when value impact > threshold
Validity	Expiry date or quantity cap, whichever first; auto-expiry job blocks further use and alerts (W.2/W.3)
Traceability	Affected serials/batches link to the deviation (genealogy evidence rows); the as-built record and certificate dossier disclose active deviations (FL-11 §4.4)
Interaction	A deviation never edits the inspection plan; results still record actuals vs released spec with the deviation reference explaining acceptance — audit-proof by construction

7.4 Installed Base and Warranty Entitlement (go-live per DEC-06)

Element	Specification
Installed Base record (IB)	One per shipped FIN serial: customer, site (from shipment), POD date, commissioning date (optional capture from field evidence), ownership transfers (resale/assignment with document), status (Shipped / Installed / In Service / Returned / Scrapped), service history links (RMA, RWO, deviations, campaigns)
Warranty entitlement rules	Start = ship date by default; customer contract may elect POD or commissioning date capped at ship + N months (config); duration from item + SO override (existing fields); extensions as dated records with approval; exclusions catalog (site conditions, installation error categories) referenced by adjudication

Element	Specification
Entitlement check service	/api/v1/warranty/entitlement (serial, claim date) → covered / not covered / conditional with rule trail; the RMA warranty toggle default (Book I A §3.4) now calls this service instead of the bare serial window — same outcome for the base case, correct outcome for extensions and transfers
Migration	Shipped serial history converts to IB records with ship-date warranty; unknown sites flagged for CS enrichment (data-quality queue)

7.5 Reporting and Analytics Governance

- KPI dictionary: one controlled row per KPI — formula, grain, filters, owner, target, source lineage, freshness class. Seed catalog: OTD %, promise reliability (Part E §5), FPY by gate, scrap value %, PPV/IPV, inventory turns, WIP aging, quote win rate, backlog (BBB), three-view order margin, DSO (ACC), close cycle days, traceability coverage %, data completeness (Book II §16.4), SLO attainment. A number that appears on two dashboards from two formulas is a defect.
- Semantic model: facts (orders, deliveries, invoices, stock movements, confirmations, inspections, genealogy events, journal lines) and conformed dimensions (material, customer, vendor, plant/entity, work center, cost/profit center, calendar) defined once in the reporting layer (replica per ADR-05); operational reports that act as subledgers (WIP, GR/IR, AR/AP aging, stock ledger) stay in the transactional system and reconcile to GL per the close checklist — analytics never becomes the system of record.
- Freshness classes: real-time (operational queues), 15-minute (dashboards), daily (analytics); late-arriving offline scans backfill with recorded_at lineage so daily numbers are restated visibly, not silently.
- Security: the same role/entity/plant scopes apply to dashboards, exports and APIs (single policy engine, Book III Part C §3); exports of customer-identifying traceability data log as disclosures.
- Acceptance: every shipped report carries a golden dataset with expected totals and drill paths (TC-RPT range).

7.6 ETO Cost Control Additions (completes Book I Part I for ETO)

- Per ETO order cost panel: estimate (CE basis) vs committed (open POs + LLR exposure §3.2 + subcontract commitments) vs actual-to-date (WIP charges) vs estimate-at-completion (actual + remaining standard) vs margin-at-completion against the quoted price; change orders re-baseline with history.
- EAC formula is deterministic: EAC = actuals + (remaining operations at standard rates) + (unissued material at current basis) + open commitments not yet consumed; variances against CE feed the quote-calibration review (Part E §5 promise calibration sibling).
- This is order-level project cost control; full project/WBS accounting stays out of scope per DEC-06/DEC-10 boundaries until a Phase 3 decision.

7.7 Workflow Engine Schema and Runtime (completes Section W)

- Definition schema (stored, versioned): { workflow_id, version, document_type, trigger, conditions[] (field, operator, value — the rule grammar), steps[] (order, type sequential/parallel, approver resolution role/user/derived, sla_hours, escalation_target), terminal_actions[], effective_from/to, owner, approval_ref }. The W.4 catalog rows are instances of this schema; an admin UI edits definitions under change control.

- Runtime guarantees: idempotent action handling (double-click and email-link replay safe), duplicate-step suppression, timeout → escalation per step, recall/resubmit preserving history, compensation on terminal-action failure (the document returns to pre-approval state with the failure surfaced), full audit per action.
- Notification templates: versioned templates per event with localization slots (English at go-live per Book III Part F §6), sensitive-data rules (no prices in shop-floor notifications; no customer PII in vendor-facing mails), signed action links expiring per policy, per-user digest preferences for non-critical severities.
- Job contracts: every W.3 job declares inputs, dependency jobs, calendar, close-criticality, retry policy, idempotency key, output artifact and reconciliation query — rows in the job catalog, monitored per Book III Part D §5.
- Workflow analytics: SLA attainment, aging by step, bottleneck approvers, rejection rates, delegation usage, override frequency — monthly review by each workflow owner (feeds the W.4 governance loop).

8. Architecture Decision Records (DEC-10 execution)

ADR	Decision	Rationale / Consequence
ADR-01 Application shape	Modular monolith with enforced bounded contexts: Sales, Engineering, Materials, Planning, Production, Quality, Traceability, Procurement, Inventory, Controlling & Costing, Posting, Workflow, Integration, Identity	One deployable at go-live scale (NFR volumes are modest for modern hardware); context boundaries enforced by module visibility rules and the interface catalog so later extraction of a hot context (e.g. Traceability) is a deployment change, not a rewrite
ADR-02 Database	Single PostgreSQL cluster; schema-per-context ownership; foreign keys within a context, events across contexts	Referential integrity where it is cheap, autonomy where it matters; the Book II/III physical rules (effectivity, append-only audit, partitioning for genealogy and scan telemetry) apply per schema
ADR-03 Eventing	Log-based broker (Kafka-class) + transactional outbox in every context	Book III Part B §2.2/2.3 standards implemented once; DLQ/replay via the integration console
ADR-04 Search	Dedicated search index (OpenSearch-class) for material, document and customer/vendor lookup incl. trigram duplicate checks	Sub-500 ms lookups at 50k+ materials without abusing the OLTP database
ADR-05 Reporting	Read replica + semantic layer for analytics; operational subledger reports stay transactional	Freshness classes per §7.5; close-relevant reports never depend on the analytics pipeline
ADR-06 Workflow engine	In-platform workflow service implementing the §7.7 schema (no external BPM product at go-live)	The 27-row catalog is configuration; an external BPM adds integration surface without capability PE needs yet
ADR-07 Scheduling engine	Finite scheduler as an internal service consuming the constraint model (Book III Part E); heuristic seeded-deterministic at go-live; solver pluggable Phase 3	Matches DEC-01 boundary; determinism NFR preserved
ADR-08 CAD connector	Server-side vault-event connector per Book III Part G (SolidWorks PDM baseline, tool-neutral contract)	No designer-workstation agents; outage behavior per IF-17

ADR	Decision	Rationale / Consequence
ADR-09 Client	Single responsive web application; kiosk mode for terminals (badge login, focus-lock scanning per Book I Part F §2); Android rugged devices run the same app	One codebase, one design system (§9 UX standards); offline queue per existing spec
ADR-10 Printing/devices	Server-rendered ZPL print service with printer registry (existing Book I F §2 design elevated to ADR); scanner input via WebSocket wedge	Deterministic labels regardless of client; reprint governance preserved

9. Design and Development Gates and Authoring Standards

9.1 Definition of Ready — Design (DG-1, per module/flow)

Gate Item	Pass Condition
Scope approved	Go-live/later/excluded boundary named per DEC register; business owner assigned
Process defined	Actors, trigger, happy path, exceptions, reversals and closure agreed (flow template §9.4)
Objects identified	Master/transaction objects, ownership, relationships known (Books I–III + this Book)
Decisions resolved	No open DEC/C-register item blocks the flow or data model
References valid	Controlling REQ IDs and cross-Book precedence confirmed (A-08 validation for INV/ACC references)

9.2 Definition of Ready — Development (DG-2, per story)

Gate Item	Pass Condition
Approved UX	Wireframe/prototype with all states (normal, empty, loading, validation, authorization, conflict, offline, degraded, read-only) approved by business owner + UX lead in the design vault
Approved state/data design	State machine, physical fields, constraints and migration impact approved by the data architect
Approved contracts	OpenAPI/AsyncAPI files + examples + security scopes + retry/reconciliation published in the contract repository with passing contract tests (Book III Part B §4)
Postings and controls	Inventory/financial effects, permissions, SoD, audit and reversal specified (matrix rows exist)
Acceptance criteria	Given/When/Then criteria + test cases + data linked in the ALM tool
Operational design	Logs, metrics, alerts, jobs, performance and failure recovery defined
Dependencies available	Upstream interfaces, configuration, environments, test doubles ready

9.3 Definition of Done — Development (DG-3, per story)

Code reviewed with static and security analysis clean; unit, property/calculation, contract, integration and UI tests passing; negative, duplicate, concurrency, timeout, cancellation and correction paths tested; schema migration and configuration transport tested; role/scope/SoD/audit tests passing; observability shipped with runbook links; reconciliation controls verified against seeded discrepancies; documentation updated; business demo/UAT evidence attached with defects within

gate policy. The go-live gate itself remains Book III Part H §7, now additionally requiring: the FL-09/FL-10/FL-11 packs passing, the conflict register at zero open rows, and the DEC register signed.

9.4 Authoring Standards and Reusable Templates

- Module authoring template: every future module document (and every revision of an existing Part) must follow this twenty section structure — purpose/exclusions, RACI, capabilities with REQ IDs, process flows, objects/master data, state machines, screen inventory, approved UX reference, field dictionary, rules/calculations with worked examples, approvals/automation, integration contracts, inventory/accounting effects, exceptions/recovery, security/controls, reports/KPIs, NFRs, migration/configuration, acceptance/tests, dependencies/decisions. Books I–III substantially conform; their gaps against this template are tracked as DG-1 items per module, not rewritten wholesale.
- Requirement format: atomic and testable — REQ-{DOMAIN}-{OBJECT}-{SEQ} with title/objective, actor+authorization, preconditions+trigger, main behavior, alternate/exception behavior, state/data/posting effects, integration effects, Given/When/Then acceptance, test+evidence links. The Book III Part H parse generates these rows; new content is authored in this format directly.
- Flow specification template: the FL table format used throughout (Book III Part A + this Book) is the standard — flow ID/objective, scope/variants, trigger/preconditions, actors/owners, objects/state changes, main path, exceptions, inventory/financial effects, interfaces, security/SoD, performance, closure/reconciliation, tests/evidence.
- Screen design template: screen ID/route/persona, task objective and success condition, layout+prototype link, fields/columns dictionary, actions with visibility/enabled rules and API calls, the nine mandatory states, navigation/drill-through, accessibility (keyboard, focus, contrast, scanner/audio feedback, touch targets), audit/security, Given/When/Then acceptance. Applies to every screen in the UX pack.
- API/event contract template: contract ID/owners, purpose/boundary/source-of-truth, protocol/version/compatibility, machine-readable schema+examples+lineage, authentication/authorization scopes, idempotency/ordering, errors/retry/replay/compensator, transaction behavior, reconciliation, observability/SLO, contract tests. Every IF catalog row publishes against this template in the contract repository.

10. Prioritized Action Register

The action register below is the program work plan. Items marked Specified are closed inside this document as specification content. Items marked Artifact must be produced in the named repository and accepted at the gates above.

ID	Pri	Action	Owner	Status / Disposition
A-01	P0	Master baseline, precedence and conflict register; scheduling contradiction resolved	Product Owner / PMO	Specified — §1 of this Book; steering signature pending
A-02	P0	Dedicated Sales-to-Delivery, Engineering-to-Purchasing and Traceability designs	Process Leads / Architect	Specified — §2/§3/§4 (FL-09/10/11); BPMN artifacts per DG-1
A-03	P0	UX design system and approved prototypes for all screens and scanner workflows	UX Lead	Artifact — design vault; standards §9.4; gate DG-2

ID	Pri	Action	Owner	Status / Disposition
A-04	P0	Bounded contexts, component/deployment diagrams and ADRs	Solution Architect	Specified — §8 ADR set; diagrams as artifacts per DG-1
A-05	P0	Physical ERDs, data dictionary and state models	Data Architect	Partially specified (Books II/III physical rules, state models throughout); full DDL artifact per DG-2
A-06	P0	OpenAPI/AsyncAPI/JSON schema contracts and test doubles	Integration Lead	Artifact — contract repository against IF catalog + §9.4 template
A-07	P0	Pricing, MRP/CTP/scheduling and costing algorithm specs with golden examples	Functional/Technical Leads	Specified — §5, §6, Book I Part I §4 worked example; golden datasets GD-* as artifacts
A-08	P0	Version-lock and cross-reference INV V3 and ACC V2	PMO / Module Owners	Register specified §1.1; validation sweep executes at DG-1
A-09	P0	Detailed test cases and RTM per requirement and flow	QA Lead	Framework specified (Book III Part H + TC ranges herein); cases as ALM artifacts
A-10	P0	Source-to-target migration workbooks and reconciliation scripts	Migration Lead	Approach specified (Book II §16.3, Book III Part H §5); workbooks as artifacts
A-11	P1	Module completion additions	Module Owners	Specified — §7 closes the specification gaps; remaining rows are artifacts/decisions mapped to DG gates
A-12	P1	Configuration workbook and seed data	Functional Leads	Artifact — seed catalogs named throughout (statuses, reasons, workflows W.4, accounts, calendars, labels) consolidate into the config workbook
A-13	P1	Security role/SoD/data-classification design and tests	Security Lead	Specified (Book III Part C); TC-SEC artifacts per DG-3
A-14	P1	Observability, reconciliation controls, job console and runbooks	DevOps / Support	Specified (Book III Parts B §4 / D §5); runbook artifacts pre-go-live
A-15	P2	Advanced target-state items (global ATP/CTP, portals/EDI, optimizer, actual costing, machine/OEE, field service, localization)	Program Steering	Phased per DEC register; activation packages pre-specified where architectural (Book III Part F, DEC rows)

Build Note: *Acceptance language. This document is the functional requirements baseline and the controlling design governance. It becomes the complete design and development package only when the named artifacts, the design packs of the three consolidated flows, the detailed test and migration evidence and the formal sign offs complete through the gates in §9.*

BOOK V

DESIGNER'S GUIDE & VISUAL ATLAS

SAP S/4HANA Capability Map & Glossary • System Structure • Master Data Atlas • Document Flow Atlas • Cross-Module Linkage Map
Module Design Briefs with Screen Inventories • Barcoding & Device Specification with Label Mockups • CAD Linkage Visuals • Design System Starter

Written for UX designers and builders: plain language, seventeen diagrams, every screen indexed to its field-level specification

Book V is written for the people who will design and build the screens: UX designers, Figma authors, and front-end developers. It uses plain language, pictures and short tables. It removes nothing from Books I–IV; every topic here points to the exact section where the full field-level rules live. Read this Book first, design from it, and open the referenced sections only when a rule detail is needed.

1. How to Use This Book

If you are…	Read in this order	Then design
A UX designer starting cold	§2 (what the system is, in SAP terms) → §3 (structure) → §5 (flows) → §7 (your module's brief) → §11 (design system)	The module's "prototype first" list, using the screen inventory and the states checklist
A designer doing the shop floor	§3 → §5.4 → §7.6 (Production) → §8 (barcoding and devices) → §11	Kiosk screens: dispatch, start/complete, material scan, QC gate, hold
A developer joining mid-build	§2 glossary → §4 (master data) → §5 (flows) → §6 (linkages) → Book IV §8 (architecture)	Against the field tables of Books I–III and the contracts of Books III–IV
A reviewer verifying capability coverage	§2 capability map	Nothing to design — the map states what exists, where, and in which phase

One rule keeps this simple: every screen in this system is specified twice. Once here, briefly, so you can find it and understand its job. Once in Books I–III, completely, with every field, validation and posting. The screen inventory tables in §7 carry both references.

2. The Enterprise Capability Map

The system must provide the enterprise capabilities catalogued below. Each row names the industry benchmark term so readers who know SAP S/4HANA can orient instantly, states the capability in plain words, cites where this document specifies it, and carries the delivery phase from the decision register (Book IV §1.2).

Capability (benchmark term)	What It Does (plain words)	Specified In	Phase
SD — Sales & Distribution	Quotes, sales orders, deliveries, billing triggers	Book I Part B; schedule lines and delivery rules Book IV §2	Go-live
SD — Billing / FI-AR	Invoices, credit notes, receivables, cash	ACC V2.0 Sections 3–5 (external volume)	Go-live
aATP — Advanced ATP	Promise dates checked against stock and capacity	ATP Book I B §4.3; CTP Book III Part E	Go-live (single plant)
MM — Purchasing	Requisitions, RFQs, POs, vendor management	Book I Part G; info records and source lists Book II §7	Go-live
MM — Inventory Management	Stock, movements, counts, valuation	INV V3.0 (external volume); gap closures Book I Part A	Go-live
MM — Material Master	One record per material, many views	Book II (entire book)	Go-live
EWM — Extended Warehouse	Task-directed warehouse work, waves, labor mgmt	Task model Book IV §7.2 (light version); full EWM-class not planned	Partial / excluded

Capability (benchmark term)	What It Does (plain words)	Specified In	Phase
PP — MRP	Nets demand against supply, proposes orders	Book I D §3; algorithm baseline Book IV §6.2	Go-live
PP-DS — Finite Scheduling	Schedules against real capacity constraints	Book III Part E; DEC-01	Go-live (heuristic; optimizer Phase 3)
PP-SFC — Shop Floor Control	Work orders, confirmations, WIP	Book I Part D	Go-live
QM — Quality Management	Inspection plans, results, NCR, CAPA	Book I Part E + Part A §6; deviation/waiver Book IV §7.3	Go-live; SPC/audit Phase 2 (DEC-09)
PM — Plant Maintenance	Maintain machines, tools, downtime	Constraint hooks Book III E §2; full PM module	Phase 3 (DEC-08 adjacent)
PS — Project System	Project/WBS cost control	ETO order cost control Book IV §7.6 (order-level)	Order-level go-live; WBS excluded
FI — GL / AP / AR / Assets	Books, payables, receivables, close	ACC V2.0 (external volume); close orchestration FL-06	Go-live
CO — Cost Centers / Profit Centers	Where money is spent; which business earns it	Book I Part H	Go-live
CO-PC — Product Costing	Standard costs, WIP, variances, settlement	Book I Part I; account determination Book II Part B	Go-live
ML — Material Ledger (actual costing)	Periodic actual unit costs	DEC-02: data captured now, ledger Phase 3	Phase 3
VC — Variant Configuration	150% BOMs, option rules, configurators	DEC-03: structured ETO attributes now; configurator Phase 3	Phase 3
PLM / ECM — Engineering Change	ECR/ECO, effectivity, BOM revisions	Book I Part C §4	Go-live
DMS — Document Management	Controlled drawings and documents	Book I Part C §5; CAD connector Book III Part G	Go-live; connector per IF-17
GTS — Global Trade	Customs, export control, screening	Foundations Book II §2.4/Book III F §6; full GTS	Phase 3
TM — Transportation Mgmt	Carrier rating, tendering, tracking	Manual at go-live; IF-21 Phase 3 (DEC-04)	Phase 3
CS — Customer Service / Field Service	Service orders, technicians, installed base	Installed base + warranty engine Book IV §7.4 now; field execution	Phase 3 (DEC-06)
Fiori — UX model	Role-based apps, worklists, one design language	Section §11 design system + workbench/queue patterns throughout	Go-live
BTP — Extensibility	Custom fields, APIs, side-by-side extensions	Book III T-06 heritage: custom fields/workflows/APIs Book III B, Book IV §7.7	Go-live core; marketplace n/a

Capability (benchmark term)	What It Does (plain words)	Specified In	Phase
Group Reporting / Consolidation	Multi-entity consolidation, eliminations	Feeds specified FL-07/Book III F §4; consolidation itself	ACC scope / Phase 2

2.1 Plain-Language Glossary (the 36 terms designers will meet)

Term	Meaning in one sentence
Material Master	The single record for anything the company buys, makes, stocks or sells; every module reads its own view of it
Material Type	The category (raw, semi-finished, finished, trading, packaging, spare, consumable, service) that decides how a material behaves everywhere
BOM / EBOM / MBOM	The parts list of a product; EBOM is how engineering designs it, MBOM is how the factory builds it
Routing	The ordered list of operations (with times and work centers) to build something
Work Center	A machine, station or crew where operations happen; carries capacity and hourly rates
Production Version	The approved pairing of one MBOM and one routing for a plant; orders always pick a version, never loose pieces
MRP	The nightly engine that compares demand with supply and proposes orders to make or buy
ATP / CTP	ATP answers “is there stock by that date?”; CTP answers “can we make it by that date?” checking materials AND capacity
Planned Order	MRP’s proposal to make something; a planner converts it into a real production order
Production Order (WO)	The document that authorizes building one item, collects its costs and carries its serial
Backflush	Components consumed automatically when an operation confirms, instead of scanning each issue
Kitting / Staging	Picking all components for an operation and delivering them to the work center before it starts
Confirmation	The operator’s report at an operation: good quantity, scrap, time; it posts labor and materials
WIP	Work in process: the money value sitting inside unfinished orders
Settlement	Closing an order’s books: comparing actual cost with standard and posting the differences (variances)
Standard Cost	The frozen planned cost of an item, rebuilt by a costing run; the yardstick everything is measured against
Moving Average	A cost that re-averages with every purchase receipt; used for bought items
Valuation Class	The tag on a material deciding which balance-sheet account its stock value sits in
PPV	Purchase price variance: what you paid minus what the standard said it should cost
GR/IR	A clearing account between goods received and invoice received; it must always net out
Three-Way Match	Invoice pays only when PO, goods receipt and invoice agree
PGI	Post Goods Issue: the moment shipped stock leaves the books and cost of goods sold posts

Term	Meaning in one sentence
Condition Technique	Pricing built from stacked elements (base price, discount, surcharge, freight, tax) resolved by precedence rules
Info Record	The commercial memory of one vendor supplying one material: price, lead time, order rules
Source List / AVL	Which vendors are allowed (AVL) and which one is fixed for a period (source list)
Handling Unit (HU)	A numbered physical package (crate, pallet, reel) with contents, weights and its own barcode
SSCC	The GS1 standard 18-digit license plate for a shipping unit; printed as GS1-128 barcode
Batch / Heat	A lot of material tracked as one identity; for steel and copper the mill heat number is the key
Serial	One physical unit tracked individually for life: build, test, customer, returns
Genealogy	The permanent family tree of a serial: every batch, child serial, operator, test and shipment it touched
ECR / ECO	Change request (asks and analyzes) and change order (executes) for engineering changes
Effectivity	When a change takes effect: by date, by serial number, or immediately
FEFO	First-expired, first-out picking for shelf-life materials
Safety Stock	A quantity floor MRP protects so demand spikes do not empty the shelf
Milestone Billing	Invoicing an ETO order in agreed slices (down payment, design approval, tested, shipped, delivered)
ASC 606	The revenue accounting standard deciding when invoiced money counts as earned revenue

3. System Structure

One picture of the whole system, then one picture of the company structure it runs on. Every module box below is a design brief in §7.

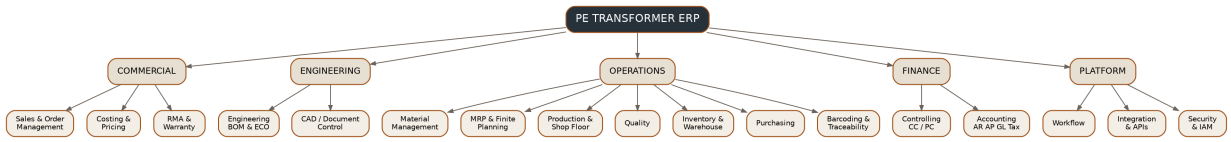


Figure V-1. Module map. Five domains, eighteen modules. Inventory and Accounting are released external volumes; everything else is specified in Books I–IV.

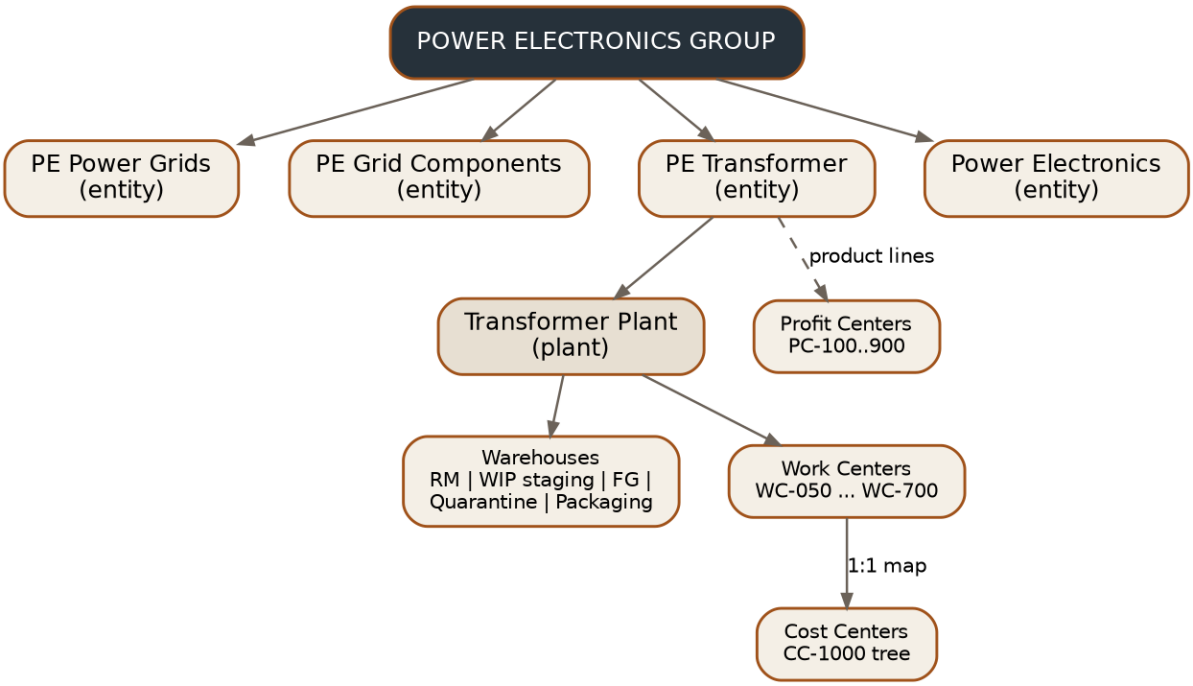


Figure V-2. Organization structure. Four legal entities; the Transformer plant carries warehouses, work centers and cost centers; profit centers segment the P&L by product line.

3.1 Application Navigation (what the sidebar contains)

Top Level	Children (each is a screen or queue)
Home / Dashboards	Executive • Inventory • Finance • Production • Quality • Sales
Sales	Leads & Opportunities (new, Book VI §4.1) • Customers • Quotations • Sales Orders • Backorders (new, Book IV §2.2) • Deliveries & Shipments • Delivery Cockpit (new) • Packing • RMA / Returns • Spares Finder (new, Book VI §5.1) • Sales Reports
Engineering	Engineering Items • Engineering BOMs • CAD Documents • Change Requests (ECR) • Change Orders (ECO) • Production Versions (new, Book IV §6.1) • Where-Used • Engineering Reports
Materials	Material Master • Creation Requests (MCR) • Change Requests (MCH) • Supersessions • Info Records • Source Lists • Account Determination • E&O Reserve • Data Quality
Production	Production Dashboard • MRP Workbench • Planned Orders • Production Orders • Dispatch List • Shop Floor Terminal • Work Centers • Routings • Rework Orders • WIP Report • Production Reports
Quality	Quality Dashboard • Inspection Plans • Incoming / In-Process / Final / Pre-Shipment • Stock Hold & Release • NCR • CAPA • SCAR • Deviations & Waivers (new) • Calibration • Recall Campaigns • CoC
Inventory	Per INV V3.0 navigation, unchanged
Purchasing	Vendors • Approved Vendor Lists • Purchase Requisitions • RFQs • Purchase Orders • Long-Lead Releases (new, Book IV §3.2) • Expedite Board • Vendor Performance
Costing & Pricing	Cost Estimates • Costing Runs • Work Center Rates • Price Lists • Customer Price Agreements • Commodity Surcharges • Margin Analysis
Controlling	Cost Centers • Profit Centers • Hierarchies • Allocation Cycles • Reports

Top Level	Children (each is a screen or queue)
Accounting	Per ACC V2.0, unchanged
Configuration	Number Series • Barcode & Label Settings • Scan Point Policies • Workflows • Tax • Periods • RMA Policies • Sales / Production / Engineering / Quality Settings

4. Master Data Atlas

Master data is the still water under everything that moves. The material is the hub; the pictures below show what hangs off it and how a real product structure looks as a tree.

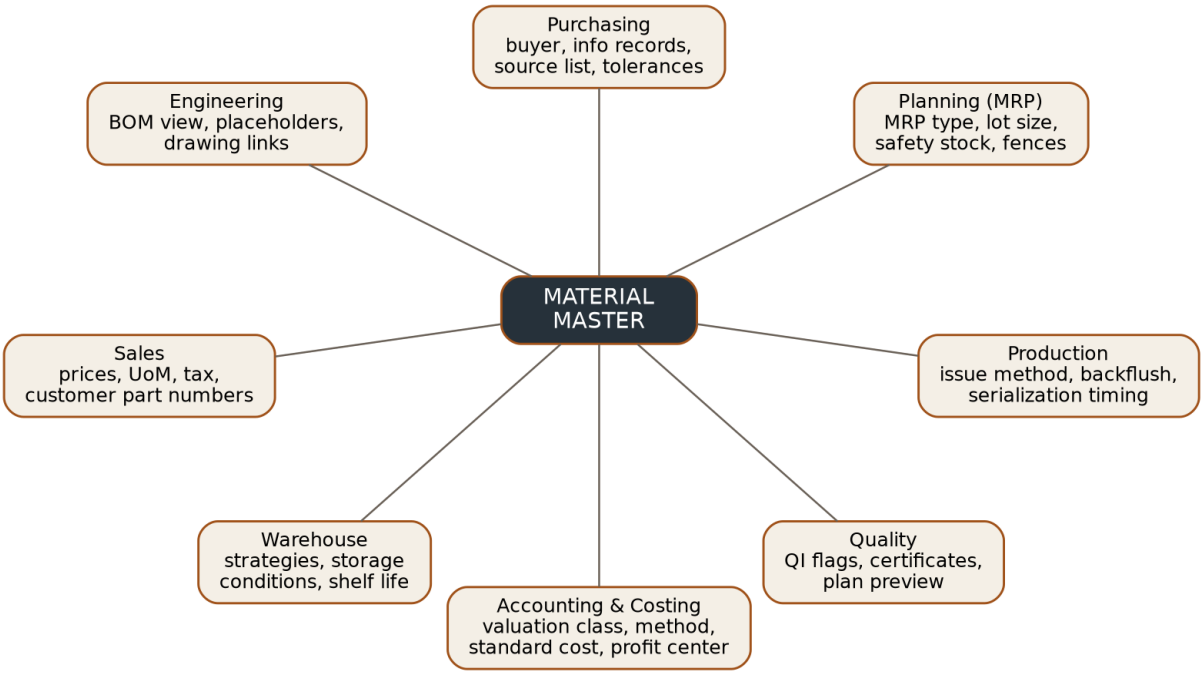


Figure V-3. The Material Master hub. Each spoke is a view owned by one role; the full field lists are Book II §3.

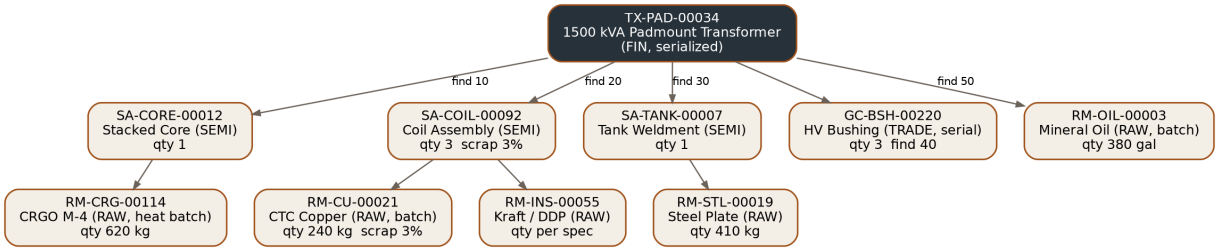


Figure V-4. A real BOM tree. A padmount transformer explodes into semi-finished assemblies and raw materials; find numbers position parts on drawings; scrap percentages feed MRP and costing.

4.1 Master Data Object Catalog

Object	What it is	Who owns it	Where specified
Material	The hub record with 9 types and per-view ownership	Master Data Steward + view owners	Book II
Customer	Commercial + credit + tax + addresses	Sales Admin (Finance owns credit)	Book I B §2

Object	What it is	Who owns it	Where specified
Vendor + AVL	Vendor record; AVL says who may supply what	Buyer + Quality	Book I G §2
Info Record	Vendor-material price and lead-time memory	Buyer	Book II §7.2
Source List	Period rule fixing or blocking sources	Purchasing Manager	Book II §7.4
EBOM / MBOM	Design and manufacturing structures	Engineering	Book I C
Routing	Operations, times, QC gates	Manufacturing Engineering	Book I D §4
Production Version	Approved MBOM + routing pairing	Engineering Manager	Book IV §6.1
Work Center	Capacity + rates + cost center link	Production + Controlling	Book I D §2
Inspection Plan	What to test, how, sampling	Quality Engineer	Book I A §6
Price List / Agreement / Surcharge Rule	Selling prices and escalators	Pricing Manager	Book I I §6
Cost / Profit Center	Spend responsibility; margin segmentation	Controller / CFO	Book I H
Batch / Serial / Serial Profile	Lot and unit identities and rules	System-generated per profile	Book II §15
Installed Base	Where each shipped transformer lives	Customer Service	Book IV §7.4
Document / Drawing	Controlled files with revisions	Engineering	Book I C §5

5. Document Flow Atlas

Six flows move everything in this business. Each picture reads left to right; dark boxes are the moments money or ownership changes; amber boxes are quality gates. Under each picture: what a designer should make effortless. Full closure rules per flow live in Book III Part A and Book IV.

5.1 Order to Cash (stocked products)

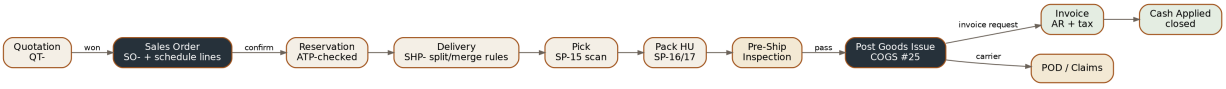


Figure V-5. Make-to-stock order to cash. FL-01 and FL-09 carry the complete rules.

Design focus	Why
One readiness view per order (supply, credit, quality, documents as four chips)	CSRs answer “can this ship?” fifty times a day; the answer must be one glance, not five screens
Promise dialog with the constraint named	Partial availability decisions (accept date / split / backorder) are the hardest moment at order entry
Delivery cockpit as the daily queue	Shipping lives in it all day; status columns and readiness flags must sort and filter fast

5.2 Engineer to Order (custom transformers)



Figure V-6. ETO quote to cash. The dashed line is milestone money arriving while the build progresses. FL-02 carries the rules.

Design focus	Why
Configuration panel that reads like a datasheet (kVA, voltages, cooling, taps)	Engineers and customers check these values repeatedly; they print on the quote
Milestone strip on the ETO order (6 slices with evidence status)	Everyone from Finance to the PM asks “which milestone are we on?”
Drawing approval loop as a visible ping-pong (sent → customer → returned)	The single most common ETO delay; make its state impossible to miss

5.3 Procure to Pay



Figure V-7. Procure to pay. FL-03 carries the rules; long-lead engineered buys follow FL-10.

5.4 Plan to Produce



Figure V-8. Plan to produce. FL-04 carries the rules; the shop floor terminal executes the middle.

5.5 Design to Source (engineering to purchasing)

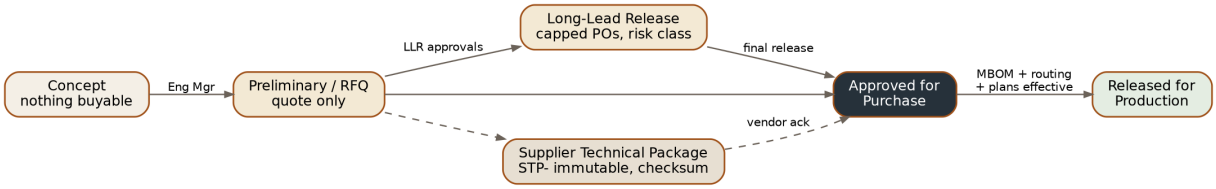


Figure V-9. Release levels let purchasing move before design is final, safely. FL-10, Book IV §3.

5.6 Returns and Warranty



Figure V-10. RMA flow. FL-05 carries the rules; four return classes share one path.

5.7 How Cost Moves (the money picture)

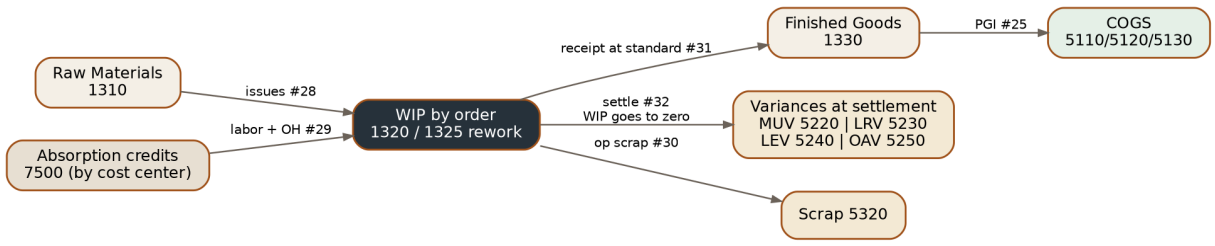


Figure V-11. Cost flow. Material value moves left to right; labor and overhead enter from cost centers; settlement empties WIP to exactly zero.

6. Cross-Module Linkage Map

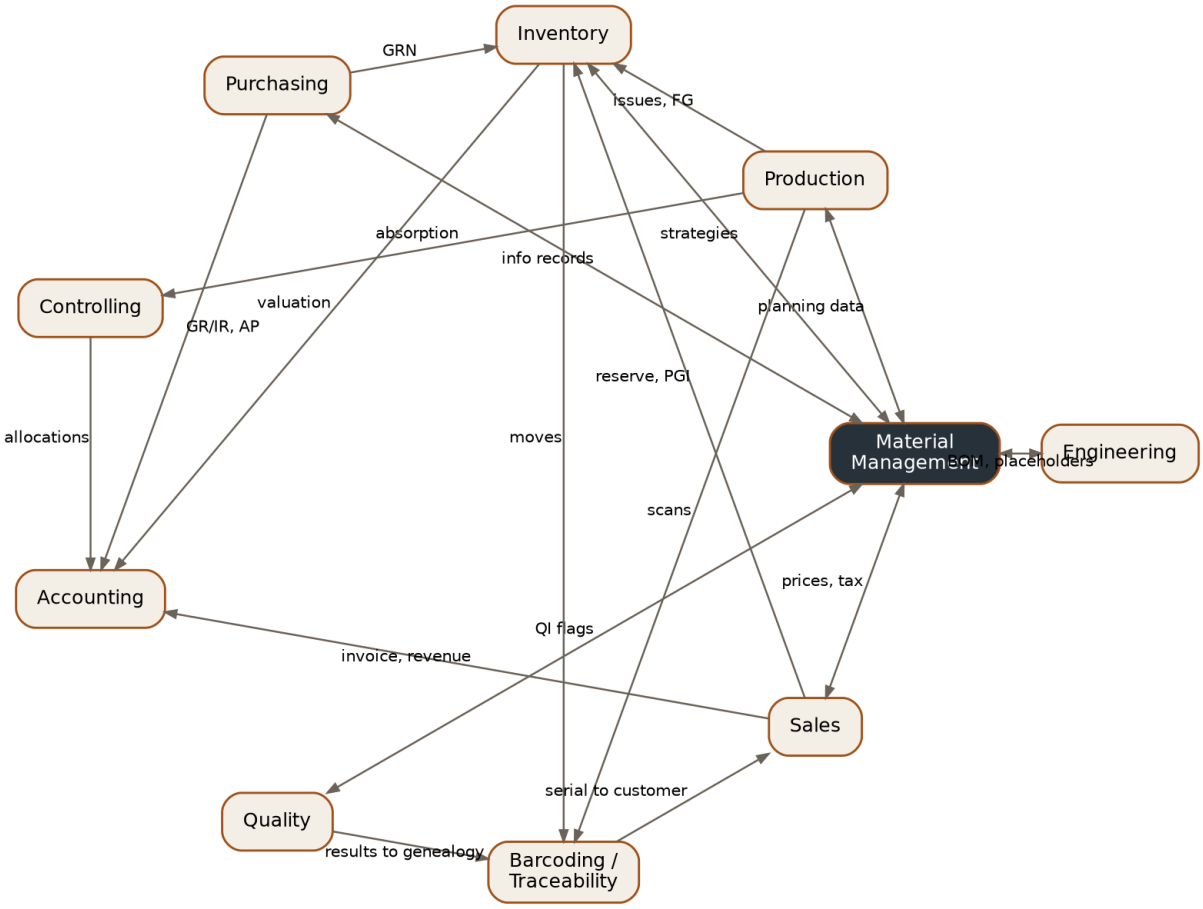


Figure V-12. Who talks to whom. Every arrow is a specified contract, not an aspiration.

Linkage	The Contract in One Line	Full Spec
Sales ↔ Inventory	Confirmations reserve stock; PGI moves it out and posts COGS; serials carry customer and site	Book I B §4–5; Book IV §2
Sales ↔ Accounting	PGI and milestones raise invoice requests; Finance posts; revenue follows ASC 606, not the billing plan	Book I B §4.4; ACC 3/24
Sales ↔ Shipping ↔ Barcoding	Pick, pack, load are scans SP-15…18; the shipment cannot PGI with an open HU or failed inspection	Book I A §5, F; Book IV §2
Engineering ↔ Materials	Placeholders reserve real numbers; activation unblocks the BOM; supersession retires parts safely	Book II §4.3/§4.6/§6
Engineering ↔ Purchasing	Release levels + supplier package versions govern what may be bought when	Book IV §3
Engineering ↔ Production	Only released production versions build; ECO effectivity decides in-flight orders	Book I C §4/§6; Book IV §6.1
Purchasing ↔ Inventory ↔ Accounting	GRN books stock and GR/IR; match clears it; PPV/IPV explain price differences	INV 5.6; ACC 6; Book I A §8

Linkage	The Contract in One Line	Full Spec
Production ↔ Inventory	Issues and backflush drain components; FG receipt fills at standard	Book I D §6/§8
Production ↔ Controlling ↔ Accounting	Confirmations absorb labor/overhead from cost centers; settlement posts variances by cause	Book I D §8, H, I §4
Quality ↔ everything	Gates block movement (receipt, operation, shipment); results write to the batch/serial forever	Book I E; Part A §6
Barcoding ↔ everything	Twenty-one scan points are the data entry; genealogy is their permanent record	Book I F; Book IV §4
Materials ↔ Accounting	Valuation class + event key = accounts; nobody types account numbers	Book II Part B

7. Module Design Briefs

One brief per module: what it is for, who uses it, every screen with its reference, and what to prototype first. Screen types: List (filterable register), Detail (tabbed record), Workbench (act-on-many queue), Wizard (guided steps), Kiosk (touch + scanner), Config (admin).

7.1 Sales & Order Management

Turns customer demand into confirmed, promised, shippable orders and carries them to invoiced closure. The heart is the sales order with schedule lines; everything else serves it.

Who uses it: Customer service, sales reps, sales admin, shipping office, credit/finance.

Screen	Type	What the user does here	Spec
Leads & Opportunities (OM-LEAD-001/2)	List/Detail	Capture demand before a customer exists; prior build lookup; convert to quotation	Book VI §4.1
Customer List / Detail (OM-CUST-001/2)	List/Detail	Maintain customers: addresses, commercial terms, credit view, tax, history	Book I B §2
Quotation List / Editor (OM-QT-001/2)	List/Detail	Build priced, versioned offers; ETO configuration panel; margin check	Book I B §3
Sales Order List / Detail (OM-SO-001/2)	List/Detail	The contract: lines, schedule lines, pricing elements, milestones, pegs, changes	Book I B §4; Book IV §2.1
Backorder Workbench (OM-BO-001)	Workbench	Fix shortages: reallocate, re-promise, split, substitute	Book IV §2.2
Delivery Cockpit (OM-DLV-001)	Workbench	Group schedules into deliveries; watch readiness; release to warehouse	Book IV §2.4
Shipment Detail (OM-SHP-002)	Detail	Carrier, permits, HUs, documents, PGI	Book I B §5
Pack Station (OM-PACK-STATION-001)	Kiosk	Scan items into handling units; weights; labels	Book I A §5
POD / Claims (OM-POD)	List/Detail	Record delivery proof; open and resolve claims	Book IV §2.4

Key lifecycles: Quote: Draft → Issued → Won/Lost. Order: Draft → Confirmed → In Fulfillment → Shipped → Invoiced/Closed (credit hold branch). Schedule line: Open → Confirmed → Delivered. Delivery: Planned → …→ PGI → Closed.

Prototype first: Sales order detail with schedule lines and the readiness chips, then the delivery cockpit, then the promise dialog.

7.2 Engineering & BOM

Owns the technical truth: items, BOMs, drawings, changes. An engineer takes an ETO order from configuration to released drawings and BOMs without leaving the module, and nothing unreleased can be bought or built.

Who uses it: Design engineers (electrical, mechanical), engineering manager, CCB members.

Screen	Type	What the user does here	Spec
Engineering Workspace (ENG-HOME-001)	Workbench	My design queue, approvals, exceptions	Book I C §1
Engineering Request (ENG-REQ-001)	Detail	The ETO design job: checklist of deliverables, due dates, customer approval	Book I C §2
EBOM Editor (ENG-BOM-002)	Detail (3-pane)	Build the structure tree; add components or placeholders; validate; release	Book I C §3
ECR / ECO (ENG-ECR/ECO)	Detail	Propose change → impact analysis → CCB → execute with effectivity	Book I C §4
MBOM Transform	Wizard	Turn a released EBOM into the plant MBOM: phantoms, operation mapping, mfg adds	Book I C §6
Production Versions (PV)	List/Detail	Pair MBOM + routing with validity; the buildable unit	Book IV §6.1
Where-Used	Report	Everything using a component, multi-level	Book I C §7

Key lifecycles: EBOM: Draft → In Review → Released → Superseded. ECO: task checklist → atomic release at effectivity. Release levels for purchasing: Concept → Preliminary → Long-Lead → Approved for Purchase → Released for Production.

Prototype first: The EBOM editor three-pane layout — it is the most-used engineering screen and the hardest to get right.

7.3 CAD & Document Control

The controlled library of drawings and technical documents, and the bridge to native CAD. The floor and customers only ever see released PDF renditions; the vault keeps the native files.

Who uses it: Design engineers, document control, shop floor (read-only), customers (transmittals).

Screen	Type	What the user does here	Spec
Document Register / Detail (ENG-DOC-001/2)	List/Detail	Controlled files: revisions, check-in/out, title block data, security class	Book I C §5
Transmittals	Wizard	Send drawing packages to customers; capture Approved / Revise & Resubmit	Book I C §5.2
CAD Sync Monitor + Conflict Queue	Workbench	Watch vault events; resolve field conflicts side-by-side	Book III G §3
EBOM Import (CAD)	Wizard	Assembly extract → mapped proposal → transactional commit	Book I C §3.3; Book III G §3

Key lifecycles: Document: WIP → In Review → Released → Superseded/Obsolete. Sequencing rule: the floor keys off ERP effectivity, never PDM state.

Prototype first: The conflict queue and the import wizard — they define whether engineers trust the integration.

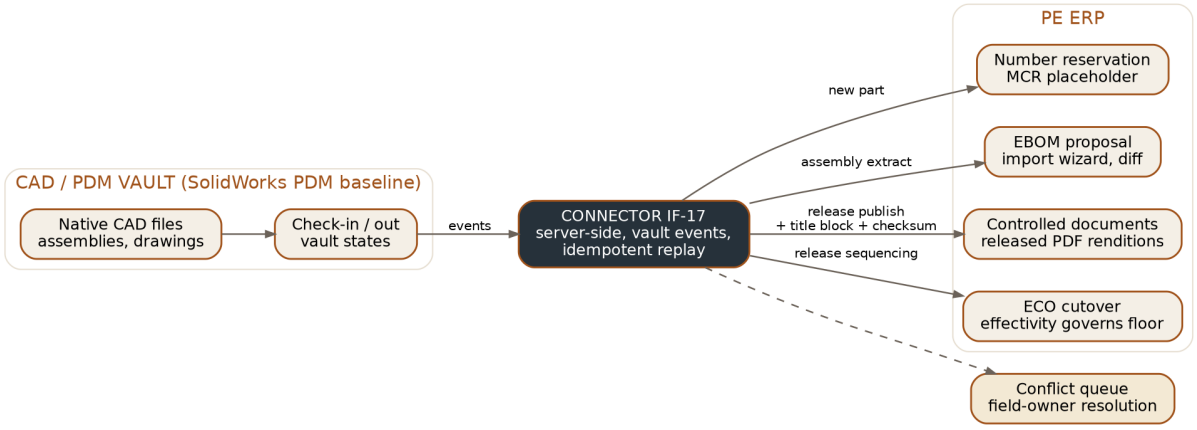


Figure V-13. CAD/PDM linkage. The connector is server-side; ERP owns numbers and released renditions; the vault owns native files.

7.4 Material Management

One record per material, born through a governed multi-department workflow, changed under control, valued by rules, retired without breaking history. The hub every other module reads.

Who uses it: Master data steward, planners, buyers, quality, cost accounting, warehouse.

Screen	Type	What the user does here	Spec
Material List (MM-MAT-001)	List	Find anything; completeness bars; lifecycle badges; bulk actions	Book II §3.1
Material Detail (INV 2.3 + views)	Detail (10 tabs)	Every view per §3.4–3.13: general, UoM, planning, purchasing, production, warehouse, quality, accounting, sales, references	Book II §3
Creation Request (MM-MCR-001/2)	Workbench	Department task checklist to activation; SLA tracking	Book II §4.2
Change Request (MM-MCH-001)	Wizard	Governed single and mass changes with simulation	Book II §4.4/§16.1
Supersession (MM-SUP-001)	Detail	Replace a material with run-out control	Book II §4.6
Info Records (MM-IR-001/2)	List/Detail	Vendor-material prices, scales, lead times	Book II §7.2
Source List (MM-SL-001)	Detail grid	Fix or block sources per period	Book II §7.4
Account Determination (MM-VAL-001)	Config	Event-key matrix with simulation; Controller-governed	Book II §5.3
Data Quality Dashboard	Dashboard	Completeness, MCR cycle time, duplicates, stale sourcing	Book II §16.4

Key lifecycles: Material: Draft → Data Completion → Active → Restricted → Blocked → Obsolete, plus function-specific blocks (purchasing, sales, production, shipping, posting).

Prototype first: The material detail tab bar and the MCR task board — creation is the workflow everyone touches first.

7.5 MRP & Finite Planning

Answers two questions forever: what should we make or buy (MRP), and when can we honestly promise (finite schedule + CTP). Planners live in the workbench; the engine runs at night.

Who uses it: Production planners, master scheduler, buyers (exceptions), sales (promises).

Screen	Type	What the user does here	Spec
MRP Workbench (PP-MRP-001)	Workbench	Exceptions EX-01…09 with one-click actions; proposal conversion	Book I D §3
Planned Orders	List	Review, firm, convert to WO or PR	Book I D §3.3
Schedule Board	Workbench	Finite schedule per work center; freeze horizon; what-if sandbox	Book III E §3
CTP / Promise Dialog	Dialog	Feasible date with the binding constraint named; soft reservation	Book III E §4
At-Risk Promises	Workbench	Recovery options when projected completion slips	Book III E §5

Key lifecycles: Proposals: created → firmed → converted. Promises: quoted → reserved (decaying) → confirmed → at-risk workflow on slip.

Prototype first: The exception workbench — MRP is invisible; its exceptions are the product.

7.6 Production & Shop Floor

Executes the build: orders, dispatch, scans, confirmations, WIP, receipt, settlement. The kiosk terminal is the factory's single screen; everything on it is reachable by touch and scanner only.

Who uses it: Production scheduler, supervisors, operators (kiosk), cost accounting (settlement).

Screen	Type	What the user does here	Spec
Production Order List / Detail (PP-WO-001/2)	List/Detail	Release gates, pegs, revisions, cost tab, serial	Book I D §5
Dispatch List	Workbench	Per work center queue in priority order; material-ready flags	Book I D §6
Shop Floor Terminal (PP-SFT-001)	Kiosk	Badge in, start op, scan materials, capture data, QC gate, complete with quantities, hold/andon	Book I D §6
WIP Board / Valuation (PP-WIP-001)	Board/Report	Where every order stands; value reconciles to GL	Book I D §7
Rework Orders (RWO)	Detail	Free-form repair with its own cost collection	Book I D §9
Settlement Run (PP-SETTLE-001)	Wizard	Month-end variance posting with preview	Book I D §8

Key lifecycles: Order: Created → Released → In Progress → Testing → Completed → Closed (TECO), with On Hold branch. Operation: queued → started → completed (good/scrap/rework must sum).

Prototype first: The kiosk terminal, full-screen at 1280×800, gloves-on touch targets, scanner-first — test it standing up.

7.7 Quality Management

Five gates (incoming, in-process, final test, pre-shipment, return), all executing from inspection plans, all writing permanent results to batches and serials, all able to escalate to NCR, CAPA, SCAR or a recall.

Who uses it: Quality inspectors, quality engineers, quality manager, calibration owner.

Screen	Type	What the user does here	Spec
Quality Dashboard (QM-DASH-001)	Dashboard	Open inspections, FPY, NCR funnel, vendor quality, cost of quality	Book I E §2
Inspection Plans (QM-PLAN-001/2)	List/Detail	Characteristics, sampling, versions, approval	Book I A §6
Inspection Execution	Detail	Results against plan rows; instruments checked against calibration	INV 6.1 pattern
Final Test & FTR (QM-FIN-001)	Detail	Transformer test catalog; runs history; certified report	Book I E §3–4
NCR / CAPA / SCAR	List/Detail	Failure records; corrective action lifecycle; supplier 8D	Book I E §5–6
Deviations & Waivers (DEV)	Detail	Temporary acceptance with expiry and customer evidence	Book IV §7.3
Calibration Register (QM-CAL-001)	List	Instruments, due dates, interlock	Book I E §7
Recall Campaigns (RCL)	Workbench	Population, holds, notifications, recovery, effectiveness	Book III A FL-08

Key lifecycles: Plan: Draft → In Review → Active → Superseded. CAPA: Open → Analysis → Actions → Implemented → Effectiveness → Closed.

Prototype first: Inspection execution with pass/fail color logic and the calibration interlock error state.

7.8 Purchasing

Turns demand into on-time, on-spec supply: sources, RFQs, POs, confirmations, expediting. The buyer's day starts on the workbench and ends when the expedite board is quiet.

Who uses it: Buyers, purchasing manager, vendor quality, receiving (read).

Screen	Type	What the user does here	Spec
Buyer Workbench	Workbench	PR queue, exceptions, unconfirmed POs	Book I G §1
Vendor / AVL (PUR-VEND-001/2)	List/Detail	Vendors, qualification, compliance, AVL rows	Book I G §2
Purchase Requisitions (PUR-PR-001)	List	Demand awaiting sourcing	Book I G §3
RFQ (RFQ-)	Detail	Sealed bids, comparison grid, technical/commercial split, award	Book I G §4; Book IV §3.4
Purchase Orders (PUR-PO-001/2)	List/Detail	The commitment: schedules, quality clauses, STP version, revisions	Book I G §5
Long-Lead Releases (LLR)	Detail	Early buying under caps with exposure tracking	Book IV §3.2
Expedite Board (PUR-EXP-001)	Workbench	Late and at-risk supply with pegged-order priority	Book I G §7

Key lifecycles: PO: Draft → Approved → Sent → Confirmed → Received → Matched/Closed with Change Pending overlay. LLR: capped → released → closed.

Prototype first: The buyer workbench and the RFQ comparison grid.

7.9 RMA & Warranty

No customer material enters without an RMA. Four return classes share one path: authorize, receive into quarantine, inspect, adjudicate warranty, disposition, settle, close when every linked document is done.

Who uses it: Customer service, receiving, quality, finance.

Screen	Type	What the user does here	Spec
RMA List (RMA-LIST-001)	List	Work queue with SLA aging	Book I A §3.3
RMA Create / Detail (RMA-CREATE-001)	Detail	Reference pull, classes, freight, lines; tabs incl. financials and linked docs	Book I A §3.4/3.7
Return Receipt	Kiosk/Detail	Scan RMA barcode; condition + photos; quarantine	Book I A §3.5
Inspection & Disposition	Detail	Teardown checklist; warranty adjudication (entitlement service); outcomes	Book I A §3.6; Book IV §7.4
Installed Base (IB)	List/Detail	Every shipped serial: site, warranty, history	Book IV §7.4
Spares Finder (SVC-SPF-001)	Workbench	Serial to as-built parts list with stock, price, supersessions; one-step spares quote	Book VI §5.1

Key lifecycles: RMA: Draft → Authorized → Received → In Inspection → Disposition → Settling → Closed.

Prototype first: The disposition screen — five outcomes with automatic document creation must feel safe, not scary.

7.10 Costing & Pricing

Costing computes what things cost (standards, rates, estimates); pricing computes what they sell for (lists, agreements, surcharges) with margin floors between them.

Who uses it: Cost accountant, controller, pricing manager, estimator, sales (read).

Screen	Type	What the user does here	Spec
Cost Estimates (CST-CE-001)	Detail	Cost trees; parametric ETO estimating by driver weights	Book I I §3.1
Costing Runs (CST-RUN-001)	Wizard	Mass rollup: errors, review, mark, release	Book I I §3.2
Work Center Rates (CST-RATE-001)	Config grid	Annual rate computation and release	Book I I §2
Price Lists / Agreements (PRC-PL-001)	List/Detail	Rates, breaks, validity, approvals	Book I I §6
Surcharge Rules + Index Values	Config	Copper/CRGO escalators with immutable history	Book I I §6.4
Margin Analysis	Report	Quote register, three-view order margin, floor breaches	Book I I §7

Screen	Type	What the user does here	Spec
Price Simulation	Dialog	The waterfall exactly as documents resolve it	Book IV §5

Key lifecycles: Estimates: Draft → Released → Superseded. Costing run: scoped → rolled → marked → released.

Prototype first: The price simulation waterfall — it is the explainability surface for every pricing argument.

7.11 Controlling & Accounting Links

Controlling gives every dollar two homes: the cost center that spent it and the profit center that earns from it. Accounting (external volume) posts everything through the shared Posting Service; nobody types account numbers anywhere.

Who uses it: Controller, controlling analyst, cost center managers, CFO.

Screen	Type	What the user does here	Spec
Cost / Profit Center Masters (CO-CC/PC)	List/Detail	Hierarchies, owners, validity	Book I H §2
Allocation Cycles (CO-ALC-001/2)	Config/Wizard	Period-end shared-cost distribution with preview	Book I H §5
Cost Center Report / PC P&L (CO-R1/R2)	Report	Actuals vs budget; product-line P&L reconciling to entity	Book I H §6
GL Determination Preview	Panel	Simulate any posting's accounts and dimensions	Book II §3.11/§5.3

Key lifecycles: Every posting carries CC and PC by derivation rules 1–7; exceptions land on CO-R4 at target zero.

Prototype first: The GL preview panel — it turns accounting from mystery into a visible simulation.

7.12 Workflow, Jobs & Admin

One engine for every approval, alert and scheduled job. Twenty-seven governed workflows are configuration rows, not code.

Who uses it: Admins, process owners, everyone (approvals inbox).

Screen	Type	What the user does here	Spec
Approvals Inbox	Workbench	Everything awaiting me, with SLA colors; act from list	Section W; Book IV §7.7
Workflow Definitions	Config	Rules, chains, SLAs, versions (W.4 catalog)	Book IV §7.7
Alert Rules	Config	Events, scopes, channels, dedupe	Section W.2
Job Console	Workbench	Runs, SLAs, dependencies, failures	Section W.3; Book III D §5
Integration Console	Workbench	Message monitor, DLQ, replay, correlation search	Book III B §4

Key lifecycles: Approval instances: pending → approved/rejected/recalled with delegation and escalation.

Prototype first: The approvals inbox — the one screen literally every manager uses daily.

8. Barcoding and Devices — the Physical Layer

Barcoding is how the factory talks to the system. This section gives designers and IT what Book I Part F assumed: the stations, the exact device classes and reference models, the labels as pictures, and the interaction rules a scan-first screen must obey. The 21 scan points (SP-01…SP-21), symbologies and validation rules remain exactly as specified in Book I Part F.

8.1 Station Map



Figure V-14. Where scanning happens. Each station names its scan points and its device kit from the table below.

8.2 Hardware Specification (device classes with reference models)

Reference models name the class precisely; equivalents from Honeywell or Datalogic meeting the same column specs are acceptable at procurement. All scanners must read Code128 and GS1-128 and 2D (QR/DataMatrix) per Book I F §1.

Device Class	Reference Model	Key Specs	Used At	Qty (Phase 1)
Rugged handheld 2D imager, corded	Zebra DS3608-SR	SR optics 5–95 cm, IP67, 2.4 m drop, USB-HID via wedge service	Receiving, QC benches, returns	8
Rugged handheld 2D, long range	Zebra DS3608-ER	ER optics 0.2–17 m (reads rack labels and flatbed HUs), IP67	Ship dock, bulk storage, yard	4
Hands-free presentation scanner	Zebra DS9308	Countertop, auto-sense	Pack bench, nameplate station, test lab kiosks	6
Wearable ring scanner + wrist terminal	Zebra RS5100 + WT6300	Bluetooth ring, glove-friendly	Pickers, kit builders	6
Rugged mobile computer	Zebra TC77 (Android)	4.7 in, IP67, hot-swap battery, runs the responsive web app	Warehouse moves, counting, putaway	10
Fixed kiosk terminal	15.6 in PCAP touch, fanless mini-PC, sealed IP54 front	Kiosk browser, badge scan login, wedge service client	Every work center (12) + pack + test	15
Industrial label printer 4-in	Zebra ZT411 (300 dpi)	Thermal transfer, ZPL, cutter option	Receiving, quarantine, pack, warehouse office	5
Desktop label printer	Zebra ZD621 (300 dpi)	Thermal transfer, compact	QC bench, engineering office	3
Metal plate marking	Existing nameplate engraver + PE-supplied barcode plates	Serial plate barcode per Book I F §1	Nameplate station	existing
Bench scale w/ interface	150 kg class, RS232/USB	Feeds gross weight at HU close (SP-17 sanity ±2%)	Pack bench	2

Device Class	Reference Model	Key Specs	Used At	Qty (Phase 1)
Floor scale	3,000 kg class	Pallet/crate weights	Pack / ship dock	1
Wedge service	PE software component	WebSocket scan events, focus-lock, per-station registry (ADR-10)	All stations	—

8.3 Label Stock and Print Specification

Label	Size / Material / Ribbon	Printer	Content Spec
Item shelf label	2×1 in polypropylene, resin ribbon	ZD621	Book I F §1 (Code128 material code)
Batch label	4×2 in polypropylene, resin	ZT411	Book I F §1; mockup Figure V-16
Serial label (components)	4×2 in polyester, resin (durable)	ZT411	Book I F §1
Location/rack label	4×1 in reflective for long-range reads	ZT411	Book I F §1
HU / shipping label	4×6 in coated paper, wax-resin	ZT411 (pack)	Book I A §5.4; mockup Figure V-15
Traveler sheets	Letter laser	Office laser	Book I D §5.4
Calibration asset label	1×0.5 in polyester	ZD621	Book I E §7

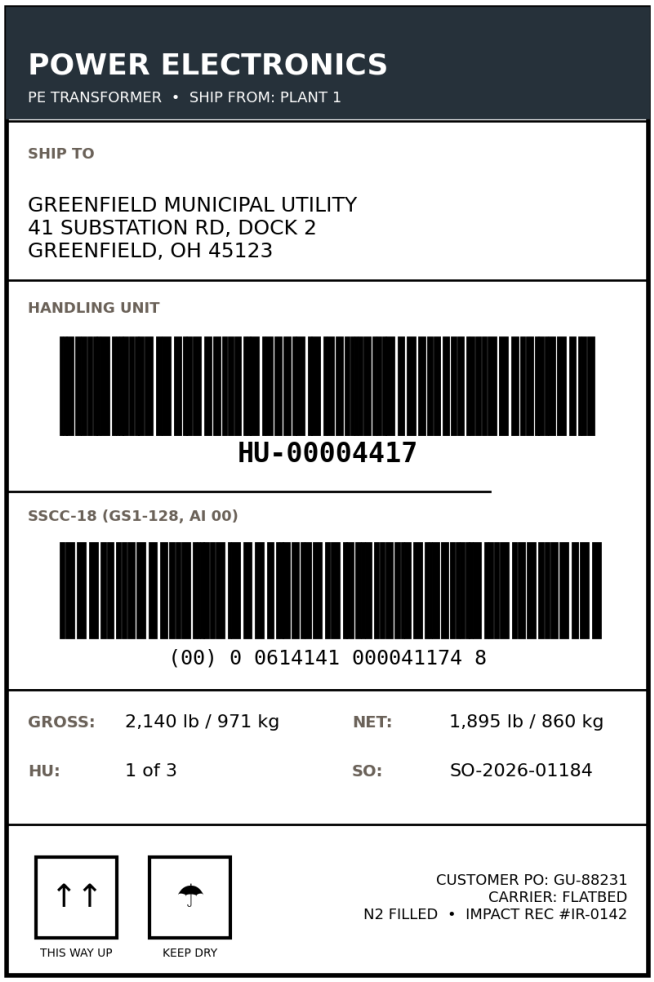


Figure V-15. Handling-unit label, 4×6 in, as printed at HU close: HU barcode, SSCC-18, weights, HU count, handling marks, transformer addenda.

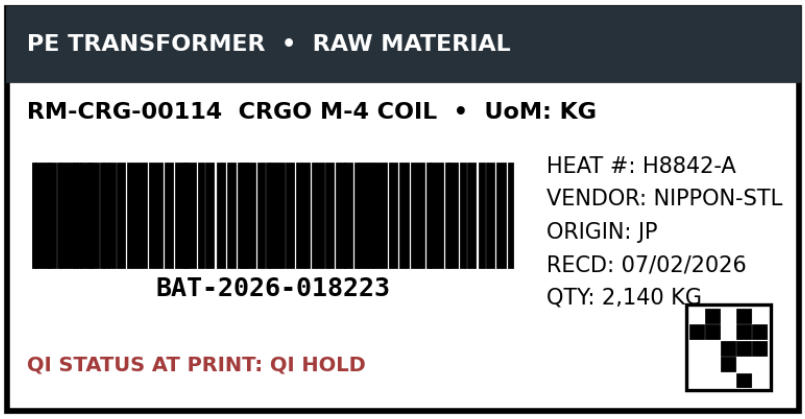


Figure V-16. Batch label, 4×2 in, as printed at receiving: batch barcode, heat number, vendor, origin, quantity, QI status at print.

8.4 Scan Interaction Rules (design law for every scan screen)

- Focus lock: the scan field owns input; a scan is an event, never keystrokes into whatever field has focus (wedge service, ADR-10). Designers never place a scan target inside a form field.

- Feedback in under 400 ms: success = single beep + green flash of the scanned row; failure = double low beep + red banner stating what was scanned, what was expected, the E-code, and the next action (Book I F §4). No silent failures, no toast that disappears before a gloved operator looks up.
- One action per scan where possible: scanning a traveler op barcode starts the operation; scanning it again opens completion. Menus are for exceptions.
- Big targets: minimum 48 px touch targets, 16 px base font on kiosks, high-contrast state colors from §11; test with gloves at arm's length.
- Offline visibly: when the terminal queues offline (up to 4 h, Book I F §2), a persistent amber banner shows queue depth; replay conflicts land in the supervisor exception queue, never silently dropped.
- Reprint is governed: every reprint asks a reason; serial/nameplate reprints need Quality approval; duplicates watermark DUPLICATE (Book I F §4).

9. Traceability at a Glance

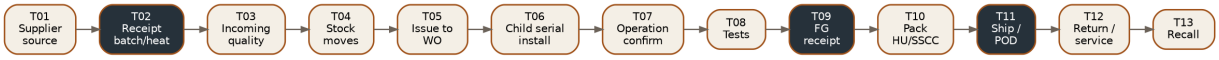


Figure V-17. The genealogy chain T01–T13. Every link is an append-only event; corrections add reversal + replacement events, never edits (Book IV §4).

For designers, traceability is two screens: the trace explorer (enter a serial or a heat number, see the tree in both directions with drill-down into every node's evidence) and the completeness dashboard (coverage, orphans, unresolved corrections). Both are specified in Book IV §4.4; the tree must render in under 10 seconds and read like a family tree, not a log file.

10. Design System Starter

10.1 Tokens

Token Group	Values
Brand	PE Blue #1F3864 (headers, primary), PE Accent #2E5FA3 (links, section heads), Light #EDF1F8 (fills), Mid #D9E2F3 (table heads)
Status colors (used consistently across every lifecycle in Books I–IV)	Draft/neutral gray #6B7280 • In progress/review amber #B58221 • Blocked/error red #A33B3B • Active/passed green #3E7D4E • Info/released blue #2E5FA3 • Terminal dark green #1F5F3F
Typography	UI: Inter or system sans; data grids 13–14 px; kiosk base 16 px; documents print in Calibri per the letterhead standard
Spacing & radius	8 px grid; 6 px radius on cards and badges; tables dense (32 px rows) with a comfortable mode (40 px)
Elevation	Flat lists; single shadow level for dialogs and the scan feedback flash

10.2 Component Inventory (build these once, reuse everywhere)

Component	Where it appears	Behavior source
Data grid (sortable, saved filters, column config, bulk bar, export)	Every List screen	INV/ACC Section 2 global pattern

Component	Where it appears	Behavior source
Status badge + lifecycle strip	Every Detail header	Lifecycle tables throughout; colors §10.1
Entity context bar	Every screen	ACC Section 2
Audit trail panel (immutable)	Every Detail	ACC Section 2
Attachment panel with checksum note	Every Detail	ACC Section 2; documents Book I C §5
Readiness chips (supply / credit / quality / docs)	SO, delivery, WO release	Book IV §2; Book I D §5.2
Approval strip (chain, current step, SLA clock, act buttons)	Everything in W.4	Section W; Book IV §7.7
Price waterfall panel	Quote/SO lines, simulation	Book I A §4.4; Book IV §5
Scan input + feedback flash	All kiosk/mobile screens	§8.4
Tree explorer (BOM, where-used, genealogy)	EBOM, traceability, where-used	§9; Book I C §3
Milestone strip	ETO orders	Book I B §4.4
Diff viewer (revisions, change orders, MCH simulation)	EBOM compare, PO revisions, mass change	Book I C §3.3; Book II §16.1
GL preview panel	All posting screens	Book I H §3 Dev Note; Book II §3.11
Exception worklist card (count, aging, owner, one-click action)	All workbenches	MRP EX-, expedite, data quality, CO-R4

10.3 The Nine Mandatory Screen States

Every screen ships with all nine states designed: normal, empty (first-use with a helpful action), loading (skeletons, no spinners over 300 ms), validation error (inline + summary), authorization error (explains the missing role, never a blank page), conflict (someone saved first; show diff), offline (kiosk banner + queue depth), degraded (dependency down; what still works), read-only (terminal statuses and view-only roles). The screen template in Book IV §9.4 makes these a deliverable, not an afterthought.

10.4 Figma Organization

- One library file: tokens, components (§10.2), icons, status colors — published; no local styles in feature files.
- One file per module brief (§7.1–7.12), pages inside per screen, frames per state; frame names carry the screen ID (OM-SO-002 / detail / conflict) so RTM linkage is mechanical.
- Flows as Figma prototypes follow the atlas diagrams (§5) — one prototype per flow, wired from the same frames, used directly in UAT scenario reviews.
- Handoff: every frame links its spec section in the description; a frame without a spec link fails design review (gate DG-2).

11. Reader Orientation Map

A Reader Looking For	Must Open
A fast, plain language route into the system	This Book: the briefs (§7), the pictures (§3–§5), the reading paths (§1), the design system (§10)
The enterprise capability catalogue and vocabulary	§2 capability map (26 capabilities with phase decisions) + the 36 term glossary
How the modules connect	§6 map + table; the binding contracts are Book II Part C, Book III Parts A–B and Book IV
Barcoding hardware, labels and interaction rules	§8: station map, device classes with reference models and quantities, label stock specification, printed label samples, interaction law
The CAD linkage	§7.3 brief + Figure V-13; the full contract Book III Part G; connector requirements IF-17
The diagrams	Figures V-1 through V-17 across structure, master data, flows, cost, CAD, stations and traceability, plus the atlas of Book VIII
The reading order to build from scratch	§1 reading paths: this Book to understand, Books I–III to build exactly, Book IV to govern, Books VI–IX to execute

BOOK VI

SCREEN BY SCREEN IMPLEMENTATION PLAN

Every screen in this specification with its route, layout, actions, arrivals, outputs, postings, scans, build wave and field authority

Linkages stated inside every entry • Build sequence in seven waves • Costing and Pricing module consolidated with its engine diagram

Book VI is the build plan at screen level. Every screen named anywhere in this document appears below with one implementation entry. Each entry states the route, the layout to build, the actions, where the user arrives from, what the screen creates, which postings and scans it triggers, its build wave and the section that carries its complete field rules. The linkage question is answered inside every entry. The rows named Arrives from and Creates or updates are the linkage, screen by screen, with the journal template numbers and scan points beside them. Nothing earlier in the document is changed by this Book.

1. Method

- Entries are grouped by module in build order. List and detail pairs share one entry when they ship together.
- Build waves follow the roadmap in Book I Part K. Wave 1 runs weeks 1 to 4, wave 2 weeks 3 to 8, wave 3 weeks 6 to 12, wave 4 weeks 10 to 18, wave 5 weeks 14 to 20, wave 6 weeks 16 to 22, wave 7 weeks 20 to 26.
- Every entry reuses the component inventory of Book V section 10.2. A layout cell names components, not pixels. The Figma frame set per screen follows the nine states of Book V section 10.3.
- Posting references use the journal template numbers of the unified map. Scan references use the SP numbers of Book I Part F.
- A developer finishing an entry has one definition of done. The screen passes gate DG-3 of Book IV section 9.3 with its field rules from the referenced section.

2. Build Sequence at a Glance

Wave	Screens Delivered	Why This Order
1 (wks 1–4)	Controlling masters, account determination workbench, number series, period control, tax config, workflow definitions, alert rules, job console	Everything downstream posts through these. Nothing financially relevant ships before wave 1 is green
2 (wks 3–8)	Material listing and detail views, MCR, MCH, duplicate check, leads and opportunities, customer master, quotations, sales orders with schedule lines, pricing service surfaces, packing, count documents, safety stock	The commercial spine. Order to cash for stocked goods becomes demonstrable
3 (wks 6–12)	Engineering workspace, requests, EBOM editor, documents and transmittals, ECR and ECO, MBOM transform, production versions, CAD import	Design cycle before production. Placeholder path live so materials are born governed
4 (wks 10–18)	Work centers, routings, MRP workbench, planned orders, production orders, dispatch, shop floor terminal, WIP, schedule board, CTP dialog, trace explorer	The factory goes live in the system with scans SP-07 to SP-13
5 (wks 14–20)	Quality dashboard, plans, executions, final test, CAPA, SCAR, deviations, calibration, RMA end to end, installed base, spares finder, recall campaigns	Gates and returns on top of a working floor
6 (wks 16–22)	Vendor and AVL, PR, RFQ, PO, LLR, expedite board, info records, source lists, cost estimates, costing runs, rates, price lists, agreements, surcharges, CVA, margin analysis, allocations, E&O	Procurement and the money layer complete

Wave	Screens Delivered	Why This Order
7 (wks 20–26)	Backorder workbench, delivery cockpit, POD and claims, completeness dashboards, integration console hardening, close orchestration additions	Flow closure, reconciliation and go live readiness

3. Materials Module Screens

MM-MAT-001 • Material Listing

Aspect	Implementation
Route, type, users	/inventory/items • List • Steward, buyers, planners, all read
Build the layout	Standard data grid plus type badges, lifecycle badges, completeness bars, plant chips, quick stat pills
Primary actions	Open detail. Bulk assign owners. Set block. Add to change request. Export. New material starts an MCR, never a bare record
Arrives from	Sidebar Materials. Search everywhere resolves here
Creates or updates	MCR documents from the New action. MCH rows from bulk actions
Postings and scans	None
Build wave and field spec	Wave 2 • Fields and rules in Book II §3.1

INV 2.3 + MM views • Material Detail (ten views)

Aspect	Implementation
Route, type, users	/inventory/items/{id} • Detail, tabbed • View owners edit their view, all read
Build the layout	Tab bar with General, Units, Planning, Purchasing, Production, Warehouse, Quality, Accounting and Costing, Sales, references. Field group edit rights render per role. GL preview panel on the accounting view
Primary actions	Save per view. Activate. Set block. Supersede. Extend to plant. Copy from. Simulate GL. Print label
Arrives from	MM-MAT-001, any material link anywhere in the system
Creates or updates	Governed field changes raise MCH rows. Activation completes the MCR
Postings and scans	None directly. Valuation fields drive every later posting
Build wave and field spec	Wave 2 • Fields and rules in Book II §3.4 to §3.13

MM-MCR-001/2 • Material Creation Requests

Aspect	Implementation
Route, type, users	/materials/creation-requests • Workbench + detail • All departments, Materials Manager activates

Aspect	Implementation
Build the layout	Queue with SLA aging colors. Detail shows the task checklist per department with per task status and owner
Primary actions	Complete my task. Reassign. Activate. Reject to task with comments
Arrives from	EBOM placeholder creation, buyer initiation, sales initiation, imports
Creates or updates	Active materials. Unblocks the originating EBOM line, PR or quote on activation
Postings and scans	None
Build wave and field spec	Wave 2 • Fields and rules in Book II §4.2, §4.3

MM-MCH-001 • Material Change Requests

Aspect	Implementation
Route, type, users	/materials/change-requests • Wizard • Owners request, class rules approve
Build the layout	Selection step, change definition step, simulation grid with per row validation results, apply step with result log
Primary actions	Simulate. Submit for approval. Apply now or effective dated. Download log
Arrives from	Material detail, listing bulk bar, ECO tasks
Creates or updates	Applied field changes with full audit. Linked ECO task completion
Postings and scans	None
Build wave and field spec	Wave 2 • Fields and rules in Book II §4.4, §16.1

MM-SUP-001 • Supersessions

Aspect	Implementation
Route, type, users	/materials/supersessions • Detail • Materials Manager
Build the layout	Predecessor and successor cards, run out policy selector, affected document worklist, progress bar on remaining stock
Primary actions	Approve. Watch run out. Close
Arrives from	Material detail Supersede button, ECO tasks, duplicate merges
Creates or updates	Predecessor moves to Restricted then Obsolete. Buyer tasks on open POs. MRP flip to successor
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book II §4.6, Book IV §7.1

MM-IR-001/2 • Purchasing Info Records

Aspect	Implementation
Route, type, users	/purchasing/info-records • List + detail • Buyers, Purchasing Manager approves manual prices
Build the layout	Grid by material and vendor. Detail with price rows, scales, surcharge link, rolling actuals panel
Primary actions	New row from RFQ award or manual with approval. Expire. Export
Arrives from	Material purchasing view, vendor detail, RFQ award writeback
Creates or updates	The price and lead time basis MRP and PO pricing consume
Postings and scans	None. Invoice deviation over five percent raises a buyer alert
Build wave and field spec	Wave 6 • Fields and rules in Book II §7.2

MM-SL-001 • Source Lists

Aspect	Implementation
Route, type, users	/purchasing/source-lists • Detail grid • Purchasing Manager
Build the layout	Per material grid of vendor rows with validity, fixed flag, blocked flag, MRP relevance
Primary actions	Add row. Fix source for period. Block source
Arrives from	Material purchasing view, vendor detail
Creates or updates	The source determination input of Book II §7.5
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book II §7.4

MM-VAL-001 • Account Determination Workbench

Aspect	Implementation
Route, type, users	/materials/account-determination • Config • Controller, step up MFA on release
Build the layout	Matrix grid by event key and valuation class, version diff view, simulation panel with full journal preview, completeness check results
Primary actions	Draft rows. Mark. Release with effective date. Simulate
Arrives from	Configuration area. GL preview panels link here read only
Creates or updates	The account resolution every posting uses. Journal lines store the matrix row that resolved them
Postings and scans	None itself. Everything else posts through it
Build wave and field spec	Wave 1 • Fields and rules in Book II §5.2, §5.3

MM-EOR-001 • Excess and Obsolete Reserve Run

Aspect	Implementation
Route, type, users	/materials/eo-reserve • Wizard • Controller
Build the layout	Policy table, run preview grid with per material reserve deltas and drill to slow moving evidence
Primary actions	Compute. Review. Post
Arrives from	Period close calendar, quarterly
Creates or updates	Reserve build or release journal
Postings and scans	DR 6320 CR 5340 on build, reverse on release
Build wave and field spec	Wave 6 • Fields and rules in Book II §17.3

MM-DQ-001 • Data Quality Dashboard

Aspect	Implementation
Route, type, users	/materials/data-quality • Dashboard • Materials Manager, stewards
Build the layout	KPI tiles for completeness, MCR cycle time, duplicates, stale sourcing, governance exceptions, PC-900 count, each with drill lists
Primary actions	Open exception lists. Assign owners
Arrives from	Materials sidebar, monthly review ritual
Creates or updates	Worklist assignments
Postings and scans	None
Build wave and field spec	Wave 7 • Fields and rules in Book II §16.4

4. Sales Module Screens

OM-LEAD-001/2 • Leads and Opportunities

Aspect	Implementation
Route, type, users	/sales/leads • List + detail • Sales reps own their records, Sales Director reads all
Build the layout	Pipeline board grouped by stage with value and age on every card, plus a register view with the standard filter bar. Detail with a contact block, a requirement capture panel using the same configuration attributes as the quotation (kVA rating, primary and secondary voltage, vector group, cooling class, impedance, applicable standards), a prior build lookup panel that searches the design library on those attributes, a qualification checklist and an activity timeline
Primary actions	Qualify to opportunity. Run the prior build search. Convert to quotation, which creates the prospect customer when none exists and carries every captured attribute forward. Disqualify with a mandatory reason
Arrives from	Sales sidebar, inbound web and email capture, trade contact import

Aspect	Implementation
Creates or updates	Opportunities. Prospect customers on conversion. Quotations pre filled with the captured attributes and the prior build reference so nothing is retyped
Postings and scans	None. No credit exposure exists before a quotation
Build wave and field spec	Wave 2 • Fields and rules in This Book §4.1

4.1 Lead and Opportunity Object, Field Rules

The lead is the first formal record of demand and precedes the customer master. It must let sales answer the repeat build question before engineering spends a single hour. The field rules below are the complete authority for this object.

Field Group	Fields and Rules
Identity	lead_id auto assigned from the LD number series. Source dropdown with the values Referral, Trade Show, Website, Direct, Agent, Other. Owner defaults to the creator and reassigns only by the Sales Director. Region from the shared region table
Contact	Company name mandatory, contact name, title, email, phone, address block. Duplicate check on save against company name and email domain across leads and customers, same dialog pattern as the material duplicate check
Requirement	The quotation configuration attribute set, every field optional at this stage: kVA rating, primary voltage, secondary voltage, phases, frequency, vector group, cooling class, impedance percent, tap arrangement, applicable standards, special requirements text. Values entered here flow to the quotation unchanged
Prior build	Read only panel populated by the prior build search. Stores the matched design reference, match score and the decision New Design, Variant of Existing or Repeat Build. The decision copies to the engineering request when the order later confirms
Commercial	Estimated value, estimated award date, competitor field, win probability percent defaulted by stage and manually adjustable, currency from the entity
Stage	Open, Qualified, Quoting, Won, Lost, Disqualified. Open to Qualified requires the qualification checklist complete. Qualified to Quoting happens automatically at quotation creation. Won and Lost set only by the win or loss recording on the linked quotation and never by hand. Disqualified requires a reason code

- Conversion must create the customer with status Prospect, no credit limit and no financial views until Finance activates it through the normal customer master path. A prospect customer must not be selectable on a sales order.
- The pipeline report must show count and value by stage, conversion rate stage to stage, win rate by source and by product class, and average age per stage. The existing Quote Funnel report gains the lead stages ahead of its first column and reads from this object.
- Validation register. V-OM-40 duplicate warning on company and domain. V-OM-41 stage transitions only along the defined path. V-OM-42 conversion blocked while mandatory contact fields are empty. E-OM-43 hard stop on deleting a lead that owns a quotation, disqualify instead.

Build Note: *Linkage. The quotation created from a lead carries the lead reference for funnel reporting. The prior build panel reads the same design library index that ENG-REQ-001 uses, so sales and engineering answer the repeat question from one source. Installed base records surface in the prior build search when the searching attribute set matches a shipped unit.*

OM-CUST-001/2 • Customer Master

Aspect	Implementation
Route, type, users	/sales/customers • List + detail • Sales edits commercial, Finance edits credit
Build the layout	Detail tabs for general, addresses, contacts, commercial, credit, tax, history, documents. Credit tab renders read only for sales roles
Primary actions	Create. Activate with address validation. Credit hold toggle for Finance
Arrives from	Sales sidebar, quote and order headers
Creates or updates	The customer every sales document requires
Postings and scans	None
Build wave and field spec	Wave 2 • Fields and rules in Book I B §2

OM-QT-001/2 • Quotations

Aspect	Implementation
Route, type, users	/sales/quotations • List + detail • Sales reps, estimators on ETO
Build the layout	Line grid with price elements, ETO configuration panel that reads like a datasheet, margin panel against the linked estimate, version chain
Primary actions	Issue PDF. Revise to new version. Convert to order. Record win or loss with reason
Arrives from	Customer detail, pipeline lists
Creates or updates	Sales orders on conversion with frozen prices. Engineering estimates linked on ETO
Postings and scans	None. Tax estimate on save
Build wave and field spec	Wave 2 • Fields and rules in Book I B §3, Book IV §5

OM-SO-001/2 • Sales Orders with Schedule Lines

Aspect	Implementation
Route, type, users	/sales/orders • List + detail • Sales admin, CS, Finance for credit actions
Build the layout	Header, line grid, expandable schedule lines per line, readiness chips for supply, credit, quality and documents, milestone strip on ETO, price waterfall per line, pegging panel
Primary actions	Submit runs credit and price checks. Confirm runs ATP or CTP per branch. Change order dialog after confirm. Cancel with fee logic
Arrives from	Quotation conversion, customer detail, direct entry
Creates or updates	Reservations for MTS. Engineering requests for ETO. Planned demand for MTO. Invoice requests per billing plan. Revenue contract on milestone plans
Postings and scans	None directly. Everything downstream pegs here
Build wave and field spec	Wave 2 • Fields and rules in Book I B §4, Book IV §2.1

OM-BO-001 • Backorder Workbench

Aspect	Implementation
Route, type, users	/sales/backorders • Workbench • Planners, sales admin, Sales Director approves reallocations
Build the layout	Exception cards per shortage with supply options, priority class columns, mass re promise run panel after MRP
Primary actions	Reallocate with dual audit. Re promise. Split. Substitute per approved alternates
Arrives from	Schedule line backorder flags, MRP run completion
Creates or updates	Updated confirmations and promises on schedule lines
Postings and scans	None
Build wave and field spec	Wave 7 • Fields and rules in Book IV §2.2

OM-DLV-001 • Delivery Cockpit

Aspect	Implementation
Route, type, users	/sales/deliveries • Workbench • Shipping office
Build the layout	Status pipeline columns from Planned to Closed, readiness flags per delivery, split and merge rule explanations on grouping, load planning side panel
Primary actions	Run delivery creation. Release to warehouse. Assign carrier and equipment. Print documents. Post goods issue when green
Arrives from	Confirmed schedule lines due inside the horizon
Creates or updates	Deliveries, picklists, packing orders, shipment documents, invoice requests at PGI
Postings and scans	PGI posts templates #25, #26, #27. Loading scans SP-18
Build wave and field spec	Wave 7 • Fields and rules in Book I B §5, Book IV §2.3, §2.4

OM-PACK-* • Packing Queue and Pack Station

Aspect	Implementation
Route, type, users	/sales/packing • Queue + kiosk • Packers
Build the layout	Kiosk with left panel of items to pack, right panel HU tree, center scan input with focus lock, scale readout
Primary actions	Create HU. Scan items in. Close HU with weight sanity. Complete pack order. Reopen with reason
Arrives from	Completed picklists
Creates or updates	Handling units with labels, packing list, shipment assignment
Postings and scans	Packaging backflush #27 at HU close. Scans SP-16, SP-17

Aspect	Implementation
Build wave and field spec	Wave 2 • Fields and rules in Book I A §5

OM-POD-001 • POD and Claims

Aspect	Implementation
Route, type, users	/sales/pod • List + detail • Shipping office, CS on claims
Build the layout	Expected POD queue with aging, claim detail with type, value, carrier reference and resolution path
Primary actions	Record POD. Open claim. Resolve through credit memo request, redelivery or repair paths
Arrives from	PGI creates the expected POD row
Creates or updates	Delivery closure. Final milestone trigger. Claim resolutions through existing credit and RMA paths
Postings and scans	Freight claim recovery per Book I A §3.9 when applicable
Build wave and field spec	Wave 7 • Fields and rules in Book IV §2.4

5. Returns, Warranty and Service Screens

RMA-LIST/CREATE • RMA Register and Authorization

Aspect	Implementation
Route, type, users	/sales/rma • List + detail • CS creates, managers approve per matrix
Build the layout	Register with class badges and SLA aging. Create form pulls the reference document and fills lines with shipped serials and prices. Class RC3 exposes oil and transport fields
Primary actions	Submit for approval. Print return instructions with barcode. Email customer. Cancel before receipt
Arrives from	Customer detail, invoice or serial lookup, POD claims
Creates or updates	Authorized RMA with barcode. Warranty adjudication task downstream
Postings and scans	None until receipt
Build wave and field spec	Wave 5 • Fields and rules in Book I A §3.3, §3.4

RMA-RECEIPT • Return Receipt

Aspect	Implementation
Route, type, users	scan driven • Kiosk • Receiving
Build the layout	Scan RMA barcode to open. Lines prefilled. Condition dropdown with mandatory photos when damaged

Aspect	Implementation
Primary actions	Receive full or partial. Book unauthorized extras with supervisor override
Arrives from	Arriving crate with the RMA label. Scan SP-19
Creates or updates	Quarantine stock, return inspections, serial status Returned
Postings and scans	Template #17 when ownership returns, none for customer property
Build wave and field spec	Wave 5 • Fields and rules in Book I A §3.5

RMA-DISP • Inspection and Disposition

Aspect	Implementation
Route, type, users	/sales/rma/{id}/disposition • Detail • Inspectors propose, Quality Manager approves
Build the layout	Findings with teardown checklist for RC3, entitlement service verdict displayed with its rule trail, per line disposition selector with outcome preview
Primary actions	Adjudicate warranty. Approve dispositions. Convert denial to billable
Arrives from	Completed return inspections
Creates or updates	Credit memo requests, rework orders, replacement orders, scrap postings, NCR and SCAR when attributable
Postings and scans	Templates #18 to #23 per outcome
Build wave and field spec	Wave 5 • Fields and rules in Book I A §3.6, Book IV §7.4

IB-001 • Installed Base

Aspect	Implementation
Route, type, users	/service/installed-base • List + detail • CS maintains, all read
Build the layout	One record per shipped serial with customer, site, dates, warranty state from the entitlement service, service history timeline
Primary actions	Record ownership transfer. Capture commissioning date. Extend warranty with approval
Arrives from	Created automatically at PGI. RMA screens read it
Creates or updates	The entitlement basis for every warranty decision
Postings and scans	None
Build wave and field spec	Wave 5 • Fields and rules in Book IV §7.4

SVC-SPF-001 • Spares Finder

Aspect	Implementation
Route, type, users	/service/spares-finder • Workbench • Customer service, service desk, spare parts sales

Aspect	Implementation
Build the layout	One search input accepting a transformer serial number, a customer name or a sales order number. Result header shows the unit from the installed base with site, commissioning date and warranty state. Below it the as built structure from the trace record rendered as an indented parts list, each line carrying current stock availability, the active spares price, the lead time from the info record when not stocked, and a supersession chip wherever the built part has since been replaced
Primary actions	Select lines and create a spares quotation in one step. Open the full trace record. Print the parts list with the serial in the footer. Filter to wear parts and safety critical items per the engineering marking
Arrives from	Customer detail, installed base record, RMA screens, direct search from the service sidebar
Creates or updates	Spares quotations through the standard OM-QT path with the transformer serial carried on every line
Postings and scans	None. The downstream quotation, order and delivery carry the money
Build wave and field spec	Wave 5 • Fields and rules in This Book §5.1

5.1 Spares Finder, Resolution Rules

- Resolution order. The serial resolves to the installed base record, the installed base record resolves to the as built genealogy, and each genealogy line resolves to the current material master state. The screen therefore shows what is inside that specific unit, not what the current BOM revision says.
- Supersession. Where a built part has a successor in the supersession register, both appear, the built part grayed with its identity intact and the successor as the default orderable line. The user may force the original where stock remains and policy allows.
- Pricing and entitlement. Prices come from the spares price list through the condition model of Book IV §5. Before quoting, the entitlement service runs. When the unit is under warranty and the requested part relates to an open or attributable failure, the screen must warn and offer the RMA path instead of a sale.
- Availability. Stocked SPARE materials show ATP. Non stocked lines show the info record lead time. Lines with neither must route to an engineering review task before the quotation can issue, so no promise is invented.
- Compliance. Lines flagged export controlled or hazardous carry their badges and block quotation release until the compliance checklist on the quotation is complete.
- Traceability forward. Every spares quotation line stores the serial reference, so the trace explorer later answers which spares were supplied to which unit, in both directions.

> **Build Note:** *Linkage. This screen composes three existing authorities and adds none. The installed base of Book IV §7.4, the trace record of Book IV §4.4 and the material, pricing and info record masters of Books II and IV. The field service module of Book IX consumes the same resolution rules for van stock and service order parts, so the logic must be built once as a service and called from both places.*

6. Engineering Module Screens

ENG-HOME-001 • Engineering Workspace

Aspect	Implementation
Route, type, users	/engineering • Workbench • Engineers, Engineering Manager
Build the layout	Panels for my design queue, open changes, awaiting my approval, recent releases, exceptions such as placeholder aging and EBOM ahead banners
Primary actions	Open any queued object. Approve from the list
Arrives from	Engineering sidebar landing
Creates or updates	Navigation only
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §1

ENG-REQ-001 • Engineering Request

Aspect	Implementation
Route, type, users	/engineering/requests • Detail • Engineering Manager assigns, engineers execute
Build the layout	Configuration attributes read only from the order, deliverables checklist with per row status and links, release level indicator, due date back scheduled from the promise
Primary actions	Assign engineers. Toggle customer approval loop. Complete when the checklist is done
Arrives from	Created automatically when an ETO order confirms
Creates or updates	The design job container linking EBOM, drawings, MBOM, estimate and the sales milestone
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §2, Book IV §3.1

ENG-BOM-002 • EBOM Editor

Aspect	Implementation
Route, type, users	/engineering/boms/{id} • Detail, three panes • Assigned engineers, manager releases
Build the layout	Left structure tree with level badges, center line grid keyboard first, right context panel with item card, drawing thumbnail, where used and alternates. Header bar with validate, compare, import, release
Primary actions	Add component or create placeholder inline. Map consume at operation. Validate. Release package
Arrives from	Engineering request checklist, CAD import proposals
Creates or updates	Released EBOM revisions. Placeholder MCRs. The basis for MBOM transform
Postings and scans	None

Aspect	Implementation
Build wave and field spec	Wave 3 • Fields and rules in Book I C §3

ENG-ECR/ECO • Change Requests and Change Orders

Aspect	Implementation
Route, type, users	/engineering/change-* • Detail pair • Anyone raises ECR, CCB decides, Engineering Manager releases ECO
Build the layout	ECR shows the computed impact panel with open orders, stock, POs and cost delta simulation. ECO shows the task checklist, effectivity selector, old stock and open order disposition grids
Primary actions	Submit ECR. Vote as CCB. Release ECO atomically at effectivity
Arrives from	Any object Propose Change action, quality NCRs, supplier deviations
Creates or updates	New revisions across EBOM, MBOM, routings, documents and plans in one cutover. MCH rows and supersessions where materials change. LLR commercial decisions when long lead items are touched
Postings and scans	Disposition outcomes post through their own documents
Build wave and field spec	Wave 3 • Fields and rules in Book I C §4, Book IV §3.2

MBOM-TRF • MBOM Transform

Aspect	Implementation
Route, type, users	wizard from a released EBOM • Wizard • Manufacturing engineer
Build the layout	Steps for phantom resolution preview, operation mapping validation, manufacturing additions, scrap review, release
Primary actions	Resolve unmapped lines. Release which sets the costing invalidation flag
Arrives from	Released EBOM revisions
Creates or updates	Active MBOM revisions feeding production versions
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §6

PV-001 • Production Versions

Aspect	Implementation
Route, type, users	/engineering/production-versions • List + detail • Engineering Manager releases
Build the layout	Grid per material and plant with MBOM revision, routing revision, lot range, validity and overlap validation
Primary actions	Pair and release. Supersede through ECO cutover
Arrives from	MBOM and routing releases

Aspect	Implementation
Creates or updates	The selectable build definition every WO and MRP explosion uses
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book IV §6.1

6.1 Sales Kit BOM

Accessory kits sold as one catalog line, commissioning kits, installation hardware sets, recommended spares packages, must be formal BOM objects, not order notes. The rules below are the complete authority for the sales kit and they reuse the existing editors and postings without exception.

Aspect	Rule
Kit header	A material of type KIT. Non stocked and non valuated by default, so the header is a commercial wrapper. A stocked variant is permitted for prepacked kits, in which case the header behaves as a normal finished material and the rules below do not apply
Kit structure	A BOM with usage Sales, single level, components only, no phantoms and no nested kits. Maintained in the EBOM editor with the usage selector set to Sales and released through the same review path as any BOM revision
Order entry	Entering the kit on a quotation or order explodes it into component sub lines under the kit line. Pricing defaults to the header through the condition model, with a component priced option per price list where the commercial team wants itemized kits. The margin panel evaluates the kit as one unit either way
Availability	ATP runs on the components, never on the non stocked header. The kit line confirms at the date the last component confirms
Delivery and packing	Components pick individually. The pack station groups them under one handling unit and the packing list prints the kit header with its components indented. Serialized components scan at SP-16 against the kit line so genealogy holds
Postings	Cost of goods posts per component at PGI through template #25. The non stocked header never carries value, so no kit level valuation, variance or count exists
Returns	RMA lines are raised at component level and carry the kit and order reference, so a partial kit return is normal processing
Costing view	A report only rollup shows the kit standard as the sum of component standards for margin analysis. No cost estimate object is created for a non stocked header

> **Build Note:** *Linkage. Where used from any component lists its kit memberships. The recommended spares package for a transformer design is itself a sales kit referenced from the engineering request deliverables checklist, which connects this object to the Spares Finder in §5.1 of this Book.*

7. CAD and Document Control Screens

ENG-DOC-001/2 • Controlled Documents

Aspect	Implementation
Route, type, users	/engineering/documents • List + detail • Engineers author, manager releases

Aspect	Implementation
Build the layout	Register with type and security class filters. Detail with revision history, native and rendition files, title block metadata, linked objects, check out state
Primary actions	Check out and in. Submit for review with redlines. Release which stamps the rendition. Obsolete with where used warning
Arrives from	Engineering sidebar, material documents tab, BOM line drawing links
Creates or updates	Released renditions the floor and customers consume
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §5

TR-001 • Transmittals

Aspect	Implementation
Route, type, users	/engineering/transmittals • Wizard + register • Engineers, document control
Build the layout	Document picker restricted to released revisions, recipient and purpose fields, response capture with the three customer outcomes
Primary actions	Send package. Record response which drives the design approval milestone
Arrives from	Engineering request checklist, document register
Creates or updates	Customer approval evidence on the ETO order
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §5.2

CAD-SYNC • CAD Sync Monitor and Conflict Queue

Aspect	Implementation
Route, type, users	/engineering/cad-sync • Workbench • Engineering IT, engineers resolve conflicts
Build the layout	Vault event stream with retry states, conflict cards showing both values side by side with the owning role highlighted, publish failure queue
Primary actions	Resolve conflict per field ownership. Replay failed events. Open the linked object
Arrives from	Connector events per IF-17
Creates or updates	Synchronized attributes, published renditions, EBOM proposals in the import wizard
Postings and scans	None
Build wave and field spec	Wave 3 with the connector • Fields and rules in Book III G §3

CAD-IMP • CAD BOM Import

Aspect	Implementation
Route, type, users	wizard inside the EBOM editor • Wizard • Engineers
Build the layout	Mapped proposal grid with match states per line, unmatched rows offered as placeholder proposals, transactional commit with error log
Primary actions	Map columns. Accept matches. Commit all or nothing
Arrives from	Vault assembly extraction or CSV upload
Creates or updates	EBOM lines and placeholder MCRs
Postings and scans	None
Build wave and field spec	Wave 3 • Fields and rules in Book I C §3.3, Book III G §3

8. Planning Module Screens

PP-MRP-001 • MRP Workbench

Aspect	Implementation
Route, type, users	/production/mrp • Workbench • Planners, buyers see their exceptions
Build the layout	Exception cards EX-01 to EX-09 grouped by type with counts and aging, proposal grid with pegging drill, run history with input snapshot IDs
Primary actions	One click actions per exception type. Firm proposals. Mass convert with preview
Arrives from	Nightly run completion, planning sidebar
Creates or updates	Production order drafts and purchase requisitions with pegs preserved
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book I D §3, Book IV §6.2

PLO-001 • Planned Orders

Aspect	Implementation
Route, type, users	/production/planned-orders • List • Planners
Build the layout	Grid with firmed flag, pegs, conversion state
Primary actions	Firm. Convert. Cancel proposals
Arrives from	MRP workbench
Creates or updates	Production orders and PRs on conversion
Postings and scans	None

Aspect	Implementation
Build wave and field spec	Wave 4 • Fields and rules in Book I D §3.3

SCH-BOARD • Schedule Board

Aspect	Implementation
Route, type, users	/production/schedule • Workbench • Master scheduler
Build the layout	Finite buckets per work center with load bars, freeze horizon shading, sandbox mode banner, publish diff dialog
Primary actions	Reschedule inside guardrails. Open what if sandbox. Publish with diff preview
Arrives from	Planning sidebar, at risk records
Creates or updates	Published schedules with snapshot IDs that promises reference
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book III E §3

CTP-DLG • Promise Dialog

Aspect	Implementation
Route, type, users	dialog on quotes and orders • Dialog • Sales with planner visibility
Build the layout	Feasible date result with the binding constraint named in words, alternative options, soft reservation countdown
Primary actions	Accept which reserves capacity. Decline. Request planner review
Arrives from	Quote lines and order confirmation for MTO and ETO
Creates or updates	Soft capacity reservations converting to firm load at confirmation
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book III E §4

AT-RISK • At Risk Promises

Aspect	Implementation
Route, type, users	/production/at-risk • Workbench • Scheduler and sales owner jointly
Build the layout	Cards per endangered promise with recovery options and decision log
Primary actions	Choose recovery. Trigger the sales change order path when a re promise is unavoidable
Arrives from	Automatic revalidation events
Creates or updates	Recovery decisions logged on the order

Aspect	Implementation
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book III E §5

9. Production Module Screens

PP-WC-001/2 • Work Centers

Aspect	Implementation
Route, type, users	/production/work-centers • List + detail • Manufacturing engineering, Controlling links rates
Build the layout	Capacity model fields, cost center link, batch resource settings, alternate work center
Primary actions	Maintain capacity. Set maintenance status which warns the scheduler
Arrives from	Production sidebar, routing rows
Creates or updates	The capacity and rate basis for scheduling and costing
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book I D §2

PP-RTG-001/2 • Routings

Aspect	Implementation
Route, type, users	/production/routings • List + detail • Manufacturing engineering
Build the layout	Operation grid with times, QC checkpoint toggles, work instruction links, component mapping view
Primary actions	Author. Release through the review lifecycle. Change through ECO
Arrives from	Engineering releases, production versions
Creates or updates	Standard times feeding costing and the schedule
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book I D §4

PP-WO-001/2 • Production Orders

Aspect	Implementation
Route, type, users	/production/orders • List + detail • Scheduler, supervisors, cost accounting reads
Build the layout	Header with frozen revisions and serial, release gate checklist with red and green rows, operations tab, materials tab, cost tab with live WIP, genealogy tab

Aspect	Implementation
Primary actions	Release when gates pass or short with the shortage flag. Print traveler. Hold. Close which runs settlement checks
Arrives from	Planned order conversion, ETO design release
Creates or updates	Dispatch entries, reservations firmed, serial records, traveler documents
Postings and scans	Postings happen through terminal confirmations and receipt, not here
Build wave and field spec	Wave 4 • Fields and rules in Book I D §5

PP-DSP-001 • Dispatch List

Aspect	Implementation
Route, type, users	/production/dispatch • Workbench • Supervisors
Build the layout	Per work center queue in priority order with material ready flags and hold markers
Primary actions	Resequence within guardrails. Assign crew. Open terminal view
Arrives from	Released orders, schedule publishes
Creates or updates	The operator queue the kiosk shows
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book I D §6

PP-SFT-001 • Shop Floor Terminal

Aspect	Implementation
Route, type, users	/production/terminal • Kiosk • Operators by badge, supervisors override
Build the layout	Full screen queue, big start and complete buttons, per operation component list with scan states, capture template fields, QC gate button, hold and andon, offline banner with queue depth
Primary actions	Badge in SP-07. Start SP-10. Scan materials SP-08. Install child serials SP-12. Complete with quantities SP-11. Request inspection. Hold
Arrives from	Dispatch queue at the station
Creates or updates	Confirmations, genealogy events, rework orders on rework quantities
Postings and scans	Templates #28, #29, #30 per confirmation events
Build wave and field spec	Wave 4 • Fields and rules in Book I D §6

PP-WIP-001 • WIP Board and Valuation

Aspect	Implementation
Route, type, users	/production/wip • Board + report • Supervisors, cost accounting
Build the layout	Kanban mirror by work center plus the valuation report reconciling to the GL balance
Primary actions	Drill to orders. Export the reconciliation for close step 6c
Arrives from	Production sidebar, close checklist
Creates or updates	Reconciliation evidence
Postings and scans	None
Build wave and field spec	Wave 4 • Fields and rules in Book I D §7

RWO-001 • Rework Orders

Aspect	Implementation
Route, type, users	/production/rework • List + detail • Supervisors, quality on retest gates
Build the layout	Free form operation list, ad hoc materials, settlement receiver banner naming where costs will land
Primary actions	Add operations. Complete with mandatory re inspection when the origin demands it
Arrives from	Terminal rework quantities, test failures, NCR dispositions, RMA repairs
Creates or updates	Repair cost collection in 1325 flowing to the receiver on close
Postings and scans	Templates #21, #22, #33 per origin at settlement
Build wave and field spec	Wave 4 • Fields and rules in Book I D §9

PP-SETTLE-001 • Settlement Run

Aspect	Implementation
Route, type, users	/production/settlement • Wizard • Cost accountant, Controller reopens
Build the layout	Eligible order list, per order variance preview split by cause, post step with journal links
Primary actions	Preview. Post which drives every order WIP balance to zero
Arrives from	Close calendar, order close actions
Creates or updates	Variance journals and closed orders
Postings and scans	Template #32 with the split of Book I A §8
Build wave and field spec	Wave 4 • Fields and rules in Book I D §8, Book I I §4

10. Quality Module Screens

QM-DASH-001 • Quality Dashboard

Aspect	Implementation
Route, type, users	/quality/dashboard • Dashboard • Quality team
Build the layout	Widgets for open inspections by gate, first pass yield, NCR funnel, vendor quality, cost of quality, hold inventory
Primary actions	Drill to each queue
Arrives from	Quality sidebar landing
Creates or updates	Navigation only
Postings and scans	None
Build wave and field spec	Wave 5 • Fields and rules in Book I E §2

QM-PLAN-001/2 • Inspection Plans

Aspect	Implementation
Route, type, users	/quality/inspection-plans • List + detail • Quality engineers author, manager approves
Build the layout	Header with type, scope and sampling settings, ordered characteristics grid, revision compare view
Primary actions	Author. Approve to Active. New revision. Copy plan. Where used
Arrives from	Quality sidebar, material quality view preview
Creates or updates	The test rows every inspection materializes
Postings and scans	None
Build wave and field spec	Wave 2 config, Wave 5 full • Fields and rules in Book I A §6

QM-EXEC • Inspection Execution

Aspect	Implementation
Route, type, users	per gate queues • Detail • Inspectors
Build the layout	Characteristics from the plan with pass and fail coloring, instrument fields validated against calibration, sample instructions, attachment drop
Primary actions	Record results. Accept or reject with disposition routing. Create NCR
Arrives from	GRN QI rows, routing checkpoints, shipment gates, RMA receipts
Creates or updates	Permanent results on the batch or serial, stock status changes, NCRs
Postings and scans	Stock status moves per gate. Scans SP-05, SP-14
Build wave and field spec	Wave 5 • Fields and rules in INV 6.1 pattern, Book I E §1

QM-FIN-001 • Final Test and FTR

Aspect	Implementation
Route, type, users	/quality/final-test • Detail • Test technicians, Quality Manager countersigns
Build the layout	Transformer test catalog with capture templates per test, run history preserved, certified report generator
Primary actions	Record runs across sessions. Pass generates the FTR and fires the milestone. Fail routes to rework with retest scope
Arrives from	Orders at the Testing status
Creates or updates	FTR documents attached to serial, order and shipment set
Postings and scans	Milestone trigger. Scan SP-14 instrument binding
Build wave and field spec	Wave 5 • Fields and rules in Book I E §3, §4

QM-CAPA/SCAR • CAPA and SCAR

Aspect	Implementation
Route, type, users	/quality/capa and /scar • List + detail • Quality engineers, manager closes
Build the layout	CAPA with sources, root cause method, action table, effectiveness gate. SCAR in 8D structure with vendor response tracking
Primary actions	Progress the lifecycle. Enforce AVL restriction on overdue SCARs
Arrives from	NCRs, RMA failure analysis, audits
Creates or updates	Corrective actions with evidence, vendor status changes
Postings and scans	None
Build wave and field spec	Wave 5 • Fields and rules in Book I E §5, §6

QM-DEV-001 • Deviations and Waivers

Aspect	Implementation
Route, type, users	/quality/deviations • List + detail • Quality and Engineering managers approve
Build the layout	Scope selector, specification versus accepted values, validity by date or quantity, customer evidence slot, affected serial links
Primary actions	Approve with expiry. Auto expiry blocks further use
Arrives from	Inspection failures worth accepting, supplier deviation requests
Creates or updates	Traceable acceptances disclosed on as built records
Postings and scans	None

Aspect	Implementation
Build wave and field spec	Wave 5 • Fields and rules in Book IV §7.3

QM-CAL-001 • Calibration Register

Aspect	Implementation
Route, type, users	/quality/calibration • List • Calibration owner
Build the layout	Instrument grid with due date colors, certificate uploads, measurements at risk report
Primary actions	Record calibrations. Out of service toggle which interlocks tests
Arrives from	Quality sidebar, test screens validate against it
Creates or updates	The interlock every instrument scan checks
Postings and scans	None
Build wave and field spec	Wave 5 • Fields and rules in Book I E §7

RCL-001 • Recall Campaigns

Aspect	Implementation
Route, type, users	/quality/recalls • Workbench • Quality Manager, executive approves notifications
Build the layout	Population panel from forward trace with version history, containment status, notification list export, recovery tracker, effectiveness gate
Primary actions	Derive population. Mass hold. Approve notifications dual. Track recovery to one hundred percent accounted
Arrives from	Suspect batches, field failures, supplier notices
Creates or updates	Mass holds, recall class RMAs, evidence dossier
Postings and scans	Through the RMAs it creates
Build wave and field spec	Wave 5 • Fields and rules in Book III A FL-08, Book IV §4

11. Purchasing Module Screens

PUR-WB-001 • Buyer Workbench

Aspect	Implementation
Route, type, users	/purchasing/workbench • Workbench • Buyers
Build the layout	PR queue by buyer code, MRP exceptions for buy items, unconfirmed PO list, source gaps
Primary actions	Source and group PRs into POs. Jump to RFQ where policy demands

Aspect	Implementation
Arrives from	Purchasing landing, MRP outputs
Creates or updates	POs from grouped PRs
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I G §1

PUR-VEND-001/2 • Vendors and AVL

Aspect	Implementation
Route, type, users	/purchasing/vendors • List + detail • Buyers, Quality on qualification
Build the layout	Operational tabs with compliance panel, AVL grid with rank, qualification, FAI and quality rating
Primary actions	Onboard through the workflow. Restrict or disqualify with cascade to sources
Arrives from	Purchasing sidebar, PO headers
Creates or updates	The vendor eligibility every source decision reads
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I G §2

PUR-PR/RFQ • Requisitions and RFQs

Aspect	Implementation
Route, type, users	/purchasing/requisitions and /rfqs • List + detail • Buyers, Engineering on technical scoring
Build the layout	PR grid with pegs and budget results. RFQ with sealed responses, technical pass fail before commercial opening, landed cost comparison grid
Primary actions	Approve expense PRs. Issue RFQ with STP attachments. Award which writes info records
Arrives from	MRP, manual demand, FL-10 stage 8
Creates or updates	Awarded price bases, POs on award
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I G §3, §4, Book IV §3.4

PUR-PO-001/2 • Purchase Orders

Aspect	Implementation
Route, type, users	/purchasing/orders • List + detail • Buyers within limits, ladder approves

Aspect	Implementation
Build the layout	Header with quality clauses and Buy America state, line grid with delivery schedules, STP version chip, LLR flag, revision history
Primary actions	Approve per ladder. Send. Capture confirmation and STP acknowledgement. Revise with change summary
Arrives from	PR grouping, RFQ award, LLR releases
Creates or updates	The commitment receiving and matching work against
Postings and scans	None at creation. GRN and match post downstream
Build wave and field spec	Wave 6 • Fields and rules in Book I G §5, Book IV §3.3

LLR-001 • Long Lead Releases

Aspect	Implementation
Route, type, users	/purchasing/long-lead • Detail • Engineering and Purchasing managers, Controller above threshold
Build the layout	Item list with caps, risk class, exposure tracker bar, linked POs, ECO impact panel
Primary actions	Approve with exposure acceptance. Close on final release or full disposition
Arrives from	FL-10 stage 5 decisions on ETO orders
Creates or updates	Authority for early POs with the preliminary revision flag
Postings and scans	Exposure feeds ETO committed cost
Build wave and field spec	Wave 6 • Fields and rules in Book IV §3.2

PUR-EXP-001 • Expedite Board

Aspect	Implementation
Route, type, users	/purchasing/expedite • Workbench • Buyers
Build the layout	Cards for reschedule and past due exceptions with pegged order exposure, action log per vendor contact
Primary actions	Record promises. Push and pull dates. Escalate
Arrives from	MRP exceptions, shortage released orders
Creates or updates	Updated supply dates feeding promise revalidation
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I G §7

12. Costing and Pricing Module

This section carries the module in one place, as a designer and builder would consume it. The picture shows how the two halves connect. Costing produces the standard that valuation and variances measure against. Pricing produces the selling price through fixed condition steps. Margin floors sit between them and margin analysis closes the loop after settlement.

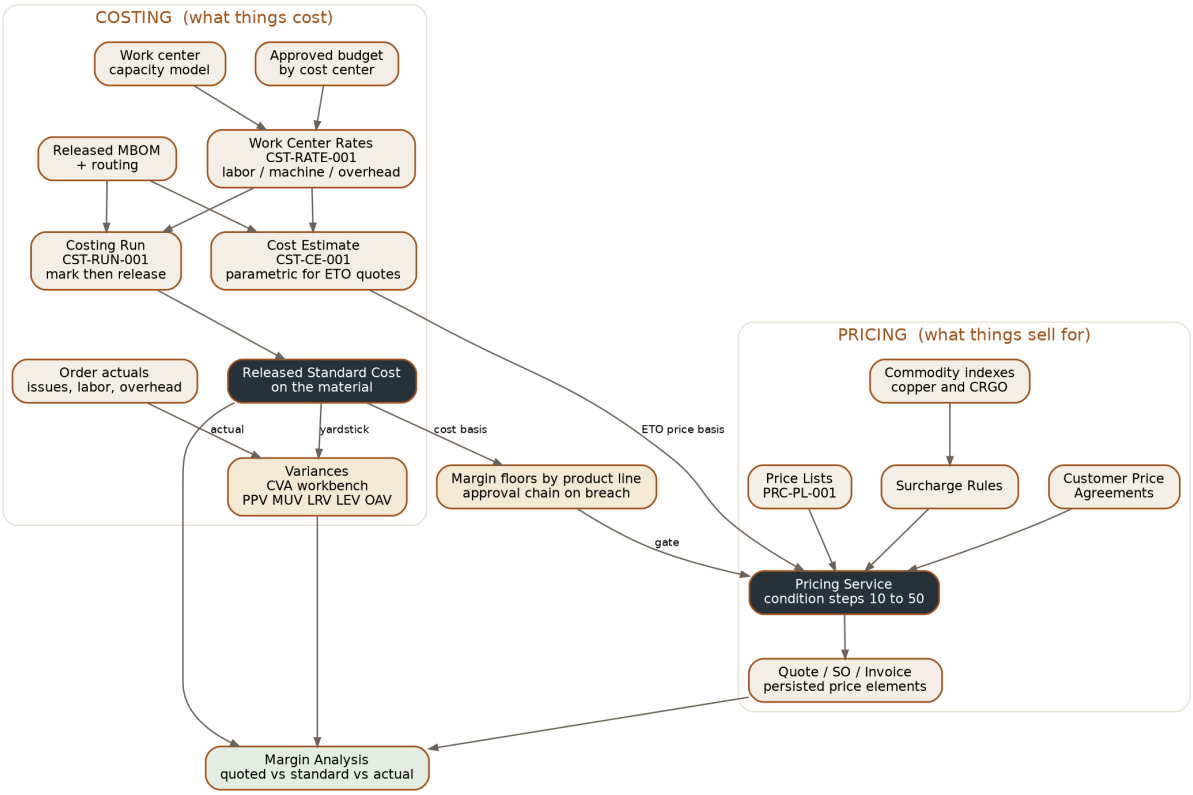


Figure V-18. The costing and pricing engine room. Left builds cost truth, right builds price truth, the floor gate and margin analysis connect them.

12.1 The Calculation Contracts in Brief

Calculation	Rule in One Line	Full Spec
Work center rates	Planned cost center money divided by planned productive hours, released yearly with the budget	Book I I §2
Standard cost rollup	Material per the MBOM with scrap, labor and overhead per the routing at released rates, rolled bottom up by low level code	Book I I §1, §3.2
ETO estimate	Driver quantities such as core and copper weight times maintained driver rates when no design exists yet	Book I I §3.1
Price determination	Condition steps 10 to 50. Base price by precedence, one discount, surcharge, freight treatment, tax. Elements persist on the document	Book I A §4.4, Book IV §5
Commodity surcharge	Index at pricing date minus base index, times content per unit, times pass through percent	Book I I §6.4
Margin floor	Product line floor percent with an escalation chain on any breach	Book I I §6.3

Calculation	Rule in One Line	Full Spec
Variance split	Actuals against standard split into purchase price, usage, labor rate, efficiency and overhead causes with signs defined	Book I A §8
Order margin three ways	Quoted against estimate, standard at PGI, actual at settlement	Book I B §6, Book IV §7.6

CST-RATE-001 • Work Center Rates

Aspect	Implementation
Route, type, users	/costing/work-center-rates • Config grid • Cost accountant drafts, Controller releases
Build the layout	Grid per work center with planned costs pulled from the budget, planned hours from capacity, computed rates, prior year and delta columns, simulation column
Primary actions	Compute. Simulate. Release with effective date
Arrives from	Annual budget cycle, Controlling reports
Creates or updates	The rates every rollup, confirmation and estimate uses
Postings and scans	None. Confirmations absorb at these rates
Build wave and field spec	Wave 6 • Fields and rules in Book I I §2

CST-CE-001 • Cost Estimates

Aspect	Implementation
Route, type, users	/costing/estimates • Detail • Estimators, cost accountant
Build the layout	Basis selector for released structures, draft structures or the parametric model, itemized cost tree with source badges, comparison view against standard and last actuals
Primary actions	Estimate. Release which makes it quotable. Supersede
Arrives from	Quote lines demand them, costing sidebar
Creates or updates	The cost basis under every firm ETO price
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I I §3.1

CST-RUN-001 • Costing Runs

Aspect	Implementation
Route, type, users	/costing/runs • Wizard • Cost accountant marks, Controller releases
Build the layout	Scope step, rollup with error list, review grid sorted by value impact with drill to itemization, two step mark and release
Primary actions	Roll. Review. Mark. Release which revalues stock

Aspect	Implementation
Arrives from	Annual cycle, weekly new item runs, MBOM invalidation flags
Creates or updates	Released standards stamped on materials
Postings and scans	Revaluation through the REVAL event key at release
Build wave and field spec	Wave 6 • Fields and rules in Book I I §3.2, Book II §5

PRC-PL-001 • Price Lists and Agreements

Aspect	Implementation
Route, type, users	/pricing/price-lists • List + detail • Pricing Manager, Controller on floor breaches
Build the layout	List headers with basis and rounding rules, rate rows with validity and margin columns, agreement container with scope and rebate terms
Primary actions	Maintain rates through dated rows. Mass update creating new rows. Approve
Arrives from	Pricing sidebar, material sales view grids
Creates or updates	The rates the pricing service resolves
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I I §6.1, §6.2

PRC-SUR-001 • Surcharge Rules and Index Values

Aspect	Implementation
Route, type, users	/pricing/surcharges • Config • Costing maintains indexes, Pricing Manager rules
Build the layout	Index series with immutable monthly values, rule rows per family with base index, content and pass through
Primary actions	Post monthly values. Maintain rules
Arrives from	Pricing sidebar, monthly commodity ritual
Creates or updates	Surcharge elements on documents and purchase side per info records
Postings and scans	Ship date adjusted contracts true up at PGI through template #38
Build wave and field spec	Wave 6 • Fields and rules in Book I I §6.4, Book IV §5

PRC-SIM • Price Simulation

Aspect	Implementation
Route, type, users	dialog and /api/v1/pricing/simulate • Dialog • Sales, pricing, audit read

Aspect	Implementation
Build the layout	Inputs for customer, item, quantity and date. Output waterfall listing every access tried, the winning rows and the floor verdict
Primary actions	Simulate. Copy result reference
Arrives from	Material sales view, quote lines, pricing arguments
Creates or updates	Explanations. The same service documents call, so simulation equals reality
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book IV §5.2

INV-CVA-001 • Cost Variance Workbench

Aspect	Implementation
Route, type, users	/inventory/valuation/cost-variance • Workbench • Cost accountant, Controller
Build the layout	Summary tiles per variance type with trends, grouped grid with drill to transaction and journal in three clicks, pareto and trend charts
Primary actions	Filter. Drill. Export the four report layouts
Arrives from	Costing sidebar, close rituals, PPV questions
Creates or updates	Explanations and the exception list for missing standards
Postings and scans	None. It reads what others posted
Build wave and field spec	Wave 6 • Fields and rules in Book I A §8

PRC-MARG • Margin Analysis

Aspect	Implementation
Route, type, users	/pricing/margin • Report set • Pricing, sales leadership, Controller
Build the layout	Quote margin register, three view order margin, price realization decomposition, floor breach register with aging
Primary actions	Drill to documents. Export
Arrives from	Pricing sidebar, monthly reviews
Creates or updates	The closed loop from quote to actual
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I I §7

13. Controlling Screens

CO-CC/PC-001/2 • Cost and Profit Center Masters

Aspect	Implementation
Route, type, users	/controlling/* • List + detail + tree • Controlling analyst drafts, Controller approves
Build the layout	Hierarchy tree editor with effective dated moves, master fields with owner and validity, where used check before deactivation
Primary actions	Maintain. Move with history. Block for posting
Arrives from	Controlling sidebar, wave 1 seed load
Creates or updates	The dimensions every posting derives
Postings and scans	None. Rule 7 fallbacks land on the exception report
Build wave and field spec	Wave 1 • Fields and rules in Book I H §1, §2

CO-ALC-001/2 • Allocation Cycles

Aspect	Implementation
Route, type, users	/controlling/allocations • Config + wizard • Analyst defines, Controller posts
Build the layout	Ordered segments with senders, receivers and drivers, statistical key figure grid, run preview with the rounding rule visible
Primary actions	Define. Preview. Post. Rerun replaces with both retained
Arrives from	Close checklist step 5b
Creates or updates	Distributed shared costs
Postings and scans	Template #36, net zero at entity level
Build wave and field spec	Wave 6 • Fields and rules in Book I H §5

CO-R1..R4 • Controlling Reports

Aspect	Implementation
Route, type, users	/controlling/reports • Report set • Managers see their centers, Controller sees all
Build the layout	Cost center report with rollups, profit center P and L with the reconciliation column, rate review, exception report
Primary actions	Drill to journal items. Export
Arrives from	Controlling sidebar, monthly reviews
Creates or updates	The spend and margin accountability views
Postings and scans	None
Build wave and field spec	Wave 6 • Fields and rules in Book I H §6

14. Traceability and Barcode Administration Screens

TRC-EXPL • Trace Explorer

Aspect	Implementation
Route, type, users	/traceability/explorer • Tree explorer • Quality, engineering, service, audit read
Build the layout	Input accepts a serial or a batch or a heat number. Tree renders both directions with evidence drill per node. Export produces the certificate dossier
Primary actions	Trace forward. Trace backward. Export dossier. Feed a recall population
Arrives from	Serial and batch links everywhere, recall campaigns
Creates or updates	Evidence packages
Postings and scans	None. Reads the append only event store
Build wave and field spec	Wave 4 • Fields and rules in Book I F §5, Book IV §4.4

TRC-DASH • Traceability Completeness Dashboard

Aspect	Implementation
Route, type, users	/traceability/completeness • Dashboard • Quality IT, quality manager
Build the layout	Coverage percent, orphan and missing event counts, delayed offline events, unresolved corrections, each with assigned owner lists
Primary actions	Assign. Resolve through the correction pair flow
Arrives from	Daily control jobs
Creates or updates	Worklists keeping the chain trustworthy
Postings and scans	None
Build wave and field spec	Wave 7 • Fields and rules in Book IV §4.3

BC-ADMIN • Label and Scan Administration

Aspect	Implementation
Route, type, users	/configuration/barcode • Config • Plant IT
Build the layout	Printer registry per station, label template versions, scan point policy toggles, reprint audit log, scan exception queue for offline conflicts
Primary actions	Register devices. Publish templates. Resolve scan exceptions
Arrives from	Configuration area
Creates or updates	The print and scan infrastructure state
Postings and scans	None

Aspect	Implementation
Build wave and field spec	Wave 4 • Fields and rules in Book I F §2, §4, Book V §8

15. Workflow and Administration Screens

WF-INBOX • Approvals Inbox

Aspect	Implementation
Route, type, users	/approvals • Workbench • Everyone with pending steps
Build the layout	Unified queue with SLA clocks, document context cards, act buttons inline, delegation banner when active
Primary actions	Approve. Reject with comments. Delegate. Recall own submissions
Arrives from	Every W.4 workflow routes here plus email actions
Creates or updates	Approval records on the immutable trail
Postings and scans	Terminal actions release the gated postings
Build wave and field spec	Wave 1 • Fields and rules in Section W, Book IV §7.7

WF-DEF • Workflow Definitions

Aspect	Implementation
Route, type, users	/configuration/workflows • Config • Process owners, Controller on posting gates
Build the layout	Catalog grid of the twenty seven workflows, versioned definition editor with condition builder and chain designer
Primary actions	Version definitions. Effective date them
Arrives from	Configuration area
Creates or updates	The rules the engine runs
Postings and scans	None
Build wave and field spec	Wave 1 • Fields and rules in Book IV §7.7

OPS-JOBS/INT • Job and Integration Consoles

Aspect	Implementation
Route, type, users	/operations/* • Workbench pair • Operations, on call
Build the layout	Job runs with SLA timers and dependency chains. Message monitor with lag, DLQ depth, correlation search, replay tools
Primary actions	Retry. Replay after fix. Escalate per severity

Aspect	Implementation
Arrives from	Alerts, on call rotation
Creates or updates	Operational recovery actions with audit
Postings and scans	None
Build wave and field spec	Wave 1 skeleton, Wave 7 hardened • Fields and rules in Book III B §4, D §5

CFG-CORE • Core Configuration Set

Aspect	Implementation
Route, type, users	/configuration/* • Config screens • Per owner column of each register
Build the layout	Number series, fiscal periods, tax determination, RMA policies, sales and production and engineering and quality settings, each per its specification
Primary actions	Maintain under change control
Arrives from	Configuration area
Creates or updates	The seed data catalog of the configuration workbook
Postings and scans	Period control gates every posting date
Build wave and field spec	Wave 1 • Fields and rules in Book I A §9, §11, INV 11.x, Book IV A-12

16. The Linkage Spine in One Table

Reading the entries above end to end shows the same chain from every angle. This table states it once, forward. Any screen not advancing this spine is either master data feeding it or a report reading it.

This Document	Creates This Next	Money and Identity Effects
Lead	Opportunity, then quotation on qualification	No money. Prospect customer born without credit exposure. Prior build decision attached
Quotation	Sales order on win	Prices frozen as elements. Estimate linked on ETO
Sales order schedule line	Reservation, engineering request, planned order or PR per branch	Credit exposure. Peg identity everything downstream carries
Engineering request	EBOM, drawings, transmittals	Design approval milestone money on ETO
Released EBOM	MBOM through transform, production version	Costing invalidation flag
MCR	Active material	Valuation class and profit center set before first use
PR	RFQ or PO per source policy	Budget check on expense demand

This Document	Creates This Next	Money and Identity Effects
PO	Vendor confirmation, GRN	Commitment. LLR exposure when preliminary
GRN	QI inspection, stock, match basis	Templates for stock and GR/IR. Batch and heat identity born. Scans SP-01 to 03
Production order	Confirmations, genealogy, FG receipt	WIP charges #28 #29, scrap #30, receipt #31, settlement #32. Serial born at release
Delivery	Pick, pack, PSI, PGI	COGS #25, freight #26, packaging #27. Serial to customer at SP-18
Invoice request	Invoice, cash application	Revenue and tax per the external accounting volume
RMA	Receipt, inspection, disposition documents	Templates #17 to #24. Serial history continues, never restarts
Recall campaign	Holds, notifications, recall RMAs	Population accounted to one hundred percent before closure

Build Note: *Build discipline for this Book. Each entry becomes one epic in the backlog with its wave as the target increment. The Arrives from and Creates rows become the integration test list for that epic. The referenced section is the acceptance authority. When an entry and its referenced section ever disagree, the referenced section wins and the entry is corrected through change control.*

BOOK VII

GO-LIVE READINESS SUPPLEMENT

Gap Register Closure Crosswalk • Supplemental Screen Entries • Multi Plant Planning Setup • Build Package Manifest
Data Migration Package • Complete Barcode Sample Book • Go Live Readiness Statement and Execution Directive

Book VII closes the final readiness findings. It proves, bullet by bullet, that the nine items in the Post Inventory Design Review gap register are specified in this document and shows exactly where. It adds the supplemental screen entries those items were missing in the implementation plan. It specifies multi plant planning with the setup screens the single plant model lacked. It records the build package manifest decision and the manifest itself. It describes the delivered migration workbook with its cleansing rules, transformation logic, opening balance rules and reconciliation scripts. It completes the barcode sample book with every label in the system and a usage matrix stating where and when each is printed and scanned. It ends with the go live readiness statement and the execution directive. Nothing earlier in the document is removed or altered.

1. Findings and Dispositions

Finding Raised	Disposition
MRP, finite planning and CTP specified for a single plant only, with no place to set up more	Closed in §4. The plant master, transfer lanes, stock transport orders and network planning screens are specified design ready now. The network engine activates per DEC-07 with its test packs, and the setup path exists from day one
Not design ready for every screen	Book VI carries 78 entries. The six screens it covered only inside bundles now have their own entries in §3, so every navigation item resolves to a named entry. Wireframes themselves are Figma artifacts accepted at gate DG-2, which no text document can substitute
Not standalone, external Inventory and Accounting volumes	Closed in §5 by the controlled build package manifest, option two of the finding. The rationale for not merging the released volumes is recorded there
Missing source to target workbooks, cleansing rules, transformation logic, opening balance rules and reconciliation scripts	Closed in §6. The migration workbook is delivered as a separate controlled file with fourteen sheets, and §6 governs it
Not SAP equivalent in every module	The capability map in Book V §2 and the DEC register in Book IV §1.2 record exactly which capabilities ship at go live and which are phased. The finding itself lists those rows correctly. Equivalence claims follow the recommended submission language and nothing else
All barcoding samples missing, where and when used, what should be there	Closed in §7. Nine label samples now render in this document, Figures V-15 and V-16 in Book V and V-19 through V-25 here, with the usage matrix stating where each is printed, when, what it carries and which scan points read it
The nine gap register items still not covered	Answered in §2. Every bullet of the designers gap document maps to a Book I Part A section that specifies it in full. The crosswalk makes the coverage checkable line by line

2. Gap Register Closure Crosswalk

The designers gap document of June 2026 lists nine features and, for each, the specific missing details. Book I Part A was written as the closure of that register and the coverage matrix in Book I §1.1 records it. The crosswalk below goes one level deeper and answers every identified gap bullet with its exact section. A reviewer can open each reference and find the demanded content, not a summary of it.

Item	The Gaps the Designers Named	Where Each Is Specified
1. Return from Customer	End to end flow. Screen and field definitions with roles. Errors and exceptions including partial, non stock and no reference returns	Book I A §3.2 flow. §3.3 to §3.7 screens and fields. §3.8 lifecycle. §3.9 exception register covering partial, non stock and no reference paths. §3.10 roles. Flow closure FL-05
2. Sales Tab	Field descriptions, actions and pricing rules. Inheritance and override logic with validations. Downstream document interaction	Book I A §4.1 fields. §4.2 and §4.3 price list and customer price grids with validations V-SAL-02 and 03. §4.4 the price determination contract with override display and repricing rules. Book IV §5 the condition model
3. Packing	Process flows, states, statuses. Label and packing list requirements. Field definitions and integration points	Book I A §5.1 HU model with statuses. §5.2 and §5.3 screens and fields. §5.4 labels and packing list content plus the shipment, inventory, invoicing and carrier integrations. Sample Figure V-15
4. Inspection Plan	Configuration models with types, statuses, versioning. Business rules, approvals, sampling. No plan exception handling	Book I A §6.1 types and triggers. §6.2 header with sampling incl. ANSI Z1.4 switching. §6.3 characteristics. §6.4 lifecycle and approval. §6.5 the no plan policy options
5. Count Documents	Lifecycle and statuses. Role based access and reconciliation logic. Tolerances, approvals, posting rules	Book I A §7.2 lifecycle. §7.3 roles with server side blind enforcement. §7.4 reconciliation incl. interim movements and third count rule. §7.5 tolerance bands. §7.6 posting rules with the journal entries
6. Cost Variance Analysis	Formulas and drill downs. Report layouts and postings by type. Edge cases such as missing cost and back dated rows	Book I A §8.1 all eight formulas with sign convention. §8.2 workbench with three click drill. §8.3 four report layouts. §8.4 edge cases covering exactly the named scenarios
7. Fiscal Period Control	Open and close sequences. Module behaviors and back posting. Roles and error messages	Book I A §9.2 the module behavior matrix. §9.3 the open close sequence. §9.4 back posting policy. §9.5 permissions and the E-PER message catalog
8. Safety Stock	Interaction with planning runs. Exceptions, alerts, lead times. Overrides per warehouse or customer	Book I A §10.2 planning arithmetic incl. MRP floor behavior and lead time interplay. §10.3 warehouse, customer buffer and seasonal overrides. §10.4 alerts and exception reports
9. Tax Configuration	Determination logic and region handling. Rate maintenance and customer or vendor profiles. Retroactive change behavior	Book I A §11.1 the determination precedence. §11.2 configuration fields. §11.3 the profiles. §11.4 nexus and regions. §11.5 rate maintenance and the retroactivity doctrine

Build Note: *This crosswalk must travel with the design kickoff package so the gap register reads as closed evidence rather than open work.*

3. Supplemental Screen Entries

Book VI covered these six inside larger bundles. Each now has its own entry so every sidebar item resolves to a named implementation entry.

INV-RETURN • Return from Customer (transaction stub, superseded)

Aspect	Implementation
Route, type, users	/inventory/transactions/return • Redirect • All
Build the layout	No standalone screen ships. The navigation item opens the RMA register filtered to Awaiting Receipt. A banner explains that customer material enters only through an authorized RMA
Primary actions	Open RMA register. Start an RMA
Arrives from	Legacy navigation habit, inventory sidebar
Creates or updates	Nothing. The controlled path creates everything
Postings and scans	The old direct journal is superseded by template #17 inside the RMA receipt
Build wave and field spec	Wave 5 • Fields and rules in Book I A §3, decision recorded in §3.5

INV-ITEM-SALES • Material Master Sales Tab

Aspect	Implementation
Route, type, users	/inventory/items/{id} Sales view • Detail view • Sales Admin edits, Pricing Manager approves
Build the layout	Three blocks. Sales data fields, price list assignment grid with margin columns, customer specific price grid. Price waterfall simulation button
Primary actions	Add assignments and customer prices as dated rows. Mass update creating new rows. Simulate
Arrives from	Material detail tab bar, price list detail
Creates or updates	The rates the pricing service resolves at steps of the condition model
Postings and scans	None. Below floor rows route approval per W.4
Build wave and field spec	Wave 2 • Fields and rules in Book I A §4, Book II §3.12

INV-COUNT-DOC • Count Documents

Aspect	Implementation
Route, type, users	/inventory/counting/count-documents • List + detail + mobile entry • Counters restricted, supervisor reconciles, Finance approves bands
Build the layout	Register with lifecycle badges. Detail with line grid, interim movements column, recount flags. Mobile entry shows no expected quantities on blind plans, enforced server side
Primary actions	Assign counters. Enter counts SP-20. Force recount. Reconcile. Approve per tolerance bands. Post
Arrives from	Generate Count Sheets on a count plan
Creates or updates	Stock adjustments posted per approved variances, plan completion
Postings and scans	Surplus and shortage entries per Book I A §7.6. Scan SP-20

Aspect	Implementation
Build wave and field spec	Wave 2 • Fields and rules in Book I A §7

INV-PLAN-SAFETY • Safety Stock Maintenance

Aspect	Implementation
Route, type, users	/inventory/planning/safety-stock • Config grid • Planners edit, Planning Manager approves large deltas
Build the layout	Grid per material and warehouse with inherited defaults grayed, customer buffer rows, seasonal multiplier columns, reserved suggested columns for the statistical phase
Primary actions	Edit inline with reason codes. Deltas above a quarter route approval
Arrives from	Material planning view, planning sidebar
Creates or updates	The floor MRP protects and the projection line reports draw
Postings and scans	None. Breach alerts per W.2
Build wave and field spec	Wave 2 • Fields and rules in Book I A §10

INV-CONFIG-PERIOD • Fiscal Period Control

Aspect	Implementation
Route, type, users	/inventory/configuration/fiscal-periods • Config • Controller, dual control with CFO on reopen
Build the layout	Period grid per entity with the three status presentation, pre close validation list with document links and owners, module behavior matrix reference panel
Primary actions	Soft close. Run inventory close. Hard close after checklist. Reopen dual control with tagged postings
Arrives from	Close calendar, configuration sidebar
Creates or updates	The posting date gate every transaction validates against
Postings and scans	Closing journal at inventory close. E-PER messages govern rejections
Build wave and field spec	Wave 1 • Fields and rules in Book I A §9

INV-CONFIG-TAX • Tax Configuration

Aspect	Implementation
Route, type, users	/inventory/configuration/tax • Config • Finance, Controller approves manual rates
Build the layout	Engine mode selector, default code panel, trigger point checkboxes, nexus register grid with the economic threshold monitor, manual rate table grayed in engine mode
Primary actions	Maintain nexus rows. Review reconciliation monthly. Manual mode edits as dated rows with approval
Arrives from	Configuration sidebar, tax questions from any document

Aspect	Implementation
Creates or updates	The determination inputs every taxable line resolves through
Postings and scans	None here. Documents commit tax per the precedence of Book I A §11.1
Build wave and field spec	Wave 1 • Fields and rules in Book I A §11

4. Multi Plant Planning, Setup and Operation

The single plant model was a delivery decision, never a design ceiling. This section specifies the objects and screens that make the network real, so the second plant is configuration plus testing, not development. The screens below build in wave 4 alongside their single plant siblings. The network engine behaviors activate when DEC-07 fires, and until then the screens run with one plant and one implicit lane, which keeps the go live path unchanged.

4.1 Setup Path (the ordered checklist)

Step	Action	Where
1	Create the plant with entity, calendar and address	MP-PLANT-001 below
2	Create its warehouses, zones and bins	INV warehouse model
3	Create work centers, cost center nodes and rates for the plant	Book I D §2, H, I §2
4	Extend materials to the plant with plant level data completion	MM Extend action, Book II §3.3
5	Pair production versions per material and plant	Book IV §6.1
6	Define transfer lanes in both directions with transit times	MP-LANE-001 below
7	Set sourcing rules per material and plant	Planning view special procurement plus §4.4 sourcing table
8	Configure intercompany pairing where the plants sit in different entities	Book III F §3, FL-07
9	Run the multi plant test packs and the cross plant recall drill	FL-04, FL-07, FL-08 in network configuration

MP-PLANT-001 • Plant Master

Aspect	Implementation
Route, type, users	/configuration/plants • List + detail • IT admin drafts, Controller approves
Build the layout	Plant code, owning entity, address, working calendar link, default receiving and shipping warehouses, shipping points, planning parameters block, status
Primary actions	Create. Activate after the setup checklist steps 2 to 5 verify. Deactivate blocks new documents only
Arrives from	Configuration sidebar, DEC-07 activation project
Creates or updates	The plant key every plant level table already carries

Aspect	Implementation
Postings and scans	None
Build wave and field spec	Wave 4 build, activation per DEC-07 • Fields and rules in Book III F §2, Book II §3.3

MP-LANE-001 • Transfer Lanes

Aspect	Implementation
Route, type, users	/configuration/transfer-lanes • Config grid • Planning Manager
Build the layout	Rows per from plant and to plant with transit days, transport mode, lane calendar, cost estimate per unit weight, active flag, direction explicit
Primary actions	Maintain lanes. Deactivate stops new transfer proposals only
Arrives from	Configuration sidebar
Creates or updates	The lead time offsets network MRP and global ATP use between plants
Postings and scans	None
Build wave and field spec	Wave 4 build • Fields and rules in Book III F §5, this section

STO-001 • Stock Transport Orders

Aspect	Implementation
Route, type, users	/inventory/transfers/sto • List + detail • Planners create, warehouses execute
Build the layout	Header with lane, dates and priority. Lines with material, quantity, batch or serial selection at issue. Status path Created, Released, Issued, In Transit, Received, Closed
Primary actions	Release. Issue at source with scans. Receive at destination. Short close with reason
Arrives from	Network MRP proposals, manual planner action
Creates or updates	In transit stock in bucket 1350 between issue and receipt. Cross entity cases pair through FL-07 instead
Postings and scans	Issue DR 1350 CR 13xx at source. Receipt DR 13xx CR 1350 at destination. Scans SP-06 both ends. Genealogy continues with plant stamped per event
Build wave and field spec	Wave 4 build, live use at activation • Fields and rules in Book III F §5, INV 5.3 pattern

4.4 Network Planning Behavior (binding once active)

- Sourcing per material and plant. Each plant level planning view states how that plant obtains the material. Own production, transfer from a named plant over a lane, or purchase. The sourcing table allows a ranked sequence, and MRP consumes it in order.
- Network netting. MRP still runs by low level code, and a transfer source turns demand at the receiving plant into lane offset demand at the supplying plant. Proposals carry both plants and the lane. Exception EX-10 names unworkable network demand, such as an inactive lane or a supplying plant without a valid production version.

- Global ATP and CTP. The availability check walks the sourcing sequence. Local stock first, then supplying plant stock plus lane transit, then production CTP at the plant owning the production version. The promise names the sourcing plant and lane so the commitment is auditable, and the plan snapshot rule of Book III Part E applies unchanged.
- Cross plant genealogy needs no new design. Events already stamp plant, and STO issue and receipt write movement events on the same chain, so forward and backward traces cross plants natively. The recall population derivation of FL-08 is network correct by construction.
- Finite scheduling stays per plant. Each plant schedules its own work centers. The network connects them through dated transfer demand, not through one global schedule, which keeps the engine deterministic and the freeze horizons local.

5. Controlled Build Package Manifest

The finding offered two remedies. Merge the released Inventory and Accounting volumes into this document, or issue a controlled manifest. The manifest is the correct remedy and is recorded here. The two volumes are released specifications already under change control with a build in progress against them. Absorbing them would reset their approval state, break the section references the build teams use daily, and double the size of this document. The manifest gives the single controlled baseline the finding actually needs. Any change to any manifest row issues a new manifest revision through change control.

#	Controlled Item	Version	Owner	Repository	SHA-256 (first 16)
1	Enterprise ERP Unified Build Specification (this document)	V1.0, Jul 2026	Product Owner	Controlled documents vault, /erp/specifications	Recorded at release signing, a file cannot carry its own final hash
2	Inventory & Warehouse Module Spec	V3.0, Mar 2026	Inventory Module Owner	Controlled documents vault, /erp/specifications	Recorded at manifest signing from the released binary
3	Accounting Module Implementation Spec	V2.0, Mar 2026	Finance Module Owner	Controlled documents vault, /erp/specifications	Recorded at manifest signing from the released binary
4	Data Migration Workbook	V1.0, Jul 2026	Data Migration Lead	Vault, /erp/migration	Recorded at manifest signing
5	Contract repository (OpenAPI, AsyncAPI, schemas per IF catalog)	Tagged releases	Integration Lead	Source control, /erp/contracts	Git tags carry commit hashes
6	Figma design library and module files	Published versions	UX Lead	Figma workspace per Book V §10.4	Figma version history
7	Configuration workbook and seed data	V per release	Functional Leads	Source control, /erp/config	Git tags

6. Data Migration Package

The migration workbook ships beside this document as a controlled file, manifest row 7. It contains fourteen sheets. A ReadMe with the column dictionary and fill in legend. Ten object mapping sheets

covering materials, customers, vendors, BOMs, routings, info records with AVL, stock with batches and serials, open sales orders, open purchase orders and open production orders, each row carrying legacy field, example value, target table and field, verification screen, transformation logic, cleansing rule reference, validation and owner. A cleansing rule catalog of twenty four numbered rules the mapping rows reference. An opening balance sheet with twelve OB rules covering trial balance, stock value, cost basis, receivables and payables, GR and IR, WIP, warranty reserve, revenue contracts, consignment, customer property, the migration clearing account and open long lead exposures. A reconciliation sheet with twelve scripts, R-01 through R-12, in reference SQL, run after every mock load and at cutover.

- The workbook implements the migration approach already specified in Book II §16.3 and Book III Part H §5. Those sections remain the governing text. The workbook is the executable form the finding asked for.
- Two full mock conversions remain mandatory. Mock two must complete without manual data fixes, with every R script tying, before the go live gate opens.
- The workbook is version controlled with the manifest. Mapping changes after extract freeze follow the same change control as specification changes, because a silent mapping edit is a silent requirement change.

7. Barcode Sample Book, Complete

Every label class in the system now has a rendered sample in this document. The handling unit and batch labels appear in Book V, Figures V-15 and V-16. The remaining seven follow, with the usage matrix stating where each is printed, when, what it must carry and which scan points read it. Symbologies, print governance and hardware remain as specified in Book I Part F and Book V §8.

7.1 Label Usage Matrix

Label	Printed Where and When	Must Carry	Read At	Sample
Item shelf label	Receiving or warehouse office, on new bin assignment	Material code barcode, description, UoM, home bin	SP-04 putaway, SP-06 transfers, SP-15 picks	Figure V-19
Batch label	Receiving at GRN, one per batch container	Batch barcode, item, vendor batch or heat, origin, dates, quantity, QI state at print	SP-05, SP-06, SP-08, SP-15, SP-20	Figure V-16
Serial component label	Receiving for serialized buys, durable stock	Serial barcode, item, vendor serial, receipt references	SP-12 install, SP-15, SP-19 returns	Figure V-20
Location label	Warehouse setup and changes, reflective stock	Location barcode, zone description, long range readable	Every movement scan naming a bin	Figure V-21
Traveler operation strip	Order release, laser printed in the shop packet	Order and operation composite barcode, WC, standard hours, QC gate marker, QR for human assist	SP-10 start, SP-11 complete	Figure V-22
Handling unit label	Pack station at HU close	HU barcode, SSCC-18 where the customer demands GS1, ship to, weights, HU count, handling marks, transformer addenda	SP-18 loading	Figure V-15

Label	Printed Where and When	Must Carry	Read At	Sample
RMA return label	Emailed to the customer inside the return instructions	RMA barcode, ship to dock, class specific handling instructions	SP-19 return receipt	Figure V-23
Operator badge	HR issuance, reprint under governance	Operator barcode, name, work center, qualification note	SP-07 terminal login, supervisor overrides	Figure V-24
Calibration label	Calibration close, applied to the instrument	Instrument barcode, due date, calibrating lab	SP-14 at every test	Figure V-25
Nameplate serial plate	Nameplate station, engraved plate with barcode	Nameplate serial per the design data	SP-13 verification against the order serial	Physical plate, format per Book I F §1

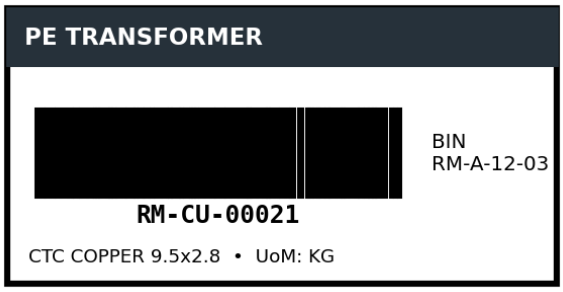


Figure V-19. Item shelf label, 2 by 1 inch polypropylene.

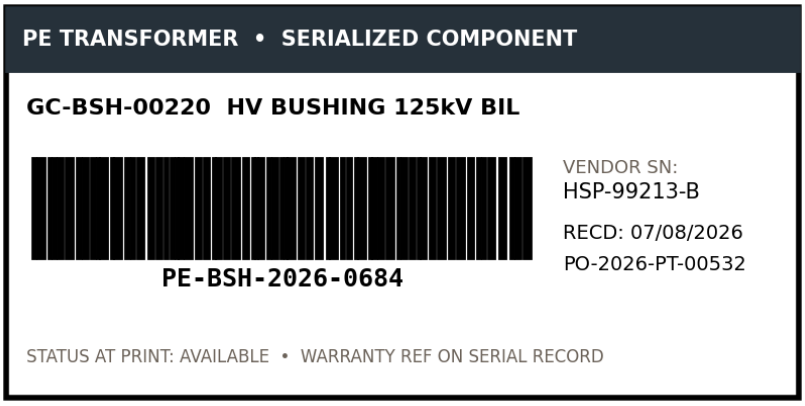


Figure V-20. Serialized component label, 4 by 2 inch polyester.



Figure V-21. Location label, 4 by 1 inch reflective for long range reads.

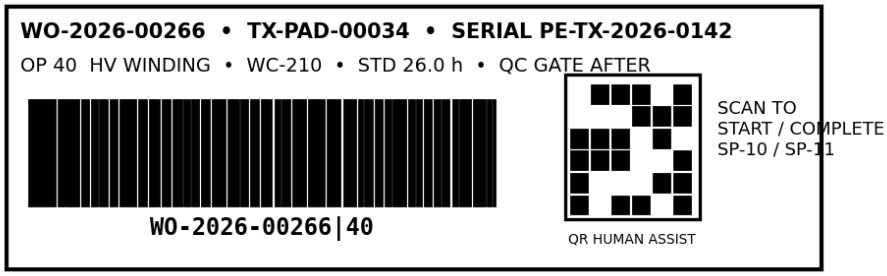


Figure V-22. Traveler operation strip as printed in the shop packet.

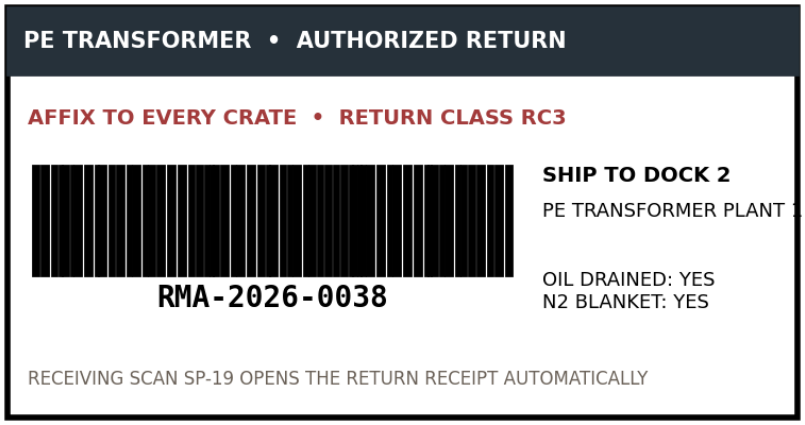


Figure V-23. RMA return label the customer affixes to every crate.



Figure V-24. Operator badge.

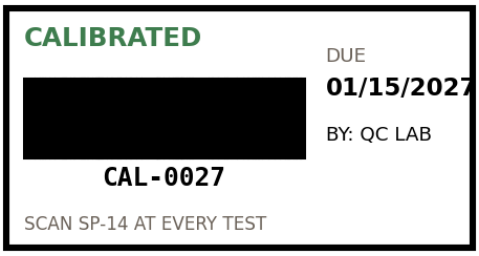


Figure V-25. Calibration label.

8. Go Live Readiness Statement and Execution Directive

- Go live ready means the gate of Book III Part H §7 passes with the Book IV additions. Every requirement traced to passing evidence, the eight closure flows plus FL-09 to FL-11 demonstrated, mock two clean, security and resilience drills filed, training complete and the sign off executed. The specification set is now sufficient for that path. What remains is execution and evidence, which documents cannot substitute.
- Capabilities the phase decisions defer are recorded in Book IV §1.2 and mapped in Book V §2. The list in the finding matches those records. That boundary is the honest reading of SAP scale at first go live, and every deferred row carries its activation path.
- The directive stands. This document stops growing by narrative revision. This edition is the content baseline. Subsequent changes ride change control against the manifest, and effort moves to the DG gates, the Figma library, the contract repository, the configuration workbook and the migration mocks.

BOOK VIII

DESIGN EXECUTION HANDBOOK

The Eight Screen Archetypes • The Business on One Page • BPMN and Sequence Diagrams for the Five Governing Flows
Inventory, Warehousing and Accounting Consolidated • Target State Capability Specifications • Reconciliation Control Register • The Implementation Package

Book VIII is the operating manual for turning this specification into designed screens and a delivered system. It must be the first Book a designer opens. It supplies the eight screen archetypes every screen in Book VI maps to, the one page business flow from lead to delivery, the process diagrams the earlier reviews demanded, the consolidated warehousing and accounting detail, the requirement specifications for the ten target state capabilities, the reconciliation control register and the implementation package. Nothing elsewhere in the document is removed or altered by this Book.

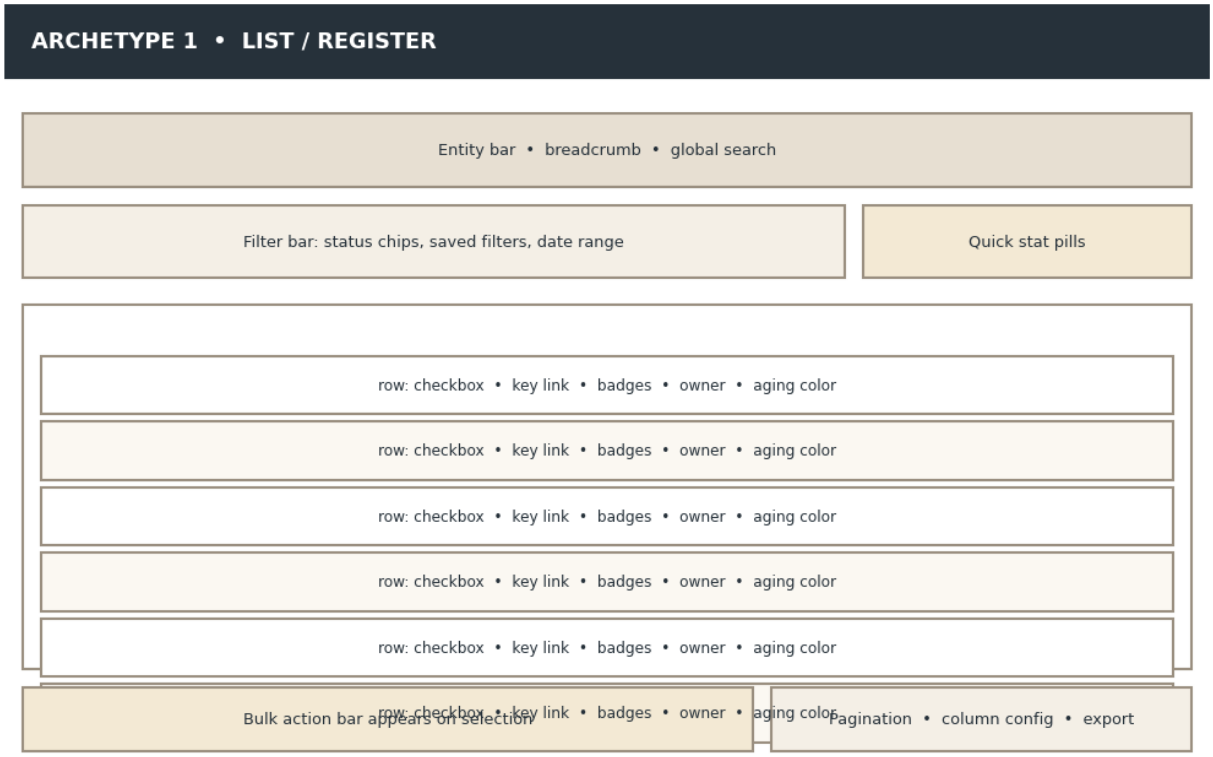
1. The Design Method

Every screen in this system must be designed as one of eight archetypes. An archetype fixes the skeleton. The Book VI entry for the screen fills in the organs. The referenced field section supplies the cells. A designer therefore needs exactly three inputs per screen and no research. The archetype wireframe below, the seven row implementation entry, and the field table the entry cites. A Figma frame must exist per screen per state, named with the screen ID, and every frame must link its specification section. A frame without that link must fail design review at gate DG-2.

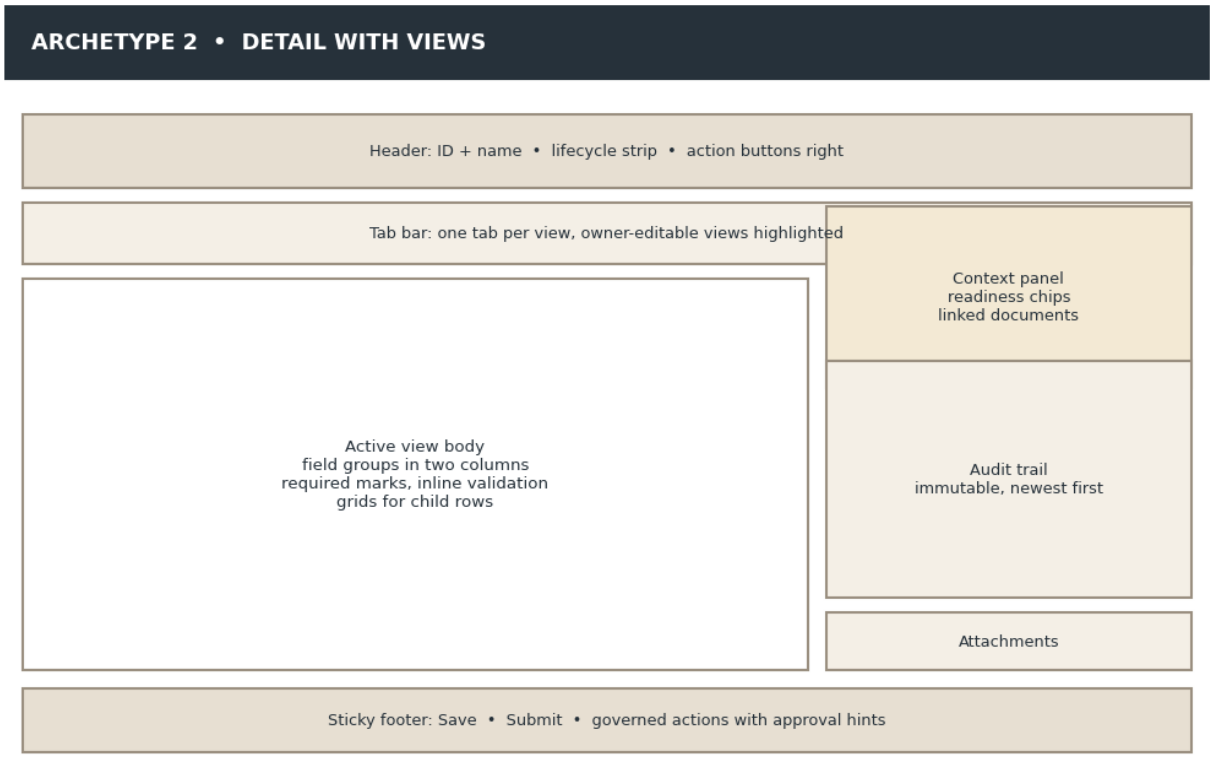
Step	The Rule
1. Pick the archetype	The Type cell of the Book VI entry names it. List, Detail, Workbench, Wizard, Kiosk, Dashboard, Config or Dialog
2. Copy the archetype frame	The wireframes in §2 are the skeletons. Zones must not be invented per screen. A screen needing a zone the archetype lacks is a change request against this Book, not a local improvisation
3. Fill zones from the entry	The Build the layout cell names what goes in each zone. The Primary actions cell names the buttons and their consequences. The Arrives from and Creates cells define the navigation in and out
4. Pull fields from the cited section	Field names, types, requiredness, validations and error codes come from the cited section verbatim. Labels must not be paraphrased
5. Design all nine states	Normal, empty, loading, validation error, authorization error, conflict, offline, degraded, read only. The states are deliverables, not variants
6. Wire the flow prototype	Each of the five BPMN flows in §4 must exist as one Figma prototype wired from the same frames, used directly in UAT scenario reviews

2. The Eight Screen Archetypes

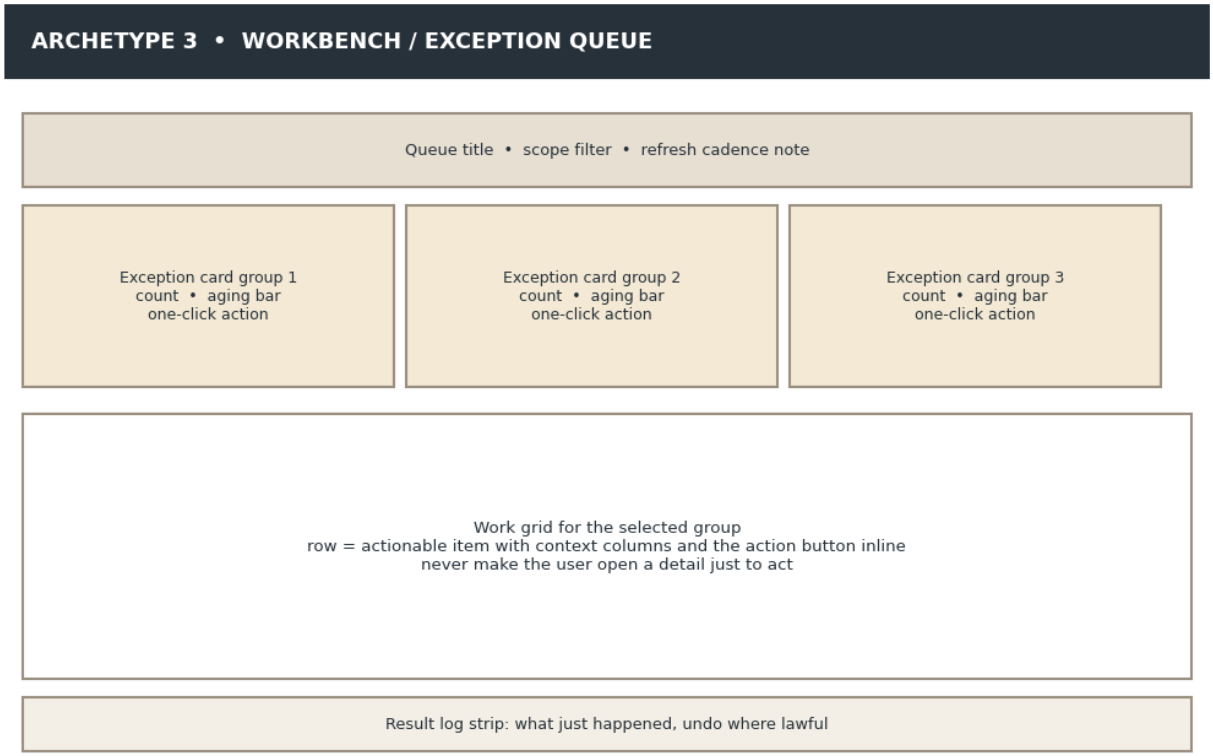
These wireframes are binding skeletons. The counts beside each name state how many Book VI screens map to it, so the reuse mathematics is visible. Build each archetype once as a Figma component set, then produce screens by instantiation.



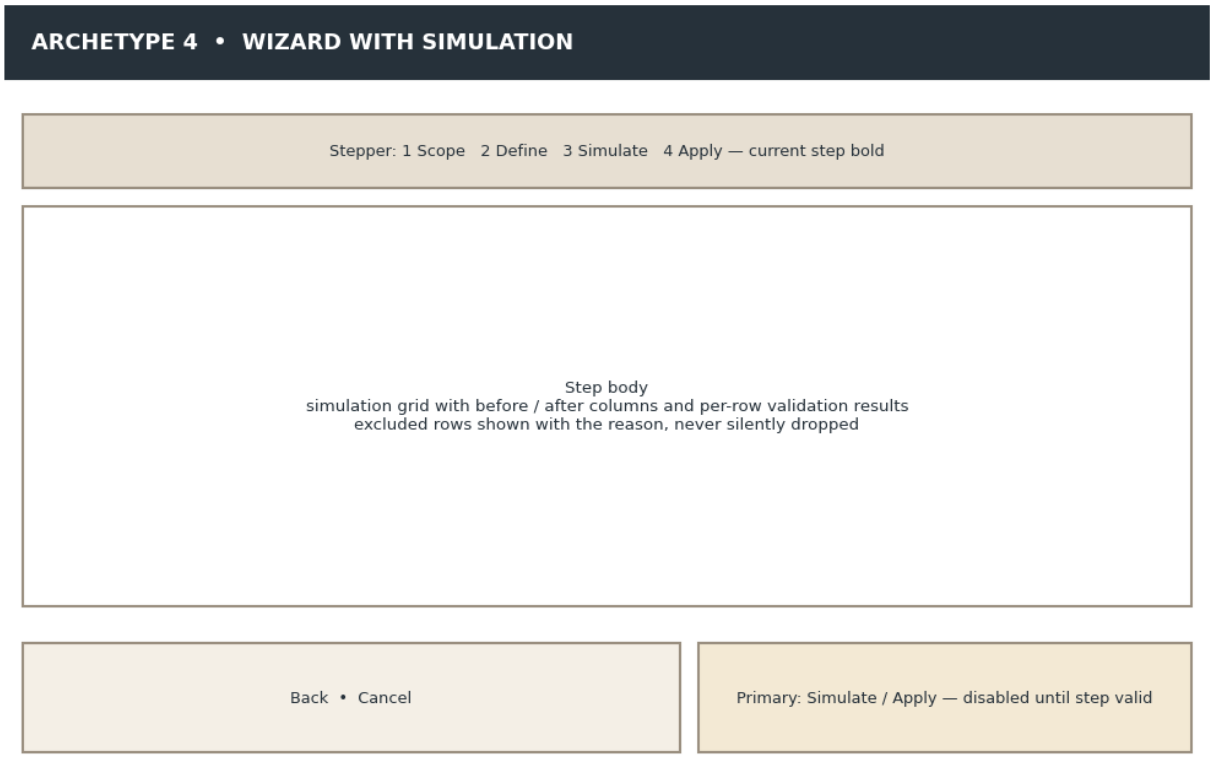
Archetype 1, the list. Every register screen. Roughly eighteen screens instantiate it.



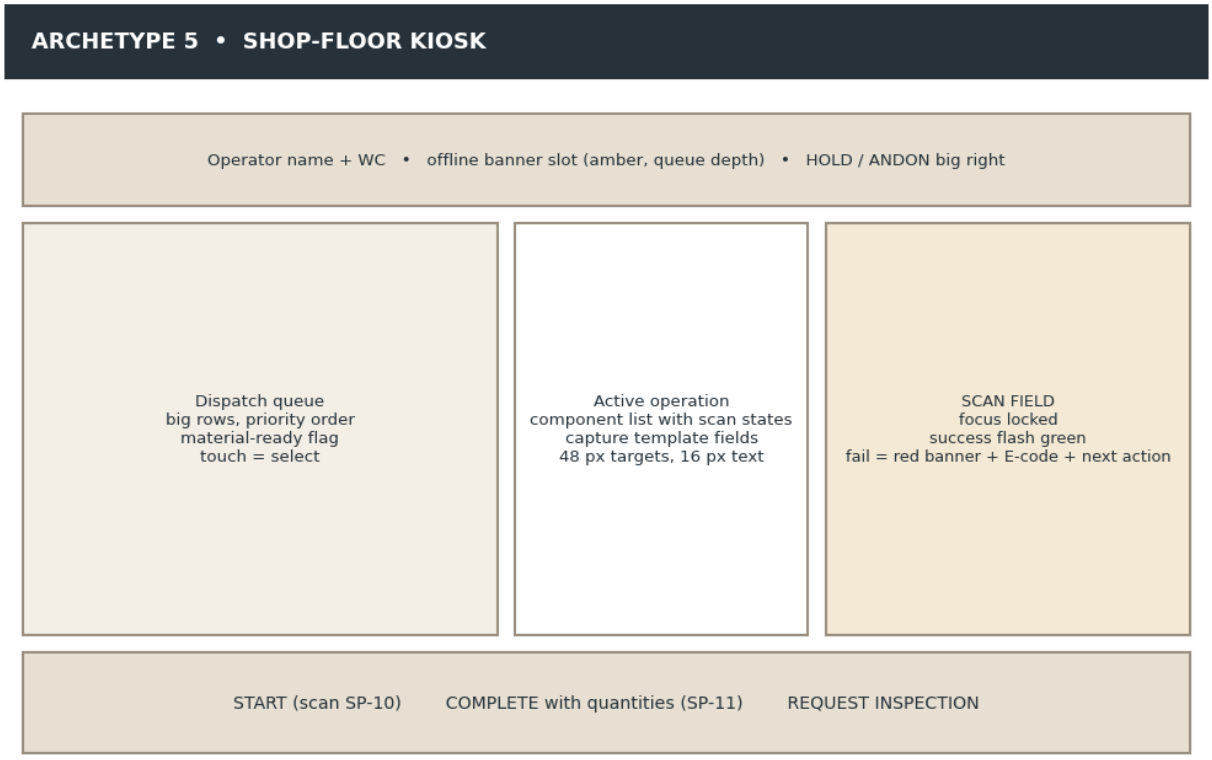
Archetype 2, the detail with views. Every master and document record. Roughly twenty screens instantiate it.



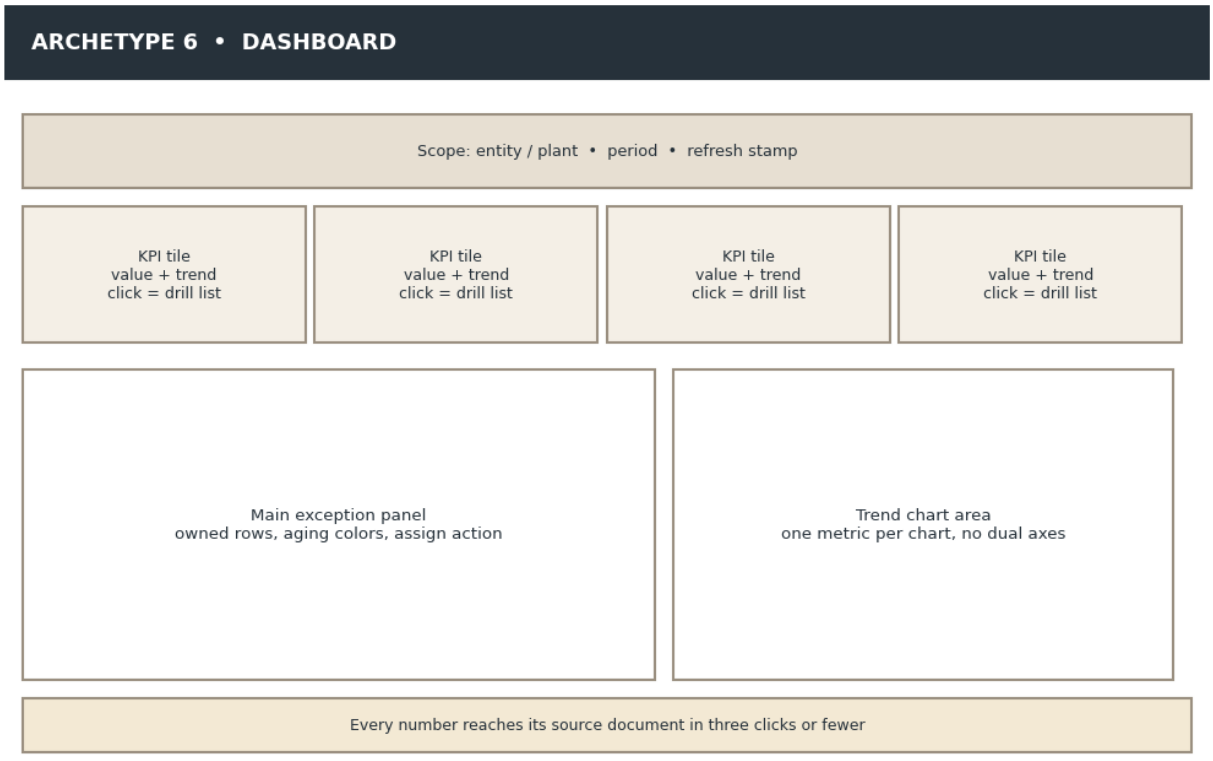
Archetype 3, the workbench. Every exception queue. Roughly twelve screens instantiate it. The rule that matters most, the user must act from the row without opening a detail.



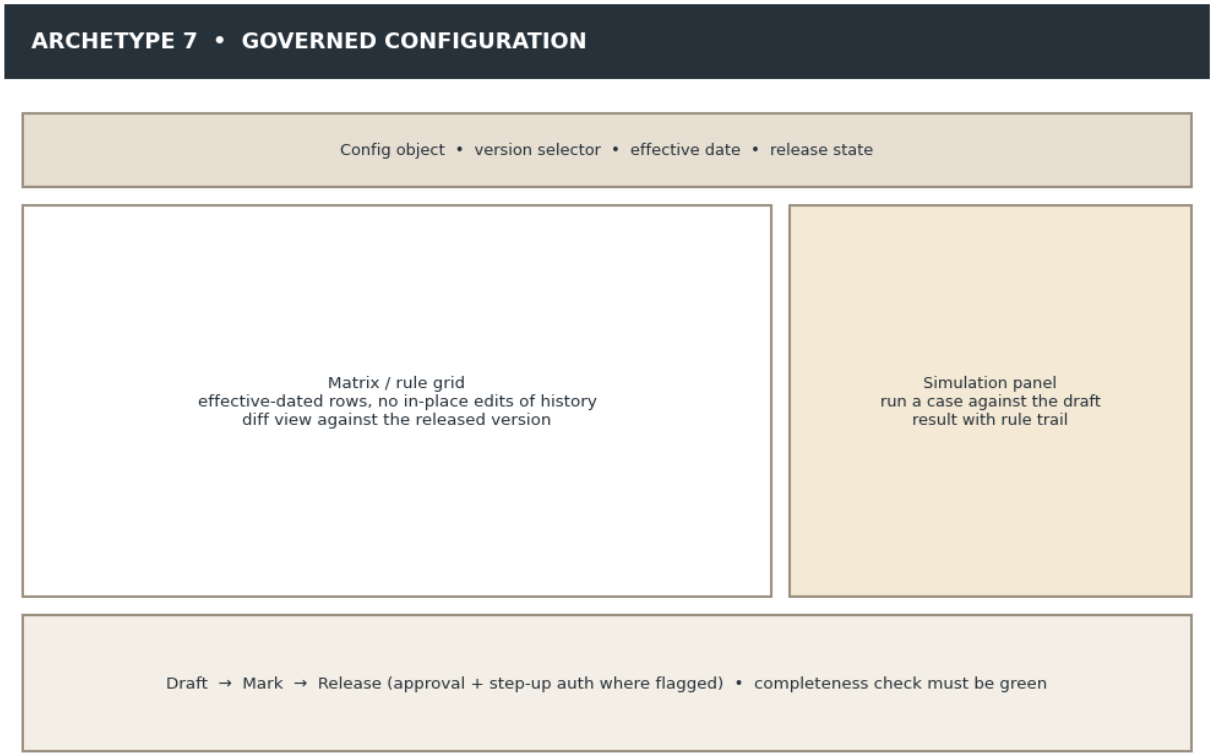
Archetype 4, the wizard with simulation. Mass change, costing runs, settlement, delivery creation, migration steps. Simulation before apply is not optional.



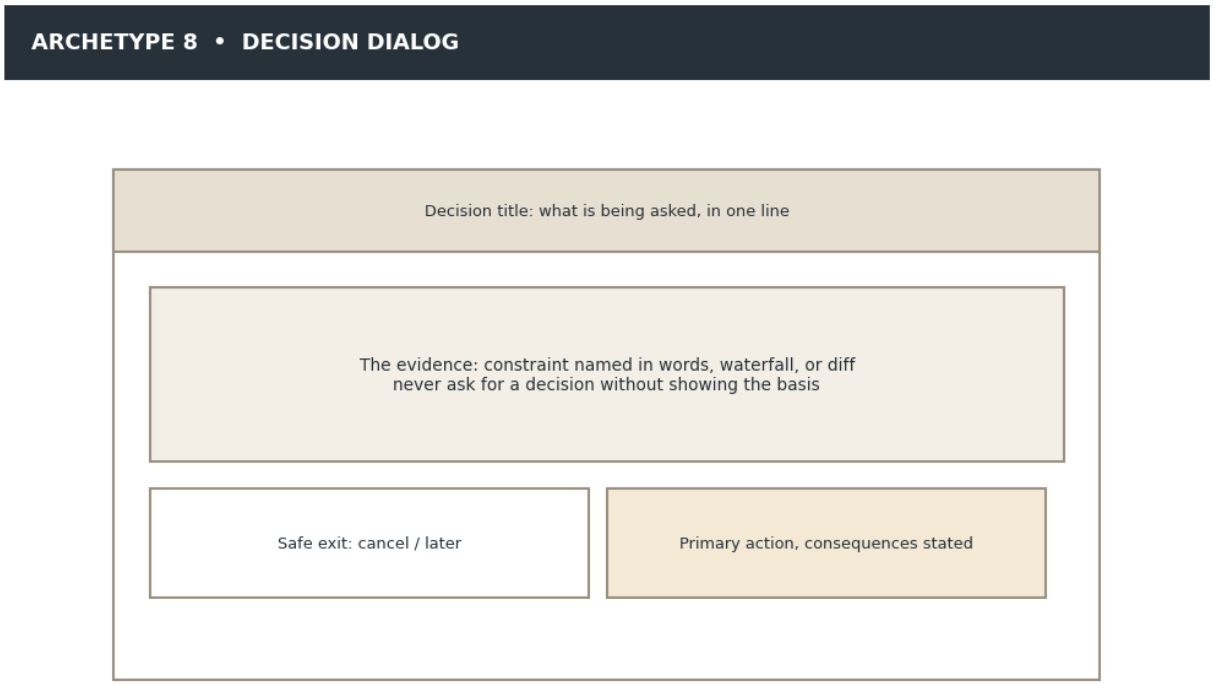
Archetype 5, the shop floor kiosk. Terminal, pack station, return receipt. Gloves, glare and distance govern every measurement.



Archetype 6, the dashboard. Six landing pages. Every number must reach its source document in three clicks or fewer.



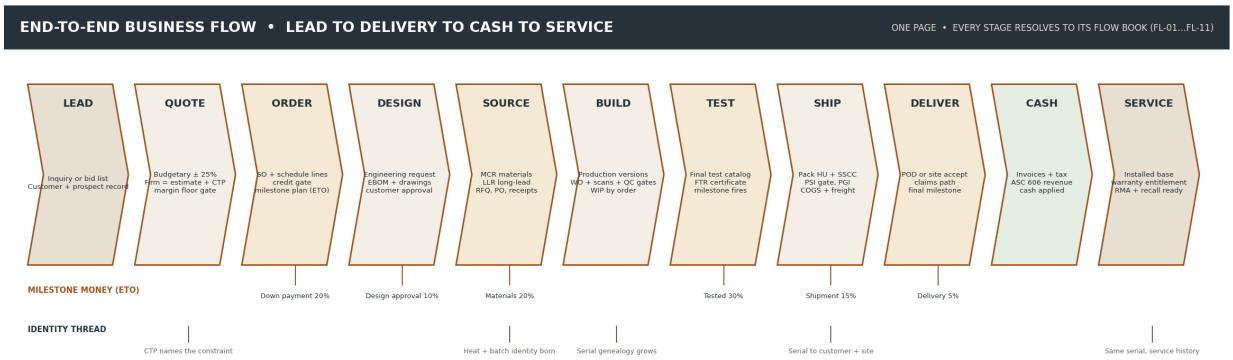
Archetype 7, governed configuration. Account determination, workflows, tax, periods, lanes. History is immutable and release is a ceremony.



Archetype 8, the decision dialog. Promise dates, credit releases, dispositions, reversals. The evidence must appear with the question.

3. The Business on One Page

The sheet below is the whole business from first contact to cash and service. It must hang beside every designer and every developer. Each chevron resolves to its flow book and its screens. The milestone band shows when ETO money arrives. The identity band shows the thread that never breaks, the serial and its genealogy. The same sheet ships as a standalone one page file beside this document.

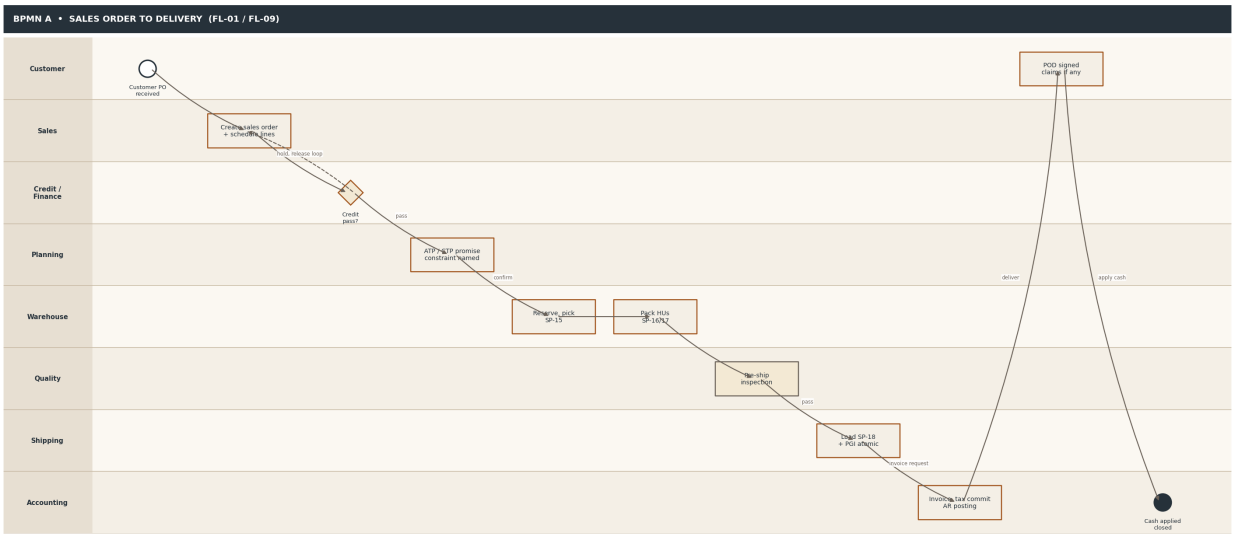


The end to end business flow. Lead to quote to order to design to source to build to test to ship to deliver to cash to service.

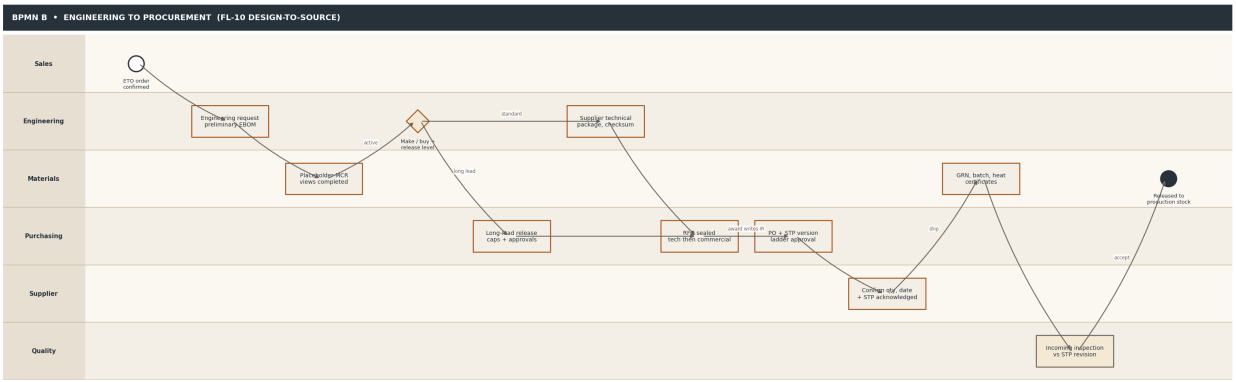
Stage	Governing Flow	Primary Screens (Book VI)
Lead and quote	FL-01 / FL-02 openings	Customer master, quotations, promise dialog, price simulation
Order	FL-01 / FL-02	Sales order with schedule lines, backorder workbench
Design	FL-02 / FL-10	Engineering request, EBOM editor, documents, transmittals
Source	FL-03 / FL-10	MCR, info records, RFQ, PO, LLR, expedite board
Build	FL-04	MRP workbench, schedule board, production orders, kiosk terminal
Test	FL-04 quality gates	Final test and FTR, calibration, deviations
Ship and deliver	FL-09	Delivery cockpit, pack station, POD and claims
Cash	FL-01 §7 / FL-06	Invoice requests to the accounting volume, close checklist
Service	FL-05 / FL-08	Installed base, RMA, recall campaigns, trace explorer

4. Process Diagrams, BPMN and Sequence

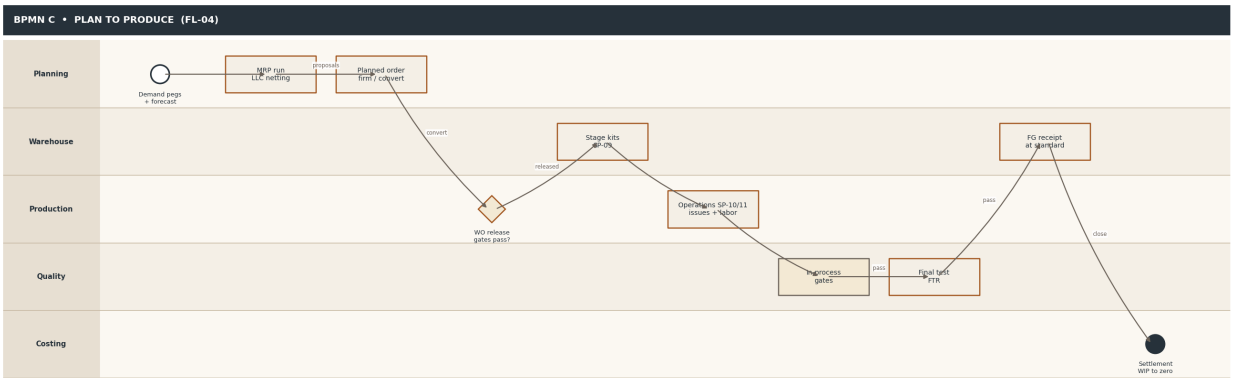
The five governing flows follow as swimlane diagrams. Lanes are the accountable roles. Sand boxes are quality or decision gates. Each diagram must be read beside its closure table, which carries the postings, exceptions, reversals and reconciliations the picture cannot hold. Two sequence diagrams follow for the interactions where transaction boundaries matter most.



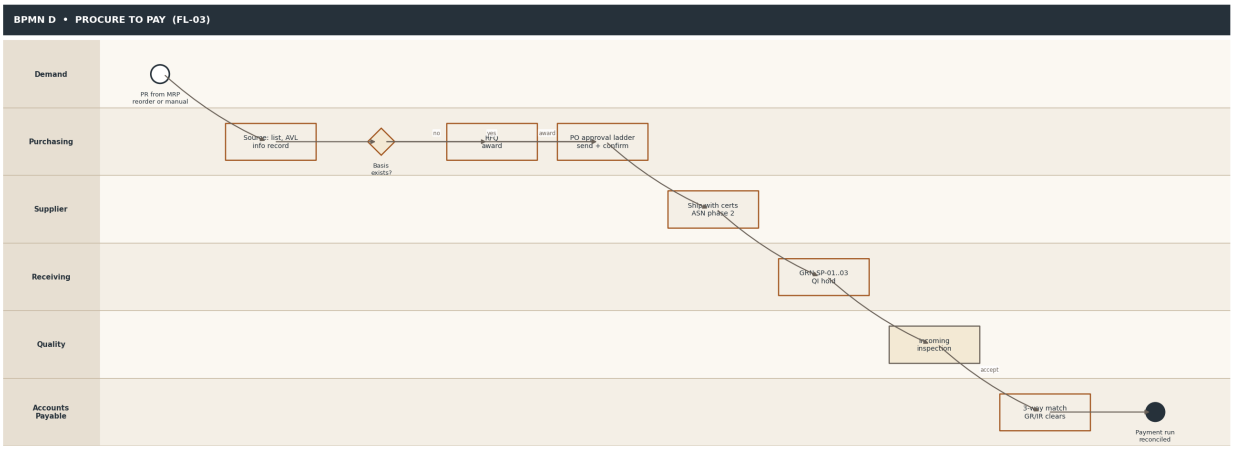
BPMN A. Sales order to delivery. Closure table FL-01 and FL-09.



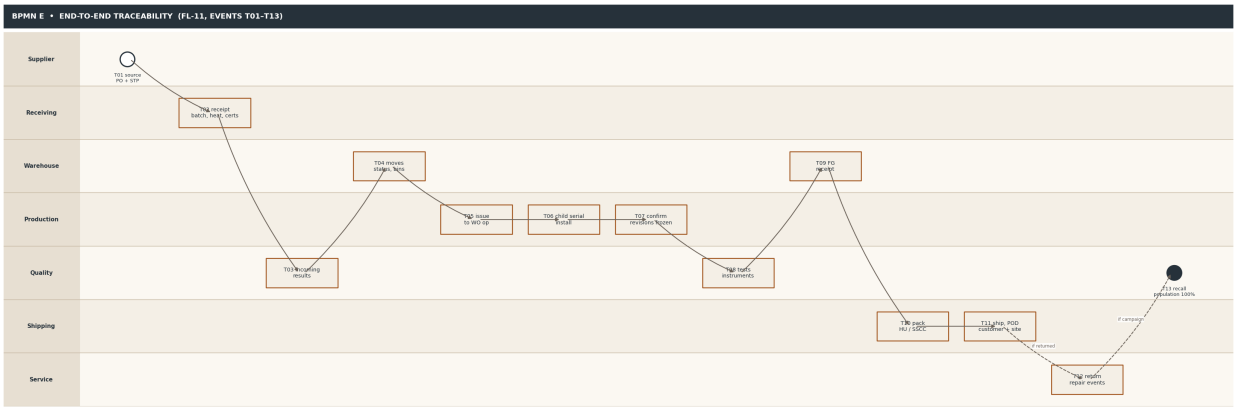
BPMN B. Engineering to procurement. Closure table FL-10.



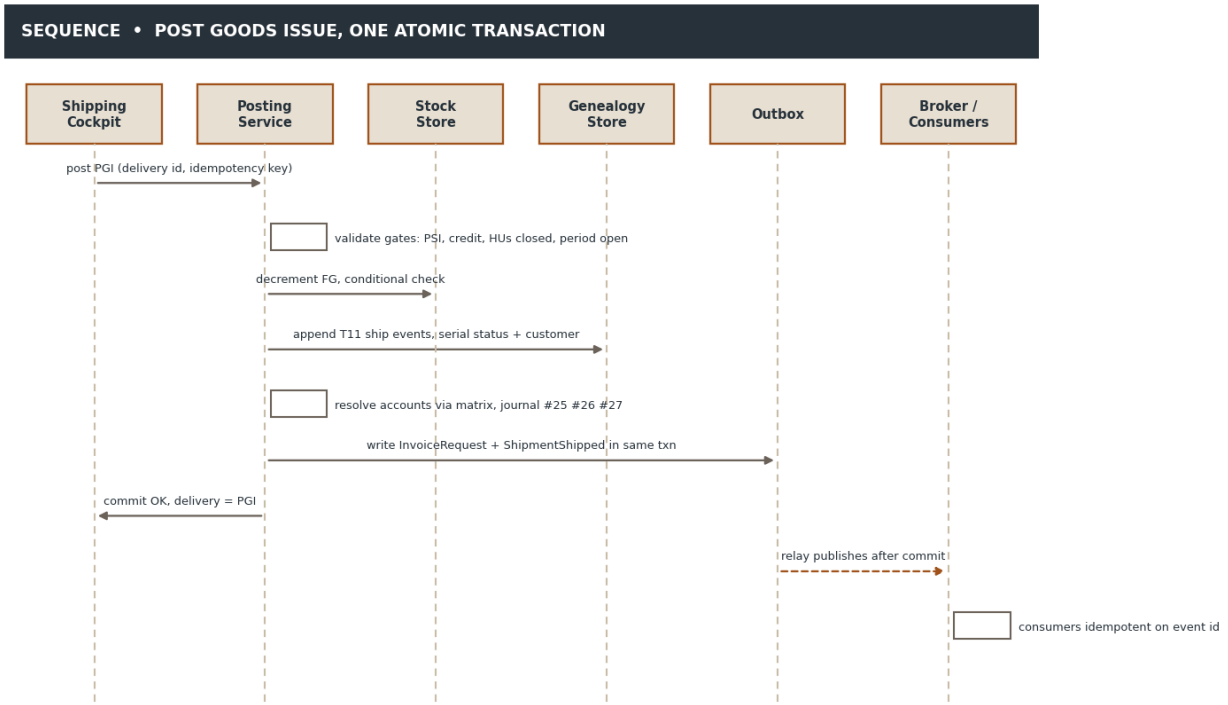
BPMN C. Plan to produce. Closure table FL-04.



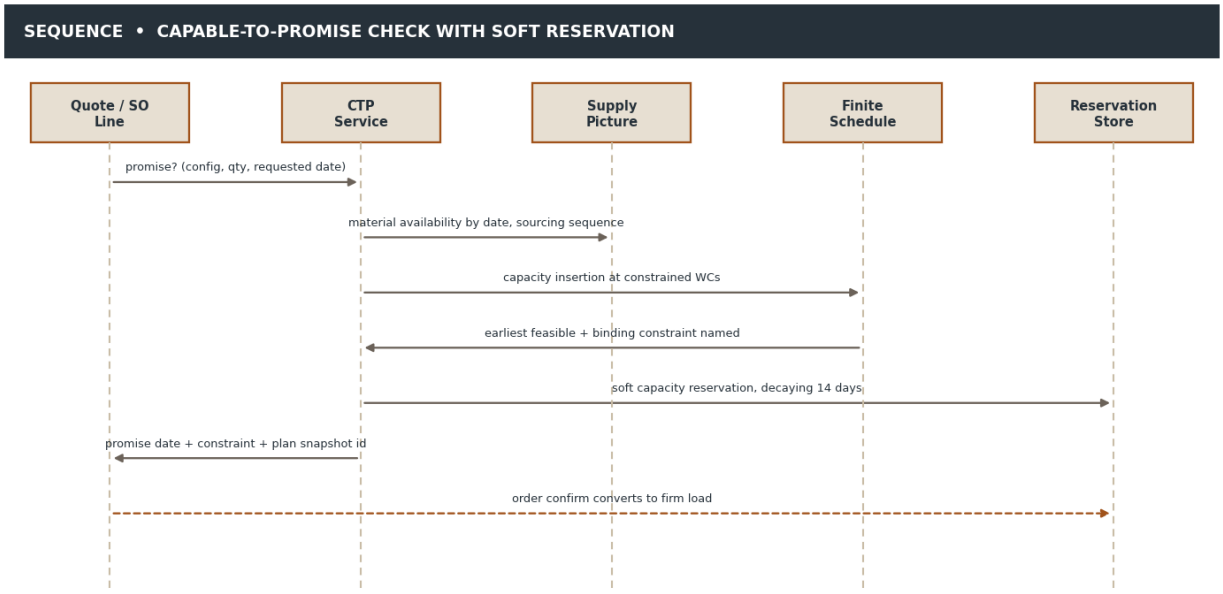
BPMN D. Procure to pay. Closure table FL-03.



BPMN E. End to end traceability. Contract in Book IV §4, events T01 to T13.



Sequence 1. Post goods issue must commit as one transaction. Stock, genealogy, journal and outbox together or not at all.



Sequence 2. Capable to promise must name its binding constraint and must reserve soft capacity so parallel quotes cannot double promise one bottleneck slot.

5. Inventory, Warehousing and Accounting, Consolidated

The external Inventory and Accounting volumes remain the field level authorities under the build package manifest. This section consolidates their operating core into this document so a reader needs no other file to understand how stock and money behave. Where a number or rule below differs from the external volume, the external volume controls and the difference must be raised as a conflict register entry.

5.1 The Warehouse Physical Model

Object	It Must Carry	Rules
Warehouse	Type (RM, WIP staging, FG, Quarantine, Packaging), owning plant, cost center link, capabilities (climate control, flammable, oil containment, outdoor)	Quarantine warehouses must be the only landing zone for returns and QI failures. Capabilities must validate against material storage conditions at putaway
Zone	Zone code, purpose (bulk, pick face, staging, dock), temperature class	Putaway strategies target zones before bins
Bin	Bin code with barcode label, capacity class, fixed material where assigned	One barcode per bin. Long range reflective labels on rack ends
Dock and staging	Dock doors, STAGE-OUT locations, appointment slots on the plant calendar	A shipment must not post while its HUs sit outside STAGE-OUT

5.2 The Movement Catalog, Every Way Stock Moves

Every stock change must be one of these movements, each with its scan point and its posting behavior. There must be no movement type outside this catalog.

Movement	Trigger and Scan	Value Effect
Goods receipt against PO	GRN at receiving, SP-01 to 03	DR stock account by valuation class, CR GR/IR. PPV recognized for standard cost materials
Goods receipt from production	FG receipt at order completion	DR finished goods at standard, CR WIP
Return receipt from customer	RMA scan SP-19	DR RMA quarantine account, CR COGS reversal. Customer property receives at zero value
Putaway and bin transfer	SP-04, SP-06	None. Location only
Issue to production	SP-08 or backflush at confirmation	DR WIP by order, CR stock
Issue to subcontractor	Provision document	DR subcontract stock at vendor, CR stock
Kit staging	SP-09	None. Staged status only
Pick, pack, load	SP-15, 16, 17, 18	None until PGI. Packaging backflush at HU close
Post goods issue	Shipment posting	DR COGS, CR finished goods. Freight accrual where owed
Stock transport between plants	STO issue and receipt	Through in-transit. Cross entity cases pair through the intercompany flow
Adjustment from count	Approved count posting, SP-20	DR or CR inventory adjustment against stock
Scrap	SP-21 with approved disposition	DR scrap expense, CR stock or WIP
Status change	Hold, release, QI decisions	None. Availability changes only
Revaluation	Costing run release only	DR or CR revaluation against stock. Never manual

5.3 Warehouse Task Execution

- Receiving must run dock to stock inside the GR processing time on the material. The sequence is fixed. Identify against the PO, capture batch and heat and certificates, print labels, quality route, putaway proposal, bin confirm by scan.
- Putaway and picking must follow the material strategies. FEFO is mandatory for shelf life stock, serial directed picking for serialized stock. Every override must log.
- Replenishment must keep fixed pick bins between their min and max from bulk, generated as tasks, confirmed by scan.
- Counting must follow the count document lifecycle with blind enforcement server side, recount rules, tolerance bands and approval chains, and the location freeze or the interim movement replay.
- Every task must be a record with a queue, an owner, a device and a status, so supervisors balance load and the mobile app is a work queue, not a menu.

5.4 Valuation, In One Table

Question	Binding Answer
What carries value	Every stock account resolves from the valuation class through the account determination matrix. No screen and no developer may name a GL account

Question	Binding Answer
Bought materials	Moving average by default. Receipts re-average, issues post at current average, invoice differences correct stock while coverage exists
Made materials	Standard cost, released only by costing runs. Receipts at standard, variances split by cause at settlement, revaluation only at release
Special ownership	Consignment carries no value until withdrawal. Customer property never carries value. Subcontract stock and in-transit stock stay on the books in their own accounts
Reserves	Excess and obsolete reserve by aging policy, quarterly, value adjustment only, never a quantity change

5.5 Accounting Essentials, the Money Map

Area	What Must Hold True
Document spine	Sales invoices, credit notes, vendor invoices, payments, journals. Every operational posting arrives through the shared posting service with its journal template, its dimensions and its source link. Nothing posts to the ledger directly
Receivables	Invoice requests from PGI, milestones and RMA credits become invoices under finance review. Cash application clears open items. Credit exposure recomputes continuously and gates orders and shipments
Payables	Three way match is the only road to payment. GR/IR must clear item by item. Vendor bank detail changes demand dual approval
Revenue	Recognition follows the contract standard, never the billing plan. Milestone cash and earned revenue diverge by design and reconcile through deferred and unbilled schedules
Tax	Determination follows the precedence chain, commits at invoice, survives engine outages through the failover queue, and never restates posted documents from configuration changes
Close	Soft close warns, hard close blocks. Subledgers must tie to control accounts before the gate. The checklist runs the operational steps by machine, allocations, warranty accrual, inventory close, WIP tie, settlement aging, reconciliation set
Dimensions	Every P&L posting must carry one cost center and one profit center by the derivation rules. Fallback postings are defects with a report that must read zero

6. Target State Capability Specifications

Ten capabilities are phased beyond first go live by the decision register. Deferral is a date, not a blank. Each specification below is binding for its activation project. Each must be built on hooks that already exist in this document, and each names its activation evidence. A capability activates only when its evidence passes, under change control, without redesign of the go live core.

Capability	Binding Specification	Activation Evidence
Periodic actual costing and material ledger	The ledger must compute periodic actual unit costs per material and plant from the already captured actuals, order charges, receipt prices, invoice corrections. It must support multi level actual rollup, revaluation of inventory and COGS at period end as a parallel valuation view beside standard, and a reconciliation report tying actual to standard plus variances to the cent. Standard remains the operational yardstick. The ledger must never post mid period	A parallel quarter computed on production data reconciling to the existing GL within rounding

Capability	Binding Specification	Activation Evidence
Advanced scheduling optimizer	A solver must replace the seeded heuristic behind the same constraint model, setup families, alternate resources, batch cycles, skills, tools, lanes. It must return reproducible solutions under a fixed seed, respect freeze horizons absolutely, improve makespan or lateness against the heuristic on the benchmark set by a measured margin, and expose every override to the planner with reasons	Benchmark pack results and a month of shadow running beside the heuristic with decision logs
Global multi plant ATP and CTP	The availability walk must extend across the sourcing sequence of every plant with lane transit, per Book VII §4. Promises must name plant and lane. Capacity checks must run at the owning plant. Substitution across plants must demand the same approvals as local substitution	FL-04 and FL-07 network test packs green in the two plant configuration
Supplier and customer portals	The supplier portal must expose POs with STP versions for acknowledgement, deviation requests, ASN creation and invoice status against the same APIs and the same authorization engine as internal screens, external identity tenant only. The customer portal must expose order status, drawing transmittals with approve and comment, shipment tracking and document dossiers. No portal action may bypass a single validation that internal screens enforce	Pilot with two suppliers and one customer executing their full loops without email fallback
Full EDI automation	Transactions 850, 855, 856 and 810 must map to the existing PO, confirmation, ASN and invoice objects with the interface catalog contracts, idempotent processing, duplicate rejection, and a reprocessing console entry per failed envelope. Partner onboarding must be configuration, mapping profile and identifiers, never code	Three month parallel run with the top volume vendor reconciling to paper flow with zero unexplained differences
Machine integration, OEE and downtime	Machine signals must post through the device gateway as confirmation events bound to work center and order, with manual fallback always available. Downtime must capture reason coded, planned separated from unplanned. OEE must compute from availability, performance and quality per work center per shift, from posted data only, no side spreadsheets	One pilot work center reporting OEE from signals for one month, reconciled against operator confirmations
Full field service management	Service orders must dispatch technicians against the installed base, consume van stock and spares through the same movement catalog, capture labor and expenses, respect warranty entitlement at pricing, and settle to service COGS and revenue. Scheduling must respect skills and travel. The RMA loop stays authoritative for depot repair	Ten field orders executed end to end with correct entitlement pricing and genealogy continuity
Plant maintenance and EAM	Equipment records must extend work centers and tools with hierarchies, counters and plans. Preventive plans must generate maintenance orders that reserve capacity in the finite schedule. Breakdowns must flow from the terminal hold reason to a maintenance order and into downtime capture. Spares must be materials with the SPARE type, planned by reorder point	A quarter of preventive execution with schedule integration and spare consumption through standard movements
International tax and localization	Each new country must enter through the compliance matrix, registrations, calendars, statutory books. VAT and GST must run through the same determination precedence with the engine, e-invoicing mandates as adapters on the tax interface. Document templates must localize through the language layer already keyed in the text tables. No parallel tax logic may exist per country	First country closes a statutory period with reconciled VAT returns generated from the system

Capability	Binding Specification	Activation Evidence
Advanced warehousing at EWM level	Wave planning, task interleaving, labor standards and slotting must build on the task model of §5.3 without changing a single posting. Waves must group picks by route and cutoff. Interleaving must feed putaway to pickers returning empty. Slotting must propose bin moves from velocity. RF screens stay on the kiosk and mobile archetypes	Pick productivity baseline and a measured improvement over two months in the pilot zone

7. The Reconciliation Control Register

Every reconciliation in the package, in one table. Each control must run on its stated cadence, must alert its owner through the standard rules, and must block what its gate column names when it fails. An unlisted reconciliation must not exist, and a listed one must not be optional.

Control	Ties Together	Cadence	Gate on Failure	Owner
Stock value by valuation class	Stock quantities times basis against the inventory control accounts	Daily and at close	Hard close blocks	Controller
WIP tie	Open order WIP report against WIP accounts	Close step	Hard close blocks	Cost accountant
GR/IR aging	Received not invoiced items against the clearing account	Weekly	Items over thirty days escalate	AP lead
AR and AP subledger tie	Open items against control accounts	Close step	Hard close blocks	Controller
Revenue schedule	Contract deferred and unbilled against invoiced and recognized	Close step	Hard close blocks	Controller
Intercompany pair match	Seller IC receivable against buyer IC payable per pair	Monthly before hard close	Close blocks on unexplained pairs	Finance IT
Transaction to event parity	Trace relevant postings against genealogy events, both directions	Daily job	Go live gate and recall readiness	Quality IT
Serial and batch integrity	Single active location per serial, batch quantity balance	Daily job	Traceability dashboard reds block recall closure	Quality IT
Scan to confirmation counts	Terminal scan telemetry against posted confirmations	Daily	Investigate before settlement	Production IT
Interface counts	Producer counts against consumer counts per interface per the catalog	Per interface schedule	DLQ escalation, incident on breach	Integration owner
Migration set R-01 to R-12	Loaded data against legacy subledgers and counts	Every mock and at cutover	Go live gate blocks	Migration lead
Budget to absorption	Cost center actuals against absorbed labor and overhead	Monthly	Rate review inputs	Controller

8. The Implementation Package

8.1 What the Package Must Contain

#	Deliverable	Form	Acceptance
1	This specification, eight Books	Controlled document, manifest row 1	Signed baseline
2	Inventory and Accounting specifications	Manifest rows 2 and 3, version locked	Cross reference validation at DG-1
3	Data migration workbook	Controlled workbook, fourteen sheets	Mock two clean, R scripts tie
4	Figma library and module files	Archetype components plus frames per screen per state, IDs in frame names	DG-2 per screen
5	Contract repository	OpenAPI, AsyncAPI and schema files per interface catalog entry	Contract tests green in CI
6	Configuration workbook	Statuses, reasons, workflows, tolerances, accounts, calendars, labels as versioned seed data	Environment transport proven
7	Test repository and RTM	Requirement rows linked to cases, data and evidence	Zero unlinked go live requirements
8	Runbooks and operations pack	Degraded modes, incidents, jobs, DR drills	Production readiness review
9	Training pack	Role curricula, kiosk drills, super user certification	Completion thresholds at gate
10	Cutover runbook	Sequenced tasks, checkpoints, rollback boundary, fallback kit	Rehearsed in mock two

8.2 How the Work Must Run

- Seven build waves per Book VI §2, each wave exiting through DG-3 for its stories and demonstrable end to end. A wave must not start feature work while its predecessor holds open S1 or S2 defects.
- Five standing workstreams. Application build, integration and platform, data and migration, design and UX, quality and evidence. Each workstream owns its manifest rows and its reconciliation controls.
- Weekly the RTM coverage, the conflict register, the DLQ depth and the data quality KPIs must publish to one program page. Trends, not snapshots, are the management signal.
- Change rides change control against the manifest. A specification change without a manifest revision must be treated as unauthorized scope.
- The three gates stand as defined. DG-1 before design, DG-2 before development, DG-3 before done, and the go live gate of Book III Part H with the Book IV and Book VII additions closes the program.

Build Note: *Precedence note for this Book. Where Book VIII summarizes, the cited Books govern. Where Book VIII adds, the additions bind, the target state specifications of §6, the reconciliation register of §7 and the archetype discipline of §1 and §2.*

BOOK IX

TARGET STATE MODULE SPECIFICATIONS

Periodic Actual Costing & Material Ledger • Advanced Scheduling Optimizer • Global Multi-Plant ATP/CTP • Supplier & Customer Portals • Full EDI
Direct Machine Integration, OEE & Downtime • Full Field Service Management • Plant Maintenance / EAM • Broader International Localization

Book IX raises the nine target state capabilities from decision register rows to module grade specifications. Each section below must be sufficient to design and build its module without further requirements work. Each module must run on hooks that already exist in Books I through VIII, must add no parallel logic beside an existing engine, and must activate only when its evidence section passes under change control. The activation summaries in Book VIII §6 remain valid and are superseded in depth by this Book. Delivery timing stays governed by the decision register. Depth of definition no longer waits for it.

1. Periodic Actual Costing and the Material Ledger

The ledger must compute what every material actually cost each period and must revalue inventory and cost of sales to that truth at period end, while standard cost remains the operational yardstick during the period. Nothing may post at actual mid period.

1.1 Objects

Object	Required Content
Ledger period record, one per material, plant and period	Beginning quantity and value at standard and at actual. Receipt lines by category, purchase receipts corrected to invoice price, production receipts at settled order actuals, transfer receipts at sending plant actual. Consumption lines by category, production issues, sales issues, scrap, transfers out. Collected single level differences. The periodic unit price. Distribution results, to consumption, to ending inventory, not distributed, not included
Difference categories	Price differences from PPV and IPV, exchange differences, settlement variances by cause, transfer differences. Each category must remain reportable separately through the rollup
Valuation views	Two parallel views per material and period, legal at standard with variances expensed as today, and periodic actual. The actual view must never leak into operational documents. Reports must name their view
Rollup graph	Consumption relationships per period by low level code, built from the genealogy and order records already captured, so multi level distribution follows real consumption, not the static BOM

1.2 Calculation, Binding Order

- Step one, single level. Periodic unit price equals beginning actual value plus receipt actual value, divided by beginning quantity plus receipt quantity. Differences allocate pro rata between consumption and ending inventory. Zero or negative denominators must park the material in the not distributed category with a worklist entry, never a division fault.
- Step two, multi level. By ascending low level code, differences allocated to consumption roll into the receiving materials of that consumption and re enter their own single level pass. Circular consumption within a level must resolve iteratively to a stated tolerance of one cent per material.
- Step three, revaluation preview. Inventory delta per material equals ending quantity times the difference between periodic unit price and standard. Cost of sales delta follows the allocated consumption. The preview must show both by valuation class before anything posts.
- Step four, posting. One revaluation journal per entity at period close, reversing option for the actual view where the Controller elects reversal accounting. New event keys MLINV and MLCOGS must resolve through the account determination matrix.

1.3 Screens

Screen	Type	Must Contain
ML-COCKPIT-001, period cockpit	Wizard	Steps for single level, multi level, preview, post. Step status per material group. Error worklist with owners. The run must be restartable per step without recomputation of finished steps
ML-MAT-001, material price analysis	Detail	The full period record of §1.1 for one material, every receipt and consumption line with its source document link, the periodic unit price derivation shown as arithmetic a cost accountant can check by hand
ML-CONFIG-001	Config	View activation per entity, difference category mapping to matrix rows, reversal election, tolerance values, all effective dated with release ceremony

Posting at close, actual view revaluation

Debit	Credit	Amount Basis	Notes
13xx by valuation class or MLCOGS expense	MLINV offset per matrix	Preview deltas per §1.2 step three	One journal per entity, close checklist step 8b, reversal per election

Activation evidence. One full quarter computed in parallel on production data, reconciling to the existing ledger within rounding, with the ML-MAT-001 arithmetic audited for twenty sampled materials.

2. The Advanced Scheduling Optimizer

A solver must replace the seeded heuristic behind the identical constraint model. The optimizer changes solution quality. It must never change what is feasible, what is frozen, or who decides.

2.1 Requirements

- Model class. Constraint programming or mixed integer over operations, work centers, batch cycles, setup families, crews and skills, tools, lanes and maintenance windows, exactly the constraint model already specified. A constraint absent from that model must not be encoded privately in the solver.
- Objective. A weighted sum the planner can see and set per horizon, weighted lateness against promises first by default, then setup time, then work in process, then overtime. Weight sets must be named profiles under change control.
- Determinism. Fixed seed, fixed time box, best found solution returned with its bound. The same inputs and profile must reproduce the same schedule.
- Guardrails. Freeze horizons are absolute. Published promises may not degrade without an at risk record. Every solver move that overrides a planner preference must list in the run log with its objective contribution.
- Benchmark. A maintained dataset of real order books must score every solver version against the heuristic on lateness, setup hours and makespan. A version that does not beat the heuristic on the weighted objective must not publish.

2.2 Screens

Screen	Type	Must Contain
OPT-RUN-001, optimization runs	Wizard	Profile selection, horizon, time box, run progress with incumbent objective, compare view against current published schedule with KPI deltas and a per order promise impact list, publish with the standard diff ceremony

Screen	Type	Must Contain
OPT-PROF-001, objective profiles	Config	Named weight sets, effective dating, approval by the planning manager
OPT-BENCH-001, benchmark board	Dashboard	Version scores over the benchmark set, trend per KPI, the gate verdict per version

Activation evidence. Benchmark pack results beating the heuristic, plus one month of shadow running beside it with planner decision logs showing no guardrail breach.

3. Full Global Multi Plant ATP and CTP

The single plant promise machinery and the network objects already exist. This module must make one availability service answer across every plant, lane and entity, and must make every cross network promise as auditable as a local one.

3.1 Requirements

- Check sequences must be configuration per material group and customer class. Each sequence is an ordered walk, own plant stock, named partner plant stock plus lane transit, production CTP at each plant holding a valid production version, purchase lead time as the last resort where policy allows.
- Every step of the walk must respect the availability definition already binding, and cross entity steps must respect intercompany pairing rules including transfer price validity before a promise may rely on them.
- The promise record must name the sourcing plant, the lane, the plan snapshot and the check sequence version. Soft reservations must hold capacity and stock at the sourcing plant, decaying exactly as local reservations decay.
- A global backorder run must re promise across the network after every planning cycle, honoring priority classes globally, with reallocation between plants demanding the same approvals as reallocation between orders.
- Substitution across plants must follow approved alternates only, and customer visible substitutions keep their confirmation rule.
- Performance. A single line global check must return within ten seconds. The nightly global re promise must complete inside the batch window at the two plant volumes.

3.2 Screens

Screen	Type	Must Contain
GATP-SEQ-001, check sequences	Config	Sequence rows per material group and customer class, effective dated, simulation of a sample check against the draft
GATP-SIM-001, global availability cockpit	Workbench	One material across all plants, stock, inbound, capacity slots, lanes, the walk result with each step accepted or rejected and why
Promise dialog, extended	Dialog	The existing dialog must additionally show sourcing plant and lane, and must let sales prefer a plant where commercial terms demand it, logged

Activation evidence. The network test packs for planning, intercompany and recall green in the two plant configuration, plus a promise audit of fifty cross plant confirmations tracing to their snapshots.

4. Supplier and Customer Portals

The portals must be external skins over the same services, the same authorization engine and the same validations as internal screens. A portal action that could bypass one internal rule disqualifies the build.

4.1 Architecture Rules

- Separate external identity tenant, invitation based onboarding, per company accounts with named users, multi factor enforced. Internal and external identities must never share a namespace.
- Every portal call must carry the partner scope. Row level security must confine a vendor to its own POs and a customer to its own orders, structurally, not by filtering in the page.
- All writes must flow through the public APIs of the contract repository with idempotency keys. Rate limits and full audit as third party disclosures apply.
- Availability tier two. Portal outage must never block internal operations, and every portal transaction must have an internal fallback path.

4.2 Supplier Portal Screens

Screen	Must Contain	Writes
PO worklist and detail	Open POs with lines, schedules, STP version chip and drawing package download with checksum shown	Acknowledgement per line, quantity, date, price, and explicit STP version acceptance or a deviation request
Deviation request	Structured alternate or concession proposal against a PO line and STP characteristic	Creates the supplier deviation record routed to engineering and quality
ASN creation	Shipment header, lines against open PO schedules, batch or heat and certificate upload per line, package structure	Creates the inbound ASN that pre populates the GRN
Invoice and payment status	Match state per invoice, discrepancy reasons in plain words, remittance history	None. Read only
Scorecard	On time and quality ratings, open SCARs with due dates	SCAR responses in the 8D structure

4.3 Customer Portal Screens

Screen	Must Contain	Writes
Order status	Orders with schedule lines, promise dates, milestone strip for ETO, at risk flags with plain language reasons	None
Drawing approvals	Transmittal packages with the released renditions, comment and markup capture	Approved, approved as noted, or revise and resubmit, driving the design milestone exactly as the internal capture does
Shipment tracking	Deliveries, HU list, carrier references, POD state, document dossier download, packing list, CoC, FTR	POD acknowledgement where the customer signs electronically
Returns	RMA request initiation against shipped serials with reason codes	Creates the RMA request in Draft for internal approval, never an authorized RMA directly

Activation evidence. Two suppliers and one customer completing their full loops for one month without email fallback, with the disclosure audit reviewed.

5. Full EDI

EDI must automate the vendor paper flow on the existing objects. A transaction that would create a state a screen could not create must be rejected at the boundary.

5.1 Transaction Set and Mapping

Transaction	Direction	Binding Behavior
850 purchase order	Outbound	Generated at PO send from the PO object with the STP reference in the notes segment. A change order must send an 860 with the change summary
855 acknowledgement	Inbound	Must write the vendor confirmation per line including dates, quantities and the STP acceptance flag. Deviations must raise the exception state, never silently overwrite
856 advance ship notice	Inbound	Must create the inbound ASN with packages, batches or heats and quantities, pre populating the GRN. Receipt still posts only at the dock scan
810 invoice	Inbound	Must create the vendor invoice into three way match. Duplicate numbers per vendor must reject with a 997 level error, matching tolerance rules unchanged
997 acknowledgement	Both	Every envelope must be functionally acknowledged, and a missing 997 beyond the partner window must alert

5.2 Platform Rules and Screens

- Partner onboarding must be configuration, a mapping profile, identifiers, channel and certificates, envelope options, never code. Channels AS2 and SFTP at minimum.
- Every envelope must log with control numbers, parse state, created documents and errors. Failed envelopes must land in the reprocessing console with edit rights restricted and audited, the same discipline as the integration DLQ.
- Screens. EDI-PART-001 partner profiles as governed configuration, EDI-MON-001 envelope monitor inside the integration console with control number continuity checks per partner.

Activation evidence. Three months of parallel running with the top volume vendor reconciling to the manual flow with zero unexplained differences, then cutover partner by partner.

6. Direct Machine Integration, OEE and Downtime

Machines must report what operators today key and scan, without ever removing the manual path. Signals must become the same confirmation and status events the terminal produces, so no downstream consumer knows or cares which source spoke.

6.1 Objects and Gateway

Element	Binding Requirement
Device gateway	An OPC UA and MQTT capable service at the plant edge, registered per station like printers and scanners, buffering four hours offline exactly as the terminal queue does, publishing signed events with device identity

Element	Binding Requirement
Signal map	Configuration per work center, machine tag to system meaning, cycle count, state run idle down setup, alarm code, measured value to a capture template field. Unmapped tags must log and never post
Machine state model	One current state per machine with a full state history, timestamps at the edge, reason required for down states per §6.2
Test set interface	Result files or direct reads from the test bench must fill the final test capture templates with instrument identity bound, the existing calibration interlock unchanged

6.2 Downtime and OEE Rules

- A down state beyond a configured grace must open a downtime record automatically. The operator must code the reason at the terminal within the shift, from the maintained reason tree, planned separated from unplanned. Uncoded downtime must escalate to the supervisor and must not silently age out.
- OEE must compute per work center per shift from posted data only. Availability equals run time over planned production time from the capacity calendar. Performance equals ideal cycle count over actual against the routing standard. Quality equals good units over total from confirmations. No side spreadsheets and no manual OEE entry may exist.
- Machine counted quantities must reconcile daily against operator confirmations, differences beyond tolerance land on the production reconciliation worklist before settlement.
- A machine in a down state must flag its dispatch queue and warn the scheduler, and a hard down beyond a threshold must raise a maintenance order once module eight of this Book is active.

6.3 Screens

Screen	Type	Must Contain
MI-MAP-001, signal mapping	Config	Tag rows per machine with meaning, template field, tolerance, effective dating and a live tag value test panel
DT-LOG-001, downtime log	Workbench	Open uncoded downtime first with aging, reason coding inline, shift and machine filters
OEE-DASH-001	Dashboard	OEE with its three factors per work center per shift, loss waterfall from the reason tree, drill to downtime and confirmation records in three clicks

Activation evidence. One pilot work center reporting a full month from signals, OEE reconciled against operator confirmations, uncoded downtime under five percent of down minutes.

7. Full Field Service Management

Depot repair keeps the RMA loop. Work at the customer site must gain its own order, its own dispatch and its own stock, settling through the same accounts and writing the same genealogy as the plant.

7.1 Objects

Object	Required Content
Service order, SVO series	Installed base serial reference, site, problem coding, entitlement verdict with rule trail stamped at creation, priority per contract response class, tasks with skills, planned

Object	Required Content
	parts, planned labor, status path New, Triaged, Scheduled, Dispatched, On Site, Work Complete, Customer Signed, Settled, Closed
Service contract	Coverage per customer or serial set, response and resolution targets, covered services, billing terms for non covered work, validity. Entitlement must read contracts before warranty
Technician resource	Skills and certifications from the qualification register, service territory, calendar, assigned van stock
Van stock	A warehouse of type van per technician, replenished by transfer orders against min and max per spare, counted monthly by the standard count documents

7.2 Process Rules

- Triage must call the entitlement service, warranty work settles against the reserve, contract work against the contract, everything else quotes at service rates before dispatch, the customer approves in writing or in the portal.
- Scheduling must respect skills, territory and travel, and must reserve technician capacity like a work center slot. The dispatch board is the workbench archetype.
- Mobile execution must run offline on the kiosk rules, labor start and stop, parts consumed by scan from van stock, findings with photos, removed and installed serials writing the T12 events so the as maintained record stays true, customer signature capture on the device.
- Settlement per origin, warranty to the reserve with overflow to warranty expense, contract to contract cost, billable through a service invoice at the service revenue and service COGS accounts already defined. A service order must settle to zero open cost like a production order.
- Every visit must update the installed base history and may raise an RMA when the unit must return to the plant, linked both ways.

7.3 Screens

Screen	Type	Must Contain
SVO-001, service orders	List and detail	The object of §7.1 with the entitlement verdict visible and immutable, cost tab live like a production order
DSP-BOARD-001, dispatch board	Workbench	Technician lanes by day, unassigned queue with SLA clocks, drag assignment within skill guardrails, travel warnings
Mobile execution set	Kiosk archetype	My day list, order execution with parts scan, forms, photos, signature, offline banner with queue depth
Van stock replenishment	Workbench	Below min lines per van, proposed transfers, one click create

Activation evidence. Ten field orders end to end across warranty, contract and billable cases, entitlement priced correctly, genealogy continuity audited, van stock counts clean.

8. Plant Maintenance and EAM

Machines, ovens, cranes, test benches and tools must become maintained assets whose care is planned, whose breakdowns flow from the andon button, and whose windows the finite schedule respects.

8.1 Objects

Object	Required Content
Equipment master, EQ series	Hierarchy location to machine to component, link to work center or tool, criticality class, counters hours and cycles fed by confirmations or signals, documents, spare parts list of SPARE materials
Maintenance plan	Time based or counter based cycles per equipment, task lists with labor, parts and instructions, tolerance windows, next due computed continuously
Maintenance order, MO series	Type preventive, corrective, breakdown, inspection. Tasks, parts reservations, permits note, planned window, status path parallel to production orders, settlement to the owning cost center, never to product WIP
Downtime linkage	Breakdown orders must create from the terminal hold reason machine down and from gateway down states per module six, closing the loop reason to work to restoration

8.2 Rules

- Preventive plans must generate orders inside the planning horizon and must reserve the equipment window in the finite schedule as blocked capacity, visible to the scheduler in the what if sandbox before publishing.
- Spares must plan by reorder point as SPARE materials, consumed to the maintenance order by the standard issue movement, valued like any stock issue, settling with the order to the cost center.
- Calibration stays in the quality register and must link to the equipment record so one asset view shows both.
- MTBF and MTTR must compute per equipment from the state history and order durations, reported on the maintenance dashboard, feeding criticality reviews.
- A work center whose governing equipment holds an open breakdown order must show Maintenance status automatically, warning dispatch exactly as the manual status does today.

8.3 Screens

Screen	Type	Must Contain
EQ-001, equipment register	List and detail	Hierarchy tree, counters, plan list, open orders, history, spares
MPLAN-001, maintenance plans	Config	Cycles, task lists, next due preview, generation horizon
MO-001, maintenance orders	List and detail	The order of §8.1 with parts, labor, window and settlement tabs
MAINT-BOARD-001, backlog board	Workbench	Due and overdue orders by equipment criticality, schedule conflicts with production highlighted

Posting, maintenance settlement

Debit	Credit	Amount Basis	Notes
Owning cost center expense per matrix	Spares stock and labor absorption credits	Order actual cost at close	Never touches product WIP, feeds the work center rate pools next cycle

Activation evidence. One quarter of preventive execution with schedule integration proven, breakdown flow from andon to restoration timed, spares consumption through standard movements only.

9. Broader International Localization

Each new country must be an activation kit assembled from one framework, never a fork. The framework must hold five layers, compliance, tax, documents, language and currency, and a country goes live only when all five pass its checklist.

9.1 The Five Layers

Layer	Binding Requirement
Compliance matrix	Registrations, statutory books and filing calendar per country, approved before the first transaction, owned by the Controller. The matrix row is the activation trigger
Tax regime	VAT and GST regimes must run through the existing determination precedence with the tax engine, registrations in the nexus register, reverse charge and import tax rules as regime configuration, e invoicing mandates as adapters on the tax interface, clearance models before invoice issue where the country demands it
Documents	Invoices, delivery notes, labels and statutory forms must localize through template variants bound to the entity and country, sequential numbering rules per jurisdiction through the number series engine, no hard coded formats
Language	The text tables already keyed by language must carry translations for customer and vendor facing content, screens remain English at first activation unless the steering decision funds full localization, printed documents must follow the partner language
Currency	Functional currency per entity as today, daily rates, statutory revaluation runs per country rules, group reporting currency in consolidation

9.2 Screens and the Playbook

- LOC-COUNTRY-001, the country activation cockpit, a governed configuration screen holding the five layer checklist per country with owners, evidence links and the go decision record.
- Regime configuration extends the existing tax screens, effective dated, simulated before release like every governed configuration.
- The first activation must follow the written playbook, entity and books, registrations, regime configuration, template variants, translated texts, opening balances, a statutory close simulation, then the gate.

Activation evidence. The first country closes one statutory period with returns generated from the system and reconciled, e invoicing accepted by the authority where mandated, and the checklist archived as the template for the next country.

10. Activation Governance for This Book

- Each module above must enter the backlog as its own epic set with the DG gates unchanged, its screens instancing the eight archetypes, its interfaces entering the catalog with contracts, and its workflows entering the register.
- No module may activate outside change control against the manifest, and no module may weaken a go live invariant, the posting matrix, the availability definition, the traceability rules and the reconciliation register bind these modules exactly as they bind the core.
- Sequencing must follow dependency, not appetite. Machine integration before OEE claims, plant maintenance before predictive ambitions, global availability before global promises are sold, the ledger before any margin statement built on actuals.